S3FBP5A

Bio-Processor

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User's Manual

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List of Acronyms

Acronyms	Descriptions
ADC	Analog to Digital Converter
AES	Advanced Encryption Standard
AFE	Analog Front End
AHB	Advanced High-performance Bus
APB	Advanced Peripheral Bus
BIA	Body Impedance Analysis
CLK	Clock
CMRR	Common Mode Rejection Ratio
DAC	Digital to Analog Converter
DC-DC	DC to DC Voltage Converter
DSP	Digital Signal Processing
DTRNG	True Random Number Generator
DVFS	Dynamic Voltage and Frequency Scaling
ECG	Electrocardiogram
ENOB	Effective Number Of Bit
fCLK	Modulator Clock
Fin	Input Frequency
FPU	Floating-Point Unit
FRC	Free-Running Counter
Fs	Sampling Frequency
HPF	High Pass Filter
LDO	Low-Dropout Regulator
LPF	Low Pass Filter
MA	Motion Artifact
MCU	Micro Control Unit
MSPS	Mega Sample Per Second
OSC	Oscillator
PFM	Pulse Frequency Modulation
PPG	Photo Plethysmography
PWM	Pulse Width Modulation
RX	Receiver
SAR	Successive Approximation
SFR	Special Function Register
SPI	Serial Peripheral Interface

Acronyms	Descriptions
SPS	Sample Per Second
SRP	Samsung Reconfigurable Processor
Ta	Ambient Temperature
Tstg	Storage Temperature
Tj	Junction Temperature
TX	Transmitter
ULP	Ultra Low Power
VCMI	Input Common Mode Voltage
WFE	Wait For Event
WFI	Wait For Interrupt

1 Product Overview

1.1 Introduction

S3FBP5A is a Bio-Processor that able to detect body signals and process them using embedded Analog Front Ends(AFE), Micro Controller Unit (MCU) and Digital Signal Processor(DSP). S3FBP5A contains ECG, PPG and BIA signal sensing dedicated AFEs along with MCU and DSP. S3FBP5A also has embedded Power Management IPs to provide various powers to analog and digital blocks. For analog to digital data conversion, five SAR type and a high resolution Sigma Delta Analog to Digital Converter (ADC) are included. Among them, 1 SAR ADC and the Sigma Delta ADC can be used for general purpose sensor readout depending on resolution requirements.

MCU in S3FBP5A is based on the ARM Cortex-M4 with Floating Point Processing Unit (FPU). ARM Cortex-M4 processor is the most widely-used embedded processor with low-power and low cost. For the DSP, Ultra-low Power Samsung Reconfigurable Processor (ULP-SRP) is applied. The ULP-SRP is the one of reconfigurable processor developed as a low-power subsystem and optimized for low to medium performance applications. S3FBP5A also has embedded 512KB flash memory.

The AFEs in S3FBP5A are used for reading out Electro Cardio Gram (ECG), Photo PlethysmoGram (PPG), and Body Impedance Analysis (BIA). Functional flexibilities of the AFEs ensure S3FBP5A to support extended applications such as Galvanic Skin Response (GSR), and Skin Temperature. They use 1.1V power supply and manage detected body signals to optimal level for appropriate digital signal processing. S3FBP5A also has on-chip power management IPs such as DC-DC converter and LDOs. They provide various power levels to analog and digital blocks in S3FBP5A . Five 12-bit successive approximation (SAR) ADCs are applied to convert 10 channels of sensed body signals to digital data coincidentally. S3FBP5A also has a 24 bit high resolution Sigma Delta ADC for general purposes such as body temperature sensing, fuel gauging and so forth.

The technologies applied to implement S3FBP5A with 45nm low power CMOS process are:

- Clock gating
- Power gating with fine-grained power domains
- Multiple VDD design
- Embedded Flash Memory (512KB)
- Embedded SRAM (522KB)
- Low Power Analog Front Ends for ECG, PPG, BIA applications
- Embedded PMIPs
- High resolution Sigma Delta ADC
- Low Power 12-bit SAR type ADC

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

1.2 Key Features

- **ECG (Electro-Cardiogram)**
  - Input-Referred Noise: 1uVrms(0.5 ~ 150Hz integrated)
  - Programmable Gain Amplifier: IA (0~55dB, 4-bit control), PGA(1~15.6 dB, 3-bit control)
  - Wide Output Dynamic Range: 55 dB (Max Gain)
  - Built-in HPF for dc offset filtering and enabling high input impedance
  - Built-in LPF with controllable cut-off frequency:75 ~ 250Hz
  - Built-in Right Leg Driver Amplifier
  - 12-Bit SAR type ADC
  - Lead-off detection and weak-contact detection
  - AFE Operating Current: 60uA

- **BIA (Bio-electrical Impedance Analysis)**
  - Supports 2 modes (Rectify mode and IQ mode)
  - Calibration scheme using external resistors
  - Supports wide frequency range of injected current from 1kHz to 255kHz
  - 6-Bit 1-MSPS Sine-Wave Generation Digital-to-Analog Converter
  - Driving Current Controllability: 100uA - 450uA
  - Impedance Test Range: 100ohm - 2kohm
  - 12-Bit SAR type ADC
  - AFE Operating Current: 910uA

- **PPG (Photo plethysmogram)**
  - LED driver to drive red, infra-red and green LEDs
  - PPG LED Driver
    - Integrated LED Driver
    - LED Current Range: 0 ~ 250mA with 7-Bit Current Resolution
    - LED Driver Operating Current: 150uA
  - PPG Receiver AFE
    - 25uA input current range
    - Ambient offset cancellation
    - Supports AC and DC value measurement for oxygen saturation level calculation
    - 12-Bit SAR ADC
    - Receiver Operating Current: 135uA
  - Integrated Foreground / Background Fault Detection
    - LED On / Off Detection

- **GSR (Galvanic Skin Response)**
  - Supports GSR signal readout by containing both tonic and phasic components
  - tonic component (DC - 0.05Hz), phasic component (0.05 - 1.5Hz)
  - Supports low-power or high-resolution signal readout

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- Low-power mode: 12bit, 1kS/s with 3uA current consumption
- High-resolution mode: 24bit, up to 1kS/s with 602uA current consumption
- Supports DC/AC-voltage source as the excitation signal
- AC (square-wave) voltage source: rail-to-rail square wave signal with adjustable frequency (2Hz~16Hz)
- DC voltage source: constant voltage signal with programmable value (1.4V - 2.175V)
- GSR signal sensing range: 200nS ~ 61uS(3V AC-Source and SD-ADC)

- **Skin Temperature**
  - Supports skin temperature using thermistor
  - Sensing range: 20℃ ~ 45℃(86℉ ~ 113℃)

- **ADC**
  - Five 12-bit SAR ADCs for bio-signal data conversion
  - Sampling rate up to 1kHz
  - One 24 bit resolution sigma-delta ADC for general purposes

- **Power Management IPs**
  - 1 DC-DC converter
  - 3 LDOs and 2 keep-alive LDOs (1.1V and 1.8V)
  - Built in POR

- **Microcontroller**
  - ARM 32bit Cortex-M4 CPU with FPU(Floating Point Unit)
  - Operating frequency: Maximum 87MHz
  - 512KB embedded flash memory and 256KB SRAM
  - 32 channel uDMA (Micro Direct Memory Access)
  - JTAG/SW interface for debug
  - PMU (Power Management Unit) and CMU (Clock Management Unit)
  - DTRNG (True Random Number Generator)
  - AES (Advanced Encryption Standard)
  - SPI bootloader

- **Digital Signal Processor**
  - Ultra low power 32bit Samsung Reconfigurable Processor
  - 266KB embedded SRAM
  - Interface with Cortex-M4 CPU through AHB-Lite bus
  - Support VLIW, CGA2X2 and CGA3X3 modes for power optimization
  - Separate JTAG interface
  - Interrupt generation after completing assigned processing

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

- **Interface**
  - 5 I2C bus interfaces
  - 5 SPIs (Serial Peripheral Interface)
  - 3 UARTs (Universal Asynchronous Receiver and Transmitter)
  - 32 GPIOs

- **Package**
  - 158 pin FOWFP 5.68 ×4.88 × 0.75mm, 0.4 mm pin pitch

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

1.3 Block Diagram

Figure 1-1 illustrates the block diagram of S3FBP5A

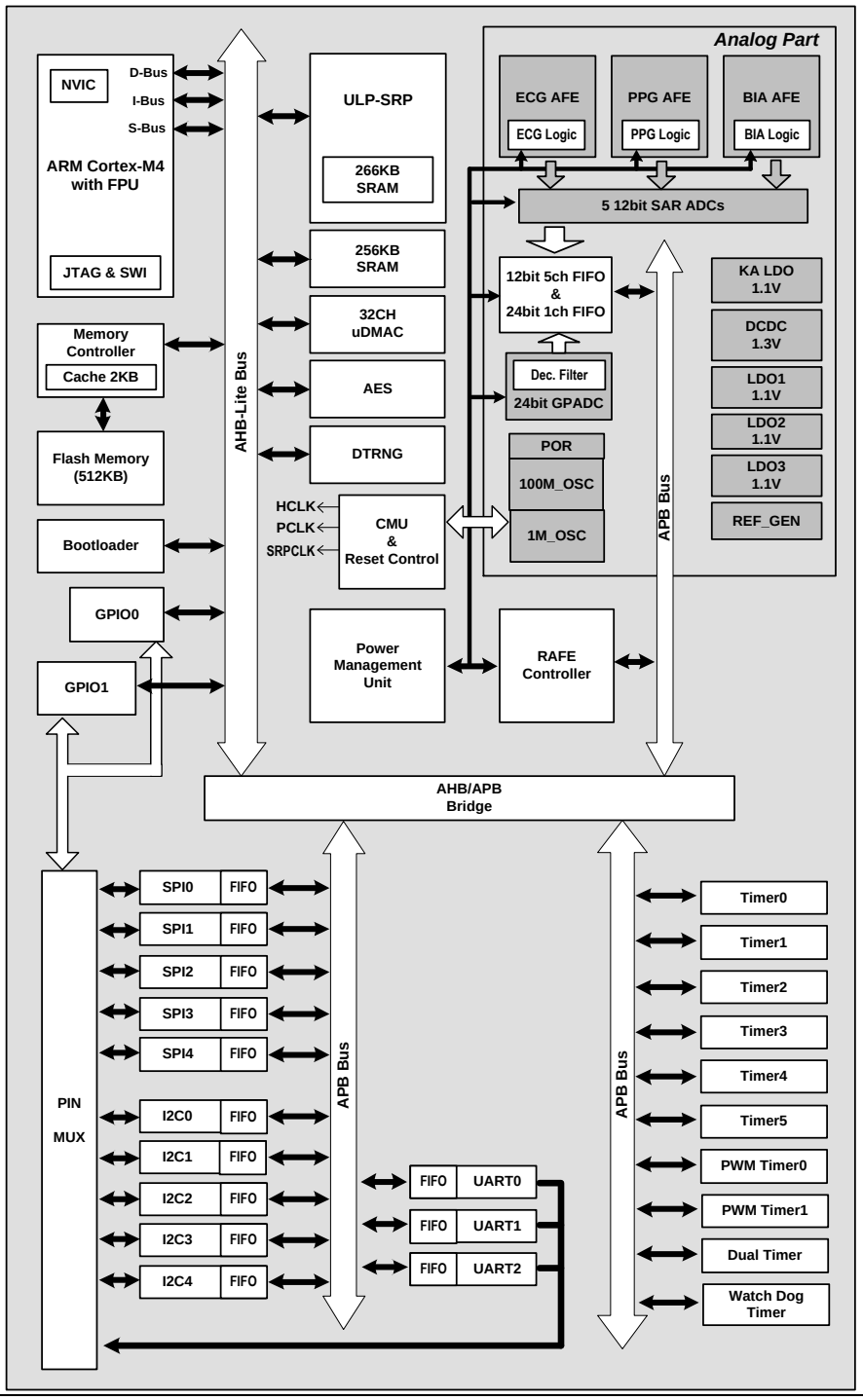


Figure 1-1 Block Diagram of S3FBP5A

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2 Operating Conditions and Electrical Characteristics

2.1 Overview

This chapter describes operating conditions and electrical specifications of digital blocks and Analog Front Ends implemented in S3FBP5A.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2.2 Absolute Maximum Ratings

Parameter	Value	Unit
Analog Power to Analog Ground	-0.3 to 3.8	V
Analog Ground to Digital Ground	-0.3 to 0.3	V
Electrostatic Discharge Ratings: Human Body Model (HBM)	TBD	V
Electrostatic Discharge Ratings: Charged Device Model (CDM)	TBD	V
Storage Temperature: Tstg	TBD	℃
Maximum Junction Temperature: Tj	Max 125	℃

NOTE: These ratings are associated only with stress. It does not imply any functional operation of the condition listed in 2.2 Absolute Maximum Ratings. Exposure to the absolute maximum rated conditions for long duration affects the reliability of the device.

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2.3 Recommended Operating Conditions

Parameter		Min.	Typ.	Max.	Unit
Power Supply of S3FBP5A	AVDD33	2.7	3	3.3	V
DC-DC Converter Switching Transistor Supply	AVDD33_SW	2.7	3	3.3	V
Green LED Driver Supply Voltage	VDD5_LED		5	5.5	V
Analog Ground	AVSS33		0		V
DC-DC Converter Switching Transistor Ground	AVSS33_SW		0		V
Digital Ground	AVSS_DSP		0		V
Operating Ambient Temperature	Ta	-25		85	°C

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2.4 Electrical Characteristics of MCU & DSP

Parameter		Conditions	Min	Typ	Max	Unit
Frequency	AHB clock	87MHz OSC / divider value (1 to 128)	0.68	-	87	MHz
	APB clock	87MHz OSC / divider value (1 to 128)	0.68	-	87	MHz
	DSP clock	87MHz OSC / divider value (1 to 128)	0.68	-	87	MHz
	AFE control clock	1MHz OSC	-	1.048	-	MHz
	Free-running clock	Free-running clock = 1MHz OSC / 32	-	32.768	-	kHz
Current consumption	Standby mode	Refer to *Table 4-4*	-	0.77	-	mW
	Run mode(1)	AHB clock = 87MHz	-	28.61	-	mW
		AHB clock = 43.5MHz	-	10.48	-	mW
		AHB clock = 21.75MHz	-	5.67	-	mW
Wakeup time		MCU core wakes up from low power modes	1	-	2	us

NOTE: MCU core processes 256-point FFT application. ULP-SRP is power-off.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

2.5 Electrical Characteristics of AFE

2.5.1 ECG

Parameter	Condition	Min	Typ	Max	Unit
Operating Voltage(VDD)			1.1		V
Operating Current	VDD=1.1V(TYP)		60		uA
IA Gain	4-bit programmable control	31	44	55	dB
PGA Gain	3-bit programmable control	0	6	15.6	dB
HPF Frequency			0.5	1	Hz
2nd LPF	3-bit programmable control	75		250	Hz
CMRR			-80		dB
Input Referred Noise	0.5Hz ~ 150Hz		4		uVrms
Input Impedance	without IMP channel operation, @Room Temperature		30M		ohms

2.5.2 PPG

Parameter		Condition	Min	Typ	Max	Unit
LED Driver	Operating Voltage(IR, RED LED)			3		V
	Operating Voltage(Green)			5	5.5	V
	Operating Current			150		uA
	LED Driving Current		0		250	mA
	LED Driving Current Resolution			7		bits
PPG Receiver	Operating Voltage			1.1		V
	Operating Current			135		uA
	Input Current Range				25	uA
	Noise-Free Bits	@320 kohm of TIA Resistance	-	8	-	Bit
	ADC Resolution		-	12	-	Bit

2.5.3 BIA

Parameter		Condition	Min	Typ	Max	Unit
TOP	Operating Voltage (VDD)			1.1		V
	Total Power Consumption			1		mW
	ADC Resolution			12		Bit
TX	DAC			6		Bit
	DAC Voltage		0.3	0.5	0.6	Vpp

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Parameter		Condition	Min	Typ	Max	Unit
	Driving Current		90	150	450	uA
	DAC Sample Rate			1		MSPS
	Current Control Resistor			1.667		kohm
	Filter Bandwidth		100		250	KHz
RX	Impedance Dynamic Range	VDD=1.1V	0.1		2	kohm
	Gain(PGA)		0.25		12.25	V/V
	Gain(DC amp)			2		V/V
	Output Bandwidth	Internal 5 kohm, External 4.7uF		6.7		Hz
	Input Impedance			1		Mohm
	Dynamic Current			250		uA
	Nonlinearity	At 200 ohm ~ 2 kohm		1		%

2.5.4 GSR

Parameter		Condition	Min	Typ	Max	Unit
Source-1 (AC)	Output Voltage	Square Wave Generator	0.0	-	3.0V	V
	Frequency		2, 4, 8, and 16			Hz
	Output Driving Current				10	uA
Source-2 (DC)	Output Voltage	Source: LDO 1.8V	1.4	1.8	2.175	V
	Source Frequency		DC			Hz
	Output Driving Current				10	mA
Readout-1 (SD-ADC)	DC input impedance		500			Mohm
	Data rate				125, 250, 500 and 1000	SPS
	Current Consumption			600		uA
	Resolution				24	Bit
	Effective Number Of Bit(ENOB)	Fs=500Hz		15		Bit
	Reference (Vref)			2.5		V
	Sensing Range	DC Source w/ External 2MΩ	50n		61u	S
		AC Source w/ External 2MΩ	200n		61u	S
Readout-2 (SAR-ADC)	DC input impedance	Vin=VDD/2		30		Mohm
	Data rate			1000		SPS
	Current Consumption			2		uA

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Parameter		Condition	Min	Typ	Max	Unit
	Resolution			12		Bit
	Effective Number Of Bit(ENOB)			10		Bit
	Reference			1.1		V
	Sensing Range	DC Source w/ External 2MΩ	0.625u		1.9m	S
		AC Source w/ External 2MΩ	1.6u		1.9m	S

2.5.5 Skin Temperature

Parameter		Condition	Min	Typ	Max	Unit
Low Power Mode	Sensing Range	Depend on thermistor	20		45	℃
	Reference Voltage	LDO_AFE		1.1		V
	ADC Resolution			12		Bits
	Current Consumption	RX + External Component		90		uA
	Temperature Resolution			0.05		℃
High Accuracy Mode	Sensing Range	Depend on thermistor	20		45	℃
	Reference Voltage	SD_REG_OUT		2.5		V
	ADC Resolution				24	Bits
	Current Consumption	RX + External Component		800		uA
	Temperature Resolution			0.0015		℃

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2.6 Electrical Characteristics of ADCs

2.6.1 SAR ADC

Parameter	Condition	Min	Typ	Max	Unit
Operating Voltage(VDD)			1.1		V
Analog Input Channel			10		ch
Conversion Rate				2	kHz
Analog Input Range	differential	0		2VDD	V
Analog Input Frequency	Fs=2kS/s	DC		1	kHz
DC Electrical Characteristics					
Resolution			12		bits
Differential Non-Linearity			±1.0	±2.0	LSB
Integral Non-Linearity			±2.0	±4.0	LSB
AC Electrical Characteristics					
Sampling Clock Duty Ratio	Fs=2kS/s		6.25		%
Normal Operation Current Consumption (1)	Fs=2kS/s		1.2		uA
Signal-to-Noise Ratio	Fs=2kS/s, Fin=50Hz		61		dB
Signal-to-Noise and Distortion Ratio	Fs=2kS/s, Fin=50Hz		60		dB
Spurious-Free Dynamic Range	Fs=2kS/s, Fin=50Hz		69		dB

NOTE: (1) : Current Consumption per a SAR ADC

2.6.2 Sigma Delta ADC

Parameter	Condition	Min	Typ	Max	Unit
AVDD			3		V
DVDD			1.1		V
Input common mode (VCMI)			1.25		V
Reference voltage (VREF)	Output of internal regulator		2.5		V
Maximum input range	VCMI=1.25, differential	-1.25		1.25	V
Effective input range	VCMI > 1.25	±2·(1.875-VCMI)			V
	VCMI < 1.25	±2·(VCMI-0.625)			V
PGA Gain		1, 2, 3, 4, 6, 7, 11 & 16			V/V
Data rate		125, 250, 500 and 1000			SPS

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Parameter	Condition	Min	Typ	Max	Unit
Input frequency(Fin)				250	Hz
Resolution			24		Bit
Effective Number Of Bit(ENOB)	Fs=500Hz, Fin=240Hz		TBD (target=15b)		Bit
Input-referred noise	Gain=1, 10 seconds of data at 500 SPS data rate		TBD (target=8)		uVpp
Modulator CLK (fCLK)			64		kHz
Analog Current Consumption			600		uA
Digital Current Consumption	Including only level shifters not the decimation filter		200		nA

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2.7 Electrical Characteristics of PMIPs & Clock Generators

2.7.1 PMIP

Parameter		Condition	Min	Typ	Max	Unit
DC-DC Converters						
Operating Voltage (VIN)			2.7	3	3.3	V
Output Voltage	DCDC			1.45		V
Output Voltage Step	DCDC			25		mV
Output Ripple	DCDC	PFM@ 1.45V		50		mV
Output Current	DCDC			60		mA
Power Down Current		Power down		25	150	uA
Operating Current		PFM@ all DC-Dc			150	
Inrush Current	DCDC			200		mA
Line Regulation	DCDC			10		mV/V
Load	DCDC			250		uV/mA
Startup Time	DCDC			0.5	2	ms
Over-current Protection	DCDC			200		mA
Linear Regulators						
Operating Voltage				1.3		V
Output Voltage	LDO1~3		0.5	1.1	1.21	V
	LDO_KA	VIN=3.0V	0.5	1.1	1.21	V
	LDO_KA18	VIN=3.0V		1.8		V
Output Current	LDO1~3				50	mA
	LDO_KA	VIN=3.0V			5	mA
	LDO_KA18	VIN=3.0V			50	mA
Dropout Voltage	LDO1~3			200		mV
	LDO_KA	VIN=3.0V		1.9		V
	LDO_KA18	VIN=3.0V		1.2		V
Operating Current	LDO1~3	No load at output		12	120	uA
	LDO_KA	No load at output &VIN=3.0V		12	120	uA
	LDO_KA18	No load at output &VIN=3.0V		12	120	uA

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Parameter		Condition	Min	Typ	Max	Unit
Power Down Current				10		uA
Line Regulation	LDO1~3			10	20	mV/V
	LDO_KA	VIN=3.0V		10	20	mV/V
	LDO_KA18	VIN=3.0V		10	20	mV/V
Load Regulation	LDO1~3	VIN>Vo+0.2V		200	500	uV/mA
	LDO_KA	VIN=3.0V		200	500	uV/mA
	LDO_KA18	VIN=3.0V		200	500	uV/mA
Inrush Current				200		mA
Over-voltage Protection		Vo rising		1.3		V
Startup Time				100		us

2.7.2 Clock Generators

- **1MHz / 32kHz Oscillator**

Parameter	Condition	Min	Typ	Max	Unit
Operating Current	VDD=1.1V		6		uA
Output Frequency	FSEL<1001110>		1.0488		MHz
Duty Error				0.1	%
Frequency Error				1	%
Frequency Setting Range		0.71		1.64	MHz

- **87MHz Oscillator**

Parameter	Condition	Min	Typ	Max	Unit
Operating Current	VDD=1.1V		139	168	uA
Output Frequency	FSEL<1001110>		100		MHz
Duty Error				0.2	%
Frequency Error				2	%
Frequency Setting Range		56		130	MHz

- **Clock Generator**

Parameter	Condition	Min	Typ	Max	Unit
Input Voltage range		1.0	1.1	1.2	V
Operating Current	VDD=1.1V		0.73		uA
CLK64			64		Hz
CLK1,2, CLK1,2_IMP			2		kHz
CLK_2X_1,2			4		kHz

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CLK_CS_1,2			8		kHz
CLK_IMP1,2_ORT			2		kHz

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3 Ball Map and Pin Description

3.1 Overview

This chapter describes S3FBP5A ball-map and lists the names of all the signal pins.

This chapter includes

- Ball Map Diagram of S3FBP5A
- Pin Assignments
- Pin Description

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3.2 Ball Map

Figure 3-1 illustrates the ball map of S3FBP5A device.

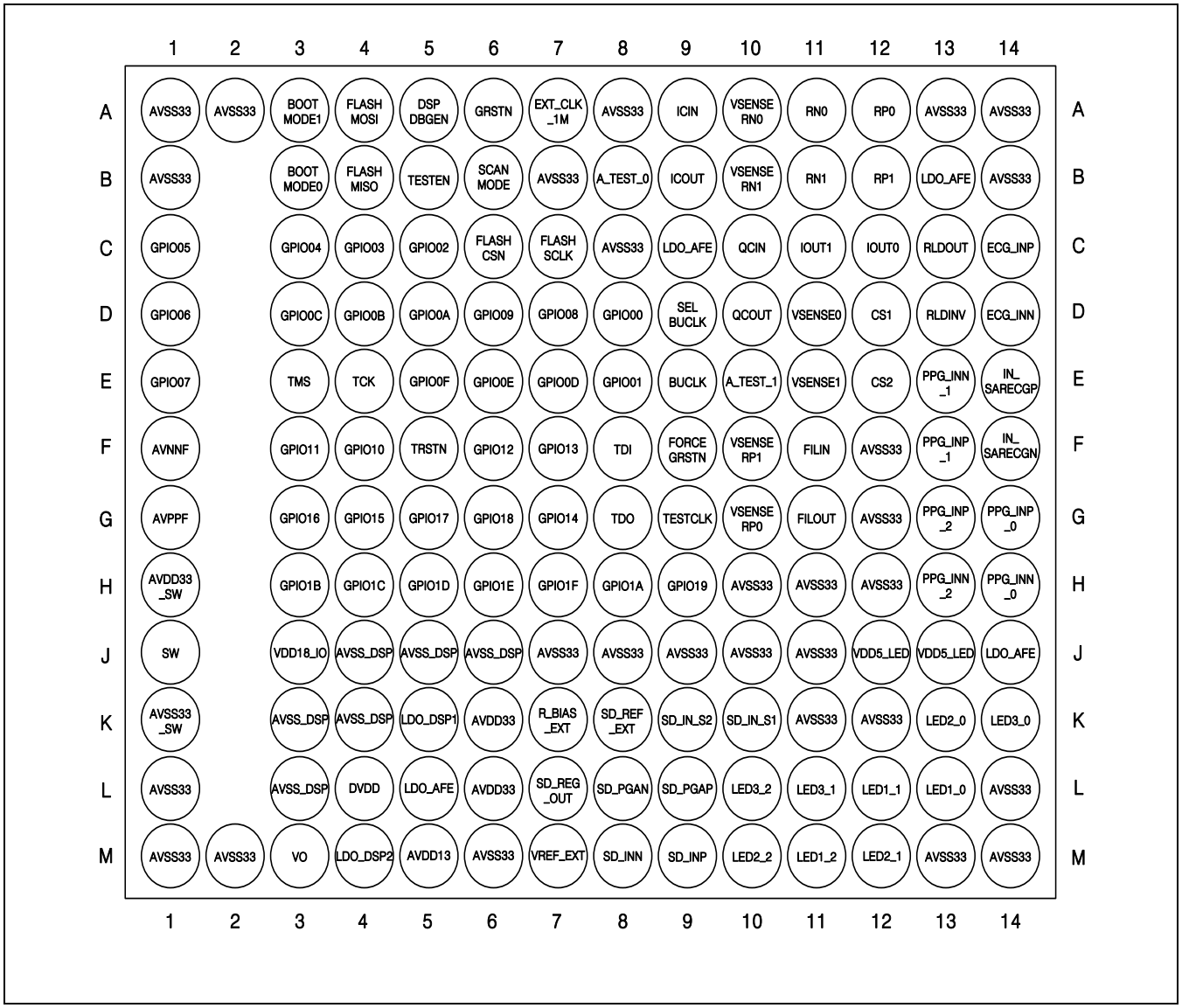


Figure 3-1 Ball Map Diagram of S3FBP5A

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3.3 Pin Description

Table 3-1 describes the pin assignment in detail.

Table 3-1 Pin Description of S3FBP5A

No.	Ball	PIN	Description	I/O
1	A14	AVSS33	Ground for 3.3V Analog Block	AG
2	B13	LDO_AFE	Analog Front Ends Input Power (1.1V Output of LDO for AFE)	P
3	C14	ECG_INP	Non-inverting Analog Signal Input for ECG	AI
4	D14	ECG_INN	Inverting Analog Signal Input for ECG	AI
5	C13	RLD_OUT	Right leg driver output	AO
6	E12	CS2	Current source pin 2 for ECG Impedance Measurement (Connect to ECG_INN Electrode)	AIO
7	D13	RLDINV	Right leg driver inverting input	AI
8	D12	CS1	Current source pin 1 for ECG Impedance Measurement (Connect to ECG_INP Electrode)	AIO
9	E14	IN_SARECGP	Non-inverting Analog Signal Input for SAR ADC test purpose	AI
10	F14	IN_SARECGN	Inverting Analog Signal Input for SAR ADC test purpose	AI
11	G14	PPG_INP_0	Photo Diode Positive(Anode) Input 0	AI
12	H14	PPG_INN_0	Photo Diode Negative(Cathode) Input 0	AI
13	F13	PPG_INP_1	Photo Diode Positive(Anode) Input 1	AI
14	E13	PPG_INN_1	Photo Diode Negative(Cathode) Input 1	AI
15	G13	PPG_INP_2	Photo Diode Positive(Anode) Input 2	AI
16	H13	PPG_INN_2	Photo Diode Negative(Cathode) Input 2	AI
17	H12	AVSS33	Ground for 3.3V Analog Block	AG
18	J14	LDO_AFE	Analog Front Ends Input Power (1.1V Output of LDO for AFE)	AP
19	L14	AVSS33	Ground for 3.3V Analog Block	AG
20	J13	VDD5_LED	LED driver Power (Typical 5 V)	AP
21	J12	VDD5_LED	LED driver Power (Typical 5 V)	AP
22	K14	LED3_0	Red LED Driver3 Output (Connect to LED)	AIO
23	K13	LED2_0	Red LED Driver2 Output (Connect to LED)	AIO
24	M14	AVSS33	Ground for 3.3V Analog Block	AG
25	L13	LED1_0	Red LED Driver1 Output (Connect to LED)	AIO
26	L12	LED1_1	IR LED Driver1 Output (Connect to LED)	AIO
27	M12	LED2_1	IR LED Driver2 Output (Connect to LED)	AIO
28	L11	LED3_1	IR LED Driver3 Output (Connect to LED)	AIO
29	M11	LED1_2	Green LED Driver1 Output (Connect to LED)	AIO
30	M10	LED2_2	Green LED Driver2 Output (Connect to LED)	AIO
31	L10	LED3_2	Green LED Driver3 Output (Connect to LED)	AIO

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No.	Ball	PIN	Description	I/O
32	L9	SD_PGAP	Sigma Delta ADC PGA Positive Output(Connect to 4.7nF)	AO
33	M9	SD_INP	Non-Inverting Analog Signal Input for Sigma Delta ADC	AI
34	M8	SD_INN	Inverting Analog Signal Input for Sigma Delta ADC	AI
35	L8	SD_PGAN	Sigma Delta ADC PGA Negative Output(Connect to 4.7nF)	AO
36	K10	SD_IN_S1	Sigma Delta ADC Analog Input 1 for Single Ended Mode	AI
37	K9	SD_IN_S2	Sigma Delta ADC Analog Input 2 for Single Ended Mode	AI
38	K8	SD_REF_EXT	Regulator (for Sigma Delta ADC) Output Voltage Control Pin	AI
39	L7	SD_REG_OUT	Sigma Delta ADC Regulator Output	AO
40	M7	VREF_EXT	External Reference Voltage Input Pin	AI
41	K7	R_BIAS_EXT	External resister connection pin used for Trimming Mode	AIO
42	L6	AVDD33	3.3V Analog Power	AP
43	M6	AVSS33	Ground for 3.3V Analog Block	AG
44	L5	LDO_AFE	Analog Front Ends Input Power (1.1V Output of LDO for AFE)	AP
45	M5	AVDD13	LDO input power	AP
46	K5	LDO_DSP1	Power for MCU (1.1V)	DP
47	M4	LDO_DSP2	Power for Flash Memory	DP
48	L4	DVDD	KA(Keep Alive) LDO Output (1.1V Power for Logic & OSC)	DP
49	L3	AVSS_DSP	Ground for DSP Block	DG
50	M3	VO	DC-DC Converter Output Sensing Pin	P
51	M2	AVSS33	Ground for 3.3V Analog Block	AG
52	K1	AVSS33_SW	Ground for Switching transistor of DC-DC Converter	G
53	J1	SW	DC-DC Converter Output(Connected to LC filter)	AO
54	H1	AVDD33_SW	Power for Switching transistor of DC-DC Converter	P
55	J3	VDD18_IO	LDO output for 1.8V IO	P
56	H7	GPIO1F	GPIO1F/UART0_RTSN	DIO
57	H6	GPIO1E	GPIO1E/ UART0_CTSN	DIO
58	H5	GPIO1D	GPIO1D/UART0_ RX	DIO
59	H4	GPIO1C	GPIO1C/UART0_ TX	DIO
60	H3	GPIO1B	GPIO1B/I2C0_ SDA	DIO
61	H8	GPIO1A	GPIO1A/I2C0_ SCL	DIO
62	H9	GPIO19	GPIO19/I2C1_ SDA	DIO
63	G6	GPIO18	GPIO18/I2C1_SCL	DIO
64	G5	GPIO17	GPIO17/SPI4_ MISO/UART1_,RTSN/I2C2_ SDA/ PTimer1_Ext.	DIO
65	G3	GPIO16	GPIO16/SPI4_ MOSI /UART1_CTSN/I2C2_SCL/Timer5_Ext	DIO
66	G4	GPIO15	GPIO15/SPI4_ CSN /UART1_RX/ Timer4_Ext.	DIO
67	G7	GPIO14	GPIO14/SPI4_ CLK /UART1_TX/ Timer3_Ext.	DIO

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No.	Ball	PIN	Description	I/O
68	F7	GPIO13	GPIO13/SPI3_ MISO/UART2_RTSN/PTimer1_Ext.	DIO
69	F6	GPIO12	GPIO12/SPI3_MOSI/UART2_CTSN/Timer2_Ext.	DIO
70	G1	AVPPF	Positive High Voltage Forcing & Monitoring Port (0V ~ 12V)	AIO
71	F1	AVNNF	Negative High Voltage Forcing & Monitoring Port (-9V ~ 0V)	AIO
72	F3	GPIO11	GPIO11/SPI3_ CSN/UART2_ RX/Timer1_Ext.	DIO
73	F4	GPIO10	GPIO10/SPI3_ CLK/UART2_ TX/ Timer0_Ext.	DIO
74	F5	TRSTN	TRSTN(Active Low)	DIO
75	G8	TDO	Test Data Output	DIO
76	F8	TDI	Test Data Input	DIO
77	E3	TMS	Test Mode Selection	DIO
78	E4	TCK	Test Clock	DIO
79	E5	GPIO0F	GPIO0F/SPI2_ MISO/I2C3_SDA .	DIO
80	E6	GPIO0E	GPIO0E/SPI2_ MOSI/I2C3_SCL.	DIO
81	E7	GPIO0D	GPIO0D/SPI2_CSN/I2C4_ SDA.	DIO
82	D3	GPIO0C	GPIO0C/SPI2_ CLK/I2C4_ SCL.	DIO
83	D4	GPIO0B	GPIO0B/SPI1_ MISO	DIO
84	D5	GPIO0A	GPIO0A/SPI1_ MOSI/TRSTN_DSP	DIO
85	D6	GPIO09	GPIO09/SPI1_CSN/TDO_DSP	DIO
86	D7	GPIO08	GPIO08/SPI1_CLK in SPI1/TDI_DSP	DIO
87	E1	GPIO07	GPIO07/SPI0_MISO/TMS_DSP.	DIO
88	D1	GPIO06	GPIO06/SPI0_MOSI/TCK_DSP.	DIO
89	C1	GPIO05	GPIO05/SPI0_ CSN	DIO
90	C3	GPIO04	GPIO04/SPI0_ CLK	DIO
91	C4	GPIO03	GPIO03/PWMOUT1/HOSTMISO	DIO
92	C5	GPIO02	GPIO02/PWMOUT0/HOSTMOSI	DIO
93	E8	GPIO01	GPIO01/HOSTCSN	DIO
94	D8	GPIO00	GPIO00/HOSTCLK	DIO
95	B3	BOOTMODE0	Boot mode[1:0] = 00: Internal flash boot mode Boot mode[1:0] = 01: External flash boot mode Boot mode[1:0] = 10: External flash write mode Boot mode[1:0] = 11: External host boot mode	DIO
96	A3	BOOTMODE1		DIO
97	B4	FLASHMISO	SPI Data Input from External Flash for System Booting	DIO
98	A4	FLASHMOSI	SPI Data Output to External Flash for System Booting	DIO
99	C6	FLASHCSN	SPI Chip Select Output to External Flash for System Booting	DIO
100	C7	FLASHSCLK	SPI Clock Output to External Flash for System Booting	DIO
101	A5	DSPDBGEN	DSP Debugger Enable Pin	DIO
102	B5	TESTEN	Test Enable Pin	DIO

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No.	Ball	PIN	Description	I/O
103	G9	TESTCLK	Test Clock Pin	DIO
104	B6	SCANMODE	Scan Mode Enable Pin	DIO
105	A6	GRSTN	Global Reset Pin	DIO
106	F9	FORCEGRSTN	Reset Force by GRSTN(1)/Use normal reset sequence(0)	DIO
107	E9	BUCLK	External Clock Source(max. 87MHz) only for back up purpose	DIO
108	D9	SELBUCLK	Clock Selection Pin (Use internal oscillator clock(0)/Use external bck clock(1))	DIO
109	A7	EXT_CLK_1M	External 1MHz clock source	AIO
110	B7	AVSS33	Ground for 3.3V Analog Block	AG
111	C9	LDO_AFE	Analog Front Ends Input Power (1.1V Output of LDO for AFE)	AP
112	B8	A_TEST_0	Test MUX output	AO
113	E10	A_TEST_1	Test MUX output	AO
114	D10	QCOUT	Q channel demodulator low pass filter, connect 4.7uF between QCOUT and Ground	AIO
115	C10	QCIN	Q channel demodulator low pass filter, connect 4.7uF between QCIN and Ground	AI
116	A9	ICIN	I channel demodulator low pass filter, connect 4.7uF between ICIN and Ground	AI
117	B9	ICOUT	I channel demodulator low pass filter, connect 4.7uF between ICOUT and Ground	AIO
118	B10	VSENSERN1	Input to differential amplifier from calibration resistors	I
119	A10	VSENSERN0	Input to differential amplifier from calibration resistors	I
120	F10	VSENSERP1	Input to differential amplifier from calibration resistors	I
121	G10	VSENSERP0	Input to differential amplifier from calibration resistors	I
122	E11	VSENSE1	Input to differential amplifier from calibration resistors	I
123	D11	VSENSE0	Input to differential amplifier from calibration resistors	I
124	B11	RN1	Calibration resistor connecting pin	IO
125	A11	RN0	Calibration resistor connecting pin	IO
126	B12	RP1	Calibration resistor connecting pin	IO
127	A12	RP0	Calibration resistor connecting pin	IO
128	C11	IOUT1	Current driving pin	AO
129	C12	IOUT0	Current driving pin	AO
130	F11	FILIN	Current generating block input	AI
131	G11	FILOUT	DAC filter out	AO
132	K6	AVDD33	3.3V Analog Power	AP
133	A1	AVSS33	Ground for 3.3V Analog Block	AG
134	B1	AVSS33	Ground for 3.3V Analog Block	AG
135	L1	AVSS33	Ground for 3.3V Analog Block	AG

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No.	Ball	PIN	Description	I/O
136	M1	AVSS33	Ground for 3.3V Analog Block	AG
137	A2	AVSS33	Ground for 3.3V Analog Block	AG
138	J7	AVSS33	Ground for 3.3V Analog Block	AG
139	A8	AVSS33	Ground for 3.3V Analog Block	AG
140	C8	AVSS33	Ground for 3.3V Analog Block	AG
141	J8	AVSS33	Ground for 3.3V Analog Block	AG
142	J9	AVSS33	Ground for 3.3V Analog Block	AG
143	H10	AVSS33	Ground for 3.3V Analog Block	AG
144	J10	AVSS33	Ground for 3.3V Analog Block	AG
145	H11	AVSS33	Ground for 3.3V Analog Block	AG
146	J11	AVSS33	Ground for 3.3V Analog Block	AG
147	K11	AVSS33	Ground for 3.3V Analog Block	AG
148	F12	AVSS33	Ground for 3.3V Analog Block	AG
149	G12	AVSS33	Ground for 3.3V Analog Block	AG
150	K12	AVSS33	Ground for 3.3V Analog Block	AG
151	A13	AVSS33	Ground for 3.3V Analog Block	AG
152	M13	AVSS33	Ground for 3.3V Analog Block	AG
153	B14	AVSS33	Ground for 3.3V Analog Block	AG
154	K3	AVSS_DSP	Ground for DSP Block	DG
155	J4	AVSS_DSP	Ground for DSP Block	DG
156	K4	AVSS_DSP	Ground for DSP Block	DG
157	J5	AVSS_DSP	Ground for DSP Block	DG
158	J6	AVSS_DSP	Ground for DSP Block	DG

NOTE: AI: Analog Input, DI: Digital Input, AO: Analog Output, DO: Digital Output, AIO: Analog Input & Output, AP: Analog Power, AG: Analog Ground, DIO: Digital Input & Output, DP: Digital Power, DG: Digital Ground, P: Power, G: Ground, NC: No Connection.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4 MCU & DSP Functional Description

4.1 Overview

Digital core of S3FBP5A consists of Micro Controller Unit (MCU) and Digital Signal Processor (DSP). MCU takes master control role of S3FBP5A such as managing Operating System (OS), body signal processing, AFE circuit control, interfacing with external devices, and other common tasks. MCU comprises ARM Cortex-M4 core as a main processor, 512KB embedded Flash and 2KB Cache, 256KB SRAM, Security IPs, and various peripherals.

Ultra-Low Power Samsung Reconfigurable Processor (ULP-SRP) as a DSP of S3FBP5A is a high performance and energy efficient core which can be used from low to high complexity applications. It utilizes a reconfigurable processing core consisting 9 functional units so that 3 different processing modes are available. MCU core controls ULP-SRP as a subsystem because ULP-SRP is connected to AHB-Lite bus with AHB-Lite Slave Interface in it. Using this interface MCU can utilize SRAM inside ULP-SRP.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.2 Functions

- MCU with Cortex-M4 core
- Memories: 512KB embedded Flash with 2KB Cache and 256KB SRAM
- System control: CMU and PMU
- 32 channel μDMAC
- Interfaces: 3 channel UARTs, 5 channel SPIs, 5 channel I2Cs, and 32 Pin GPIO (16 Pin GPIO 2 Ports)
- Various timers: 6 Timers, 2 PWM Timers, 1 Dual-input Timer, 1 Watchdog, and 1 Free-Running Counter
- Security IPs: AES and DTRNG
- ULP-SRP as a DSP: reconfigurable core, 256KB SRAM, and Debugger

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.3 Bus Architecture

S3FBP5A's MCU system consists of ARM Cortex-M4 core, µDMAC, embedded Flash with Cache, SRAM, AHB peripherals, and APB peripherals as shown in *Figure 4-1*. The bus architecture supports AHB-Lite protocol which is a simple subset of the full AHB specification for use in designs where only a single bus master is used. The Bus Matrix has 4 slave ports (S0, S1, S2 and S3) for Cortex-M4 core and uDMAC, and 2 master ports (M0 and M1) for Code Bus Mux and System Bus Mux. Cortex-M4 has 3 bus interfaces toward Bus Matrix which are I-Code, D-Code, and System Bus. I-Code and D-code bus interfaces are a 32-bit AHB-Lite bus interface. I-Code bus fetches instruction and vector from code memory space. D-Code bus performs data and debug accesses to code memory space. System Bus is connected to system components such as AHB/APB peripherals, Flash, and SRAM. uDMAC is connected to slave port of Bus Matrix and can access memories and AHB/APB peripherals.

S3FBP5A supports hard-wired remap function for system booting. Combined I-Code & D-Code Bus for master port 0 and System Bus for master port 1 of Bus Matrix select one of embedded Flash with Cache and SRAM according to booting modes. When the system starts booting on embedded Flash, embedded Flash is connected to the combined I-Code & D-Code Bus and SRAM is connected to System Bus. On the contrary, when the system starts booting using external flash device, SRAM which contains boot code from external flash at previous system reset stage is connected to the combined I-Code & D-Code Bus and embedded Flash is connected to System Bus. Detailed booting modes are described in *4.5*

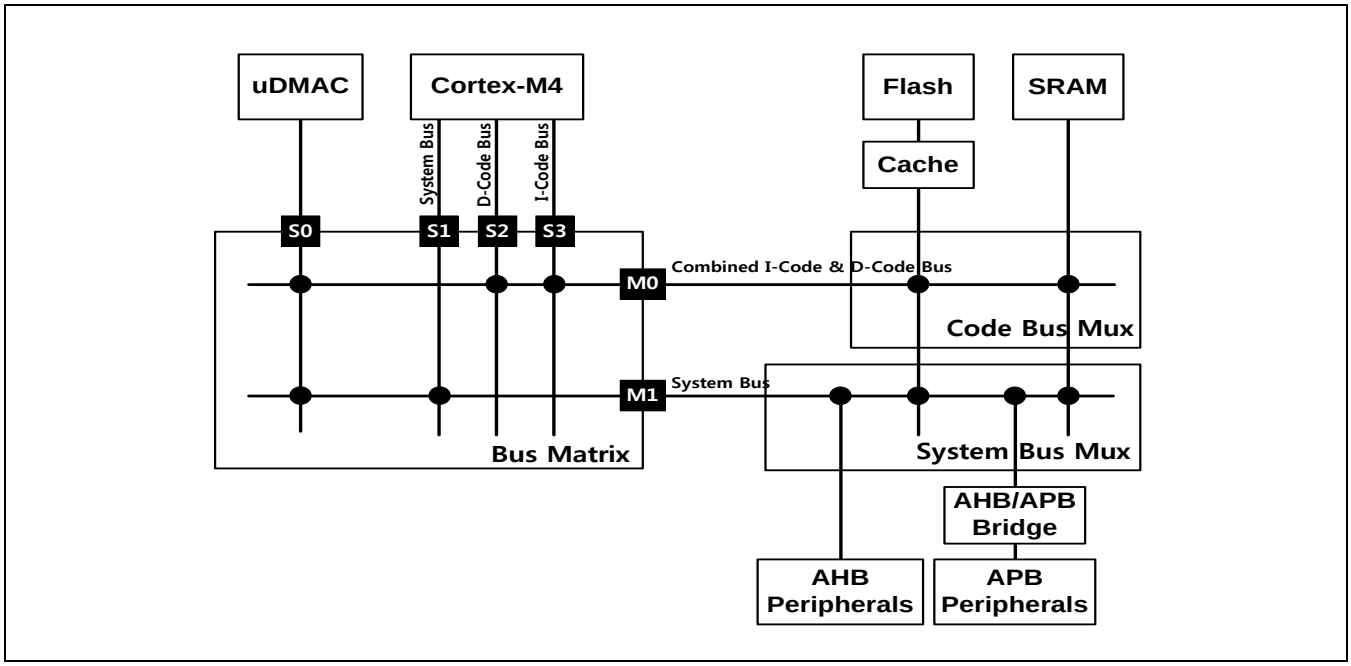


Figure 4-1 Bus Architecture

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.4 Memory Architecture

Figure 4-2 illustrates the memory architecture of S3FBP5A. The system and its peripherals are little-endian format.

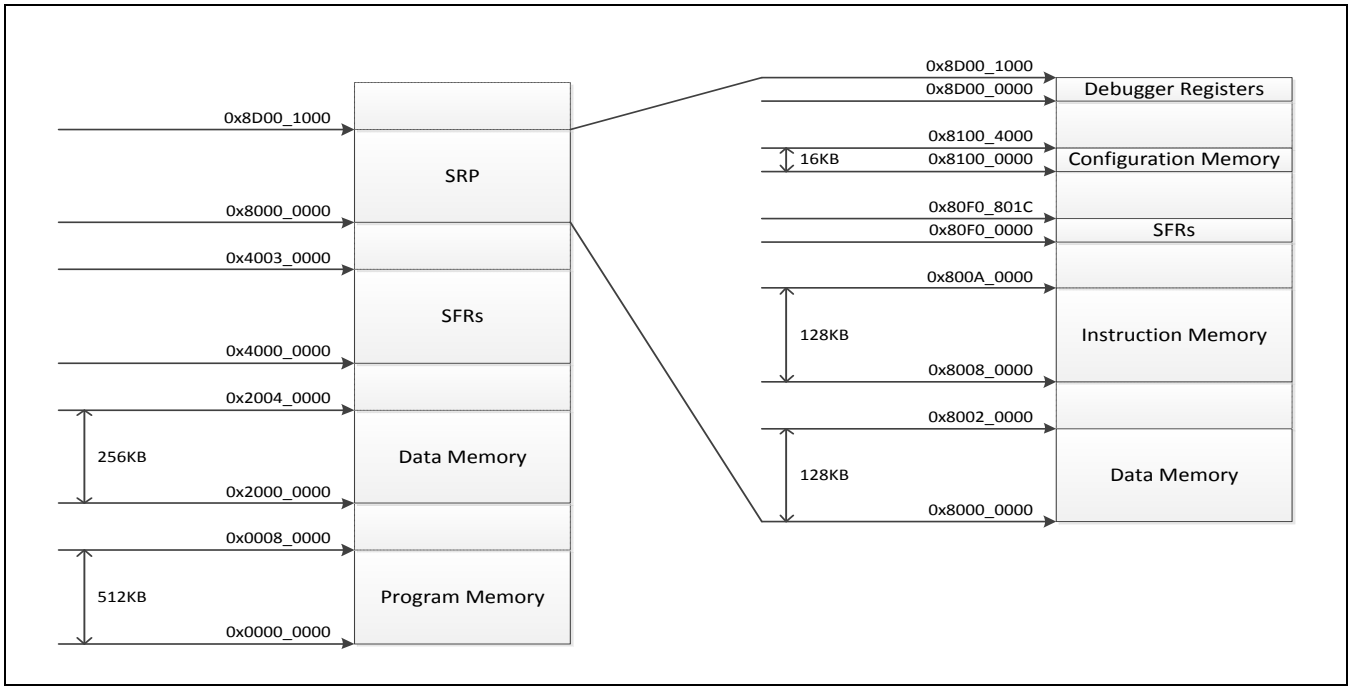


Figure 4-2 Memory Architecture of S3FBP5A

Table 4-1 lists the detailed memory allocation of S3FBP5A.

Table 4-1 Overall Memory Map

Address	Without remap	With remap
0x8D000FFF 0x80000000	SRP	SRP
0x7FFFFFFF 0x40030000	Unused	Unused
0x4002FFFF 0x4002F000	System controller registers	System controller registers
0x4002EFFF 0x40025000	Unused	Unused
0x40024FFF 0x40024000	Flash SFRs	Flash SFRs
0x40023FFF 0x40023000	DTRNG	DTRNG
0x40022FFF 0x40022000	AES	AES
0x40021FFF	AHB GPIO 1	AHB GPIO 1

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Address	Without remap	With remap
0x40021000		
0x40020FFF 0x40020000	AHB GPIO 0	AHB GPIO 0
0x4001FFFF 0x4001C000	Unused	Unused
0x4001BFFF 0x40000000	APB subsystem peripherals	APB subsystem peripherals
0x3FFFFFFF 0x20080000	Unused	Unused
0x2007FFFF 0x20040000		Data memory (Flash 512KB)
0x2003FFFF 0x20000000	Data memory (SRAM 256KB)	
0x1FFFFFFF 0x10000800	Unused	Unused
0x100007FF 0x10000000	Flash cache memory	Flash cache memory
0x0FFFFFFF 0x00080000	Unused	Unused
0x0007FFFF 0x00040000	Program memory (Flash 512KB)	
0x0003FFFF 0x00000000		Program memory (SRAM 256KB)

Table 4-2 lists APB-subsystem memory space allocation of S3FBP5A.

Table 4-2 APB Subsystem Memory Map

Address	Peripherals
0x4001BFFF 0x4001B000	µDMAC
0x4001AFFF 0x4001A000	AFE Controller
0x40019FFF 0x40019000	Free-Running Counter
0x40018FFF 0x40017000	Unused
0x40016FFF 0x40016000	I2C4
0x40015FFF 0x40015000	I2C3

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Address	Peripherals
0x40014FFF 0x40014000	I2C2
0x40013FFF 0x40013000	I2C1
0x40012FFF 0x40012000	I2C0
0x40011FFF 0x40011000	SPI4
0x40010FFF 0x40010000	SPI3
0x4000FFFF 0x4000F000	SPI2
0x4000EFFF 0x4000E000	SPI1
0x4000DFFF 0x4000D000	SPI0
0x4000CFFF 0x4000C000	UART2
0x4000BFFF 0x4000B000	UART1
0x4000AFFF 0x4000A000	UART0
0x40009FFF 0x40009000	Watchdog
0x40008FFF 0x40008000	Dual Timer
0x40007FFF 0x40007000	PWM Timer1
0x40006FFF 0x40006000	PWM Timer0
0x40005FFF 0x40005000	Timer5
0x40004FFF 0x40004000	Timer4
0x40003FFF 0x40003000	Timer3
0x40002FFF 0x40002000	Timer2
0x40001FFF 0x40001000	Timer1

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Address	Peripherals
0x40000FFF 0x40000000	Timer0

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.5 Boot Sequence

S3FBP5A has a Bootloader which deploys Serial Peripheral Interface (SPI) with external flash memory and booting host interface. SPI pins toward external flash device are FLASHSCLK, FLASHCSN, FLASHMOSI, and FLASHMISO. Similarly, HOSTSCLK, HOSTCSN, HOSTMOSI, and HOSTMISO are connected to external host interface. Host interface pins are multiplexed with GPIO00, GPIO01, GPIO02, and GPIO03, respectively. They are selected when host booting mode is enabled. There are three booting modes which are internal embedded Flash booting mode, external flash booting mode, and external host booting mode. BOOTMODE0 and BOOTMODE1 pins select one of the booting modes. *Table 4-3* lists booting modes of S3FBP5A.

Table 4-3 Booting Modes of S3FBP5A

Mode	BOOTMODE0	BOOTMODE1	Description
Internal flash	0	0	Embedded Flash booting mode.
External flash	0	1	External host writes boot code to external flash through S3FBP5A.
	1	0	S3FBP5A reads boot code from external flash.
External host	1	1	External host uploads boot code to S3FBP5A.

4.5.1 Internal Flash Mode

When BOOTMODE0=0 and BOOTMODE1=0 the system starts booting from internal embedded Flash after system reset.

4.5.2 External Flash Mode

Figure 4-3 shows the master mode booting operation of S3FBP5A. By setting BOOTMODE0=1 and BOOTMODE1=0 S3FBP5A carries out boot-loading from external flash until the end of boot code. During boot-loading S3FBP5A's internal core reset is asserted and released after completing boot loading. At this mode hard-wired remap, embedded 256KB SRAM is mapped to program memory from 0x00000000 to 0x0003FFFF and embedded 512KB Flash is mapped to Data Memory from 0x20000000 to 0x2007FFFF, is enabled so that core starts reading boot code from the first address of SRAM.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

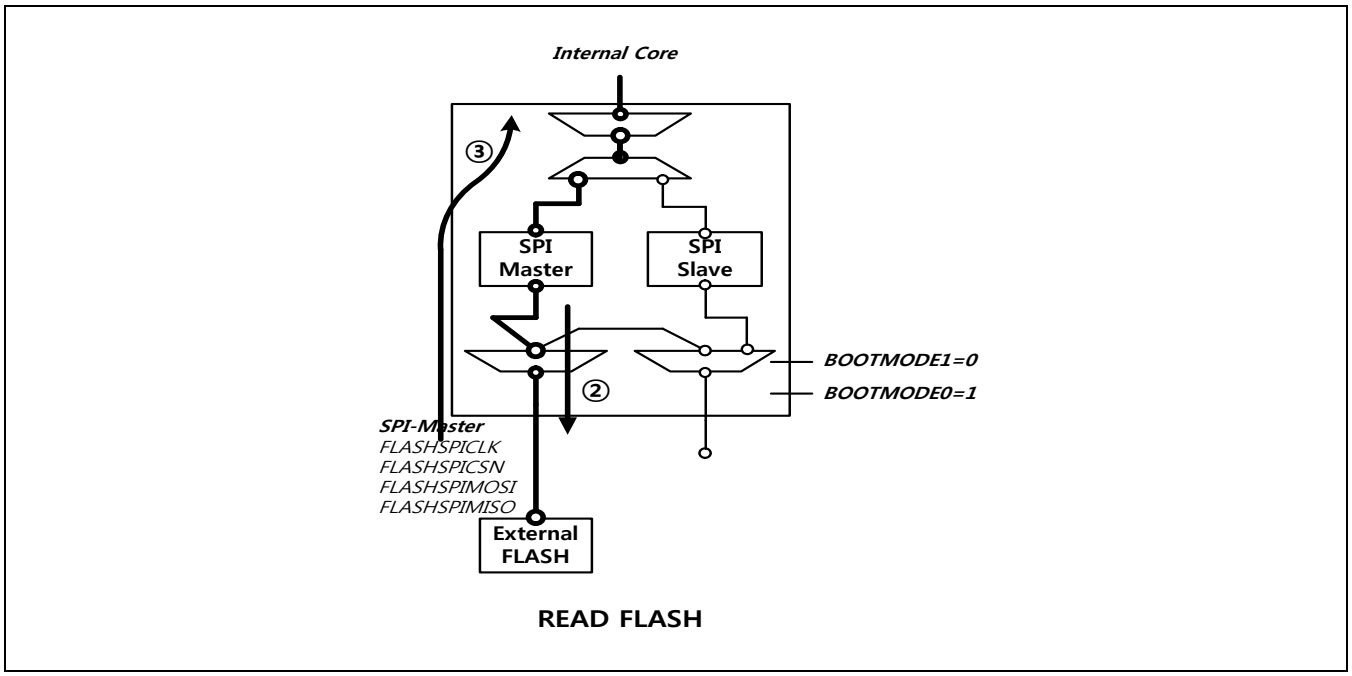


Figure 4-3 External Flash Mode Booting Operation

Figure 4-4 shows external flash write operation. When BOOTMODE0=0 and BOOTMODE1=1 external host can write boot code to external flash through S3FBP5A.

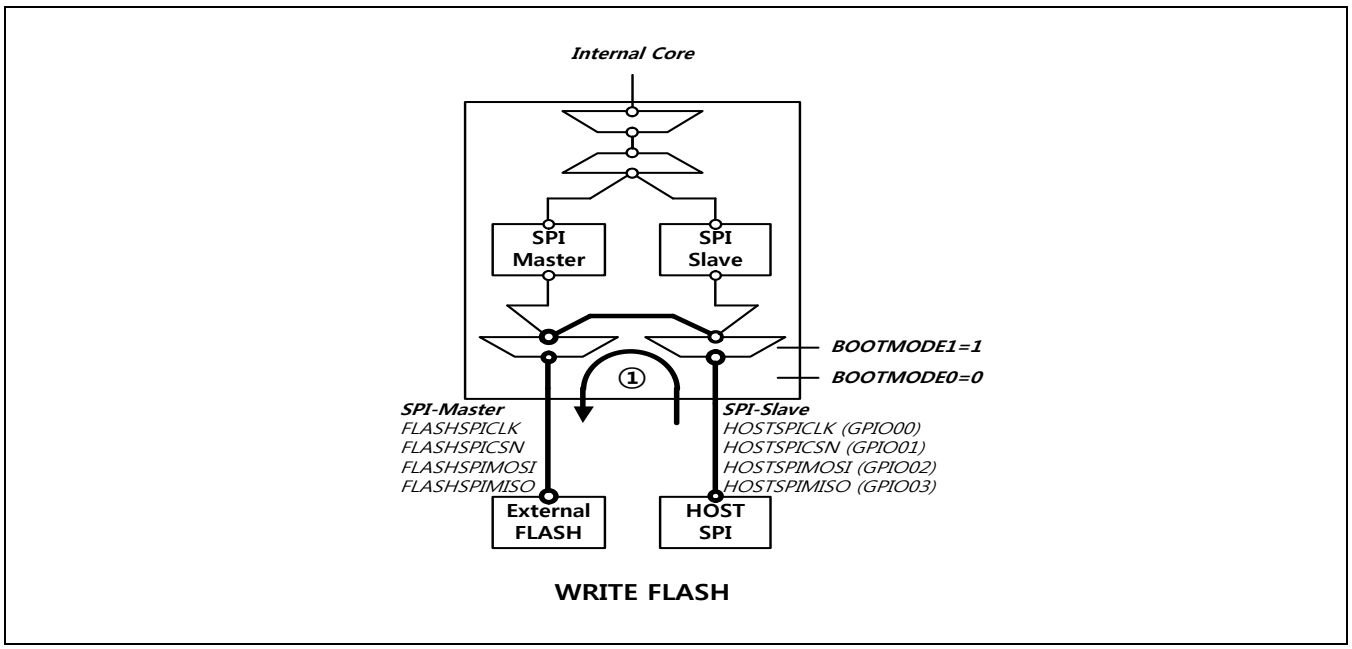


Figure 4-4 External Flash Write Operation

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.5.3 External Host Mode

Figure 4-5 shows the slave mode booting operation of S3FBP5A. When BOOTMODE0=1 and BOOTMODE1=1, external host can upload boot code to S3FBP5A using HOSTSCLK, HOSTCSN, HOSTMOSI, and HOSTMISO pins. During boot-loading S3FBP5A's internal core reset is asserted and released after completing boot-loading. At this mode hard-wired remap, embedded 256KB SRAM is mapped to program memory from 0x00000000 to 0x0003FFFF and embedded 512KB Flash is mapped to Data Memory from 0x20000000 to 0x2007FFFF, is enabled so that core starts reading boot code from the first address of SRAM.

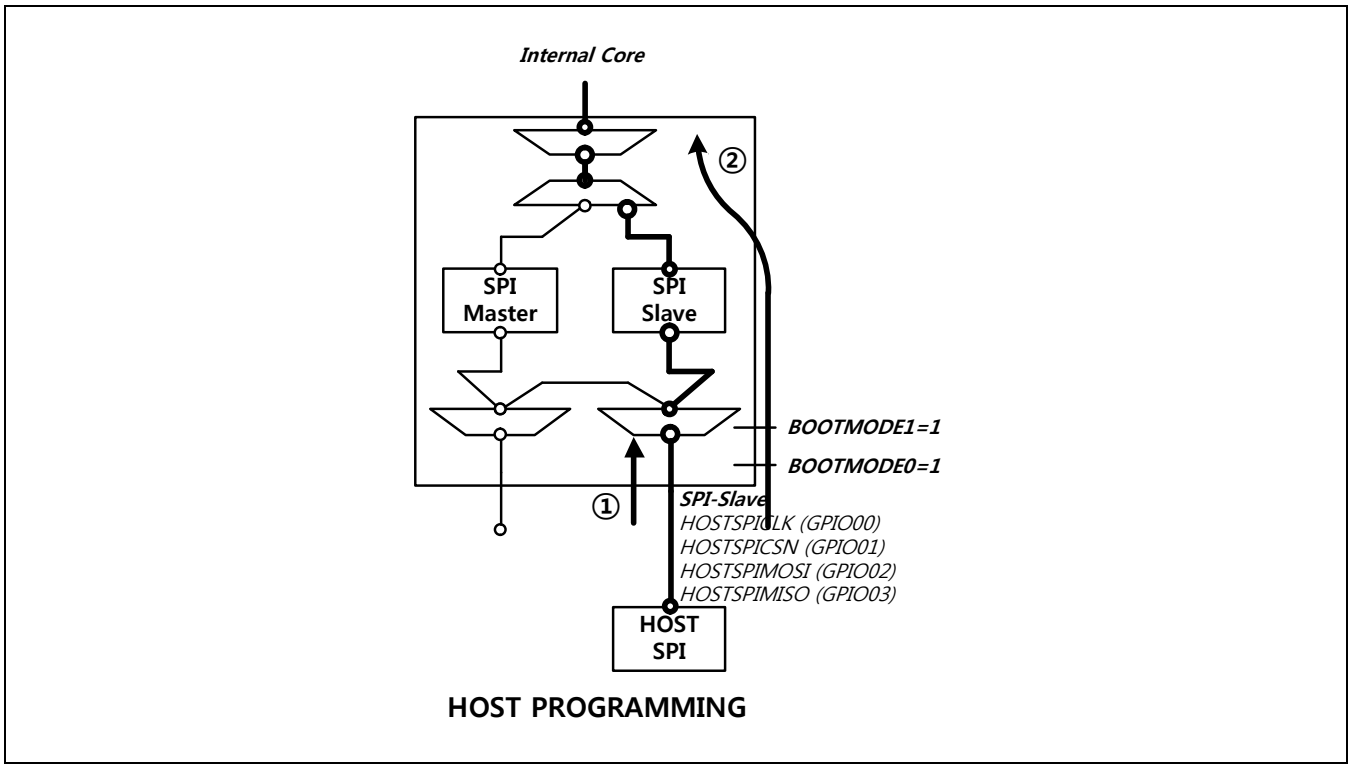


Figure 4-5 Slave Mode Booting Operation

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.6 Clock Management Unit (CMU)

CMU has 4 major functions:

- On/off control for a 87MHz Oscillator
- Clock divider of HCLK/PCLK/SRPCLK
- Clock gating of separated clock domains

A clock tree of S3FBP5A inside CMU is shown in *Figure 4-6*.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

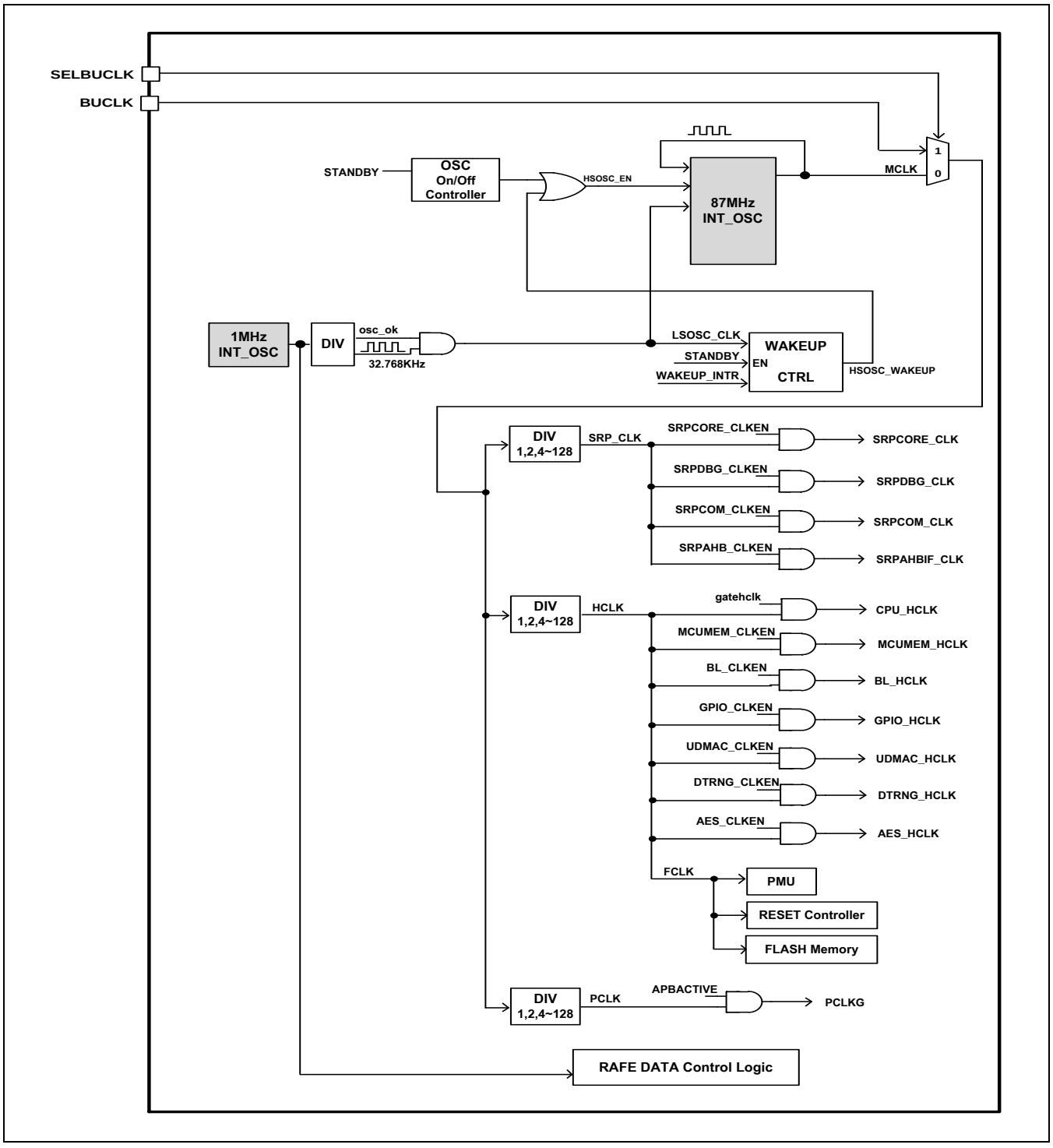


Figure 4-6 Clock Tree of S3FBP5A

4.6.1 Register Description

4.6.1.1 Register Map Summary

- Base Address: 0x4001A500

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Register	Offset	Description	Reset Value
CMU_CTRL	0x0010	CMU_CTRL	0x0018_FEDB

4.6.1.1.1 CMU_CTRL

- Address: Base Address + 0x0010, Reset Value = 0x0018_FEDB

Name	Bit	Type	Description	Reset Value
CGENB_AES	[20]	RW	* AES clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_DTRNG	[19]	RW	* DTRNG clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
UDMAC_ACG_EN	[16]	RW	* uDMAC automatic clock gating selection 1: UDMAC clock is gated by DMA request signals automatically. 0: UDMAC clock is gated by CMU_CTRL[15].	0x0000_0000
CGENB_UDMAC	[15]	RW	* UDMAC clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_GPIO	[14]	RW	* GPIO clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_BL	[13]	RW	* Bootloader clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_SRPAHB	[12]	RW	* SRP AHB Interface clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_SRPCOM	[11]	RW	* SRP common logic clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_SRPDBG	[10]	RW	* SRP debugger clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
CGENB_SRPCORE	[9]	RW	* SRP core clock gate (active low) 1: Clock enabled. 0: Clock gated.	0x0000_0001
SRPCLK_DIV_SEL	[8:6]	RW	* SRP clock divide value 0: Divided by 1 1: Divided by 2 2: Divided by 4	0x0000_0003

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Name	Bit	Type	Description	Reset Value
			3: Divided by 8 4: Divided by 16 5: Divided by 32 6: Divided by 64 7: Divided by 128 8: Divided by 256 *** SRPCLK_DIV_SEL and PCLK_DIV_SEL and HCLK_DIV_SEL should be same.	
PCLK_DIV_SEL	[5:3]	RW	* PCLK clock divide value 0: Divided by 1 1: Divided by 2 2: Divided by 4 3: Divided by 8 4: Divided by 16 5: Divided by 32 6: Divided by 64 7: Divided by 128 8: Divided by 256 *** SRPCLK_DIV_SEL and PCLK_DIV_SEL and HCLK_DIV_SEL should be same.	0x0000_0003
HCLK_DIV_SEL	[2:0]	RW	* HCLK clock divide value 0: Divided by 1 1: Divided by 2 2: Divided by 4 3: Divided by 8 4: Divided by 16 5: Divided by 32 6: Divided by 64 7: Divided by 128 8: Divided by 256 *** SRPCLK_DIV_SEL and PCLK_DIV_SEL and HCLK_DIV_SEL should be same.	0x0000_0003

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.7 Power Management Unit (PMU)

PMU has 2 major functions:

- Support 3 types of low power operation modes
- Retention, isolation, and power-down control of partitioned power domains

4.7.1 Low-Power Modes

Description of low-power modes of S3FBP5A is shown in *Table 4-4*.

Table 4-4 Low-power Modes of S3FBP5A

Mode Name	Entry	Wakeup	1M/32K OSC	87MHz OSC
Sleep	WFI	Any Interrupts	ON	CPU Clock Gating No effect on other clocks
	WFE	Wakeup event	ON	CPU Clock Gating No effect on other clocks
Stop	SLEEPDEEP =1 + WFI	Any Interrupts	ON	PD_MCU in retention mode CPU Clock Gating No effect on other clocks
	SLEEPDEEP =1 + WFE	Wakeup event	ON	PD_MCU in retention mode CPU Clock Gating No effect on other clocks
Standby	STANDBY =1 + SYSPWR_CTRL =1 + SLEEPDEEP =1 + WFI or WFE	Toggling GPIO00 pin FRC Interrupt External reset	ON	PD_MCU in retention mode PD_MCU_SYSTEM in retention mode CPU Clock Gating 87MHz OSC Power Off

4.7.1.1 Sleep Mode

S3FBP5A is entered into sleep mode by WFI or WFE system function. In sleep mode, MCU core clock is gated and S3FBP5A is waked up by any interrupts in *Table 4-6* or wakeup events.

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4.7.1.2 Stop Mode

S3FBP5A is entered into stop mode by setting SLEEPDEEP bit into 1 and executing WFI or WFE system function. In stop mode, MCU core clock is gated and PD_MCU (CortexM4 core power domain) is entered into retention mode which stores all data of flip-flops. S3FBP5A is waked up by any interrupts in *Table 4-6* or wakeup events.

4.7.1.3 Standby Mode

S3FBP5A is entered into standby mode by setting SLEEPDEEP, STANDBY, SYSPWR_CTRL registers into 1 and executing WFI or WFE system function. In standby mode, 87MHz oscillator is turned off and other condition is same as in stop mode. S3FBP5A is waked up in 3 ways, by toggling GPIO00 pin, by FRC interrupt and by external reset.

4.7.2 Power Domain Partitioning

Table 4-5 shows 27 power domains in S3FBP5A. There are 4 always-on domains and 23 power-gating domains. Always-on domains are IO pad, AFE Control logic, and other top logics. MCU has 10 power domains including Cortex-M4 core and 2 data memories. Cortex-M4 core is retained under power-off state which means all information are saved with retention mode because it needs to continue the program which was stopped before power down. All of MCU memories are power-gating and retention ones. SRP has 13 power domains which are SRP high voltage domain, SRP low power domain, 5 SRP Instruction Memory domains, 4 SRP Data Memory domains, and 2 SRP configuration domains. SRP common logic domain includes AHB Slave Interface, SRP internal logic, and memory wrapper. It does not have retention mode because SRP can restart the program without the retained information in core. SRP core domain includes SRP core logic and Debugger logic. Retention mode is not required to this domain, either. All of SRP memories can be power-down or retained.

Table 4-5 Power Domain Partitioning of S3FBP5A

Power Domains		Description	Power Sources	Power Modes	
				On/Off	Retention
Always-on	PD_IOPAD	IO pad domain, always on	IO 3V, KALDO (1.1V)	No	No
	PD_DEFAULT	Default domain, always on	LDO1 (1.1V)	No	No
	PD_FLASH	Flash memory domain	LDO2 (1.1V)		
	PD_AFE	AFE Control Logic domain, always on	KALDO (1.1V)		
MCU	PD_MCU_SYSTEM	Cortex-M4 core domain except for Cortex-M4 core	LDO1 (1.1V)	Yes	Yes
	PD_MCU	Cortex-M4 core domain	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM0	MCU Data Memory 0	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM1	MCU Data Memory 1	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM2	MCU Data Memory 2	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM3	MCU Data Memory 3	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM4	MCU Data Memory 4	LDO1 (1.1V)	Yes	Yes

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Power Domains		Description	Power Sources	Power Modes	
				On/Off	Retention
	PD_MCU_SRAM5	MCU Data Memory 5	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM6	MCU Data Memory 6	LDO1 (1.1V)	Yes	Yes
	PD_MCU_SRAM7	MCU Data Memory 7	LDO1 (1.1V)	Yes	Yes
SRP	PD_SRPH	SRP AHB I/F, internal logic, and memory wrapper	LDO1 (1.1V)	Yes	No
	PD_SRPL	SRP core domain	LDO1 (1.1V)	Yes	No
	PD_SRP_IMEM0	SRP Instruction Memory 0	LDO1 (1.1V)	Yes	Yes
	PD_SRP_IMEM1	SRP Instruction Memory 1	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_IMEM2	SRP Instruction Memory 2	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_IMEM3	SRP Instruction Memory 3	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_IMEM4	SRP Instruction Memory 4	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_DMEM0	SRP Data Memory 0	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_DMEM1	SRP Data Memory 1	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_DMEM2	SRP Data Memory 2	LDO1 (1.1V)	Yes	Yes
	PD_ SRP_DMEM3	SRP Data Memory 3	LDO1 (1.1V)	Yes	Yes
	PD_SRP_CMEM0	SRP Configuration Memory 0	LDO1 (1.1V)	Yes	Yes
	PD_SRP_CMEM1	SRP Configuration Memory 1	LDO1 (1.1V)	Yes	Yes

4.7.3 Register Description

4.7.3.1 Register Map Summary

- Base Address: 0x4001A500

Register	Offset	Description	Reset Value
PMU_PD_CTRL	0x0000	PMU_PD_CTRL	0x0000_0000
PMU_RET_CTRL	0x0004	PMU_RET_CTRL	0xE000_0000

4.7.3.1.1 PMU_PD_CTRL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHG_PWR_STATE	[31]	RW	* Change Power State - Toggling this register starts power sequence of PMU with PMU_PD_REG and PMU_RET_REG registers. Power control will be updated by changing this bit after writing power down and retention registers (PMU_PD_REG and PMU_RET_REG).	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
PD_DECIM	[23]	RW	* Power down control for memory inside decimation filter 0: power on 1: power down	0x0000_0000
PD_FCACHE	[22]	RW	* Power down control for memory inside Flash memory controller 0: power on 1: power down	0x0000_0000
PD_DSPH	[21]	RW	* Power down control for SRP AHB Interface and internal logic 0: power on 1: power down	0x0000_0000
PD_DSPL	[20]	RW	* Power down control for SRP Core 0: power on 1: power down	0x0000_0000
PD_DSP_IMEM	[19:15]	RW	* Power down control for SRP instruction memory banks (5 banks) 0: power on 1: power down	0x0000_0000
PD_DSP_DMEM	[14:11]	RW	* Power down control for SRP data memory banks (4 banks) 0: power on 1: power down	0x0000_0000
PD_DSP_CMEM	[10:9]	RW	* Power down control for SRP configuration memory banks (2 banks) 0: power on 1: power down	0x0000_0000
PD_MCU_SRAM	[8:1]	RW	* Power down control for MCU data memory banks (8 banks) 0: power on 1: power down	0x0000_0000

4.7.3.1.2 PMU_RET_CTRL

- Address: Base Address + 0x0004, Reset Value = 0xE000_0000

Name	Bit	Type	Description	Reset Value
DSPH_RESETn_REG	[31]	RW	*SRP logic reset except SRP core 1: Reset released 0: Reset asserted	0x0000_0001
DSPL_RESETn_REG	[30]	RW	* SRP core S/W reset 1: Reset release 0: Reset assert	0x0000_0001

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Name	Bit	Type	Description	Reset Value
SYSPWR_CTRL_EN	[28]	RW	* Power retention control for MCU system power domain 1: MCU System power domain enters into retention mode(according to PMU FSM) 0: MCU System power domain always on	0x0000_0000
STANDBY	[27]	RW	* Use standby mode which can turn off main 87MHz oscillator 1: Entering standby mode when MCU enter into sleep mode like WFE or WFI 0: Do not enter into standby mode	0x0000_0000
RET_DECIM	[23]	RW	* Retention mode control for memory inside decimation filter 1: Retention mode 0: Normal operation mode	0x0000_0000
RET_FCACHE	[22]	RW	* Retention mode control for cache memory inside Flash memory controller 1: Retention mode 0: Normal operation mode	0x0000_0000
RET_DSP_IMEM	[19:15]	RW	* Power retention control for SRP instruction memory banks (5 banks) 1: Retention mode 0: Normal operation mode	0x0000_0000
RET_DSP_DMEM	[14:11]	RW	* Power retention control for SRP data memory banks (4 banks) 1: Retention mode 0: Normal operation mode	0x0000_0000
RET_DSP_CMEM	[10:9]	RW	* Power retention control for SRP configuration memory banks (2 banks) 1: Retention mode 0: Normal operation mode	0x0000_0000
RET_MCU_SRAM	[8:1]	RW	* Power retention control for MCU data memory banks (8 banks) 1: Retention mode 0: Normal operation mode	0x0000_0000

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4.8 ARM Cortex-M4 with Floating-Point Unit (Cortex-M4F)

The processor core of S3FBP5A is ARM Cortex-M4 with Floating-Point Unit (Cortex-M4F). Cortex-M4 processor is a high performance 32-bit RISC processor with a 3-stage pipeline Harvard architecture, making it ideal for demanding embedded applications. Cortex-M4 processor implements a version of the Thumb instruction set based on Thumb-2 technology ensuring high code density and reduced program memory requirements. The instruction set of Cortex-M4 provides the exceptional performance expected of a modern 32-bit architecture with the high code density of 8-bit and 16-bit microcontrollers. *Figure 4-7* shows the structure of Cortex-M4F processor.

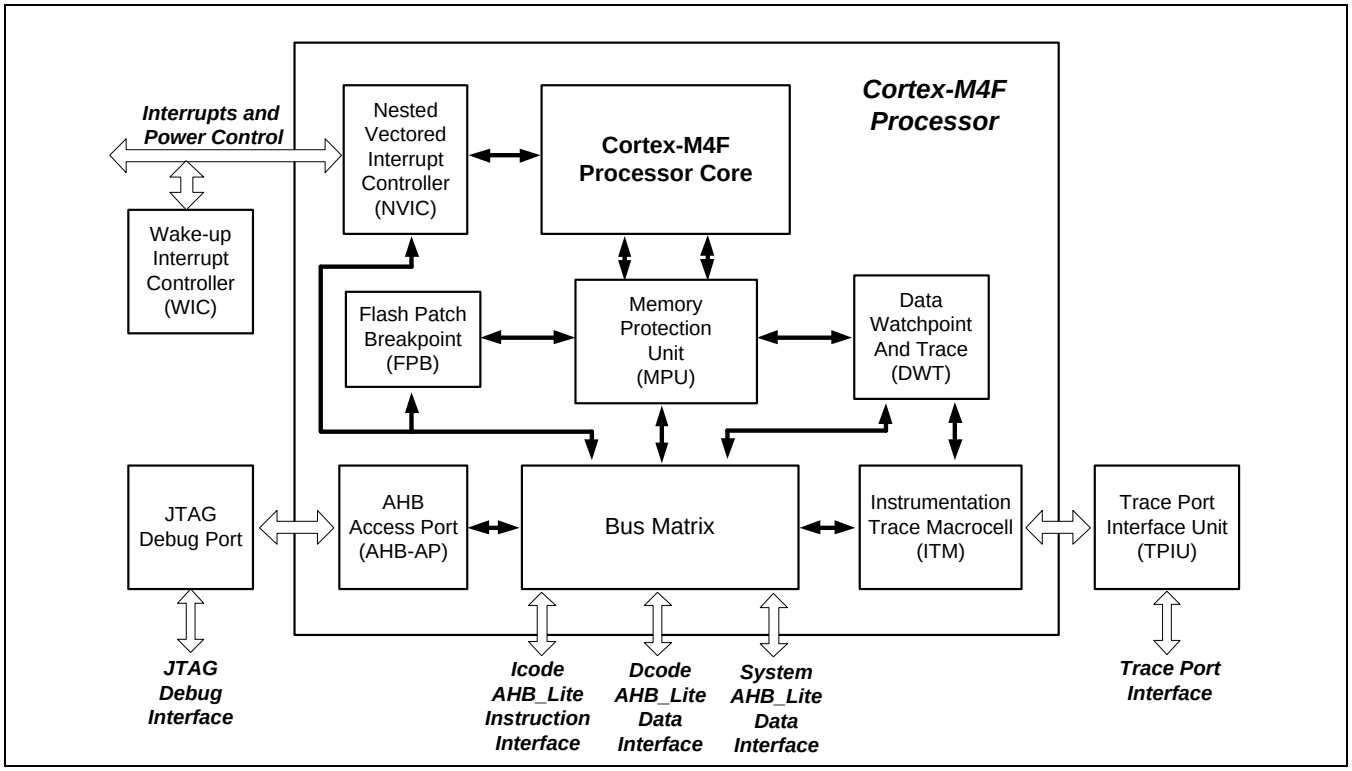


Figure 4-7 Cortex-M4F Block Diagram

ARM Cortex-M4F processor features:

- A low gate count processor core, with low latency interrupt processing that has:
  - A subset of the Thumb instruction set, defined in the ARMv7-M architecture reference manual
  - Banked stack pointer (SP)
  - Hardware integer divide instructions, SDIV and UDIV
  - Handler and thread mode
  - Thumb and debug states
  - Support for interruptible-continued instructions LDM, STM, PUSH, and POP for low interrupt latency
  - Automatic processor state saving and restoration for low latency interrupt service routine (ISR) entry and exit
  - Support for ARMv6 big-endian byte-invariant or little-endian accesses
  - Support for ARMv6 unaligned accesses

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

- Floating-Point Unit (FPU) in the Cortex-M4F processor providing:
  - 32-bit instructions for single-precision (C float) data-processing operations
  - Combined multiply and accumulate instructions for increased precision (Fused MAC)
  - Hardware support for conversion, addition, subtraction, multiplication with optional accumulate, division, and square-root
  - Hardware support for denormals and all IEEE rounding modes
  - 32 dedicated 32-bit single precision registers, also addressable as 16 double-word registers
  - Decoupled three stage pipeline

- Nested Vectored Interrupt Controller (NVIC) closely integrated with the processor core to achieve low latency interrupt processing. Features include:
  - External interrupts, configurable from 1 to 240
  - Bits of priority, configurable from 3 to 8
  - Dynamic reprioritization of interrupts
  - Priority grouping. This enables selection of preempting interrupt levels and non-preempting interrupt levels
  - Support for tail-chaining and late arrival of interrupts. This enables back-to-back interrupt processing without the overhead of state saving and restoration between interrupts
  - Processor state automatically saved on interrupt entry, and restored on interrupt exit, with no instruction overhead
  - Wake-up Interrupt Controller (WIC), providing ultra-low power sleep mode support

- Memory Protection Unit (MPU). An optional MPU for memory protection, including:
  - Eight memory regions
  - Sub Region Disable (SRD), enabling efficient use of memory regions
  - The ability to enable a background region that implements the default memory map attributes

- Bus interfaces:
  - Three AHB-Lite interfaces: I-Code, D-Code, and System Bus interfaces
  - Private Peripheral Bus (PPB) based on Advanced Peripheral Bus (APB) interface
  - Bit-band support that includes atomic bit-band write and read operations
  - Memory access alignment
  - Write buffer for buffering of write data
  - Exclusive access transfers for multiprocessor systems

- Low-cost debug solution that features:
  - Debug access to all memory and registers in the system, including access to memory mapped devices, access to internal core registers when the core is halted, and access to debug control registers even while SYSRESETn is asserted.
  - Serial Wire JTAG Debug Port (SWJ-DP) debug

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

The technical reference manual of ARM Cortex-M4 processor can be found the link below.

http://infocenter.arm.com/help/topic/com.arm.doc.ddi0439b/DDI0439B_cortex_m4_r0p0_trm.pdf

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4.9 Micro Direct Memory Access Controller (µDMAC)

S3FBP5A utilizes ARM PrimeCell µDMAC (PL230) as a Direct Memory Access Controller (DMAC) because the bus architecture is based on AMBA AHB-Lite protocol. The µDMAC is an AMBA compliant SoC peripheral that is developed, tested, and licensed by ARM. It is a very low gate count DMA controller that is compatible with the AMBA AHB-Lite protocol.

µDMAC features:

• It is compatible with AHB-Lite for the DMA transfers

• It is compatible with APB for programming the registers

• It has a single AHB-Lite master for transferring data using a 32-bit address bus and 32-bit data bus

• It has 32 DMA channels

• Each DMA channel has dedicated handshake signals

• Each DMA channel has a programmable priority level

• Each priority level arbitrates using a fixed priority that is determined by the DMA channel number

• It supports multiple transfer types:

  – Memory-to-Memory
  – Memory-to-Peripheral
  – Peripheral-to-Memory

• It supports multiple DMA cycle types

• It supports multiple DMA transfer data widths

• Each DMA channel can access a primary, and alternate, channel control data structure

• All the channel control data is stored in system memory in little-endian format

• It performs all DMA transfers using the SINGLE AHB-Lite burst type

• The destination data width is equal to the source data width

• The number of transfers in a single DMA cycle can be programmed from 1 to 1024

• The transfer address increment can be greater than the data width

• It has a single output to indicate when an ERROR condition occurs on the AHB bus.

Table 4-6 lists DMA channel assignment.

Table 4-6 DMA Channel Assignment

DMA Channel	Single Request	Request
0	SPI0 RX	-
1	SPI0 TX	-

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DMA Channel	Single Request	Request
2	SPI1 RX	-
3	SPI1 TX	-
4	SPI2 RX	-
5	SPI2 TX	-
6	SPI3 RX	-
7	SPI3 TX	-
8	SPI4 RX	-
9	SPI4 TX	-
10	I2C0 RX	-
11	I2C0 TX	-
12	I2C1 RX	-
13	I2C1 TX	-
14	I2C2 RX	-
15	I2C2 TX	-
16	I2C3 RX	-
17	I2C3 TX	-
18	I2C4 RX	-
19	I2C4 TX	-
20	UART0 RX	UART0 RX
21	UART0 TX	UART0 TX
22	UART1 RX	UART1 RX
23	UART1 TX	UART1 TX
24	UART2 RX	UART2 RX
25	UART2 TX	UART2 TX
26	AFE FIFO0	AFE FIFO0
27	AFE FIFO1	AFE FIFO1
28	AFE FIFO2	AFE FIFO2
29	AFE FIFO3	AFE FIFO3
30	AFE FIFO4	AFE FIFO4
31	AFE FIFO5	AFE FIFO5

The final version of µDMAC technical reference manual can be found the link below.

http://infocenter.arm.com/help/topic/com.arm.doc.ddi0417a/DDI0417A_udmac_pl230_r0p0_trm.pdf

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4.9.1 Register Description

4.9.1.1 Register Map Summary

- Base Address: 0x4001B000

Register	Offset	Description	Reset Value
DMA_STATUS	0x0000	DMA status	0x001F_0000
DMA_CFG	0x0004	DMA configuration	0x0000_0000
CTRL_BASE_PTR	0x0008	Channel control data base pointer	0x0000_0000
ALT_CTRL_BASE_PTR	0x000C	Channel alternate control data base pointer	0x0000_0000
DMA_WAITONREQ_STATUS	0x0010	Channel wait on request status	0x0000_0000
CHNL_SW_REQUEST	0x0014	Channel software request	0x0000_0000
CHNL_USEBURST_SET	0x0018	Channel useburst set	0x0000_0000
CHNL_USEBURST_CLR	0x001C	Channel useburst clear	0x0000_0000
CHNL_REQ_MASK_SET	0x0020	Channel request mask set	0x0000_0000
CHNL_REQ_MASK_CLR	0x0024	Channel request mask clear	0x0000_0000
CHNL_ENABLE_SET	0x0028	Channel enable set	0x0000_0000
CHNL_ENABLE_CLR	0x002C	Channel enable clear	0x0000_0000
CHNL_PRI_ALT_SET	0x0030	Channel primary-alternate set	0x0000_0000
CHNL_PRI_ALT_CLR	0x0034	Channel primary-alternate clear	0x0000_0000
CHNL_PRIORITY_SET	0x0038	Channel priority set	0x0000_0000
CHNL_PRIORITY_CLR	0x003C	Channel priority clear	0x0000_0000
ERR_CLR	0x004C	Bus error clear	0x0000_0000
PERIPH_ID_4	0x0FD0	Peripheral identification 4	0x0000_0004
PERIPH_ID_0	0x0FE0	Peripheral identification 0	0x0000_0030
PERIPH_ID_1	0x0FE4	Peripheral identification 1	0x0000_00B2
PERIPH_ID_2	0x0FE8	Peripheral identification 2	0x0000_000B
PERIPH_ID_3	0x0FEC	Peripheral identification 3	0x0000_0000
PCELL_ID_0	0x0FF0	PrimeCell identification 0	0x0000_000D
PCELL_ID_1	0x0FF4	PrimeCell identification 1	0x0000_00F0
PCELL_ID_2	0x0FF8	PrimeCell identification 2	0x0000_0005
PCELL_ID_3	0x0FFC	PrimeCell identification 3	0x0000_00B1

4.9.1.1.1 DMA_STATUS

- Address: Base Address + 0x0000, Reset Value = 0x001F_0000

Name	Bit	Type	Description	Reset Value

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
TEST_STATUS	[31:28]	R	To reduce the gate count you can configure the controller Read as: 0x0 = controller does not include the integration test logic 0x1 = controller includes the integration test logic 0x2-0xF = undefined.	0x0000_0000
Reserved0	[27:21]	R	*Reserved	0x0000_0000
CHNLS_MINUS1	[20:16]	R	Number of available DMA channels minus one. For example: b00000 = controller configured to use 1 DMA channel b00001 = controller configured to use 2 DMA channels b00010 = controller configured to use 3 DMA channels ~ b11111 = controller configured to use 32 DMA channels.	0x0000_001F
Reserved1	[15:8]	R	*Reserved	0x0000_0000
STATE	[7:4]	R	Current state of the control state machine. State can be one of the following: b0000 = idle b0001 = reading channel controller data b0010 = reading source data end pointer b0011 = reading destination data end pointer b0100 = reading source data b0101 = writing destination data b0110 = waiting for DMA request to clear b0111 = writing channel controller data b1000 = stalled b1001 = done b1010 = peripheral scatter-gather transition b1011-b1111 = undefined.	0x0000_0000
Reserved2	[3:1]	R	*Reserved	0x0000_0000
MASTER_ENABLE	[0]	R	Enable status of the controller: 0 = controller is disabled 1 = controller is enabled.	0x0000_0000

4.9.1.1.2 DMA_CFG

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	W	*Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
CHNL_PROT_CTRL	[7:5]	W	Sets the AHB-Lite protection by controlling the HPROT[3:1] signal levels as follows: Bit [7] Controls HPROT[3] to indicate if a cacheable access is occurring. Bit [6] Controls HPROT[2] to indicate if a bufferable access is occurring. Bit [5] Controls HPROT[1] to indicate if a privileged access is occurring. When bit [n] = 1 then the corresponding HPROT is HIGH. When bit [n] = 0 then the corresponding HPROT is LOW.	0x0000_0000
Reserved1	[4:1]	W	*Reserved	0x0000_0000
MASTER_ENABLE	[0]	W	Enable for the controller: 0 = disables the controller 1 = enables the controller.	0x0000_0000

4.9.1.1.3 CTRL_BASE_PTR

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CTRL_BASE_PTR	[31:9]	RW	Pointer to the base address of the primary data structure.	0x0000_0000
Reserved0	[8:0]	RW	*Reserved	0x0000_0000

4.9.1.1.4 ALT_CTRL_BASE_PTR

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALT_CTRL_BASE_PTR	[31:0]	R	Base address of the alternate data structure	0x0000_0000

4.9.1.1.5 DMA_WAITONREQ_STATUS

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
DMA_WAITONREQ_STATUS	[31:0]	R	Channel wait on request status. Read as: Bit [C] = 0 dma_waitonreq[C] is LOW. Bit [C] = 1 dma_waitonreq[C] is HIGH.	0x0000_0000

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4.9.1.1.6 CHNL_SW_REQUEST

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_SW_REQUEST	[31:0]	W	Set the appropriate bit to generate a software DMA request on the corresponding DMA channel. Write as: Bit [C] = 0 Does not create a DMA request for channel C. Bit [C] = 1 Creates a DMA request for channel C. Writing to a bit where a DMA channel is not implemented does not create a DMA request for that channel.	0x0000_0000

4.9.1.1.7 CHNL_USEBURST_SET

- Address: Base Address + 0x0018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_USEBURST_SET	[31:0]	RW	Returns the useburst status, or disables dma_sreq[C] from generating DMA requests. Read as: Bit [C] = 0 DMA channel C responds to requests that it receives on dma_req[C] or dma_sreq[C]. The controller performs 2R, or single, bus transfers. Bit [C] = 1 DMA channel C does not respond to requests that it receives on dma_sreq[C]. The controller only responds to dma_req[C] requests and performs 2R transfers. Write as: Bit [C] = 0 No effect. Use the chnl_useburst_clr Register to set bit [C] to 0. Bit [C] = 1 Disables dma_sreq[C] from generating DMA requests. The controller performs 2R transfers. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.8 CHNL_USEBURST_CLR

- Address: Base Address + 0x001C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_USEBURST_CLR	[31:0]	W	Set the appropriate bit to enable dma_sreq[] to generate requests.	0x0000_0000

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			Write as: Bit [C] = 0 No effect. Use the chnl_useburst_set Register to disable dma_sreq[] from generating requests. Bit [C] = 1 Enables dma_sreq[C] to generate DMA requests. Writing to a bit where a DMA channel is not implemented has no effect.	

4.9.1.1.9 CHNL_REQ_MASK_SET

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_REQ_MASK_SET	[31:0]	RW	Returns the request mask status of dma_req[] and dma_sreq[], or disables the corresponding channel from generating DMA requests. Read as: Bit [C] = 0 External requests are enabled for channel C. Bit [C] = 1 External requests are disabled for channel C. Write as: Bit [C] = 0 No effect. Use the chnl_req_mask_clr Register to enable DMA requests. Bit [C] = 1 Disables dma_req[C] and dma_sreq[C] from generating DMA requests. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.10 CHNL_REQ_MASK_CLR

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_REQ_MASK_CLR	[31:0]	W	Set the appropriate bit to enable DMA requests for the channel corresponding to dma_req[] and dma_sreq[]. Write as: Bit [C] = 0 No effect. Use the chnl_req_mask_set Register to disable dma_req[] and dma_sreq[] from generating requests. Bit [C] = 1 Enables dma_req[C] or dma_sreq[C] to generate DMA requests. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

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4.9.1.1.11 CHNL_ENABLE_SET

- Address: Base Address + 0x0028, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_ENABLE_SET	[31:0]	RW	Returns the enable status of the channels, or enables the corresponding channels. Read as: Bit [C] = 0 Channel C is disabled. Bit [C] = 1 Channel C is enabled. Write as: Bit [C] = 0 No effect. Use the chnl_enable_clr Register to disable a channel. Bit [C] = 1 Enables channel C. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.12 CHNL_ENABLE_CLR

- Address: Base Address + 0x002C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_ENABLE_CLR	[31:0]	W	Set the appropriate bit to disable the corresponding DMA channel. Write as: Bit [C] = 0 No effect. Use the chnl_enable_set Register to enable DMA channels. Bit [C] = 1 Disables channel C. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.13 CHNL_PRI_ALT_SET

- Address: Base Address + 0x0030, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_PRI_ALT_SET	[31:0]	RW	Returns the channel control data structure status, or selects the alternate data structure for the corresponding DMA channel. Read as: Bit [C] = 0 DMA channel C is using the primary data structure. Bit [C] = 1 DMA channel C is using the alternate data structure. Write as: Bit [C] = 0 No effect. Use the chnl_pri_alt_clr Register to set bit [C] to 0. Bit [C] = 1 Selects the alternate data structure for channel C.	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

			Writing to a bit where a DMA channel is not implemented has no effect.	

4.9.1.1.14 CHNL_PRI_ALT_CLR

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_PRI_ALT_CLR	[31:0]	W	Set the appropriate bit to select the primary data structure for the corresponding DMA channel. Write as: Bit [C] = 0 No effect. Use the chnl_pri_alt_set Register to select the alternate data structure. Bit [C] = 1 Selects the primary data structure for channel C. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.15 CHNL_PRIORITY_SET

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_PRIORITY_SET	[31:0]	RW	Returns the channel priority mask status, or sets the channel priority to high. Read as: Bit [C] = 0 DMA channel C is using the default priority level. Bit [C] = 1 DMA channel C is using a high priority level. Write as: Bit [C] = 0 No effect. Use the chnl_priority_clr Register to set channel C to the default priority level. Bit [C] = 1 Channel C uses the high priority level. Writing to a bit where a DMA channel is not implemented has no effect.	0x0000_0000

4.9.1.1.16 CHNL_PRIORITY_CLR

- Address: Base Address + 0x003C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CHNL_PRIORITY_CLR	[31:0]	W	Set the appropriate bit to select the default priority level for the specified DMA channel. Write as: Bit [C] = 0 No effect. Use the chnl_priority_set	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

			Register to set channel C to the high priority level. Bit [C] = 1 Channel C uses the default priority level. Writing to a bit where a DMA channel is not implemented has no effect.	

4.9.1.1.17 ERR_CLR

- Address: Base Address + 0x004C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:1]	RW	*Reserved	0x0000_0000
ERR_CLR	[0]	RW	Returns the status of dma_err, or sets the signal LOW. Read as: 0 = dma_err is LOW 1 = dma_err is HIGH. Write as: 0 = No effect, status of dma_err is unchanged. 1 = Sets dma_err LOW. For test purposes, use the err_	0x0000_0000

4.9.1.1.18 PERIPH_ID_4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
BLOCK_COUNT	[7:4]	R	The number of 4KB address blocks you require, to access the registers, expressed in powers of 2. These bits read back as 0x0.	0x0000_0000
JEP106_C_CODE	[3:0]	R	The JEP106 continuation code value represents how many 0x7F continuation characters occur in the manufacturer's identity code. These bits read back as 0x4.	0x0000_0004

4.9.1.1.19 PERIPH_ID_0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0030

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
PART_NUMBER0	[7:0]	R	These bits read back as 0x30	0x0000_0030

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4.9.1.1.20 PERIPH_ID_1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B2

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
JEP106_ID_3_0	[7:4]	R	JEP106 identity code [3:0]. These bits read back as 0xB because ARM is the designer of the peripheral.	0x0000_000B
PART_NUMBER1	[3:0]	R	These bits read back as 0x2.	0x0000_0002

4.9.1.1.21 PERIPH_ID_2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
REVISION	[7:4]	R	The revision status of the controller. For revision r0p0, these bits read back as 0x0.	0x0000_0000
JEDEC_USED	[3]	R	JEP106 identity code [3:0]. These bits read back as 0xB because ARM is the designer of the peripheral.	0x0000_0001
JEP106_ID_6_4	[2:0]	R	These bits read back as 0x2.	0x0000_0003

4.9.1.1.22 PERIPH_ID_3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
Reserved1	[7:4]	R	Reserved for future use. Reads are undefined.	0x0000_0000
MOD_NUMBER	[3:0]	R	The customer must update this field if they modify the RTL of the controller. ARM set this to 0x0.	0x0000_0000

4.9.1.1.23 PCELL_ID_0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
PCELL_ID_0	[7:0]	R	These bits read back as 0x0D	0x0000_000D

4.9.1.1.24 PCELL_ID_1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value

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Reserved0	[31:8]	R	*Reserved	0x0000_0000
PCELL_ID_1	[7:0]	R	These bits read back as 0xF0	0x0000_00F0

4.9.1.1.25 PCELL_ID_2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
PCELL_ID_2	[7:0]	R	These bits read back as 0x05	0x0000_0005

4.9.1.1.26 PCELL_ID_3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
PCELL_ID_3	[7:0]	R	These bits read back as 0xB1	0x0000_00B1

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4.10 Interrupts

4.10.1 Interrupt Assignment

S3FBP5A contains 97 interrupt request (IRQ) lines and one non-maskable interrupt (NMI) and one event signal. *Table 4-7* lists the interrupt assignments.

Table 4-7 Interrupt Assignments

IRQ/NMI	Device
NMI	Watchdog
0	UART0 interrupt
1	UART1 interrupt
2	UART2 interrupt
3	SPI0 interrupt
4	SPI1 interrupt
5	SPI2 interrupt
6	SPI3 interrupt
7	SPI4 interrupt
8	I2C0 interrupt
9	I2C1 interrupt
10	I2C2 interrupt
11	I2C3 interrupt
12	I2C4 interrupt
13	Timer0 interrupt
14	Timer1 interrupt
15	Timer2 interrupt
16	Timer3 interrupt
17	Timer4 interrupt
18	Timer5 interrupt
19	PWM Timer0 interrupt
20	PWM Timer1 interrupt
21	Dual Timer interrupt
22	AFE FIFO interrupt
23	DMA error interrupt
24	Embedded flash interrupt
25	FRC dual Timer interrupt
26	FPU interrupt
27	SRP user interrupt
28	SRP log interrupt
29	Reserved

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

IRQ/NMI	Device
30	AES interrupt
31-62	DMA Channel 0-31 done interrupts
63	GPIO port 0 (16 pins) combined interrupt
64	GPIO port 1 (16 pins) combined interrupt
65-80	GPIO port 0 0-15 interrupts
81-96	GPIO port 1 0-15 interrupts

4.10.2 Event

The Cortex-M4 processors have an RXEV input signal. If software uses the wait for event (WFE) instruction to put the processor to sleep, then an event received at RXEV wakes up the processor. The RXEV is connected to dma_done of the µDMA Controller. This enables the processor to wake up from WFE sleep when the DMA process finishes.

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4.11 IO Control

S3FBP5A have configurable IO drivers and receivers, supporting a wide variety of standard interface as *Figure 4-8*.

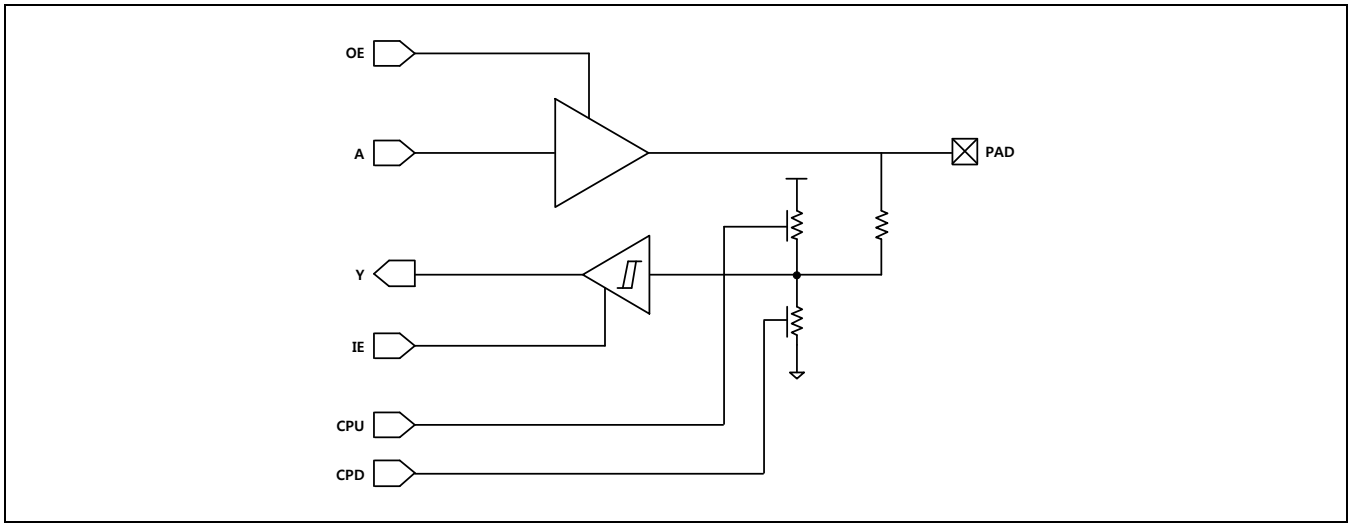


Figure 4-8 IO PAD control

Each PAD of S3FBP5A can be configured by register and configuration format is as below.

- IOCFGX Register Format

IOCFGX[5:0]		Description	
IOCFG [5]	Reserved	Reserved	Always 0
IOCFG [4]	Reserved	Reserved	Always 1
IOCFG [3]	CPD	Pull down enable	60K pull-down
IOCFG [2]	CPU	Pull up enable	60K pull-up
IOCFG [1]	IE	Input enable	0: input disable, 1: input enable
IOCFG [0]	OE	Output enable	0: output disable, 1: output enable

4.11.1 Register Description

4.11.1.1 Register Map Summary

- Base Address: 0x4001A500

Register	Offset	Description	Reset Value
IOCFG0	0x0020	IOCFG0	0x1A1A_1A16
IOCFG1	0x0024	IOCFG1	0x001A_1A1A
IOCFG2	0x0028	IOCFG2	0x1A13_1313
IOCFG3	0x002C	IOCFG3	0x1316_1716
IOCFG4	0x0030	IOCFG4	0x1A1A_1A16

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Register	Offset	Description	Reset Value
IOCFG5	0x0034	IOCFG5	0x1313_1313
IOCFG6	0x0038	IOCFG6	0x1313_1313
IOCFG7	0x003C	IOCFG7	0x1313_1313
IOCFG8	0x0040	IOCFG8	0x1313_1313
IOCFG9	0x0044	IOCFG9	0x1313_1313
IOCFGA	0x0048	IOCFGA	0x1313_1313
IOCFGB	0x004C	IOCFGB	0x1313_1313
IOCFGC	0x0050	IOCFGC	0x1313_1313

4.11.1.1.1 IOCFG0

- Address: Base Address + 0x0020, Reset Value = 0x1A1A_1A16

Name	Bit	Type	Description	Reset Value
SELBUCLK_CFG	[29:24]	RW	*IOCFG register for SELBUCLK pin	0x0000_001A
BUCLK_CFG	[21:16]	RW	*IOCFG register for BUCLK pin	0x0000_001A
FORCEGRSTN_CFG	[13:8]	RW	*IOCFG register for FORCEGRSTN pin	0x0000_001A
GRSTN_CFG	[5:0]	RW	*IOCFG register for GRSTN pin	0x0000_0016

4.11.1.1.2 IOCFG1

- Address: Base Address + 0x0024, Reset Value = 0x001A_1A1A

Name	Bit	Type	Description	Reset Value
FLASHPROG_CFG	[21:16]	RW	*IOCFG register for FLASHPROG pin	0x0000_001A
FLASHBOOT_CFG	[13:8]	RW	*IOCFG register for FLASHBOOT pin	0x0000_001A
DSPDBGEN_CFG	[5:0]	RW	*IOCFG register for DSPDBGEN pin	0x0000_001A

4.11.1.1.3 IOCFG2

- Address: Base Address + 0x0028, Reset Value = 0x1A13_1313

Name	Bit	Type	Description	Reset Value
FLASHMISO_CFG	[29:24]	RW	*IOCFG register for FLASHMISO pin	0x0000_001A
FLASHMOSI_CFG	[21:16]	RW	*IOCFG register for FLASHMOSI pin	0x0000_0013
FLASHCSN_CFG	[13:8]	RW	*IOCFG register for FLASHCSN pin	0x0000_0013
FLASHSCLK_CFG	[5:0]	RW	*IOCFG register for FLASHSCLK pin	0x0000_0013

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4.11.1.1.4 IOCFG3

- Address: Base Address + 0x002C, Reset Value = 0x1316_1716

Name	Bit	Type	Description	Reset Value
TDO_CFG	[29:24]	RW	*IOCFG register for TDO pin	0x0000_0013
TDI_CFG	[21:16]	RW	*IOCFG register for TDI pin	0x0000_0016
TMS_CFG	[13:8]	RW	*IOCFG register for TMS pin	0x0000_0017
TCK_CFG	[5:0]	RW	*IOCFG register for TCK pin	0x0000_0016

4.11.1.1.5 IOCFG4

- Address: Base Address + 0x0030, Reset Value = 0x1A1A_1A16

Name	Bit	Type	Description	Reset Value
SCANMODE_CFG	[29:24]	RW	*IOCFG register for SCANMODE pin	0x0000_001A
TESTEN_CFG	[21:16]	RW	*IOCFG register for TESTEN pin	0x0000_001A
TESTCLK_CFG	[13:8]	RW	*IOCFG register for TESTCLK pin	0x0000_001A
TRSTN_CFG	[5:0]	RW	*IOCFG register for TRSTN pin	0x0000_0016

4.11.1.1.6 IOCFG5

- Address: Base Address + 0x0034, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO03_CFG	[29:24]	RW	*IOCFG register for GPIO03 pin	0x0000_0013
GPIO02_CFG	[21:16]	RW	*IOCFG register for GPIO02 pin	0x0000_0013
GPIO01_CFG	[13:8]	RW	*IOCFG register for GPIO01 pin	0x0000_0013
GPIO00_CFG	[5:0]	RW	*IOCFG register for GPIO00 pin	0x0000_0013

4.11.1.1.7 IOCFG6

- Address: Base Address + 0x0038, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO07_CFG	[29:24]	RW	*IOCFG register for GPIO07 pin	0x0000_0013
GPIO06_CFG	[21:16]	RW	*IOCFG register for GPIO06 pin	0x0000_0013
GPIO05_CFG	[13:8]	RW	*IOCFG register for GPIO05 pin	0x0000_0013
GPIO04_CFG	[5:0]	RW	*IOCFG register for GPIO04 pin	0x0000_0013

4.11.1.1.8 IOCFG7

- Address: Base Address + 0x003C, Reset Value = 0x1313_1313

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Name	Bit	Type	Description	Reset Value
GPIO0B_CFG	[29:24]	RW	*IOCFG register for GPIO0B pin	0x0000_0013
GPIO0A_CFG	[21:16]	RW	*IOCFG register for GPIO0A pin	0x0000_0013
GPIO09_CFG	[13:8]	RW	*IOCFG register for GPIO09 pin	0x0000_0013
GPIO08_CFG	[5:0]	RW	*IOCFG register for GPIO08 pin	0x0000_0013

4.11.1.1.9 IOCFG8

- Address: Base Address + 0x0040, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO0F_CFG	[29:24]	RW	*IOCFG register for GPIO0F pin	0x0000_0013
GPIO0E_CFG	[21:16]	RW	*IOCFG register for GPIO0E pin	0x0000_0013
GPIO0D_CFG	[13:8]	RW	*IOCFG register for GPIO0D pin	0x0000_0013
GPIO0C_CFG	[5:0]	RW	*IOCFG register for GPIO0C pin	0x0000_0013

4.11.1.1.10 IOCFG9

- Address: Base Address + 0x0044, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO13_CFG	[29:24]	RW	*IOCFG register for GPIO13 pin	0x0000_0013
GPIO12_CFG	[21:16]	RW	*IOCFG register for GPIO12 pin	0x0000_0013
GPIO11_CFG	[13:8]	RW	*IOCFG register for GPIO11 pin	0x0000_0013
GPIO10_CFG	[5:0]	RW	*IOCFG register for GPIO10 pin	0x0000_0013

4.11.1.1.11 IOCFGA

- Address: Base Address + 0x0048, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO17_CFG	[29:24]	RW	*IOCFG register for GPIO17 pin	0x0000_0013
GPIO16_CFG	[21:16]	RW	*IOCFG register for GPIO16 pin	0x0000_0013
GPIO15_CFG	[13:8]	RW	*IOCFG register for GPIO15 pin	0x0000_0013
GPIO14_CFG	[5:0]	RW	*IOCFG register for GPIO14 pin	0x0000_0013

4.11.1.1.12 IOCFGB

- Address: Base Address + 0x004C, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO1B_CFG	[29:24]	RW	*IOCFG register for SELBUCLK pin	0x0000_0013

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Name	Bit	Type	Description	Reset Value
GPIO1A_CFG	[21:16]	RW	*IOCFG register for SELBUCLK pin	0x0000_0013
GPIO19_CFG	[13:8]	RW	*IOCFG register for SELBUCLK pin	0x0000_0013
GPIO18_CFG	[5:0]	RW	*IOCFG register for SELBUCLK pin	0x0000_0013

4.11.1.1.13 IOCFGC

- Address: Base Address + 0x0050, Reset Value = 0x1313_1313

Name	Bit	Type	Description	Reset Value
GPIO1F_CFG	[29:24]	RW	*IOCFG register for GPIO1F pin	0x0000_0013
GPIO1E_CFG	[21:16]	RW	*IOCFG register for GPIO1E pin	0x0000_0013
GPIO1D_CFG	[13:8]	RW	*IOCFG register for GPIO1D pin	0x0000_0013
GPIO1C_CFG	[5:0]	RW	*IOCFG register for GPIO1C pin	0x0000_0013

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4.12 General Purpose I/O

GPIO is a General Purpose I/O interface unit. It provides a 16 bits I/O interface per one GPIO port with the following properties:

- Programmable interrupt generation capability
- Bit masking support using address values
- Registers for alternate function switching with pin multiplexing support
- Thread safe operation by providing separate set and clear addresses for control registers
- Inputs are sampled using a double flip-flop to avoid meta-stability issues.

Figure 4-9 shows the control circuit and external interface of the GPIO.

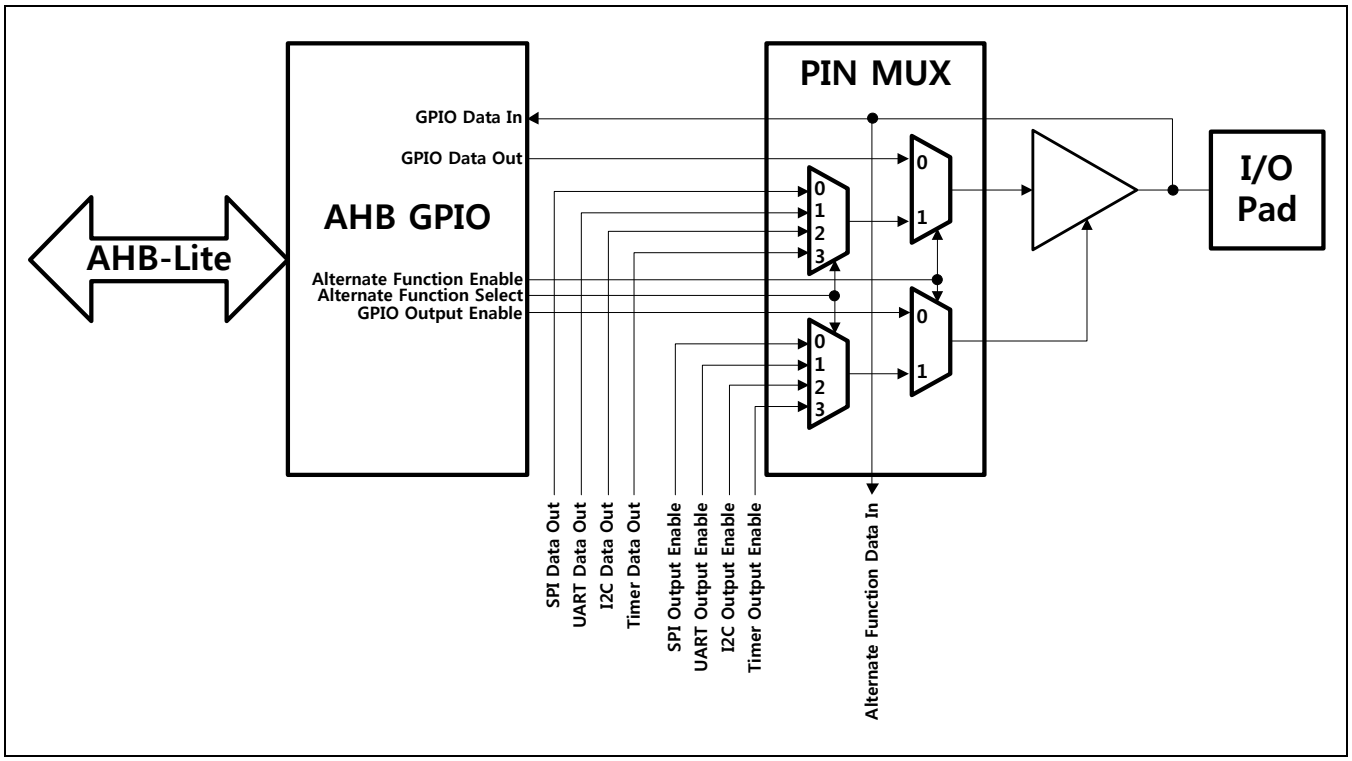


Figure 4-9 GPIO Control Circuit and External Interface

Interrupt generation

The AHB GPIO provides programmable interrupt generation features. Three registers control this, and each register has separate set and clear addresses. You can configure each bit of the I/O pins to generate interrupts based on these three registers, Interrupt enable[n], Interrupt polarity[n] and Interrupt type[n].

After an interrupt is triggered, the corresponding bit in the INTSTATUS Register is set. This also causes the corresponding bit of the GPIOINT[15:0] signal to be asserted. As a result, the combined interrupt signal, COMBINT, is also asserted. You can clear the interrupt status using an interrupt handler that writes 1 to the corresponding bit of the INTCLEAR Register, the same address as the INTSTATUS Register.

Masked access

The masked access feature permits individual bits or multiple bits, to be read from or written to in a single transfer.

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This avoids software-based read-modify-write operations that are not thread safe. With the masked access operations, the 16 bits I/O is divided into two halves, lower byte and the upper byte. The bit mask address spaces are defined as two arrays, each containing 256 words.

For example, if you must set bits[1:0] both to 1, and clear bits [7:6] in a single operation, you can carry out the write to the lower byte mask access address space. The required bit mask is 0xC3, and you can write the operation as MASKLOWBYTE[0xC3] = 0x03 as *Figure 4-10* shows.

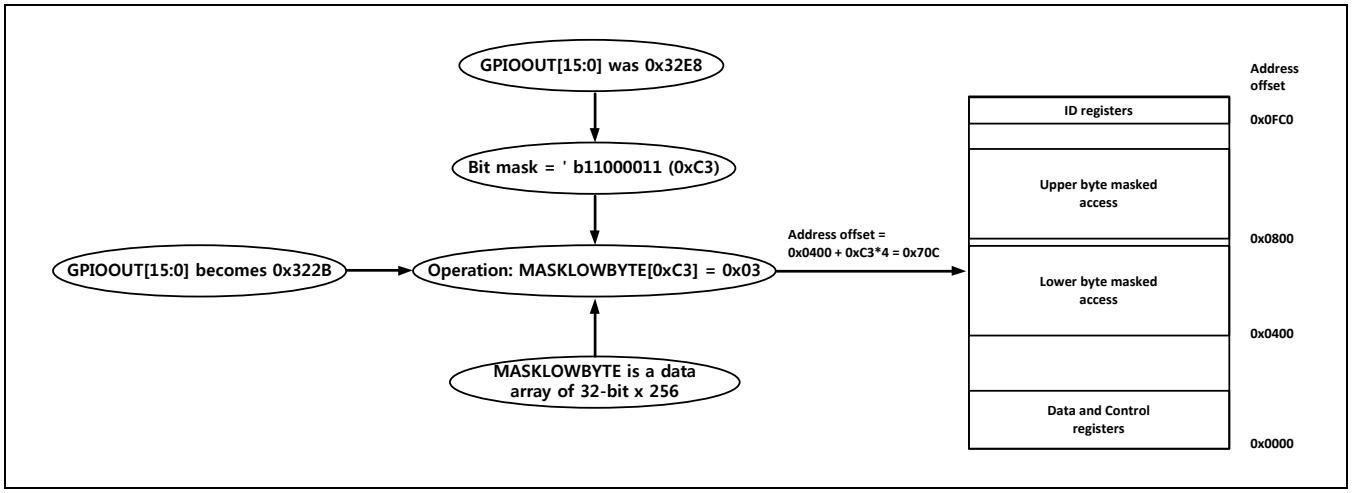


Figure 4-10 Masked access 1

Similarly, if you must update some of the bits in the upper eight bits of the GPIO port, you must use the MASKHIGHBYTE array as *Figure 4-11* shows.

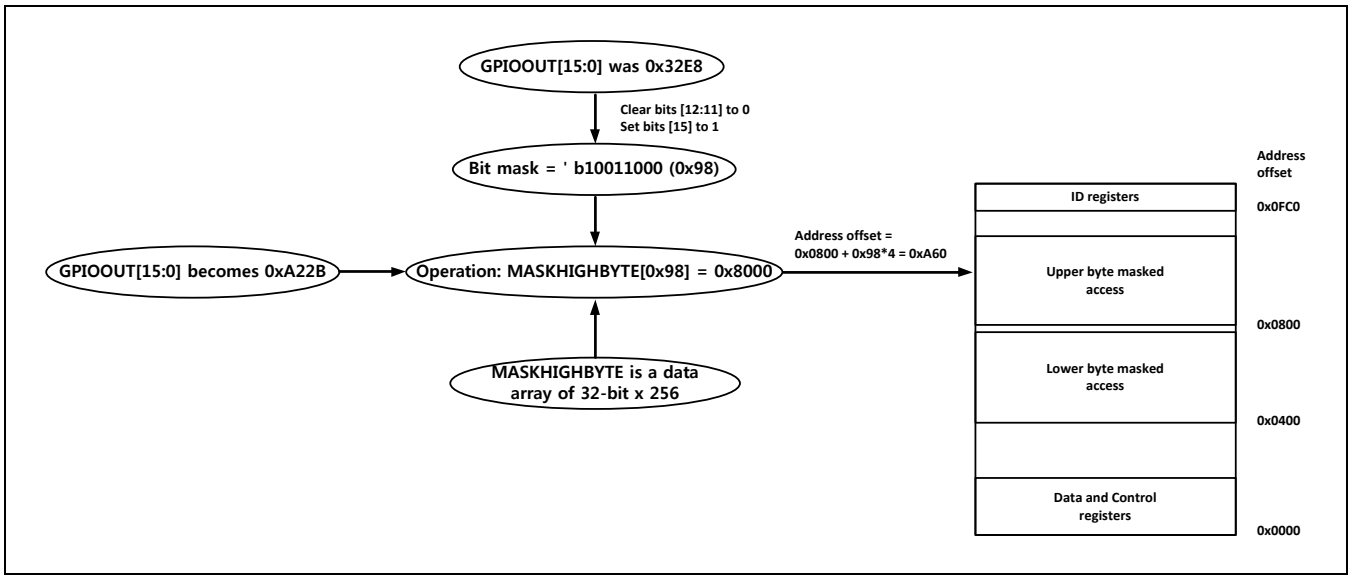


Figure 4-11 Masked access 2

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4.12.1 Register Description

4.12.1.1 Register Map Summary

- Base Address: 0x40020000

Register	Offset	Description	Reset Value
DATA	0x0000	* Data value: Read Sampled at pin Write To data output register Read back value is going through double flip-flop synchronization logic with a delay of two cycles	0x0000_0000
DATAOUT	0x0004	* Data output Register value: Read Current value of data output register Write To data output register	0x0000_0000
OUTENSET	0x0010	* Output enable set: Write 1: Set the output enable bit 0: No effect Read 0: Indicate the signal direction as input 1: Indicate the signal direction as output	0x0000_0000
OUTENCLR	0x0014	* Output enable clear: Write 1: Clear the output enable bit 0: No effect Read 0: Indicate the signal direction as input 1: Indicate the signal direction as output	0x0000_0000
ALTFUNCSET	0x0018	* Alternative function set: Write 1: Set the ALTFUNC bit 0: No effect Read 0: For I/O 1: For an alternate function	0x0000_0000
ALTFUNCCLR	0x001C	* Alternative function clear: Write 1: Clear the ALTFUNC bit 0: No effect Read 0: For I/O 1: For an alternate function	0x0000_0000
INTENSET	0x0020	* Interrupt enable set: Write 1: Set the enable bit 0: No effect Read 0: Interrupt disabled 1: Interrupt enabled	0x0000_0000
INTENCLR	0x0024	* Interrupt enable clear: Write 1: Clear the enable bit 0: No effect Read 0: Interrupt disabled 1: Interrupt enable	0x0000_0000

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Register	Offset	Description	Reset Value
INTTYPESET	0x0028	* Interrupt type set: Write 1: Set the interrupt type bit 0: No effect Read 0: For LOW or HIGH level 1: For falling edge or rising edge	0x0000_0000
INTTYPECLR	0x002C	* Interrupt type clear: Write 1: Clear the interrupt type bit 0: No effect Read 0: For LOW or HIGH level 1: For falling edge or rising edge	0x0000_0000
INTPOLSET	0x0030	* Polarity-level and edge IRQ configuration: Write 1: Set the interrupt polarity bit 0: No effect Read 0: For LOW level or falling edge 1: For HIGH level or rising edge	0x0000_0000
INTPOLCLR	0x0034	* Polarity-level and edge IRQ configuration: Write 1: Clear the interrupt polarity bit 0: No effect Read 0: For LOW level or falling edge 1: For HIGH level or rising edge	0x0000_0000
INTSTATUS_INTCLEAR	0x0038	* Write one to clear interrupt request: Write [15:0] IRQ status clear Register 1: To clear the interrupt request 0: No effect Read [15:0] IRQ status Register	0x0000_0000
ALTFUNCSEL	0x003C	* Alternative function select: Write every two bits to select one of four alternative functions	0x0000_0000
MASKLOWBYTE#	0x0400~ 0x07FF	* Lower eight bits masked access. Bits [9:2] of the address value are used as enable bit mask for the access: [15:8] Not used. Write is ignored and read as 0 [7:0] Data for lower byte access with [9:2] of address value use as enable mask for each bit	0x0000_0000
MASKHIGHBYTE#	0x0800 ~ 0x0BFF	* Higher eight bits masked access. Bits [9:2] of the address value are used as enable bit mask for the access: [15:8] Data for higher byte access with [9:2] of address value use as enable mask for each bit [7:0] Not used. Write is ignored and read as 0	0x0000_0000
PID4	0x0FD0	* Peripheral ID Register 4: [7:4] Block count [3:0] jep106_c_code	0x0000_0004

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Register	Offset	Description	Reset Value
PID5	0x0FD4	Peripheral ID Register 5	0x0000_0000
PID6	0x0FD8	Peripheral ID Register 6	0x0000_0000
PID7	0x0FDC	Peripheral ID Register 7	0x0000_0000
PID0	0x0FE0	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0020
PID1	0x0FE4	* Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8
PID2	0x0FE8	* Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B
PID3	0x0FEC	* Peripheral ID Register 3: [7:4] ECO revision number [3:0] Customer modification number	0x0000_0000
CID0	0x0FF0	Component ID Register 0	0x0000_000D
CID1	0x0FF4	Component ID Register 1	0x0000_00F0
CID2	0x0FF8	Component ID Register 2	0x0000_0005
CID3	0x0FFC	Component ID Register 3	0x0000_00B1

4.12.1.1.1 DATA

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
DATA	[15:0]	RW	* Data value: Read Sampled at pin Write To data output register Read back value is going through double flip-flop synchronization logic with a delay of two cycles	0x0000_0000

4.12.1.1.2 DATAOUT

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
DATAOUT	[15:0]	RW	* Data output Register value: Read Current value of data output register Write To data output register	0x0000_0000

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4.12.1.1.3 OUTENSET

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
OUTENSET	[15:0]	RW	* Output enable set: Write 1: Set the output enable bit 0: No effect Read 0: Indicate the signal direction as input 1: Indicate the signal direction as output	0x0000_0000

4.12.1.1.4 OUTENCLR

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
OUTENCLR	[15:0]	RW	* Output enable clear: Write 1: Clear the output enable bit 0: No effect Read 0: Indicate the signal direction as input 1: Indicate the signal direction as output	0x0000_0000

4.12.1.1.5 ALTFUNCSET

- Address: Base Address + 0x0018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALTFUNCSET	[15:0]	RW	* Alternative function set: Write 1: Set the ALTFUNC bit 0: No effect Read 0: For I/O 1: For an alternate function	0x0000_0000

4.12.1.1.6 ALTFUNCCLR

- Address: Base Address + 0x001C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALTFUNCCLR	[15:0]	RW	* Alternative function clear: Write 1: Clear the ALTFUNC bit 0: No effect Read 0: For I/O 1: For an alternate function	0x0000_0000

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4.12.1.1.7 INTENSET

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTENSET	[15:0]	RW	* Interrupt enable set: Write 1: Set the enable bit 0: No effect Read 0: Interrupt disabled 1: Interrupt enabled	0x0000_0000

4.12.1.1.8 INTENCLR

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTENCLR	[15:0]	RW	* Interrupt enable clear: Write 1: Clear the enable bit 0: No effect Read 0: Interrupt disabled 1: Interrupt enable	0x0000_0000

4.12.1.1.9 INTTYPESET

- Address: Base Address + 0x0028, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTTYPESET	[15:0]	RW	* Interrupt type set: Write 1: Set the interrupt type bit 0: No effect Read 0: For LOW or HIGH level 1: For falling edge or rising edge	0x0000_0000

4.12.1.1.10 INTTYPECLR

- Address: Base Address + 0x002C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTTYPECLR	[15:0]	RW	* Interrupt type clear: Write 1: Clear the interrupt type bit 0: No effect Read 0: For LOW or HIGH level 1: For falling edge or rising edge	0x0000_0000

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4.12.1.1.11 INTPOLSET

- Address: Base Address + 0x0030, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTPOLSET	[15:0]	RW	* Polarity-level and edge IRQ configuration: Write 1: Set the interrupt polarity bit 0: No effect Read 0: For LOW level or falling edge 1: For HIGH level or rising edge	0x0000_0000

4.12.1.1.12 INTPOLCLR

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTPOLCLR	[15:0]	RW	* Polarity-level and edge IRQ configuration: Write 1: Clear the interrupt polarity bit 0: No effect Read 0: For LOW level or falling edge 1: For HIGH level or rising edge	0x0000_0000

4.12.1.1.13 INTSTATUS_INTCLEAR

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTSTATUS_INTCLEAR	[15:0]	RW	* Write one to clear interrupt request: Write [15:0] IRQ status clear Register 1: To clear the interrupt request 0: No effect Read [15:0] IRQ status Register	0x0000_0000

4.12.1.1.14 ALTFUNCSEL

- Address: Base Address + 0x003C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALTFUNCSEL	[31:0]	RW	* Alternative function select: Write every two bits to select one of four alternative functions	0x0000_0000

4.12.1.1.15 MASKLOWBYTE#

- Address: Base Address + 0x0400~0x07FF, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
MASKLOWBYTE	[15:0]	RW	* Lower eight bits masked access. Bits [9:2] of the address value are used as enable bit mask for the access: [15:8] Not used. Write is ignored and read as 0 [7:0] Data for lower byte access with [9:2] of address value use as enable mask for each bit	0x0000_0000

4.12.1.1.16 MASKHIGHBYTE#

- Address: Base Address + 0x0800~0x0BFF, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MASKHIGHBYTE	[15:0]	RW	* Higher eight bits masked access. Bits [9:2] of the address value are used as enable bit mask for the access: [15:8] Data for higher byte access with [9:2] of address value use as enable mask for each bit [7:0] Not used. Write is ignored and read as 0	0x0000_0000

4.12.1.1.17 PID4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
PID4	[7:0]	R	* Peripheral ID Register 4: [7:4] Block count [3:0] jep106_c_code	0x0000_0004

4.12.1.1.18 PID5

- Address: Base Address + 0x0FD4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID5	[7:0]	R	Peripheral ID Register 5	0x0000_0000

4.12.1.1.19 PID6

- Address: Base Address + 0x0FD8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID6	[7:0]	R	Peripheral ID Register 6	0x0000_0000

4.12.1.1.20 PID7

- Address: Base Address + 0x0FDC, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
PID7	[7:0]	R	Peripheral ID Register 7	0x0000_0000

4.12.1.1.21 PID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0020

Name	Bit	Type	Description	Reset Value
PID0	[7:0]	R	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0020

4.12.1.1.22 PID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B8

Name	Bit	Type	Description	Reset Value
PID1	[7:0]	R	* Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8

4.12.1.1.23 PID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
PID2	[7:0]	R	* Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B

4.12.1.1.24 PID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID3	[7:0]	R	* Peripheral ID Register 3: [7:4] ECO revision number [3:0] Customer modification number	0x0000_0000

4.12.1.1.25 CID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
CID0	[7:0]	R	Component ID Register 0	0x0000_000D

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4.12.1.1.26 CID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
CID1	[7:0]	R	Component ID Register 1	0x0000_00F0

4.12.1.1.27 CID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
CID2	[7:0]	R	Component ID Register 2	0x0000_0005

4.12.1.1.28 CID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
CID3	[7:0]	R	Component ID Register 3	0x0000_00B1

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4.13 Timers

4.13.1 Timer

S3FBP5A utilizes Timer of CMSDK (ARM DDI 0479B r0p0).

The timer is a 32 bits down-counter with the following features:

- You can generate an interrupt request signal, **TIMERINT**, when the counter reaches 0. The interrupt request is held until it is cleared by writing to the INTCLEAR Register.
- If the timer count reaches 0, and at the same time, the software clears a previous interrupt status, then the interrupt status is set to 1.
- You can use the zero to one transition of the external input signal, **EXTIN**, as a timer enable.
- The external clock, **EXTIN**, must be slower than half of the peripheral clock because it is sampled by a double flip-flop and then goes through edge-detection logic when the external inputs act as a clock.
- A separate clock pin, **PCLKG**, for the APB register read or writes logic that permits the clock to peripheral register logic to stop when there is no APB activity.
- Component ID and a Peripheral ID Registers. These read-only ID registers are optional. You must modify the following in these registers:
  - Part number, 12 bits
  - JEDEC ID value, 7 bits
- The **ECOREVNUM** input signal is connected to the ECO revision number in the Peripheral ID Register 3.

Figure 4-12 shows Timer block diagram.

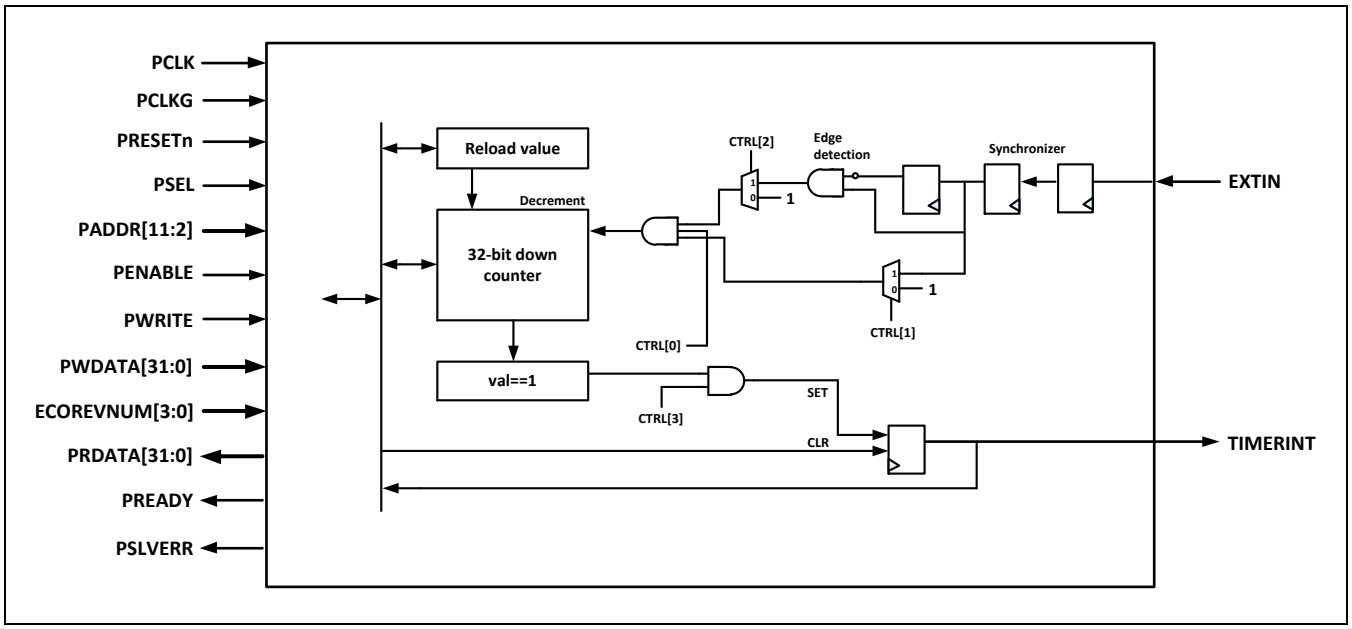


Figure 4-12 Timer block diagram

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.13.1.1 Register Description

4.13.1.1.1 Register Map Summary

- Base Address: 0x40000000

Register	Offset	Description	Reset Value
CTRL	0x0000	[3] Timer interrupt enable [2] Select external input as clock [1] Select external input as enable [0] Enable	0x0000_0000
VALUE	0x0004	Current value	0x0000_0000
RELOAD	0x0008	Reload value. A write to this register sets the current value	0x0000_0000
INTERRUPT	0x000C	Write: Timer interrupt. Write one to clear. Read: Timer interrupt status	0x0000_0000
PID4	0x0FD0	* Peripheral ID Register 4 [7:4] Block count. [3:0] jep106_c_code	0x0000_0004
PID5	0x0FD4	Peripheral ID Register 5	0x0000_0000
PID6	0x0FD8	Peripheral ID Register 6	0x0000_0000
PID7	0x0FDC	Peripheral ID Register 7	0x0000_0000
PID0	0x0FE0	* Peripheral ID Register 0 [7:0] Part number[7:0]	0x0000_0022
PID1	0x0FE4	* Peripheral ID Register 1 [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8
PID2	0x0FE8	* Peripheral ID Register 2: [7:4] Revision. [3] jedec_used. [2:0] jep106_id_6_4.	0x0000_000B
PID3	0x0FEC	* Peripheral ID Register 3 [7:4] ECO revision number [3:0] customer modification number	0x0000_0000
CID0	0x0FF0	Component ID Register 0	0x0000_000D
CID1	0x0FF4	Component ID Register 1	0x0000_00F0
CID2	0x0FF8	Component ID Register 2	0x0000_0005
CID3	0x0FFC	Component ID Register 3	0x0000_00B1

4.13.1.1.1.1 CTRL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
CTRL	[3:0]	RW	[3] Timer interrupt enable [2] Select external input as clock [1] Select external input as enable [0] Enable	0x0000_0000

4.13.1.1.1.2 VALUE

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
VALUE	[31:0]	RW	Current value	0x0000_0000

4.13.1.1.1.3 RELOAD

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RELOAD	[31:0]	RW	Reload value. A write to this register sets the current value	0x0000_0000

4.13.1.1.1.4 INTERRUPT

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTERRUPT	[0]	RW	Write: Timer interrupt. Write one to clear. Read: Timer interrupt status	0x0000_0000

4.13.1.1.1.5 PID4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
PID4	[7:0]	R	* Peripheral ID Register 4 [7:4] Block count. [3:0] jep106_c_code	0x0000_0004

4.13.1.1.1.6 PID5

- Address: Base Address + 0x0FD4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID5	[7:0]	R	Peripheral ID Register 5	0x0000_0000

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4.13.1.1.1.7 PID6

- Address: Base Address + 0x0FD8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID6	[7:0]	R	Peripheral ID Register 6	0x0000_0000

4.13.1.1.1.8 PID7

- Address: Base Address + 0x0FDC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID7	[7:0]	R	Peripheral ID Register 7	0x0000_0000

4.13.1.1.1.9 PID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0022

Name	Bit	Type	Description	Reset Value
PID0	[7:0]	R	* Peripheral ID Register 0 [7:0] Part number[7:0]	0x0000_0022

4.13.1.1.1.10 PID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B8

Name	Bit	Type	Description	Reset Value
PID1	[7:0]	R	* Peripheral ID Register 1 [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8

4.13.1.1.1.11 PID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
PID2	[7:0]	R	* Peripheral ID Register 2: [7:4] Revision. [3] jedec_used. [2:0] jep106_id_6_4.	0x0000_000B

4.13.1.1.1.12 PID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
PID3	[7:0]	R	* Peripheral ID Register 3 [7:4] ECO revision number [3:0] customer modification number	0x0000_0000

4.13.1.1.1.13 CID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
CID0	[7:0]	R	Component ID Register 0	0x0000_000D

4.13.1.1.1.14 CID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
CID1	[7:0]	R	Component ID Register 1	0x0000_00F0

4.13.1.1.1.15 CID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
CID2	[7:0]	R	Component ID Register 2	0x0000_0005

4.13.1.1.1.16 CID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
CID3	[7:0]	R	Component ID Register 3	0x0000_00B1

4.13.2 PWM Timer

Main feature of PWM (Pulse Width Modulation) Timer is the same as Timer above. PWM Timer additionally supports PWM function for motor driver.

PWM Timer supports following features:

- Period and high duration of PWM signal is programmable
- DC and AC motor signal can be selected
- Motor resonant frequency selection for AC motor

Figure 4-13 shows PWM Timer block diagram.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

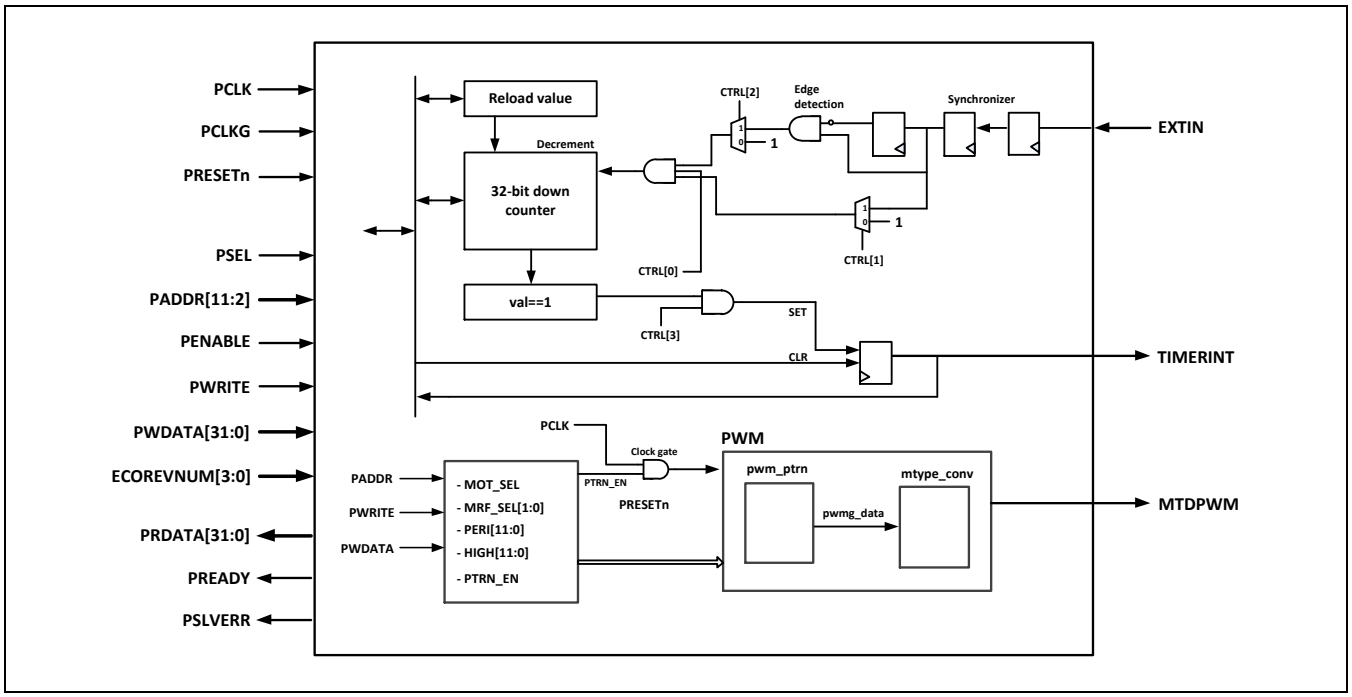


Figure 4-13 PWM Timer

4.13.2.1 Register Description

4.13.2.1.1 Register Map Summary

- Base Address: 0x40006000

Register	Offset	Description	Reset Value
CTRL	0x0000	[3] Timer interrupt enable [2] Select external input as clock [1] Select external input as enable [0] Enable	0x0000_0000
VALUE	0x0004	Current value	0x0000_0000
RELOAD	0x0008	Reload value. A write to this register sets the current value	0x0000_0000
INTERRUPT	0x000C	Write: Timer interrupt. Write one to clear. Read: Timer interrupt status	0x0000_0000
PWMCTRL	0x0C00	PWM control	0x0000_0005
PWM_PULSE	0x0C04	PWM pulse set	0x0045_B8B6
PID4	0x0FD0	* Peripheral ID Register 4 [7:4] Block count. [3:0] jep106_c_code	0x0000_0004
PID5	0x0FD4	Peripheral ID Register 5	0x0000_0000

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Register	Offset	Description	Reset Value
PID6	0x0FD8	Peripheral ID Register 6	0x0000_0000
PID7	0x0FDC	Peripheral ID Register 7	0x0000_0000
PID0	0x0FE0	* Peripheral ID Register 0 [7:0] Part number[7:0]	0x0000_0022
PID1	0x0FE4	* Peripheral ID Register 1 [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8
PID2	0x0FE8	* Peripheral ID Register 2 [7:4] Revision [3] jedec_used. [2:0] jep106_id_6_4	0x0000_000B
PID3	0x0FEC	* Peripheral ID Register 3 [7:4] ECO revision number [3:0] customer modification number	0x0000_0000
CID0	0x0FF0	Component ID Register 0	0x0000_000D
CID1	0x0FF4	Component ID Register 1	0x0000_00F0
CID2	0x0FF8	Component ID Register 2	0x0000_0005
CID3	0x0FFC	Component ID Register 3	0x0000_00B1

4.13.2.1.1.1 CTRL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CTRL	[3:0]	RW	[3] Timer interrupt enable [2] Select external input as clock [1] Select external input as enable [0] Enable	0x0000_0000

4.13.2.1.1.2 VALUE

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
VALUE	[31:0]	RW	Current value	0x0000_0000

4.13.2.1.1.3 RELOAD

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RELOAD	[31:0]	RW	Reload value. A write to this register sets the current	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			value	

4.13.2.1.1.4 INTERRUPT

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INTERRUPT	[0]	RW	Write: Timer interrupt. Write one to clear. Read: Timer interrupt status	0x0000_0000

4.13.2.1.1.5 PWMCTRL

- Address: Base Address + 0x0C00, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
PWM_EN	[3]	RW	PWM enable	0x0000_0000
MOT_SEL	[2]	RW	* Motor select 0: DC 1: AC	0x0000_0001
MRF_SEL	[1:0]	RW	* MRF(Motor Resonant Frequency) selection 00: 128 01: 256 10: 512 11: 1024 (Ex) When PWM frequency = 44.869 kHz and MRF selection = 01, MRF = 44.869 kHz / 256 = 175.27 Hz	0x0000_0001

4.13.2.1.1.6 PWM_PULSE

- Address: Base Address + 0x0C04, Reset Value = 0x0045_B8B6

Name	Bit	Type	Description	Reset Value
PWM_HIGH	[23:12]	RW	High duration of pulse. The minimum valid value for HIGH is 1	0x0000_045B
PWM_PERI	[11:0]	RW	Period of pulse. The minimum valid value for PERI is 2 (Ex) For 87 MHz(11.5ns) main clock when PERI = 1938 and HIGH = 969, PWM out = 1/(1938*11.5ns) = 44.869 KHz and PWM high duration = 969 * 11.5ns = 11.143us	0x0000_08B6

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4.13.2.1.1.7 PID4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
PID4	[7:0]	R	* Peripheral ID Register 4 [7:4] Block count. [3:0] jep106_c_code	0x0000_0004

4.13.2.1.1.8 PID5

- Address: Base Address + 0x0FD4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID5	[7:0]	R	Peripheral ID Register 5	0x0000_0000

4.13.2.1.1.9 PID6

- Address: Base Address + 0x0FD8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID6	[7:0]	R	Peripheral ID Register 6	0x0000_0000

4.13.2.1.1.10 PID7

- Address: Base Address + 0x0FDC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID7	[7:0]	R	Peripheral ID Register 7	0x0000_0000

4.13.2.1.1.11 PID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0022

Name	Bit	Type	Description	Reset Value
PID0	[7:0]	R	* Peripheral ID Register 0 [7:0] Part number[7:0]	0x0000_0022

4.13.2.1.1.12 PID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B8

Name	Bit	Type	Description	Reset Value
PID1	[7:0]	R	* Peripheral ID Register 1 [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8

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4.13.2.1.1.13 PID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
PID2	[7:0]	R	* Peripheral ID Register 2 [7:4] Revision [3] jedec_used. [2:0] jep106_id_6_4	0x0000_000B

4.13.2.1.1.14 PID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PID3	[7:0]	R	* Peripheral ID Register 3 [7:4] ECO revision number [3:0] customer modification number	0x0000_0000

4.13.2.1.1.15 CID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
CID0	[7:0]	R	Component ID Register 0	0x0000_000D

4.13.2.1.1.16 CID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
CID1	[7:0]	R	Component ID Register 1	0x0000_00F0

4.13.2.1.1.17 CID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
CID2	[7:0]	R	Component ID Register 2	0x0000_0005

4.13.2.1.1.18 CID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
CID3	[7:0]	R	Component ID Register 3	0x0000_00B1

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4.13.3 Dual-input Timer

4.13.3.1 Overview

S3FBP5A utilizes Dual-input Timer of CMSDK (ARM DDI 0479B r0p0).

The dual-input timer module is an APB slave module consists of two programmable 32 bits or 16 bits down-counters that can generate interrupts when they reach 0.

You can program a timer to implement:

- a 32 bits or a 16 bits counter, and one of the following timer modes:
  - free-running
  - periodic
  - one-shot

The operation of each timer module is identical. The dual-input timer module provides access to two interrupt-generating, programmable 32 bits Free-Running Counters (FRCs). The FRCs operate from a common timer clock, TIMCLK. Each FRC also has a prescaler that can divide down the TIMCLK rate by 1, 16, or 256. This enables the count rate for each FRC.

The system clock which is gated APB clock, PCLK, controls the programmable registers, and the second clock input which is APB clock, TIMCLK, drives the counter.

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Figure 4-14 shows a top-level block diagram of the timers.

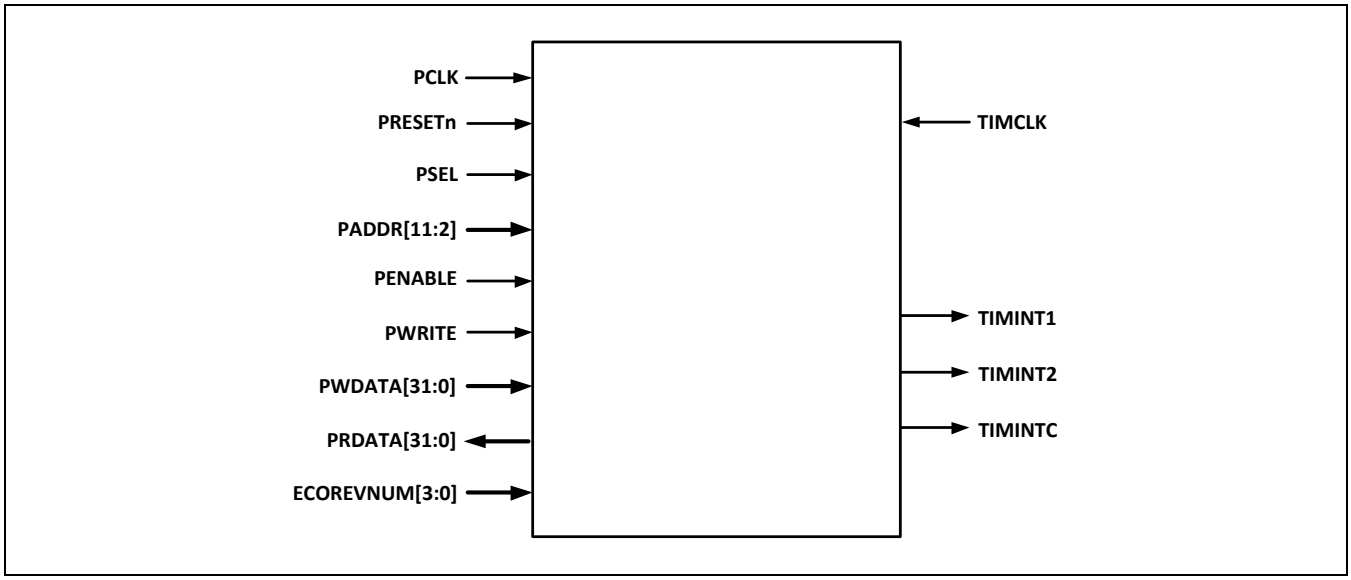


Figure 4-14 Dual-Input Timer Components

4.13.3.2 Functional Description

Two timers are defined by default, although you can easily expand this through extra instantiations of the FRC block. The same principle of simple expansion has been applied to the register configuration, to enable you to use more complex counters. For each timer, the following modes of operation are available:

Free-running mode

The counter wraps after reaching its zero value, and continues to count down from the maximum value. This is the default mode.

Periodic timer mode

The counter generates an interrupt at a constant interval, reloading the original value after wrapping past zero.

One-shot timer mode

The counter generates an interrupt once. When the counter reaches 0, it halts until you reprogram it. You can achieve this by either clearing the one-shot count bit in the control register, in which case the count proceeds according to the selection of Free-running or Periodic mode, or by writing a new value to the Load Value register.

4.13.3.3 Operation of Dual-Input Timer

Each timer has an identical set of registers as following Table shows. The operation of each timer is identical. The timer is loaded by writing to the load register and, if enabled, counts down to 0. When a counter is already running, writing to the load register causes the counter to immediately restart at the new value. Writing to the background load value has no effect on the current count. The counter continues to decrement to 0, and then recommences from the new load value, if in periodic mode, and one-shot mode is not selected.

When 0 is reached, an interrupt is generated. You can clear the interrupt by writing to the clear register. If you

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selected one-shot mode, the counter halts when it reaches 0 until you deselect one-shot mode, or write a new load value. Otherwise, after reaching a zero count, if the timer is operating in free-running mode, it continues to decrement from its maximum value. If you selected periodic timer mode, the timer reloads the count value from the Load Register and continues to decrement. In this mode, the counter effectively generates a periodic interrupt.

You select the mode using a bit in the Timer Control Register. See Table. At any point, you can read the current counter value from the Current Value Register. You can enable the counter using a bit in the Control Register.

At reset, the counter is disabled, the interrupt is cleared, and the load register is set to 0. The mode and prescale values are set to free-running, and clock divide of one respectively.

Figure 4-15 shows a block diagram of the free-running timer module.

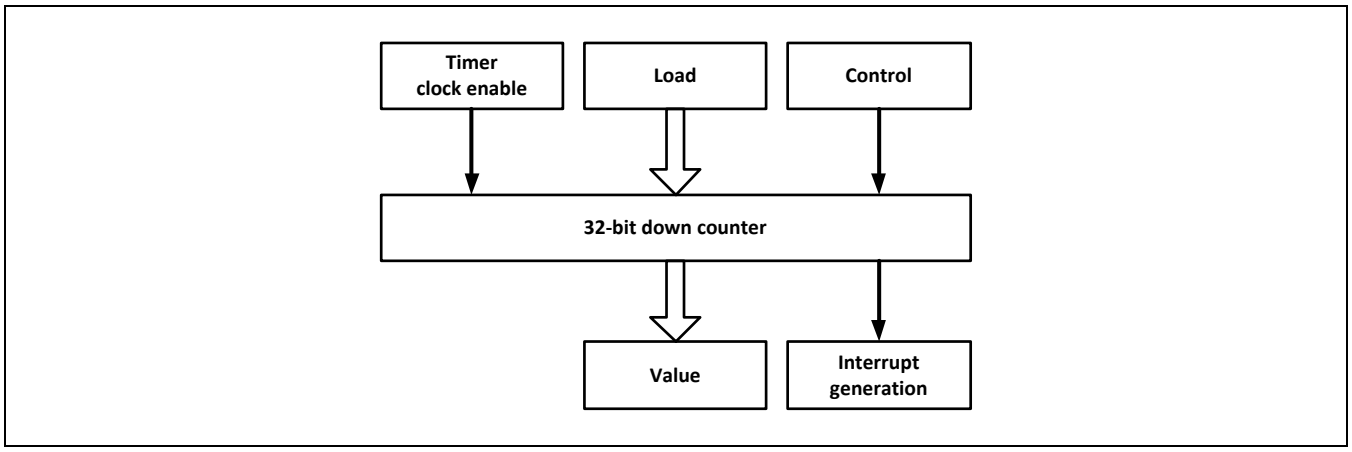


Figure 4-15 Free-Running Timer Block

The timer clock enable is generated by a prescale unit. The counter then uses the enable to create a clock with one of the following timings:

- the system clock
- the system clock divided by 16, generated by 4 bits of prescale
- the system clock divided by 256, generated by a total of 8 bits of prescale.

Figure 4-16 shows how the timer clock frequency is selected in the prescale unit. This enables you to clock the timer at different frequencies.

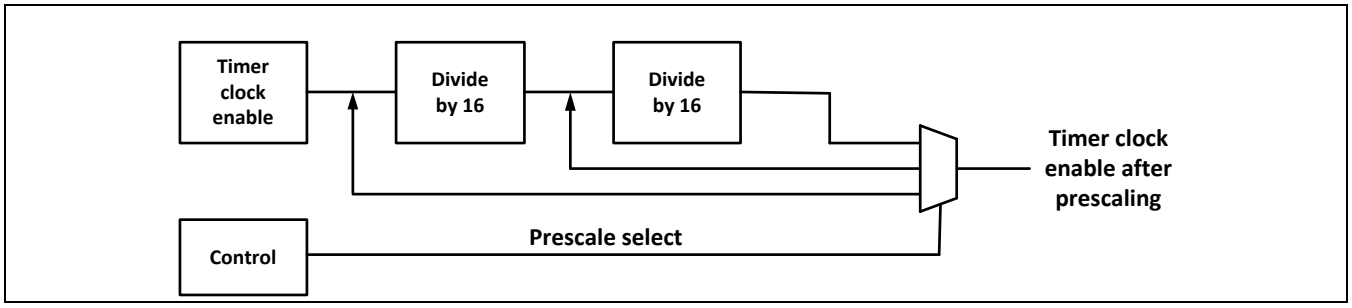


Figure 4-16 Prescale Clock Enable Generation

An interrupt is generated when the full 32 bits counter reaches 0, and is only cleared when the TimerXClear location is written to. A register holds the value until the interrupt is cleared. The most significant carry bit of the

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counter detects the counter reaching 0.

You can mask interrupts by writing 0 to the Interrupt Enable bit in the Control Register. You can read the following from status registers:

- raw interrupt status, prior to masking
- final interrupt status, after masking.

The interrupts from the individual counters, after masking, are logically ORed into a combined interrupt, TIMINTC, and this provides an additional output from the timer peripheral.

4.13.3.4 Clocking

The timers contain the PCLK and TIMCLK clock inputs, and the clocks are synchronous. PCLK is APB gated clock, and the register interface uses it. TIMCLK is the APB system clock. TIMCLK is the input to the prescale units and the decrementing counters.

4.13.3.5 Register Description

4.13.3.5.1 Register Map Summary

- Base Address: 0x40008000

Register	Offset	Description	Reset Value
TIMER1LOAD	0x0000	Timer Load value replaces the current count value	0x0000_0000
TIMER1VALUE	0x0004	It provides the current value of the decrementing counter	0xFFFF_FFFF
TIMER1CONTROL	0x0008	Timer1 control register	0x0000_0020
TIMER1INTCLR	0x000C	It clears the interrupt output from the counter	0x0000_0000
TIMER1RIS	0x0010	Raw interrupt status from the counter	0x0000_0000
TIMER1MIS	0x0014	Enabled interrupt status from the counter	0x0000_0000
TIMER1BGLOAD	0x0018	This is the value used to reload the counter when Periodic mode is enabled and the current count reaches 0	0x0000_0000
TIMER2LOAD	0x0020	Timer Load value replaces the current count value	0x0000_0000
TIMER2VALUE	0x0024	It provides the current value of the decrementing counter	0xFFFF_FFFF
TIMER2CONTROL	0x0028	The same as TIMER1CONTROL	0x0000_0020
TIMER2INTCLR	0x002C	It clears the interrupt output from the counter	0x0000_0000
TIMER2RIS	0x0030	Raw interrupt status from the counter	0x0000_0000
TIMER2MIS	0x0034	Enabled interrupt status from the counter	0x0000_0000
TIMER2BGLOAD	0x0038	This is the value used to reload the counter	0x0000_0000

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Register	Offset	Description	Reset Value
		when Periodic mode is enabled and the current count reaches 0	
TIMERPERIOHID4	0x0FD0	* Peripheral ID Register 4 : [7:4] Block count [3:0] jep106_c_code	0x0000_0004
TIMERPERIOHID5	0x0FD4	Peripheral ID Register 5	0x0000_0000
TIMERPERIPHID6	0x0FD8	Peripheral ID Register 6	0x0000_0000
TIMERPERIPHID7	0x0FDC	Peripheral ID Register 7	0x0000_0000
TIMERPERIPHID0	0x0FE0	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0023
TIMERPERIPHID1	0x0FE4	* Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8
TIMERPERIPHID2	0x0FE8	* Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B
TIMERPERIPHID3	0x0FEC	* Peripheral ID Register 3: [7:4] ECO revision number [3:0] customer modification number	0x0000_0000
TIMERPCELLID0	0x0FF0	Component ID Register 0	0x0000_000D
TIMERPCELLID1	0x0FF4	Component ID Register 1	0x0000_00F0
TIMERPCELLID2	0x0FF8	Component ID Register 2	0x0000_0005
TIMERPCELLID3	0x0FFC	Component ID Register 3	0x0000_00B1

4.13.3.5.1.1 TIMER1LOAD

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER1LOAD	[31:0]	RW	Timer Load value replaces the current count value	0x0000_0000

4.13.3.5.1.2 TIMER1VALUE

- Address: Base Address + 0x0004, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
TIMER1VALUE	[31:0]	R	It provides the current value of the decrementing counter	0xFFFF_FFFF

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.13.3.5.1.3 TIMER1CONTROL

- Address: Base Address + 0x0008, Reset Value = 0x0000_0020

Name	Bit	Type	Description	Reset Value
TimerEnable	[7]	RW	* Enable bit: 0: Timer disabled(default) 1: Timer enable	0x0000_0000
TimerMode	[6]	RW	* Mode bit: 0: Timer is in free-running mode(default) 1: Timer is in periodic mode	0x0000_0000
InterruptEnable	[5]	RW	* Interrupt Enable bit: 0: Timer Interrupt disabled 1: Timer Interrupt enable(default)	0x0000_0001
RESERVED	[4]	RW	Reserved bit. do not modify and ignore on read	0x0000_0000
TimerPre	[3:2]	RW	* Prescale bits: 00: 0 stages of prescale. clock is divided by 1 (default) 01: 4 stages of prescale. clock is divided by 16 10: 8 stages of prescale. clock is divided by 256 11: Undefined. do not use	0x0000_0000
TimerSize	[1]	RW	* Selects 16 or 32 bits counter operation: 0: 16 bits counter(default) 1: 32 bits counter	0x0000_0000
OneshortCount	[0]	RW	* Selects one-shot or wrapping counter mode: 0: Wrapping mode(default) 1: One-shot mode	0x0000_0000

4.13.3.5.1.4 TIMER1INTCLR

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER1INTCLR	[0]	W	It clears the interrupt output from the counter	0x0000_0000

4.13.3.5.1.5 TIMER1RIS

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER1RIS	[0]	R	Raw interrupt status from the counter	0x0000_0000

4.13.3.5.1.6 TIMER1MIS

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
TIMER1MIS	[0]	R	Enabled interrupt status from the counter	0x0000_0000

4.13.3.5.1.7 TIMER1BGLOAD

- Address: Base Address + 0x0018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER1BGLOAD	[31:0]	RW	This is the value used to reload the counter when Periodic mode is enabled and the current count reaches 0	0x0000_0000

4.13.3.5.1.8 TIMER2LOAD

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER2LOAD	[31:0]	RW	Timer Load value replaces the current count value	0x0000_0000

4.13.3.5.1.9 TIMER2VALUE

- Address: Base Address + 0x0024, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
TIMER2VALUE	[31:0]	R	It provides the current value of the decrementing counter	0xFFFF_FFFF

4.13.3.5.1.10 TIMER2CONTROL

- Address: Base Address + 0x0028, Reset Value = 0x0000_0020

Name	Bit	Type	Description	Reset Value
TIMER2CONTROL	[7:0]	RW	The same as TIMER1CONTROL	0x0000_0020

4.13.3.5.1.11 TIMER2INTCLR

- Address: Base Address + 0x002C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER2INTCLR	[0]	W	It clears the interrupt output from the counter	0x0000_0000

4.13.3.5.1.12 TIMER2RIS

- Address: Base Address + 0x0030, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
TIMER2RIS	[0]	R	Raw interrupt status from the counter	0x0000_0000

4.13.3.5.1.13 TIMER2MIS

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER2MIS	[0]	R	Enabled interrupt status from the counter	0x0000_0000

4.13.3.5.1.14 TIMER2BGLOAD

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER2BGLOAD	[31:0]	RW	This is the value used to reload the counter when Periodic mode is enabled and the current count reaches 0	0x0000_0000

4.13.3.5.1.15 TIMERPERIOHID4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
TIMERPERIOHID4	[7:0]	R	* Peripheral ID Register 4 : [7:4] Block count [3:0] jep106_c_code	0x0000_0004

4.13.3.5.1.16 TIMERPERIOHID5

- Address: Base Address + 0x0FD4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMERPERIOHID5	[7:0]	R	Peripheral ID Register 5	0x0000_0000

4.13.3.5.1.17 TIMERPERIPHID6

- Address: Base Address + 0x0FD8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMERPERIPHID6	[7:0]	R	Peripheral ID Register 6	0x0000_0000

4.13.3.5.1.18 TIMERPERIPHID7

- Address: Base Address + 0x0FDC, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
TIMERPERIPHID7	[7:0]	R	Peripheral ID Register 7	0x0000_0000

4.13.3.5.1.19 TIMERPERIPHID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0023

Name	Bit	Type	Description	Reset Value
TIMERPERIPHID0	[7:0]	R	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0023

4.13.3.5.1.20 TIMERPERIPHID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B8

Name	Bit	Type	Description	Reset Value
TIMERPERIPHID1	[7:0]	R	* Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8

4.13.3.5.1.21 TIMERPERIPHID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
TIMERPERIPHID2	[7:0]	R	* Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B

4.13.3.5.1.22 TIMERPERIPHID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMERPERIPHID3	[7:0]	R	* Peripheral ID Register 3: [7:4] ECO revision number [3:0] customer modification number	0x0000_0000

4.13.3.5.1.23 TIMERPCELLID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
TIMERPCELLID0	[7:0]	R	Component ID Register 0	0x0000_000D

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4.13.3.5.1.24 TIMERPCELLID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
TIMERPCELLID1	[7:0]	R	Component ID Register 1	0x0000_00F0

4.13.3.5.1.25 TIMERPCELLID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
TIMERPCELLID2	[7:0]	R	Component ID Register 2	0x0000_0005

4.13.3.5.1.26 TIMERPCELLID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
TIMERPCELLID3	[7:0]	R	Component ID Register 3	0x0000_00B1

4.13.4 Free-Running Counter (FRC)

4.13.4.1 Overview

FRC is a free running counter. It is different to FRCs in Dual-input Timer.

The FRC (Free Running Counter) is an APB slave block. It has 64bits increment free running counter and 64bits decrement alarm counter. FRC generates an interrupt request signal when the alarm counter reaches 0.

It is in always on power domain so that it always runs.

A block diagram of the FRC is shown in *Figure 4-17*. The APB interface generates decoded write signals for registers in the device. The control information written into the CONTROL and all written registers is synchronized to the LSOSC_CLK clock domain. All counters run by 32.768KHz LSOSC_CLK.

Because of synchronizer for asynchronous PCLK and LSOSC_CLK, write and read registers with 2 LSOSC_CLK period intervals.

Free running counter value can be read out. It can be used for time check.

Counter speed is programmable by CONTROL register.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

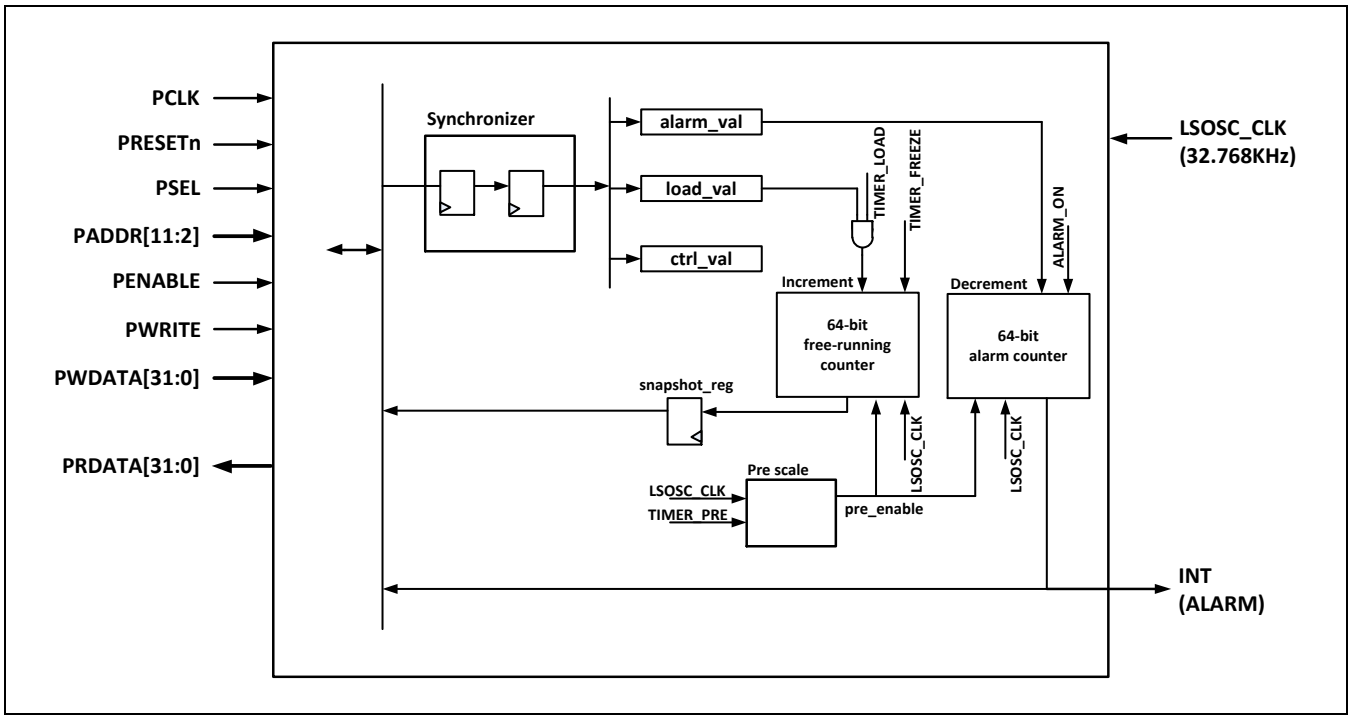


Figure 4-17 FRC Block Diagram

Figure 4-18 shows structure of 64bits value.

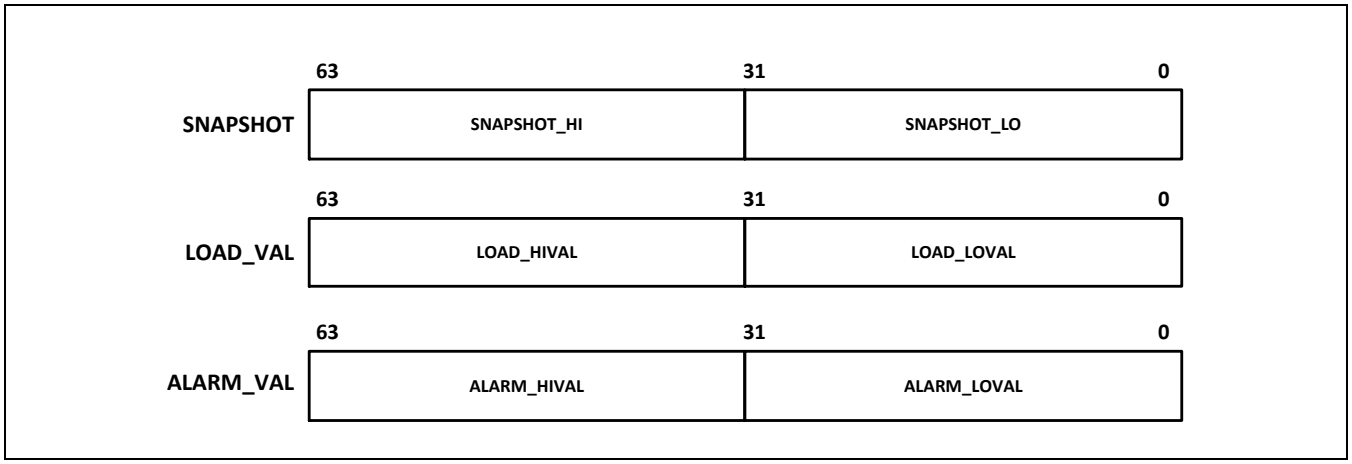


Figure 4-18 Structure of 64bits value

4.13.4.2 Functional Description

4.13.4.2.1 Get current value

Use snapshot(CONTROL[1]) function to get current timer value. The current value is captured at rising edge of snapshot. It means CONTROL[1] should be low before write 1 to CONTROL[1].

After capture the value, read 64bits value of SNAPSHOT_HI and SNAPSHOT_LO.

Figure 4-19 shows operations of getting current timer value two times.

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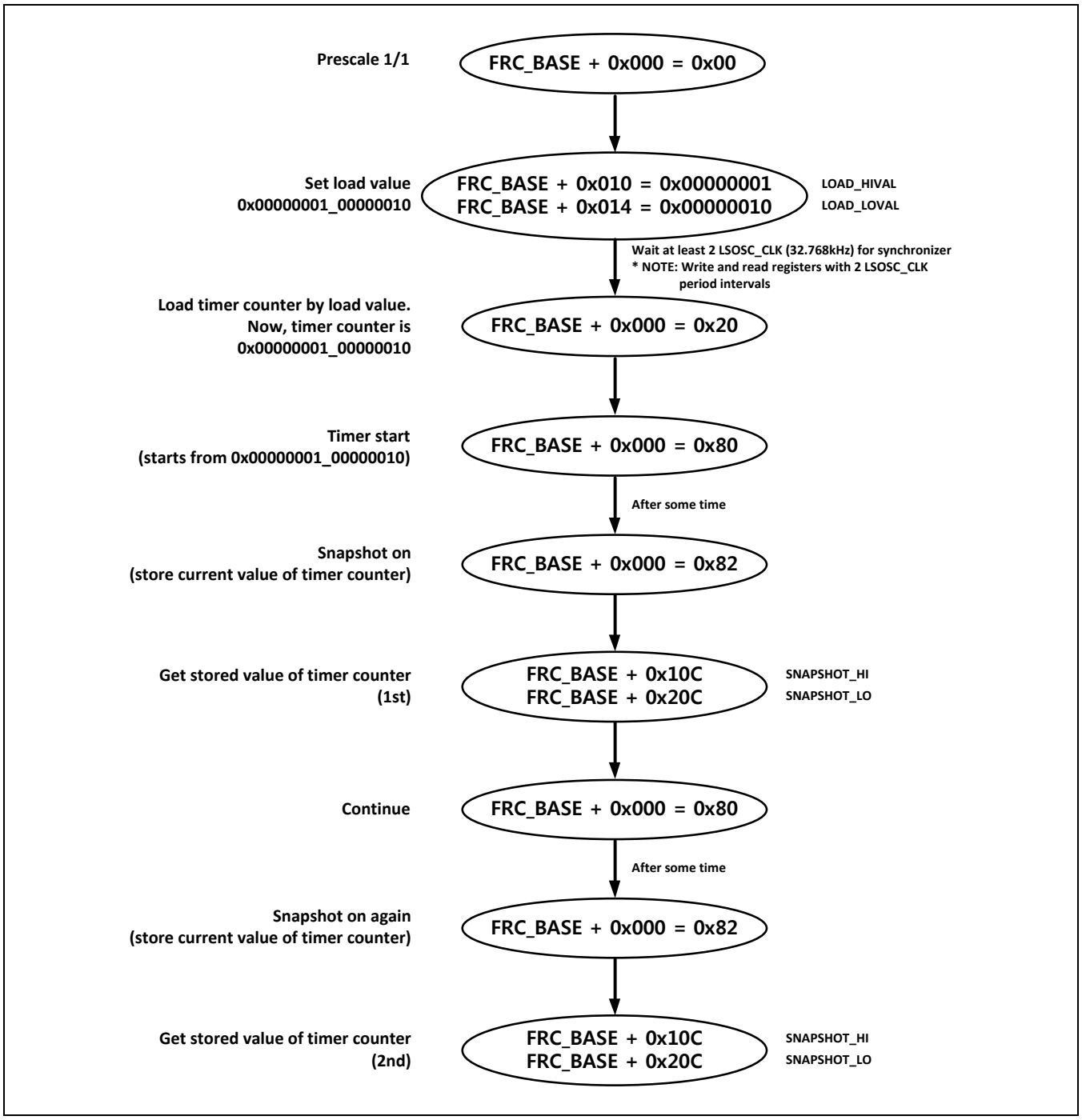


Figure 4-19 Operation of getting current timer value with snapshot function

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4.13.4.2.2 Alarm generation

Figure 4-20 shows operation of Alarm generation with 1/16 prescale LSOSC_CLK. It is possible to change timer/alarm counter speed with CONTROL[3:2] register.

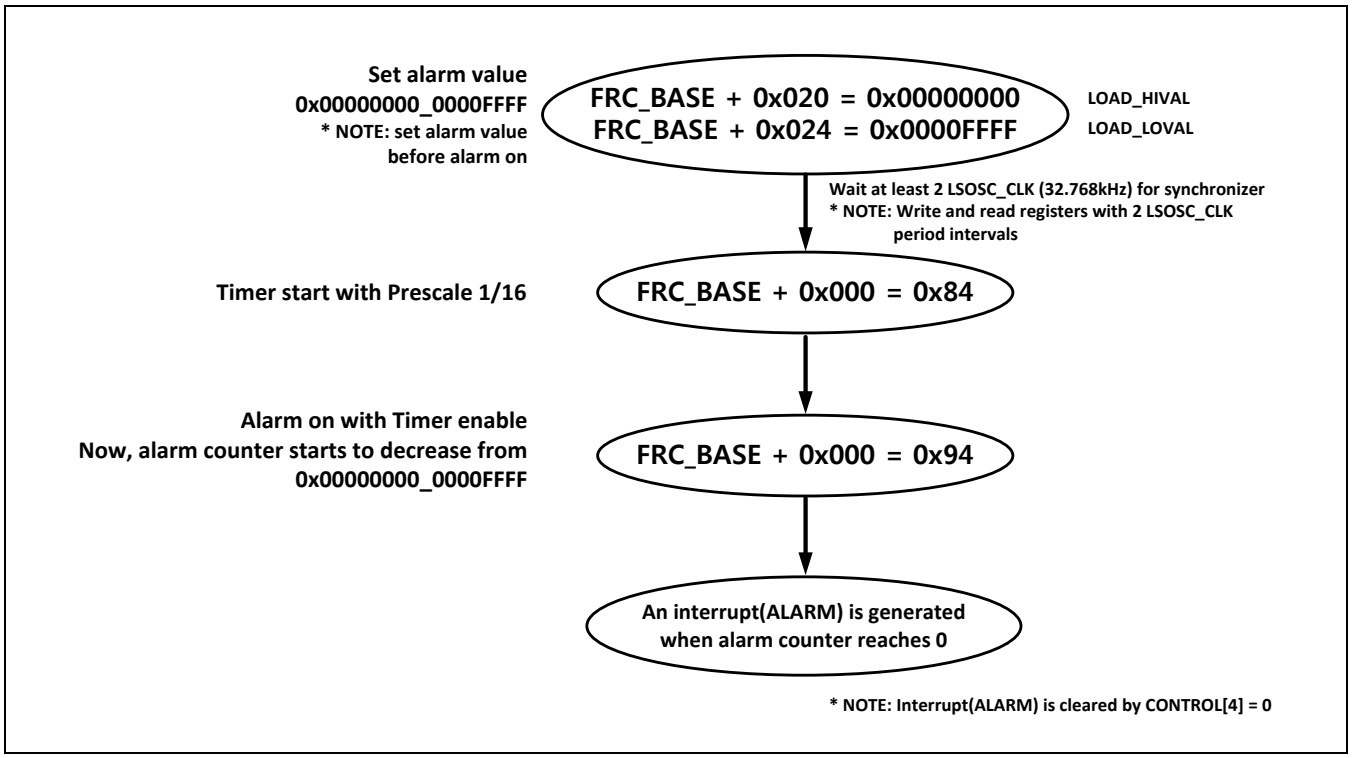


Figure 4-20 Operation of Alarm generation

Figure 4-21 shows timing diagram of ALARM generation with 0xF alarm value.

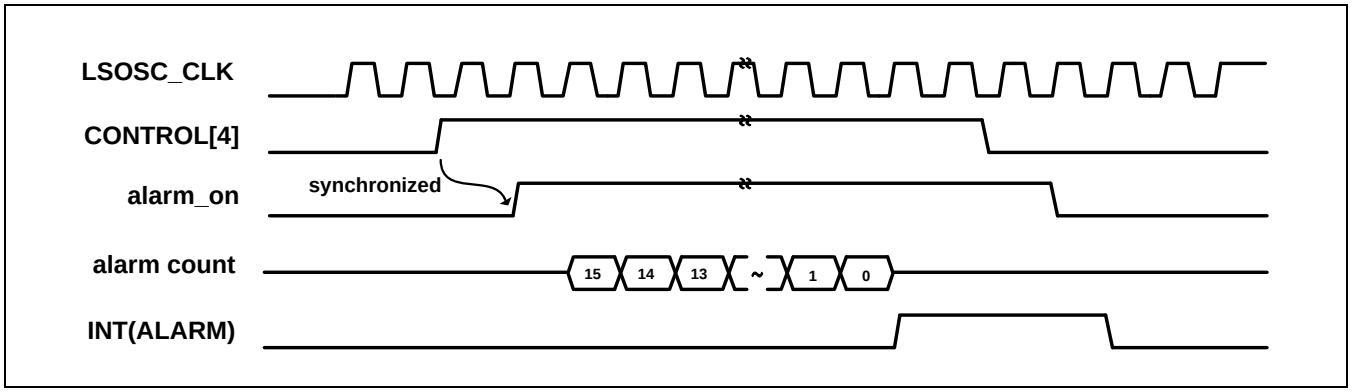


Figure 4-21 Timing diagram of Alarm generation

4.13.4.3 Register Description

4.13.4.3.1 Register Map Summary

- Base Address: 0x40019000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Register	Offset	Description	Reset Value
CONTROL	0x0000	Control register	0x0000_0000
LOAD_HIVAL	0x0010	A write to this register sets upper 32 bits of 64bits Load Value	0x0000_0000
LOAD_LOVAL	0x0014	A write to this register sets lower 32 bits of 64bits Load Value	0x0000_0000
ALARM_HIVAL	0x0020	A write to this register sets upper 32 bits of 64bits Alarm Value	0x0000_0000
ALARM_LOVAL	0x0024	A write to this register sets lower 32 bits of 64bits Alarm Value	0x0000_0000
SNAPSHOT_HI	0x010C	A read to this register gets upper 32 bits of stored 64bits Snapshot value	0x0000_0000
SNAPSHOT_LO	0x020C	A read to this register gets lower 32 bits of stored 64bits Snapshot value	0x0000_0000
ALARM	0x0400	Get ALARM status	0x0000_0000

4.13.4.3.1.1 CONTROL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TIMER_EN	[7]	RW	Timer enable. Enable free running counter.	0x0000_0000
TIMER_FREEZE	[6]	RW	Timer pause. Freeze free running counter and alarm counter	0x0000_0000
TIMER_LOAD	[5]	RW	Timer Load on. Set free running counter value to 64bit 'LOAD VALUE'. It is recommended to be low after setting. If not, counter value is loaded every clocks.	0x0000_0000
ALARM_ON	[4]	RW	Alarm On. Alarm counter is started to decrease from 64bit 'ALARM VALUE'. FRC generates an interrupt request signal when the alarm counter reaches 0.	0x0000_0000
TIMER_PRE	[3:2]	RW	* Timer counter speed 00: Clock is divided by 1 01: Clock is divided by 16 10: Clock is divided by 256 11: Clock is divided by 256	0x0000_0000
SNAPSHOT	[1]	RW	Snapshot on. Free running counter value will be stored at rising edge of this control register. After doing snapshot, you can read the stored 64bit value by 'SNAPSHOT_HI' and 'SNAPSHOT_LO'.	0x0000_0000
COUNTER_CLEAR	[0]	RW	Counter clear. Write 1 to clear 64bits free running counter, alarm counter and snapshot register. 'ALARM' signal will be cleared as well.	0x0000_0000

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4.13.4.3.1.2 LOAD_HIVAL

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LOAD_HIVAL	[31:0]	RW	A write to this register sets upper 32 bits of 64bits Load Value	0x0000_0000

4.13.4.3.1.3 LOAD_LOVAL

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LOAD_LOVAL	[31:0]	RW	A write to this register sets lower 32 bits of 64bits Load Value	0x0000_0000

4.13.4.3.1.4 ALARM_HIVAL

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALARM_HIVAL	[31:0]	RW	A write to this register sets upper 32 bits of 64bits Alarm Value	0x0000_0000

4.13.4.3.1.5 ALARM_LOVAL

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALARM_LOVAL	[31:0]	RW	A write to this register sets lower 32 bits of 64bits Alarm Value	0x0000_0000

4.13.4.3.1.6 SNAPSHOT_HI

- Address: Base Address + 0x010C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SNAPSHOT_HI	[31:0]	R	A read to this register gets upper 32 bits of stored 64bits Snapshot value	0x0000_0000

4.13.4.3.1.7 SNAPSHOT_LO

- Address: Base Address + 0x020C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SNAPSHOT_LO	[31:0]	R	A read to this register gets lower 32 bits of stored 64bits Snapshot value	0x0000_0000

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4.13.4.3.1.8 ALARM

- Address: Base Address + 0x0400, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ALARM	[0]	R	* Get ALARM status 0: Alarm is not occurred 1: Alarm is occurred Write 0 CONTROL[4] to turn off ALARM	0x0000_0000

4.13.5 Watchdog

S3FBP5A utilizes Watchdog of CMSDK (ARM DDI 0479B r0p0).

The watchdog module is based on a 32 bits down-counter that is initialized from the Reload Register, WDOGLOAD. The watchdog module generates a regular interrupt, WDOGINT, depending on a programmed value. The counter decrements by one on each positive clock edge of WDOGCLK.

The watchdog monitors the interrupt and asserts a reset WDOGRES signal, when the counter reaches 0, and the counter is stopped. On the next WDOGCLK clock edge, the counter is reloaded from the WDOGLOAD Register and the count-down sequence continues. If the interrupt is not cleared by the time that the counter next reaches 0, then the watchdog module reasserts the reset signal.

The watchdog module applies a reset to a system in the event of a software failure, providing a way to recover from software crashes. You can enable or disable the watchdog unit as required.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 4-22 shows watchdog module.

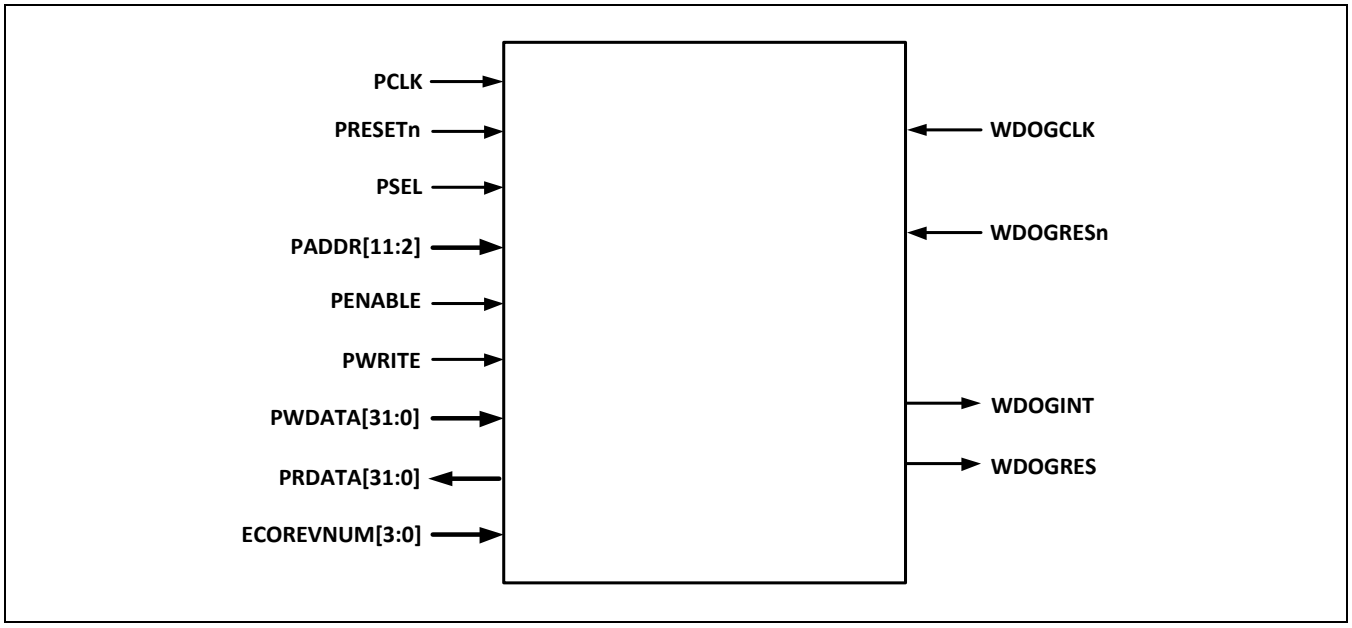


Figure 4-22 Watchdog Components

Figure 4-23 shows the flow diagram for the watchdog operation.

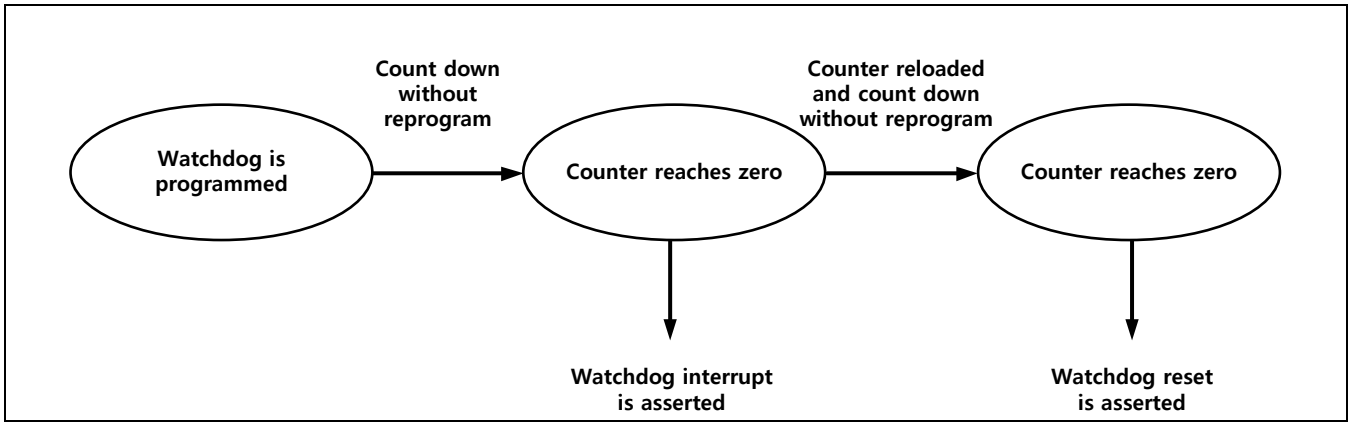


Figure 4-23 Watchdog operation flow diagram

4.13.5.1 Register Description

4.13.5.1.1 Register Map Summary

- Base Address: 0x40009000

Register	Offset	Description	Reset Value
WDOGLOAD	0x0000	The WDOGLOAD Register is a 32 bits register containing the value from which the counter is to decrement. When this register is written to the	0xFFFF_FFFF

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Register	Offset	Description	Reset Value
		count is immediately restarted from the new value. The minimum valid value for WDOGLOAD is 1	
WDOGVALUE	0x0004	The WDOGVALUE Register gives the current value of the decrementing counter	0xFFFF_FFFF
WDOGCONTROL	0x0008	The WDOGCONTROL Register is a R/W register that enables the software to control the watchdog unit.	0x0000_0000
WDOGINTCLR	0x000C	A write of any value to the WDOGINTCLR Register clears the watchdog interrupt and reloads the counter from the value in WDOGLOAD	0x0000_0000
WDOGRIS	0x0010	Raw interrupt status from the counter. This value is ANDed with the interrupt enable bit from the control register to create the masked interrupt that is passed to the interrupt output pin	0x0000_0000
WDOGMIS	0x0014	Enabled interrupt status from the counter. It indicates the masked interrupt status from the counter. This value is the logical AND of the raw interrupt status with the INTEN bit from the control register and is the same value that is passed to the interrupt output pin	0x0000_0000
WDOGLOCK	0x0C00	The WDOGLOCK Register is write-only. Using this register disables the write-accesses to all other registers. This is to prevent rogue software from disabling the watchdog functionality. Writing a value of 0x1ACCE551 enables write access to all other registers. Writing any other value disables write accesses.	0x0000_0000
WDOGPERIPHID4	0x0FD0	* Peripheral ID Register 4: [7:4] Block count [3:0] jep106_c_code	0x0000_0004
WDOGPERIPHID5	0x0FD4	Peripheral ID Register 5	0x0000_0000
WDOGPERIPHID6	0x0FD8	Peripheral ID Register 6	0x0000_0000
WDOGPERIPHID7	0x0FDC	Peripheral ID Register 7	0x0000_0000
WDOGPERIPHID0	0x0FE0	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0024
WDOGPERIPHID1	0x0FE4	Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8
WDOGPERIPHID2	0x0FE8	Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B

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Register	Offset	Description	Reset Value
WDOGPERIPHID3	0x0FEC	Peripheral ID Register 3: [7:4] ECO revision number [3:0] Customer modification number	0x0000_0000
WDOGPCELLID0	0x0FF0	Component ID Register 0	0x0000_000D
WDOGPCELLID1	0x0FF4	Component ID Register 1	0x0000_00F0
WDOGPCELLID2	0x0FF8	Component ID Register 2	0x0000_0005
WDOGPCELLID3	0x0FFC	Component ID Register 3	0x0000_00B1

4.13.5.1.1.1 WDOGLOAD

- Address: Base Address + 0x0000, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
WDOGLOAD	[31:0]	RW	The WDOGLOAD Register is a 32 bits register containing the value from which the counter is to decrement. When this register is written to the count is immediately restarted from the new value. The minimum valid value for WDOGLOAD is 1	0xFFFF_FFFF

4.13.5.1.1.2 WDOGVALUE

- Address: Base Address + 0x0004, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
WDOGVALUE	[31:0]	R	The WDOGVALUE Register gives the current value of the decrementing counter	0xFFFF_FFFF

4.13.5.1.1.3 WDOGCONTROL

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RESEN	[1]	RW	Enable watchdog reset output "WDOGRE". Acts as a mask for the reset output. Set HIGH to enable the reset and LOW to disable the reset	0x0000_0000
INTEN	[0]	RW	Enable the interrupt event "WDOGINT". Set HIGH to enable the counter and the interrupt and set LOW to disable the counter and interrupt. Reloads the counter from the value in WDOGLOAD when the interrupt is enabled and was previously disabled	0x0000_0000

4.13.5.1.1.4 WDOGINTCLR

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
WDOGINTCLR	[0]	W	A write of any value to the WDOGINTCLR Register clears the watchdog interrupt and reloads the counter from the value in WDOGLOAD	0x0000_0000

4.13.5.1.1.5 WDOGRIS

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGRIS	[0]	R	Raw interrupt status from the counter. This value is ANDed with the interrupt enable bit from the control register to create the masked interrupt that is passed to the interrupt output pin	0x0000_0000

4.13.5.1.1.6 WDOGMIS

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGMIS	[0]	R	Enabled interrupt status from the counter. It indicates the masked interrupt status from the counter. This value is the logical AND of the raw interrupt status with the INTEN bit from the control register and is the same value that is passed to the interrupt output pin	0x0000_0000

4.13.5.1.1.7 WDOGLOCK

- Address: Base Address + 0x0C00, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGLOCK	[31:0]	RW	The WDOGLOCK Register is write-only. Using this register disables the write-accesses to all other registers. This is to prevent rogue software from disabling the watchdog functionality. Writing a value of 0x1ACCE551 enables write access to all other registers. Writing any other value disables write accesses. A read from this register returns only the bottom bit: 0: Indicates that write access is enabled(not locked) 1: Indicates that write access is disabled(locked)	0x0000_0000

4.13.5.1.1.8 WDOGPERIPHID4

- Address: Base Address + 0x0FD0, Reset Value = 0x0000_0004

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Name	Bit	Type	Description	Reset Value
WDOGPERIPHID4	[7:0]	R	* Peripheral ID Register 4: [7:4] Block count [3:0] jep106_c_code	0x0000_0004

4.13.5.1.1.9 WDOGPERIPHID5

- Address: Base Address + 0x0FD4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID5	[7:0]	R	Peripheral ID Register 5	0x0000_0000

4.13.5.1.1.10 WDOGPERIPHID6

- Address: Base Address + 0x0FD8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID6	[7:0]	R	Peripheral ID Register 6	0x0000_0000

4.13.5.1.1.11 WDOGPERIPHID7

- Address: Base Address + 0x0FDC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID7	[7:0]	R	Peripheral ID Register 7	0x0000_0000

4.13.5.1.1.12 WDOGPERIPHID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0024

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID0	[7:0]	R	* Peripheral ID Register 0: [7:0] Part number[7:0]	0x0000_0024

4.13.5.1.1.13 WDOGPERIPHID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_00B8

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID1	[7:0]	R	Peripheral ID Register 1: [7:4] jep106_id_3_0 [3:0] Part number[11:8]	0x0000_00B8

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.13.5.1.1.14 WDOGPERIPHID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_000B

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID2	[7:0]	R	Peripheral ID Register 2: [7:4] Revision [3] jedec_used [2:0] jep106_id_6_4	0x0000_000B

4.13.5.1.1.15 WDOGPERIPHID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
WDOGPERIPHID3	[7:0]	R	Peripheral ID Register 3: [7:4] ECO revision number [3:0] Customer modification number	0x0000_0000

4.13.5.1.1.16 WDOGPCELLID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
WDOGPCELLID0	[7:0]	R	Component ID Register 0	0x0000_000D

4.13.5.1.1.17 WDOGPCELLID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
WDOGPCELLID1	[7:0]	R	Component ID Register 1	0x0000_00F0

4.13.5.1.1.18 WDOGPCELLID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
WDOGPCELLID2	[7:0]	R	Component ID Register 2	0x0000_0005

4.13.5.1.1.19 WDOGPCELLID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
WDOGPCELLID3	[7:0]	R	Component ID Register 3	0x0000_00B1

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4.14 Universal Asynchronous Receiver Transmitter (UART)

4.14.1 Overview

It supports three independent UART channels with asynchronous and serial input/output ports for general purpose.

Refer to the following features:

- Channel 0 to 2 with 64-byte FIFO
- All channels support Interrupt-based operation
- All channels support DMA-based or Interrupt-based operation
- All channels support Auto Flow Control with nRTS and nCTS
- Supports Handshake Transmit/Receive

Each UART channel contains two FIFOs to receive and transmit data, a Baud-rate generator, a Transmitter, a Receiver and a Control Unit. The Baud-rate generator uses EXT_UCLK. The Transmitter and the Receiver contain FIFOs and data shifters. The data that is transmitted is written to Tx FIFO, and copied to the transmit shifter. The data is then shifted out by the transmit data pin (TxDn). The received data is shifted from the Receive Data Pin (RxDn), and copied to Rx FIFO from the shifter.

Figure 4-24 shows the UART module, *Figure 4-25* shows a top-level block diagram of the UART and *Figure 4-26* shows Data flow of UART.

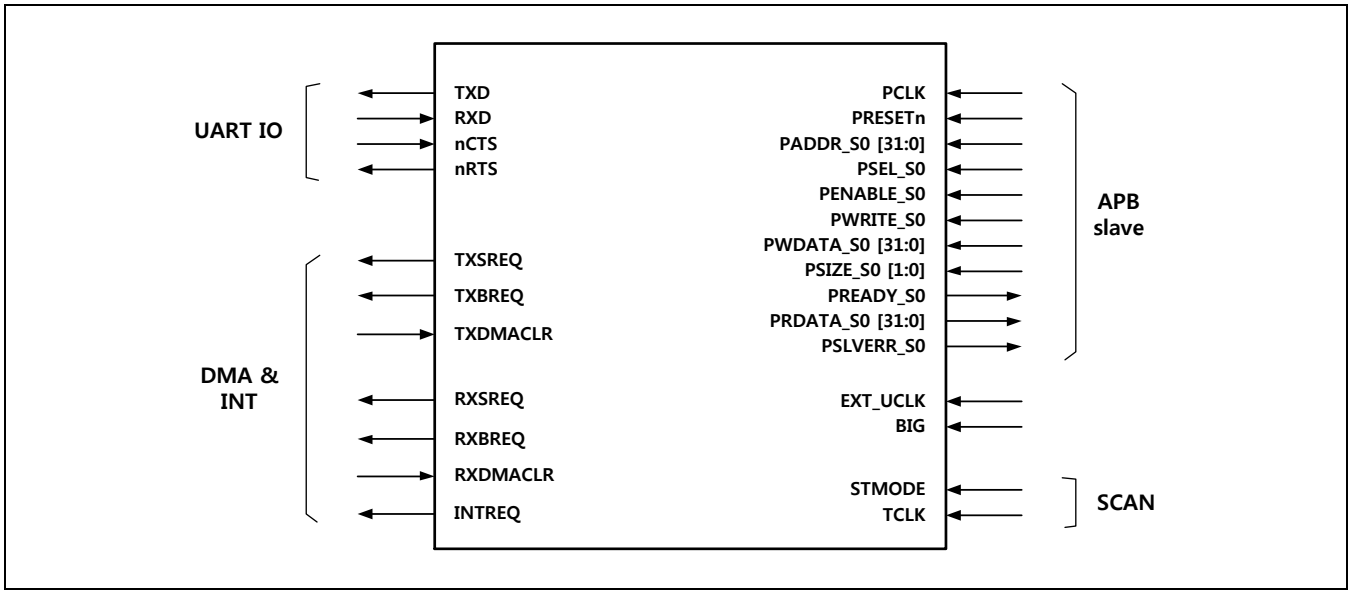


Figure 4-24 UART Components

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

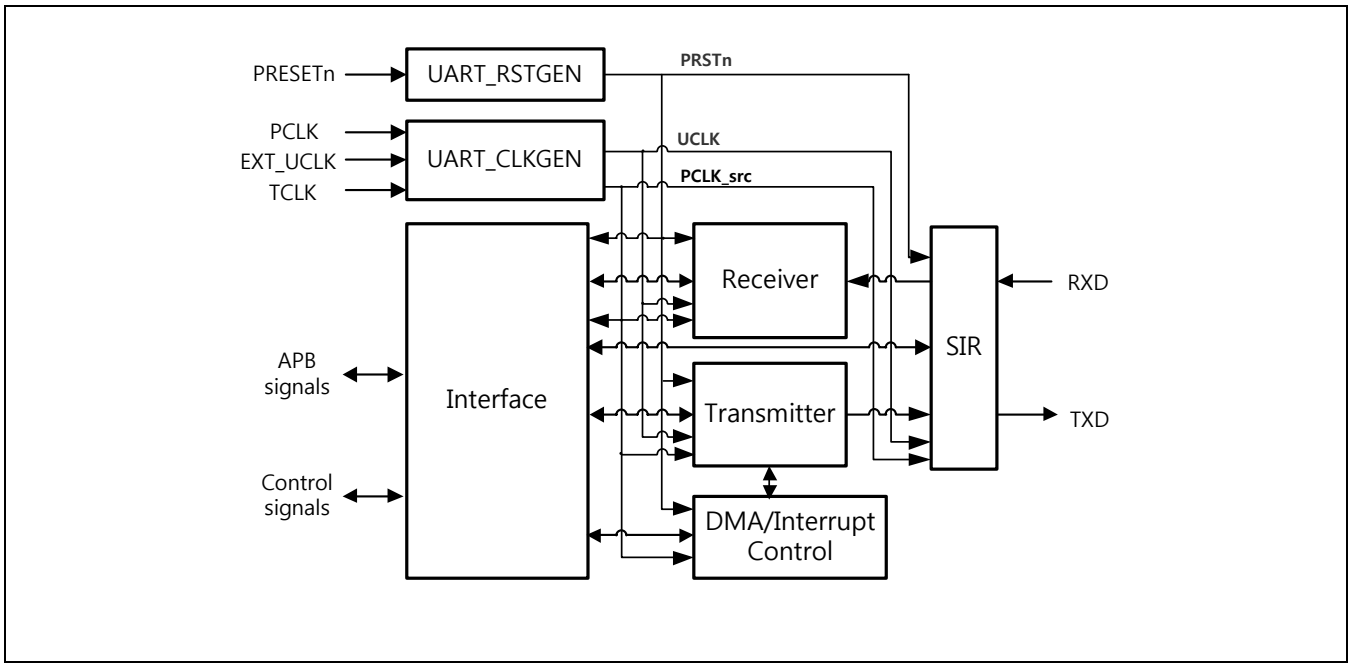


Figure 4-25 UART Block diagram

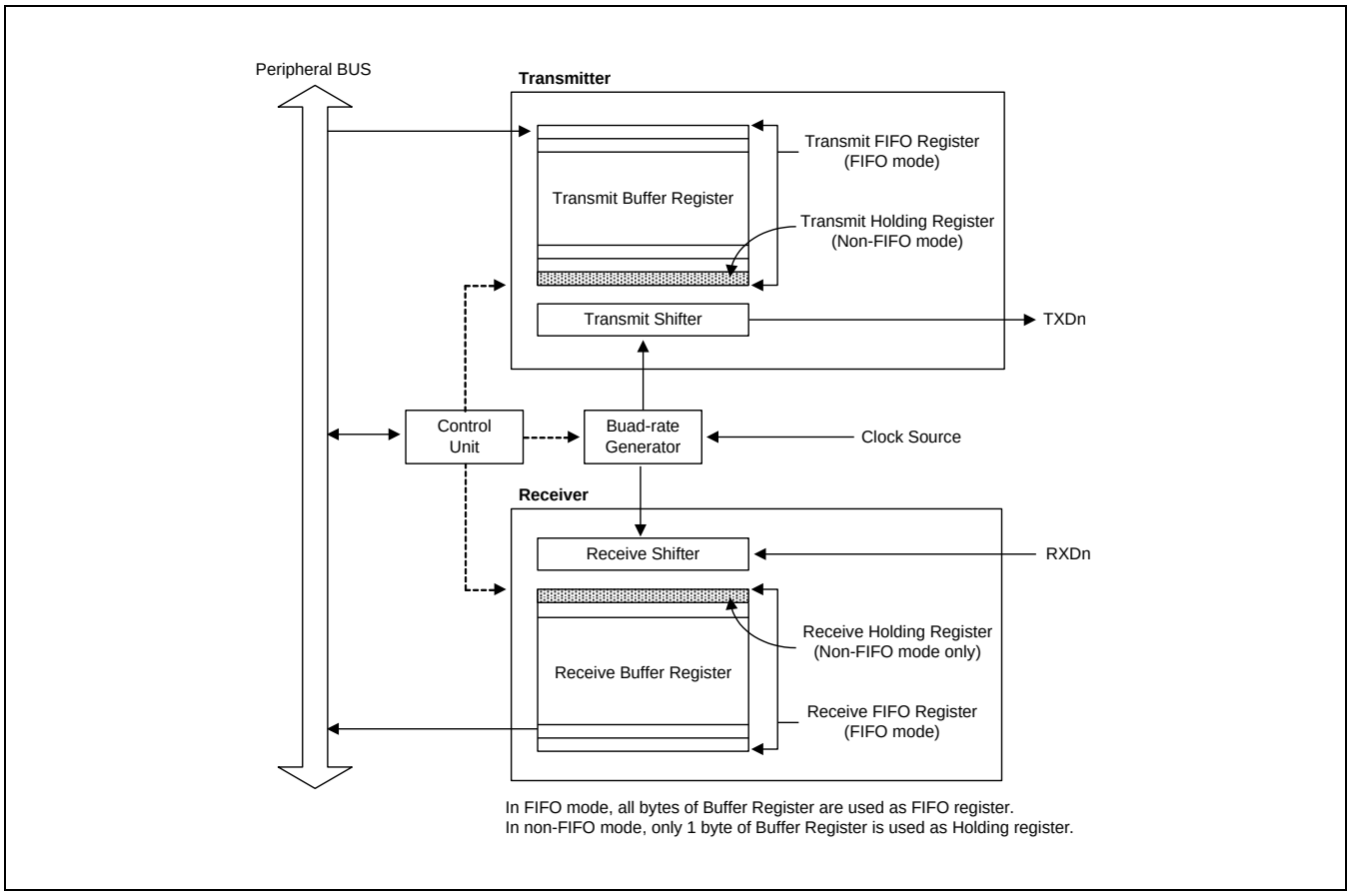


Figure 4-26 Data flow of UART

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4.14.2 Functional Description

This section describes UART operations, such as Data Transmission, Data Reception, Interrupt/DMA Request generation and Auto Flow Control.

4.14.2.1 Data Transmission

The data frame for transmission is programmable. It consists of a start bit, five to eight data bits, an optional parity bit, and one to two stop bits, specified by the line control register (ULCONn). The transmitter can also produce a break condition that forces the serial output to logic 0 state, for one frame transmission time. This block transmits the break signals, after the present transmission word is transmitted completely. After the break signal transmission, the transmitter continuously transmits data to Tx FIFO (Tx holding register, in case of Non-FIFO mode).

4.14.2.2 Data Reception

Similar to data transmission, the data frame for reception is also programmable. It consists of a start bit, five to eight data bits, an optional parity bit, and one to two stop bits in the Line Control Register (ULCONn). The receiver detects Overrun Error, Parity Error, Frame Error and Break Condition. Each of this error sets an error flag.

- Overrun Error indicates that new data has overwritten the old data before it was read
- Parity Error indicates that the receiver has detected an unexpected parity condition
- Frame Error indicates that the received data does not have a valid stop bit
- Break Condition indicates that the RxDn input is held in the logic 0 state for more than one frame transmission time

Receive time-out condition occurs when the Rx FIFO is not empty in the FIFO mode and no more data is received during the frame time specified in UCON.

4.14.2.3 Auto Flow Control (AFC)

In AFC, the nRTS signal depends on the condition of the receiver, whereas the nCTS signal controls the operation of transmitter. The transmitter of UART transfers the data to FIFO when nCTS signals are activated (in AFC, nCTS signals indicates that FIFO of other UART is ready to receive data). Before UART receives data, the nRTS signals must be activated when its Receive FIFO has more than 2-byte as spare. The nRTS signals must be inactivated when its Receive FIFO has less than 1-byte as spare (in AFC, the nRTS signals indicate that its own Receive FIFO is ready to receive data).

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 4-27 illustrates the UART AFC Interface.

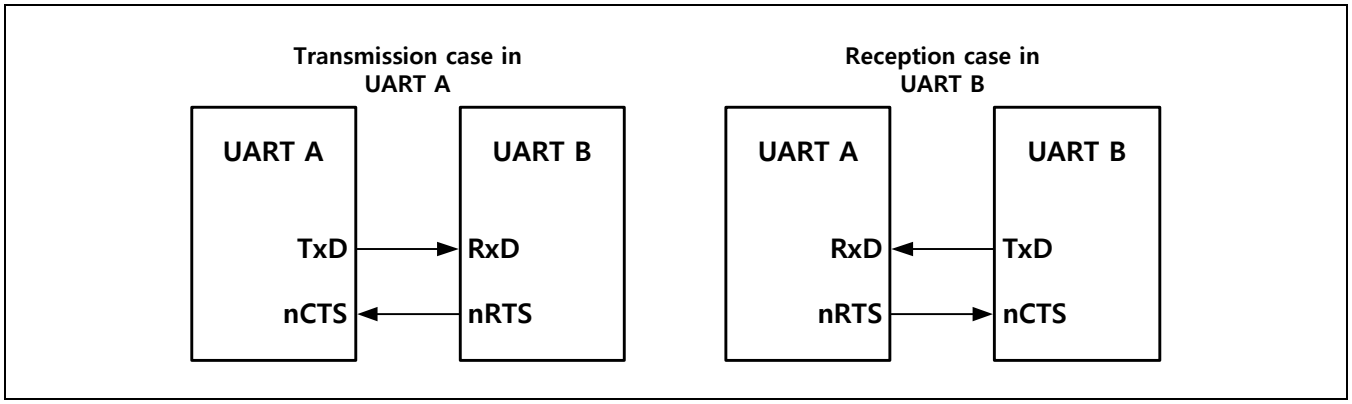


Figure 4-27 UART AFC Interface

4.14.2.4 Interrupt/DMA Request Generation

Each UART comprises of seven status (Tx/Rx/Error) signals: Overrun Error, Parity Error, Frame Error, Break, Receive Buffer Data Ready, Transmit Buffer Empty, and Transmit Shifter Empty. These conditions are indicated by the corresponding UART status register (UTRSTATn/UERSTATn).

The Overrun Error, Parity Error, Frame Error and Break Condition specify the receive error status. When the receive-error-status-interrupt-enable bit is set to "1" in the control register (UCONn), the receive error status generates receive-error-status-interrupt. When a receive-error-status-interrupt-request is detected, you can identify the source of interrupt by reading the value of UERSTATn.

When the receiver transfers data of the receive shifter to the receive FIFO register in FIFO mode, and the amount of received data is greater than or equal to the Rx FIFO Trigger Level, Rx interrupt is generated if Receive mode in control register (UCONn) is set to "1" (Interrupt request or Polling mode).

In Non-FIFO mode, transferring the data of receive shifter to receive holding register causes Rx interrupt in the Interrupt request and Polling modes.

When the transmitter transfers data from its transmit FIFO register to transmit shifter, and the amount of data left in transmit FIFO is less than or equal to the Tx FIFO Trigger Level, Tx interrupt is generated (provided Transmit mode in control register is selected as Interrupt request or Polling mode). In Non-FIFO mode, transferring the data from transmit holding register to transmit shifter, causes Tx interrupt in the Interrupt request and Polling mode.

The Tx interrupt is always requested when the amount of data in the transmit FIFO is less than the trigger level. This indicates that an interrupt is requested when you enable the Tx interrupt, unless you fill the Tx buffer. Therefore, It is recommended to fill the Tx buffer first and then enable the Tx interrupt.

The interrupt controllers are categorized as the level-triggered type. You must set the interrupt type as "Level" when you program the UART control registers.

In this situation, when Receive and Transmit modes in control register are selected as DMAn request mode, the DMAn request occurs instead of Rx or Tx interrupt.

In DMA mode, size of transfer data is greater than FIFO size. After transferred all data, DMA done will be occurred in DMAC. Then, change Tx Mode to INT/Polling mode so that Tx REQ is not occurred.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Table 4-8 describes the interrupts in connection with FIFO.

Table 4-8 Interrupts in Connection with FIFO

Type	FIFO Mode	Non-FIFO Mode
Rx interrupt	Generated when Rx FIFO count is greater than or equal to the trigger level of received FIFO. Generated when the amount of data in FIFO does not reach Rx FIFO trigger level and does not receive any data during the time specified (receive time out) in UCON.	Generated by receive holding register whenever receive buffer becomes full.
Tx interrupt	Generated when Tx FIFO count is less than or equal to the trigger level of transmit FIFO (Tx FIFO trigger Level).	Generated by transmit holding register whenever transmit buffer becomes empty.
Error interrupt	Generated when Frame Error, Parity Error, or Break Signal are detected. Generated if UART receives new data when Rx FIFO is full (overrun error).	Generated by all errors. However, when another error occurs at the same time, only one interrupt is generated.

4.14.2.5 UART Error Status FIFO

UART contains the error status FIFO besides the Rx FIFO register. The Error Status FIFO indicates the date that is received with an error, among FIFO registers. An error interrupt is issued only when the data that contains error is ready to read out. To clear the error status FIFO, URXHn with an error and UERSTATn must be read out.

For example, it is assumed that the UART Rx FIFO receives A, B, C, D, and E characters sequentially, and the Frame Error occurs when receiving "B" and the Parity Error occurs when receiving "D".

The actual UART receive error does not generate any error interrupt, because the character, which was received with an error was not read. The error interrupt occurs when the character is read out.

Table 4-9 describes Error interrupt.

Table 4-9 Error Interrupt

Time	Sequence Flow	Error Interrupt	Note
#0	When no character is read out	–	–
#1	A, B, C, D, and E is received	–	–
#2	After A is read out	Frame error (in B) interrupt occurs	The "B" has to be read out
#3	After B is read out	–	–
#4	After C is read out	Parity error (in D) interrupt occurs	The "D" has to be read out
#5	After D is read out	–	–
#6	After E is read out	–	–

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 4-28 illustrates the UART Error Status FIFO.

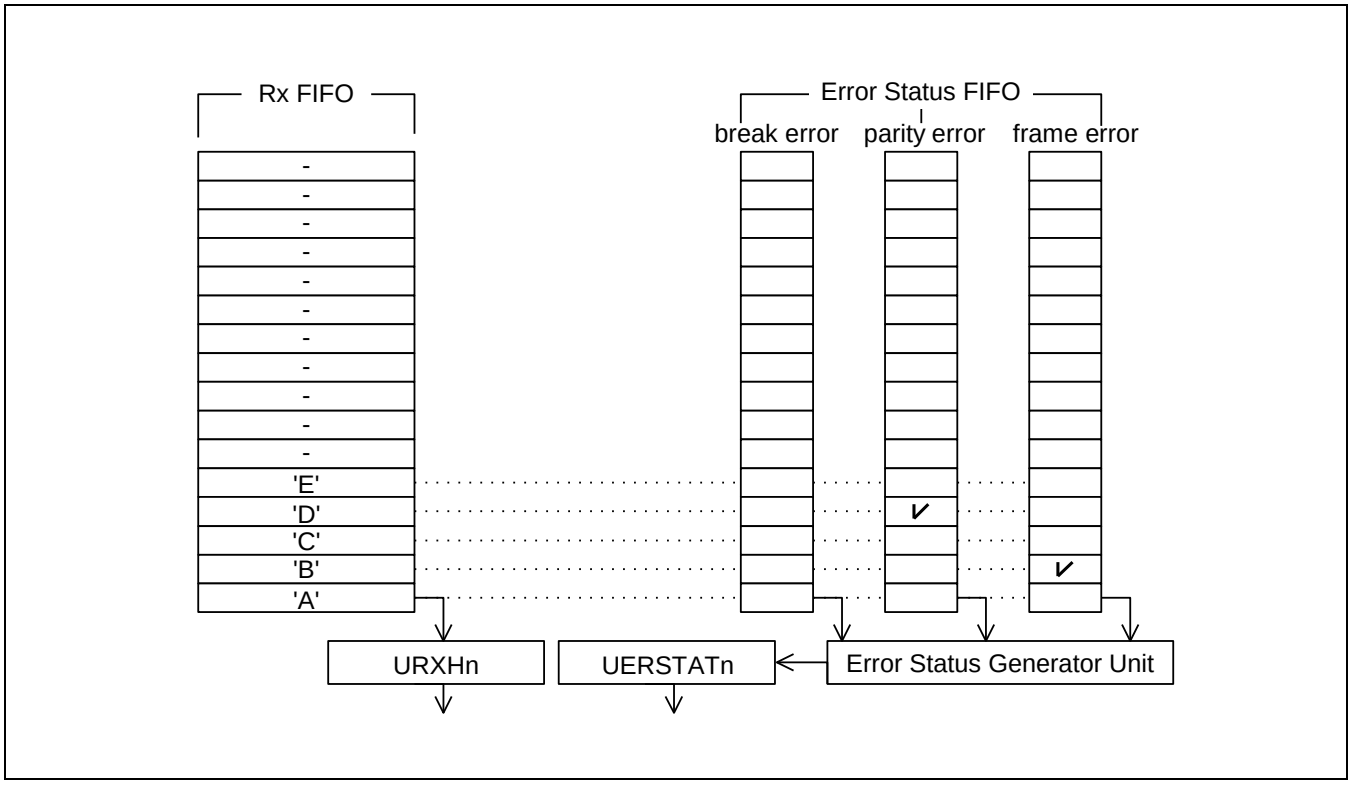


Figure 4-28 UART Receives the Five Characters Including Two Errors

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.14.3 Register Description

4.14.3.1 Register Map Summary

- Base Address: 0x4000A000

Register	Offset	Description	Reset Value
ULCON	0x0000	Specifies Line Control.	0x0000_0000
UCON	0x0004	Specifies Control	0x0000_3000
UFCON	0x0008	Specifies FIFO Control	0x0000_0000
UMCON	0x000C	Specifies AFC Control	0x0000_0000
UTRSTAT	0x0010	Specifies Tx/Rx status	0x0000_0002
UERSTAT	0x0014	Specifies Rx Error Status	0x0000_0000
UFSTAT	0x0018	Specifies FIFO Status	0x0000_0000
RSVD	0x001C	Reserved	0x0000_0000
UTXH	0x0020	Transmit Data for UART	0x0000_0000
URXH	0x0024	Receive Data for UART	0x0000_0000
UBRDIV	0x0028	Baud Rate Division Value	0x0000_0000
UFRACVAL	0x002C	Determines the fractional part of Baud Rate Divisor	0x0000_0000
UINTP	0x0030	Specifies Interrupt Pending	0x0000_0000
UINTS	0x0034	Specifies Interrupt Source	0x0000_0000
UINTM	0x0038	Specifies Interrupt Mask	0x0000_0000

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4.14.3.1.1 ULCON

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[31:6]	RW	Reserved	0x0000_0000
Parity_mode	[5:3]	RW	* Specifies the type of parity generation to be performed and checking during UART transmit and receive operation: 0xx: No parity 100: Odd parity 101: Even parity 110: Parity forced/ checked as 1 111: Parity forced/ checked as 0	0x0000_0000
Number_of_stop_bit	[2]	RW	* Specifies the number of stop bits that are used to signal end-of-frame signal: 0: One stop bit per frame 1: Two stop bit per frame	0x0000_0000
Word_length	[1:0]	RW	* Indicates the number of data bits to be transmitted or received per frame: 00: 5-bit 01: 6-bit 10: 7-bit 11: 8-bit	0x0000_0000

4.14.3.1.2 UCON

- Address: Base Address + 0x0004, Reset Value = 0x0000_3000

Name	Bit	Type	Description	Reset Value
RSVD0	[31:23]	RW	Reserved	0x0000_0000
Tx_DMA_burst_size	[22:20]	RW	* Tx DMA Burst Size Indicates the data transfer size of one DMA transaction which is triggered by a Tx DMA request. The DMA program must be programmed to transfer the same data size as this value for a single Tx DMA request: 000: 1 byte (Single) 001: 4 bytes 010: 8 bytes 011: Reserved 100: Reserved 101: Reserved 110: Reserved 111: Reserved	0x0000_0000
RSVD1	[19]	RW	Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
Rx_DMA_burst_size	[18:16]	RW	* Rx DMA Burst Size Indicates the data transfer size of one DMA transaction that is triggered by a Rx DMA request. The DMA program must be programmed to transfer the same data size as this value for a single Rx DMA request: 000: 1 byte (Single) 001: 4 bytes 010: 8 bytes 011: 16 bytes 100: Reserved 101: Reserved 110: Reserved 111: Reserved	0x0000_0000
Rx_timeout_interrupt_interval	[15:12]	RW	Rx Timeout Interrupt Interval Rx interrupt occurs when no data is received during 8 X (N+1) frame time. The default value of this field is 3 and it indicates the timeout interval is 32 frame time.	0x0000_0003
Rx_Time_out_with_empty_Rx_FIFO	[11]	RW	* Enables RX time-out feature when Rx FIFO counter is "0" This bit is valid only when UCONn[7] is "1": 0: Disables Rx time-out feature when Rx FIFO is empty 1: Enables Rx time-out feature when Rx FIFO is empty	0x0000_0000
Rx_Time_out_DMA_suspend_enable	[10]	RW	* Enables Rx DMA FSM to suspend when Rx Time-out occurs: 0: Disables suspending Rx DMA FSM 1: Enables suspending Rx DMA FSM	0x0000_0000
Tx_Interrupt_Type	[9]	RW	* Interrupt request type: 0: Pulse (interrupt is requested when the Tx buffer is empty in the Non-FIFO mode or when it reaches Tx FIFO Trigger Level in the FIFO mode) 1: Level (interrupt is requested when Tx buffer is empty in the Non-FIFO mode or when it reaches Tx FIFO Trigger Level in the FIFO mode)	0x0000_0000
Rx_Interrupt_Type	[8]	RW	* Interrupt request type: 0: Pulse (interrupt is requested when instant Rx buffer receives data in the Non-FIFO mode or when it reaches Rx FIFO Trigger Level in the FIFO mode) 1: Level (interrupt is requested when Rx buffer receives data in the Non-FIFO mode or when it reaches Rx FIFO Trigger Level in the FIFO mode)	0x0000_0000
Rx_Time_Out_Enable	[7]	RW	* Enables/Disables Rx time-out interrupts when UART FIFO is enabled. The interrupt is a receive	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			interrupt: 0: Disables Rx Time-out interrupt 1: Enables Rx Time-out interrupt	
Rx_Error_Status_Interrupt_Enable	[6]	RW	* Enables the UART to generate an interrupt upon an exception(Break or Frame Error or Parity Error or Overrun Error during a receive operation): 0: Does not generate receive error status interrupt 1: Generates receive error status interrupt	0x0000_0000
Loop_back_Mode	[5]	RW	* Setting loop-back bit to "1" triggers the UART to enter the loop-back mode. This mode is provided for test only: 0: Normal operation 1: Loop-back mode	0x0000_0000
Send_Break_Signal	[4]	RW	* Setting this bit triggers the UART to send a break during "1" frame time. This bit is automatically cleared after sending the break signal: 0: Normal transmit 1: Sends the break signal	0x0000_0000
Transmit_Mode	[3:2]	RW	* Determines which function is able to write Tx data to the UART transmit buffer: 00: Disables 01: Interrupt request or Polling mode 10: DMA mode 11: Reserved	0x0000_0000
Receive_Mode	[1:0]	RW	* Determines which function is able to read data from UART receive buffer: 00: Disables 01: Interrupt request or Polling mode 10: DMA mode 11: Reserved	0x0000_0000

NOTE:

1. DIV_VAL = UBRDIV + UFRACVAL/16. Refer to UBRDIV and UFRACVAL for more information.
2. S3FBP5A uses a level-triggered interrupt controller. Therefore, these bits must be set to "1" for every transfer.
3. When the UART does not reach the FIFO trigger level, and it does not receive data during the time specified at the "Rx Timeout Interrupt Interval" field in DMA receive mode with FIFO, the Rx interrupt is generated (receive time out). You must check the FIFO status and read out the rest.
4. UCON[11] and UCON[7] should be set to "1" to enable Rx time-out feature when Rx FIFO counter is "0".

4.14.3.1.3 UFCON

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[31:11]	RW	Reserved	0x0000_0000

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Name	Bit	Type	Description	Reset Value
Tx_FIFO_Trigger_Level	[10:8]	RW	* Determines the trigger level of Tx FIFO. When data count of Tx FIFO is less than or equal to the trigger level Tx interrupt occurs: 000: 0 byte 001: 8 bytes 010: 16 bytes 011: 24 bytes 100: 32 bytes 101: 40 bytes 110: 48 bytes 111: 56 bytes	0x0000_0000
RSVD1	[7]	RW	Reserved	0x0000_0000
Rx_FIFO_Trigger_Level	[6:4]	RW	* Determines the trigger level of Rx FIFO. When data count of Rx FIFO is greater than or equal to the trigger level Rx interrupt occurs: 000: 8 byte 001: 16 bytes 010: 24 bytes 011: 32 bytes 100: 40 bytes 101: 48 bytes 110: 56 bytes 111: 64 bytes	0x0000_0000
RSVD2	[3]	RW	Reserved	0x0000_0000
Tx_FIFO_Reset	[2]	RW	* Auto-clears after resetting FIFO: 0: Normal 1: Tx FIFO reset	0x0000_0000
Rx_FIFO_Reset	[1]	RW	* Auto-clears after resetting FIFO: 0: Normal 1: Rx FIFO reset	0x0000_0000
FIFO_Enable	[0]	RW	* FIFO Enable: 0: Disables 1: Enables	0x0000_0000

NOTE: When the UART does not reach the FIFO trigger level and does not receive data during the specified timeout interval in DMA receive mode with FIFO, the Rx interrupt will be generated (receive time out). You must check the FIFO status and read out the rest.

4.14.3.1.4 UMCON

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RTS_trigger_level	[7:5]	RW	* Determines the trigger level of Rx FIFO to control	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			nRTS signal. When AFC bit is enabled and Rx FIFO have bytes that are greater than or equal to the trigger level nRTS signal is deactivated: 000: 63 bytes 001: 56 bytes 010: 48 bytes 011: 40 bytes 100: 32 bytes 101: 24 bytes 110: 16 bytes 111: 8 bytes	
AFC	[4]	RW	* Auto flow control: 0: Disables 1: Enables	0x0000_0000
RSVD0	[3:1]	RW	These bits must be "0"	0x0000_0000
Request_to_send	[0]	RW	* When AFC bit is enabled this value is ignored. In this case it controls nRTS signals automatically. When AFC bit is disabled the software must control nRTS signal. 0: "H" level (Inactivate nRTS) 1: "L" level (Activate nRTS)	0x0000_0000

NOTE: In AFC mode, Rx FIFO trigger level should be set to lower than RTS trigger level, because transmitter stops data transfer when it gets deactivated nRTS signal.

4.14.3.1.5 UTRSTAT

- Address: Base Address + 0x0010, Reset Value = 0x0000_0002

Name	Bit	Type	Description	Reset Value
Rx_FIFO_count_in_ Rx_time_out_status	[23:16]	R	Rx FIFO counter capture value when Rx time-out occurs	0x0000_0000
Tx_DMA_FSM_state	[15:12]	R	* Current State of Tx DMA FSM: 0x0: IDLE 0x1: Burst Request 0x2: Burst Acknowledgement 0x3: Burst Next (intermediate state for next request) 0x4: Single Request 0x5: Single Acknowledgement 0x6: Single Next (intermediate state for next request) 0x7: Reserved 0x8: Reserved 0x9: Reserved 0x10: Reserved	0x0000_0000

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Name	Bit	Type	Description	Reset Value
Rx_DMA_FSM_state	[11:8]	R	* Current State of Rx DMA FSM: 0x0: IDLE 0x1: Burst Request 0x2: Burst Acknowledgement 0x3: Burst Next (intermediate state for next request) 0x4: Single Request 0x5: Single Acknowledgement 0x6: Single Next (intermediate state for next request) 0x7: Reserved 0x8: Reserved 0x9: Reserved 0x10: Reserved	0x0000_0000
RSVD0	[7:4]	R	Reserved	0x0000_0000
Rx_time_out_status_clear	[3]	RW	* Rx Time-out status when read: 0: Rx Time out did not occur 1: Rx Time out * Clear Rx Time-out status when write: 0: No operation 1: Clears Rx Time-out status *NOTE: If UCONn[10] is set to "1" writing 1 to this bit resumes Rx DMA FSM that was suspended when Rx time-out had occurred.	0x0000_0000
Transmitter_empty	[2]	R	* This bit is automatically set to "1" when the transmit buffer has no valid data to transmit and the transmit shift is empty. 0: Buffer is not empty 1: Transmitter (that includes transmit buffer and shifter) empty	0x0000_0000
Transmit_buffer_empty	[1]	R	* This bit is automatically set to "1" when transmit buffer is empty: 0: Buffer is not empty 1: Buffer is empty In Non-FIFO mode Interrupt or DMA is requested. In FIFO mode Interrupt or DMA is requested when Tx FIFO Trigger Level is set to 00 (empty). When UART uses FIFO check Tx FIFO Count bits and Tx FIFO Full bit in UFSTAT instead of this bit.	0x0000_0001
Receive_buffer_data_ready	[0]	R	* This bit is automatically set to "1" if receive buffer contains valid data which is received through the RXDn port: 0: Buffer is empty 1: Buffer has a received data (In Non-FIFO mode Interrupt or DMA is requested) When UART uses the FIFO check Rx FIFO Count	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			bits and Rx FIFO Full bit in UFSTAT instead of this bit.	

4.14.3.1.6 UERSTAT

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Break_detect	[3]	R	* This bit is automatically set to "1" to indicate that a break signal has been received: 0: No break signal is received 1: Break signal is received (Interrupt is requested)	0x0000_0000
Frame_error	[2]	R	* This bit is automatically set to "1" when a Frame Error occurs during the receive operation: 0: No Frame Error occurs during the receive operation 1: Frame Error occurs (interrupt is requested) during the receive operation	0x0000_0000
Parity_error	[1]	R	* This bit is automatically set to "1" when a Parity Error occurs during the receive operation: 0: No Parity Error occurs during receive operation 1: Parity Error occurs (interrupt is requested) during the receive operation	0x0000_0000
Overrun_error	[0]	R	* This bit is automatically set to "1" automatically when an Overrun Error occurs during the receive operation: 0: No Overrun Error occurs during the receive operation 1: Overrun Error occurs (interrupt is requested) during the receive operation	0x0000_0000

NOTE: These bits (UERSAT[3:0]) are automatically cleared to 0 when UART error status is read.

4.14.3.1.7 UFSTAT

- Address: Base Address + 0x0018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Tx_FIFO_full	[24]	R	* This bit is automatically set to "1" when the transmitted FIFO is full during transmit operation: 0: Not Full 1: Full	0x0000_0000
Tx_FIFO_count	[23:16]	R	Number of data in Tx FIFO * NOTE: This field is set to "0" when Tx FIFO is full.	0x0000_0000
RSVD0	[15:10]	R	Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
Rx_FIFO_error	[9]	R	This bit is set to "1" when Rx FIFO contains invalid data which results due to Frame Error or Parity Error or Break Signal.	0x0000_0000
Rx_FIFO_full	[8]	R	* This bit is automatically set to "1" when the received FIFO is full during receive operation: 0: Not Full 1: Full	0x0000_0000
Rx_FIFO_count	[7:0]	R	Number of data in Rx FIFO * NOTE: This field is set to "0" when Rx FIFO is full.	0x0000_0000

4.14.3.1.8 RSVD

- Address: Base Address + 0x001C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[31:0]	R	Reserved	0x0000_0000

4.14.3.1.9 UTXH

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
UTXH	[7:0]	W	Transmit Data for UART	0x0000_0000

4.14.3.1.10 URXH

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
URXH	[7:0]	R	Receive Data for UART	0x0000_0000

NOTE: When an Overrun Error occurs, the URXH must be read. If not, the next received data makes an Overrun Error, even though the overrun bit of UERSTAT had been cleared.

4.14.3.1.11 UBRDIV

- Address: Base Address + 0x0028, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
UBRDIV	[15:0]	RW	Baud Rate Division Value * NOTE: UBRDIV value must be greater than 0.	0x0000_0000

4.14.3.1.12 UFRACVAL

- Address: Base Address + 0x002C, Reset Value = 0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
UFRACVAL	[3:0]	RW	Determines the fractional part of Baud Rate Divisor	0x0000_0000

NOTE:

1. UART Baud Rate Configuration

   The value stored in the Baud Rate Divisor (UBRDIV) and Divisor Fractional value (UFRACVAL) is used to determine the serial Tx/Rx clock rate (baud rate) as:

   DIV_VAL = UBRDIV + UFRACVAL/16

   Or

   DIV_VAL = (SCLK_UART/(bps × 16)) – 1

   Where, the divisor should be from 1 to (216 – 1).

   Using UFRACVAL, you can generate the Baud Rate more accurately.

   For example, when the Baud Rate is 115200 bps and SCLK_UART is 40 MHz, UBRDIV and UFRACVAL are:

   DIV_VAL = (40000000/(115200 × 16)) – 1
   = 21.7 – 1
   = 20.7

   UBRDIV = 20 (integer part of DIV_VAL)

   UFRACVAL/16 = 0.7

   so, UFRACVAL = 11

2. Baud Rate Error Tolerance

   UART Frame error should be less than 1.87 % (3/160)

   tUPCLK = (UBRDIV + 1 + UFRACVAL/16) × 16 × 1Frame/SCLK_UART

   tUPCLK: Real UART Clock

   tEXTUARTCLK = 1Frame/baud-rate

   tEXTUARTCLK: Ideal UART Clock

   UART error = (tUPCLK – tEXTUARTCLK)/tEXTUARTCLK × 100 %

   * 1Frame = start bit + data bit + parity bit + stop bit.

3. UART Clock and PCLK Relation

   There is a constraint on the ratio of clock frequencies for PCLK to UARTCLK.

   The frequency of UARTCLK must not be more than 5.5/3 times faster than the frequency of PCLK:

   FUARTCLK ⇐ 5.5/3 × FPCLK
   FUARTCLK = baudrate × 16

   This allows sufficient time to write the received data to the receive FIFO

4.14.3.1.13 UINTP

- Address: Base Address + 0x0030, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[3]	RW	Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
TXD	[2]	RW	Generates Transmit interrupt	0x0000_0000
ERROR	[1]	RW	Generates Error interrupt	0x0000_0000
RXD	[0]	RW	Generates Receive interrupt	0x0000_0000

NOTE: Interrupt pending contains information of the interrupts that are generated.

If one of these 3 bits is logical high ("1"), each UART channel generates interrupt.

This must be cleared in the interrupt service routine after clearing interrupt pending in Interrupt Controller (INTC). Clear specific bits of UINTP by writing "1" to the bits that you want to clear.

4.14.3.1.14 UINTS

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[3]	R	Reserved	0x0000_0000
TXD	[2]	R	Generates Transmit interrupt	0x0000_0000
ERROR	[1]	R	Generates Error interrupt	0x0000_0000
RXD	[0]	R	Generates Receive interrupt	0x0000_0000

NOTE: Interrupt Source contains information of the interrupt that are generated, regardless of the value of Interrupt Mask

4.14.3.1.15 UINTM

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RSVD0	[3]	RW	Reserved	0x0000_0000
TXD	[2]	RW	Mask Transmit interrupt	0x0000_0000
ERROR	[1]	RW	Mask Error interrupt	0x0000_0000
RXD	[0]	RW	Mask Receive interrupt	0x0000_0000

Figure 4-29 illustrates the block diagram of UNITS, UINTP and UINTM.

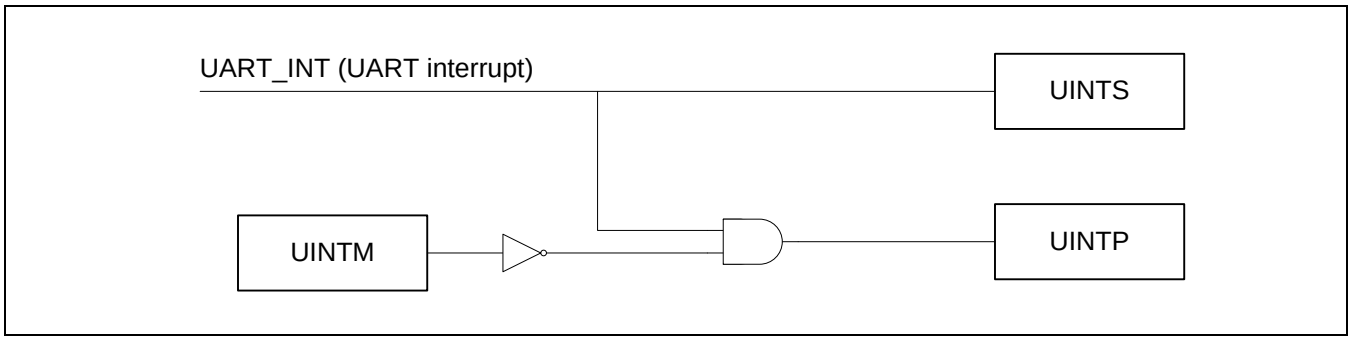


Figure 4-29 Block diagram of UINTS, UINTP and UINTM

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

NOTE: Interrupt Mask contains information about the interrupt source that is masked. When a specific bit is set to "1", interrupt request signal to the Interrupt Controller is not generated even though corresponding Interrupt Source is generated.

If the Mask bit is 0, the interrupt requests are serviced from the corresponding Interrupt Source.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.15 Inter-Integrated Circuit (I2C)

4.15.1 Overview

The high-speed I2C (HSI2C) unit supports serial data communication with the following features:

- Supports operation based on DMA or interrupt;
- Transmits and receives data separately
- Supports non-integer clock division
- Operates in two modes, namely, master and slave modes

Figure 4-30 shows a top-level block diagram of the HSI2C

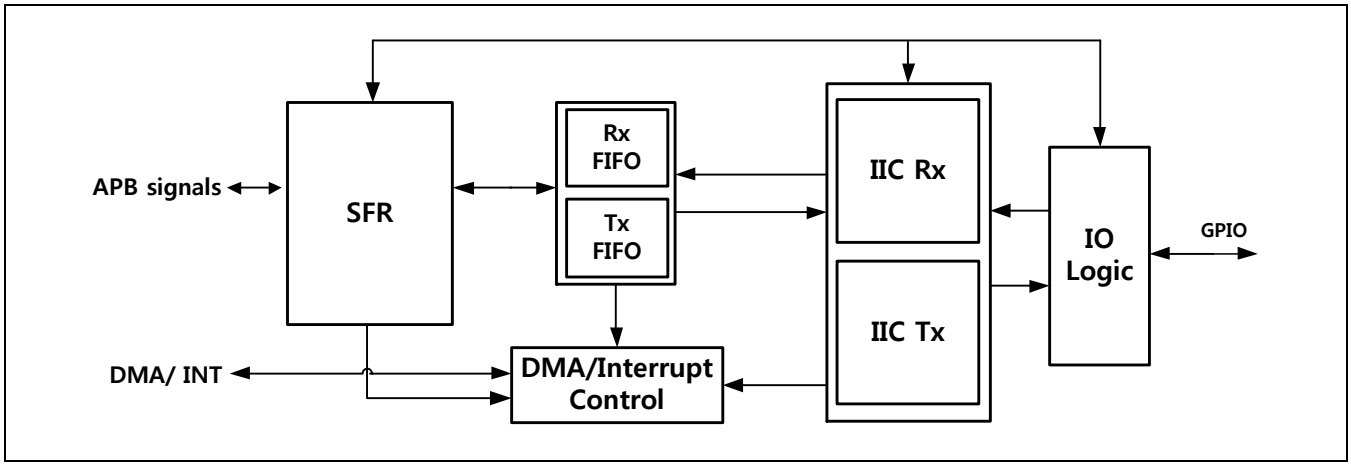


Figure 4-30 HSI2C Block diagram

4.15.2 Functional Description

The HSI2C unit supports Interrupt / DMA operation, master / slave mode, and two channels, namely, HSI2C TX and HSI2C RX channels.

To write data in TX FIFO, the MCU (or DMA) must write data in TXDATA register. The data is automatically moved to TX FIFO.

To read data from RX FIFO, the MCU (or DMA) must access the RXDATA register and data is automatically sent to RXFIFO register.

4.15.2.1 FIFO Access

The HSI2C unit supports access to FIFOs. The data size is selected from 8-bit, 16-bit or 32-bit data. The trigger level of each FIFO is set from 0byte to 16byte. The channel (TX or RX) accesses FIFO using 8-bit.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.15.2.2 Trailing Bytes

If the number of samples in RX FIFO is less than the threshold value and no additional data is received for a long time, the remaining bytes in RX FIFO is called as trailing bytes. To remove these bytes, the internal timer and interrupt signal are used. If internal timer value is zero, interrupt signal is generated and the CPU can remove the trailing bytes.

4.15.2.3 Start and Stop Conditions

The master device generates Start and Stop conditions. After initiation of Start condition, the master selects the slave device by writing the salve's 7-bit address in the first outgoing data byte. The 8th bit determines the direction of transfer (read or write). The I2C-bus gets busy if a Start condition is generated, and a Stop condition frees the I2C-bus.

Each data byte put onto the SDA line must be eight bits. There is no limit to send or receive bytes during the bus transfer operation. Data is always sent first from MSB and every byte must be followed by acknowledge (ACK) bit.

Figure 4-31 shows Start and Stop Conditions.

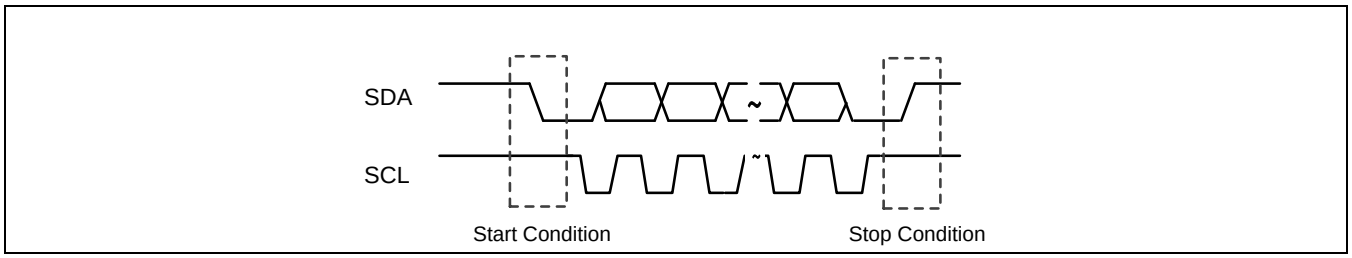


Figure 4-31 Start and Stop Conditions

4.15.2.4 ACK Signal Transmission

To complete a one-byte transfer operation, the receiver must send an ACK bit to the transmitter. The ACK pulse occurs at the ninth clock of the SCL line. Eight clocks are required for the one-byte data transfer. The master generates clock pulse required to transmit the ACK bit.

The transmitter releases the SDA line by making the SDA line High if the ACK clock pulse is received. The receiver drives the SDA line Low during the ACK clock pulse, so that the SDA keeps Low during the High period of the ninth SCL pulse.

The software (CTL) enables or disables ACK bit transmit function. However, the ACK pulse on the ninth clock of SCL is required to complete the one-byte data transfer operation.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 4-32 shows Acknowledge on I2C-bus Block Diagram

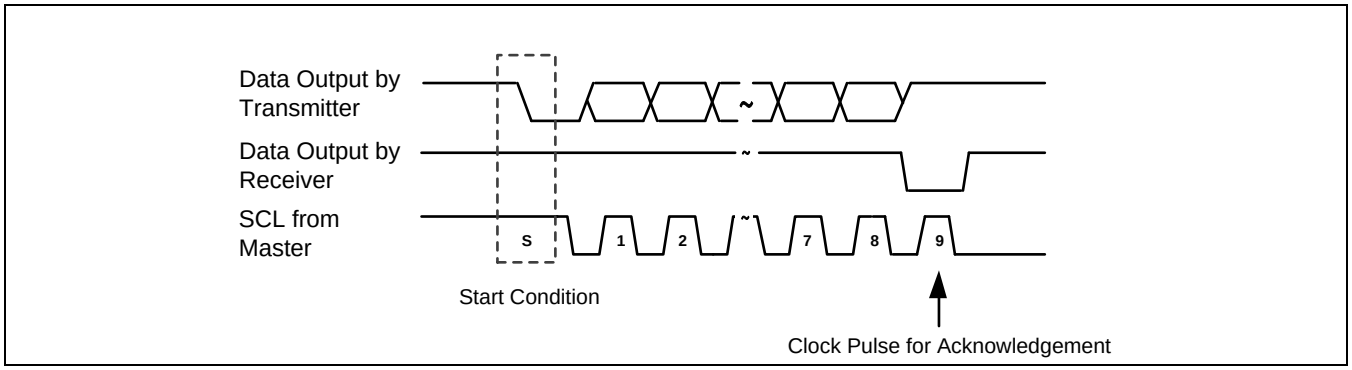


Figure 4-32 Acknowledge on I2C-bus Block Diagram

4.15.2.5 Bus Arbitration Procedures

Arbitration takes place on the SDA line to prevent the contention on the bus between two masters. If a master with SDA High level detects other master with SDA active Low level, it does not initiate a data transfer because the current level on the bus does not correspond to its own. The arbitration procedure will be extended until the SDA line turns high.

If the masters lower the SDA line simultaneously, each master evaluates whether the mastership is allocated itself or not. For the purpose of evaluation, each master detects the address bits. While each master generates the slave address, it detects the address bit on the SDA line because the SDA line is likely to get Low rather than to keep high.

Assume that one master generates a Low as first address bit, while the other master is maintaining High. In this case, both masters detect Low on the bus because the Low status has superior priority than the High status. If this happens, the master that generates a Low output (as the first bit of address) will get the mastership, while the master that generates a High output (as the first bit of address) withdraws the mastership.

If both masters generate Low as the first bit of address, there is arbitration for the second address bit again. This arbitration continues to the end of last address bit.

Assume that both masters trying to address the same device, there will be arbitration again for the data-bits.

4.15.2.6 Abort Conditions

If a slave receiver cannot acknowledge the confirmation of the slave address, it holds the level of the SDA line High. In this case, the master generates a Stop condition to abort the transfer.

If a master receiver aborts the transfer, it signals the end of the slave transmit operation by canceling the generation of an ACK after the last data byte received from the slave. The slave transmitter releases the SDA to allow a master to generate a Stop condition.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.15.2.7 Fast-Speed Mode

Each byte placed on the SDA line must be eight bits in length. There is no limit to the number of bytes transmitted per transfer. The first byte following a Start condition has the address field. The address field can be transmitted by the master when the I2C-bus is operating in Master mode. Each byte is followed by an acknowledgement (ACK) bit. The MSB bit of the serial data and addresses are always sent first.

Figure 4-33 shows Data Transfer Format in Fast-Speed Mode

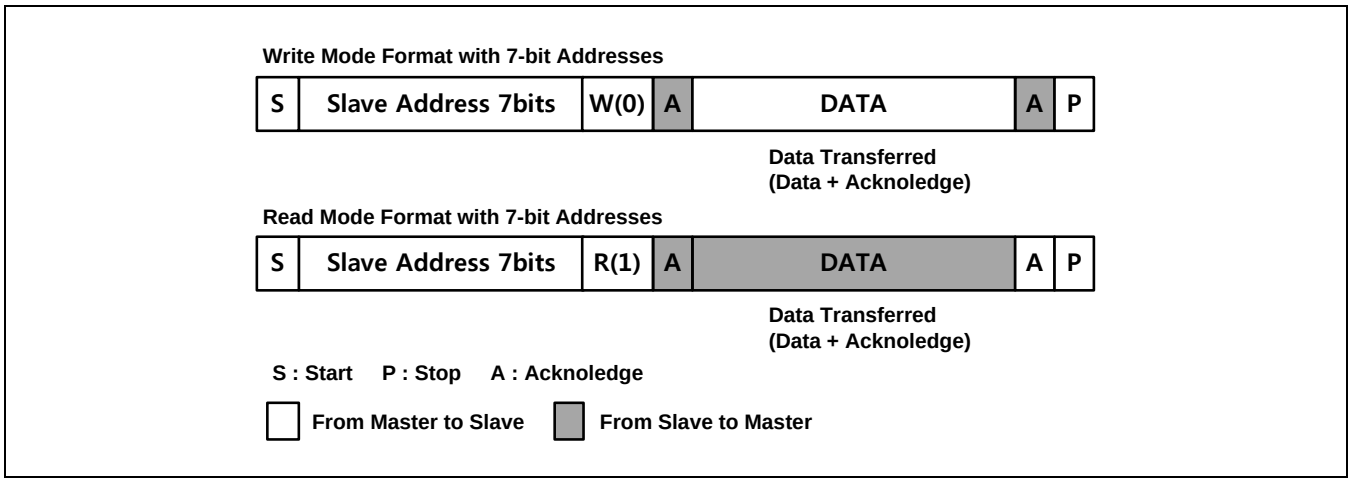


Figure 4-33 Data Transfer Format in Fast-Speed Mode

4.15.2.8 High-Speed Mode

Serial data transfer format in high-speed mode meets the standard-mode (fast-speed mode) I2C-bus specification. High-speed mode can only commence after the following conditions (all of which are in fast-speed mode):

1. START condition (S)
2. 8-bit master id code (00001xxx)
3. Non-acknowledge bit

High-speed master id codes are reserved 8-bit codes, which are not used for slave addressing or other purpose.

Figure 4-34 shows Data Transfer Format in High-Speed Mode.

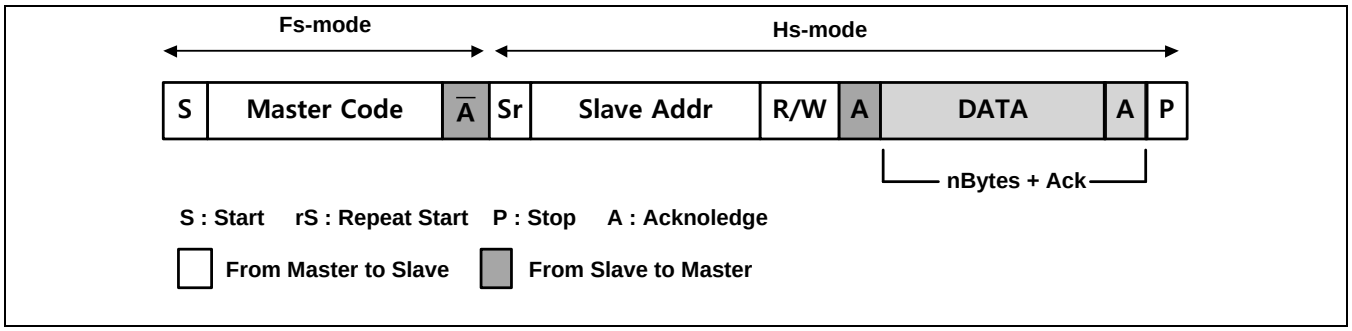


Figure 4-34 Data Transfer Format in High-Speed Mode

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.15.2.9 10-Bit Slave-Address Mode

I2C channel of HSI2C supports 10-bit slave-address mode and *Figure 4-35* and *Figure 4-36* show data-transfer using 10-bit address.

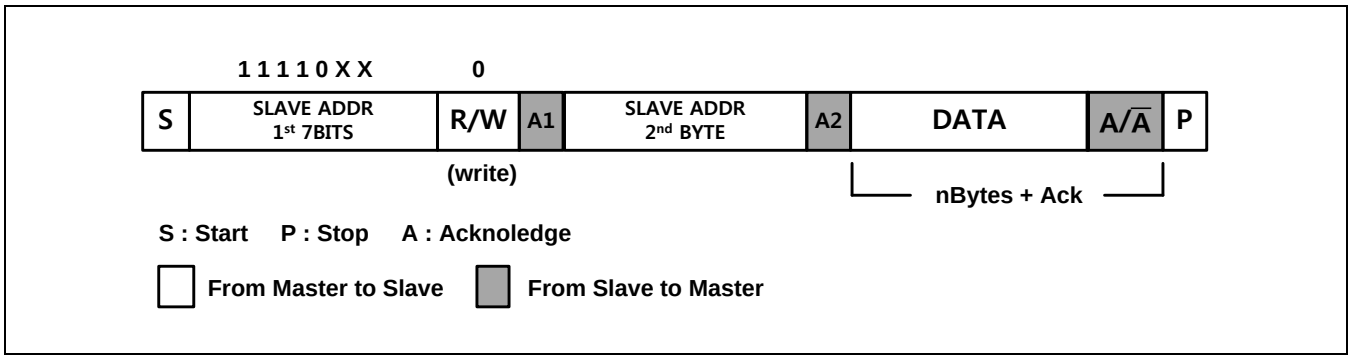


Figure 4-35 Master-Transmitter Addresses Slave Receiver with 10-bit Address

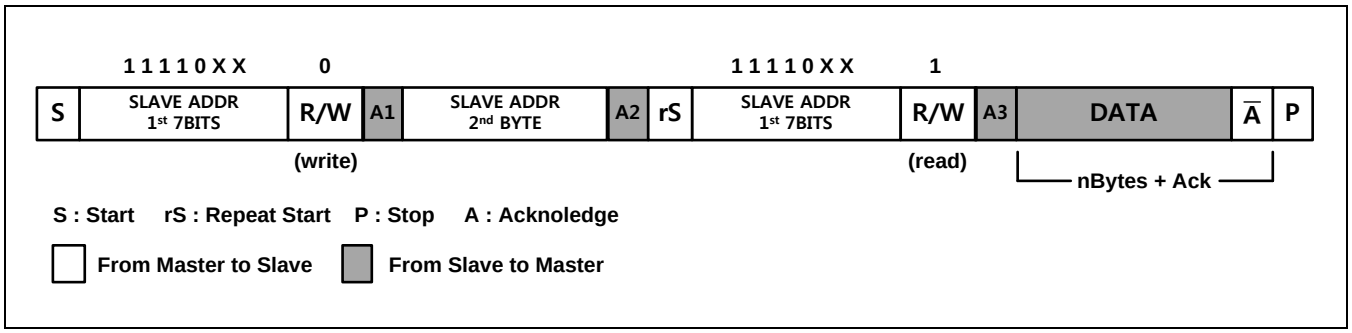


Figure 4-36 Master-Receiver Addresses Slave-Transmitter with 10-bit Address

4.15.2.10 Done Interrupt Recommendations

For auto mode and manual mode, there are several done interrupt signals like INT_TRANSFER_DONE_AUTO, INT_TRANS_ABORT_AUTO, INT_TRANSFER_DONE_MANUAL, and INT_TRANSFER_DONE_NOACK in the INT_STAT register. To use these interrupt signals in master operation mode, be careful for the SCL clock stretching. Some I2C slave device might use the SCL clock stretching to do something internally so that I2C slave device may hold the SCL line to low.

General I2C slave doesn't use the SCL clock stretching. As shown in both *Figure 4-37* and *Figure 4-38*, they have a time interval difference. To check the final completion of the transfers, HSI2C master should check the master state if using the SCL clock stretching slaves. After detecting above 3 interrupts, HSI2C master might check whether I2C_TRANS_STATUS[3:0](MASTER_ST) is 0 (IDLE) or I2C_TRANS_STATUS[17](MASTER_BUSY) is 0. If there is any SCL clock stretching by slave, both bit fields don't show 0. But these bit checking may delay the next operations of HSI2C, it should be considered to use or not according to the slave specification.

Note that this HSI2C also can stretch SCL low in slave operation mode. Refer to the I2C_CONF[9:8] (STRCH_MANUAL,STRCH_EN) bits.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

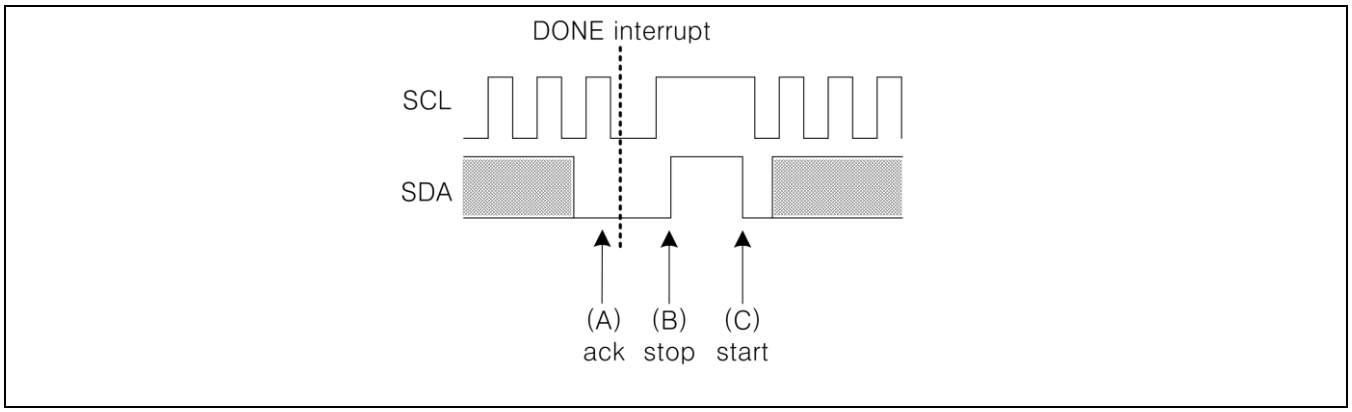


Figure 4-37 Done interrupt timing when accessing general I2C slaves

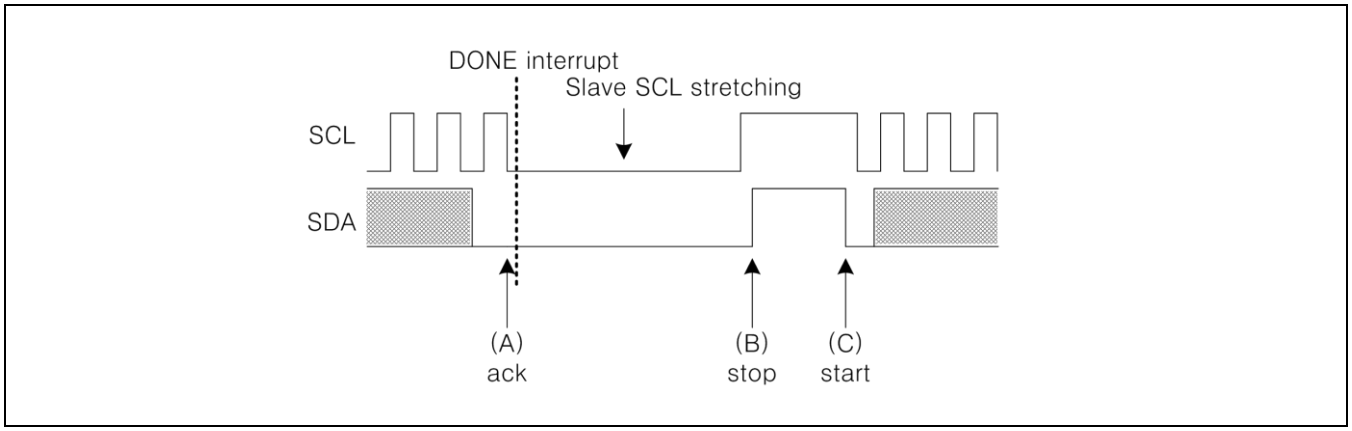


Figure 4-38 Done interrupt timing when accessing SCL clock stretching I2C slave

4.15.2.11 HSI2C Frequency Control

HSI2C Clock (SCL) Frequency is controlled by HSI2C master's registers like below equation.

Clock Generation Equation of HSI2C clock frequency:

$$Fpclk/Fi2c = (CLK_DIV+1) \times (TSCL_L+TSCL_H+2) + \{ (FLT_CYCLE_SCL+3) - (FLT_CYCLE_SCL+3)\%(CLK_DIV+1) \} \times 2$$

Fi2c is the clock frequency of I2C channel.

Fpclk is the clock frequency of PCLK.

CLK_DIV is a bit field of I2C_TIMING_HS3 (for high-speed mode) or I2C_TIMING_FS3 (for fast-mode) SFRs.

TSCL_L and TSCL_H are bit fields of I2C_TIMING_HS2 (for high-speed mode) or I2C_TIMING_FS2 (for fast-mode) SFRs

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FLT_CYCLE_SCL is a bit field of I2C_CONF SFR.

% is modulo operator which is the remainder after division.

4.15.3 Register Description

4.15.3.1 Register Map Summary

- Base Address: 0x40012000

Register	Offset	Description	Reset Value
CTL	0x0000	HSI2C control register	0x0000_0000
FIFO_CTL	0x0004	HSI2C FIFO control register	0x0008_0083
TRAILING_CTL	0x0008	Specifies the count value from writing the last data in RX FIFO to flush trailing bytes in FIFO. * Note: The counter is based on PCLK.	0xFFFF_FFFF
INT_EN	0x0020	HSI2C interrupt enable register	0x0000_0000
INT_STAT	0x0024	HSI2C interrupt register	0x0000_0000
FIFO_STAT	0x0030	HSI2C FIFO status register	0x0100_0100
TXDATA	0x0034	Transmit data buffer for HSI2C	0xDEAD_BEAF
RXDATA	0x0038	Received data buffer for HSI2C.	0x0000_0000
I2C_CONF	0x0040	I2C configuration	0x9803_C1FF
I2C_AUTO_CONF	0x0044	I2C configuration for AUTO mode	0x0003_00FF
I2C_TIMEOUT	0x0048	Time Out for I2C master	0x8000_00FF
I2C_MANUAL_CMD	0x004C	Command register for MANUAL mode	0x0000_0000
I2C_TRANS_STATUS	0x0050	HSI2C Transfer status register	0x0000_0000
I2C_TIMING_HS1	0x0054	Timing parameter for high-speed mode	0x0F0F_0F00
I2C_TIMING_HS2	0x0058	Timing parameter for high-speed mod	0x0F0F_FFFF
I2C_TIMING_HS3	0x005C	Timing parameter for high-speed mode	0x00FF_00FF
I2C_TIMING_FS1	0x0060	Timing parameter for fast-speed -mode	0x0F0F_0F00
I2C_TIMING_FS2	0x0064	Timing parameter for fast-speed -mode	0x0F0F_FFFF
I2C_TIMING_FS3	0x0068	Timing parameter for fast-speed -mode	0x00FF_00FF
I2C_TIMING_SLA	0x006C	Data hold timing for I2C slave. (data transition after SCL falling-edge) (For both auto and manual modes)	0x0000_00FF
I2C_ADDR	0x0070	* Address register: I2C slave address I2C master ID slave address for master	0xFF0F_FFFF

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.15.3.1.1 CTL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SW_RST	[31]	W	* Indicates software reset signal This bit clears the following registers and bits: Most writable registers with HSI2C_CTL(without [31] SW_RST), HSI2C_FIFO_CTL, HSI2C_TRAILING_CTL, HSI2C_INT_EN, I2C_CONF, I2C_AUTO_CONF, I2C_TIMEOUT, I2C_TIMING_HS1~3, I2C_TIMING_FS1~3, I2C_TIMING_SLA, I2C_ADDR 0: Inactive SW reset 1: Active SW reset	0x0000_0000
EXTENDED_ADDR_MODE	[30]	RW	Extended address mode (15-bit-address). If this bit is set to 1, totally 15-bit-address could be generated using I2C 10-bit-address mode. It means that 11110xx of the first SLAVE address are controllable by CTL[29] and I2C_ADDR[23:20]. This bit is meaningful only if I2C_CONF[30](ADDRMODE) is 1 (10-bit-address mode)	0x0000_0000
EXTENDED_ADDR_MSB	[29]	RW	MSB of the first SLAVE 15-bit-address In extended address mode (see bit [30]). This bit could control the MSB of 11110xx of the first SLAVE address. * Note that other bits of 11110xx of the first SLAVE address are controlled by I2C_ADDR.	0x0000_0000
NO_ARB_FASTSDA	[28]	RW	Generally, this bit should be 0. This bit deactivates the modified logic of the abnormal operation problem when SCL is falling and then SDA is driven too quickly by other I2C device during 1 PCLK period. 0: Don't check the arbitration during the first clock of SCL falling. 1: Restore original logic	0x0000_0000
RSVD0	[27:22]	RW	Reserved	0x0000_0000
TX_HWORD_SWAP	[21]	RW	* Turn on or turn off half-word swap operation on TX channel: 0: Off 1: On	0x0000_0000
TX_BYTE_SWAP	[20]	RW	* Turn on or turn off byte swap operation on TX channel: 0: Off 1: On	0x0000_0000
TX_BIT_SWAP	[19]	RW	* Turn on or turn off bit swap operation on TX channel: 0: Off 1: On	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
RX_HWORD_SWAP	[18]	RW	* Turn on or turn off half-word swap operation on RX channel: 0: Off 1: On	0x0000_0000
RX_BYTE_SWAP	[17]	RW	* Turn on or turn off byte swap operation on RX channel: 0: Off 1: On	0x0000_0000
RX_BIT_SWAP	[16]	RW	* Turn on or turn off bit swap operation on RX channel: 0: Off 1: On	0x0000_0000
RSVD1	[15:14]	RW	Reserved	0x0000_0000
BUS_WIDTH	[13:12]	RW	00: Byte 01: Halfword 10: Word 11: Reserved	0x0000_0000
CH_WIDTH	[11:10]	RW	00: Byte 01: Halfword 10: Word 11: Reserved * Note: This bit should be set 00 for I2C.	0x0000_0000
RSVD2	[9:7]	RW	Reserved	0x0000_0000
RXCHON	[6]	RW	* Specifies the Rx channel enable signal: 0: Disables 1: Enables * Note: RXCHON must be set after configurations are completed, that is, configuration cannot be set after Rx channel is working.	0x0000_0000
TX_MODE	[5]	RW	* Determines which function is used to read data from transmit buffer register: 0: Interrupt 1: DMA	0x0000_0000
RX_MODE	[4]	RW	* Determines which function is used to write data into receiver buffer register: 0: Interrupt 1: DMA	0x0000_0000
MASTER	[3]	RW	* Determines whether HSI2C is master or slave: 0: Slave 1: Master	0x0000_0000
NO_1CYC_EDGE_SDA	[2]	RW	Generally, this bit should be 0. This bit deactivates the logic to restore internal SDA edge detect logic to 2 cycle latency instead of 1 cycle latency style. SDA	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			1 cycle latency edge detect logic is inserted to remove stop condition when SDA is rising 1 cycle after SCL rising 0: 1 cycle latency 1: 2 cycle latency	
NO_TSCL_H_0	[1]	RW	* Generally, this bit should be 0. This bit deactivates the logic for the case of TSCL_H_FS=0 or TSCL_H_HS=0. 0: Enable the case of TSCL_H_FS=0 or TSCL_H_HS=0. 1: Deactivate this additional function.	0x0000_0000
CS_ENB	[0]	RW	Generally, this bit should be 0 (for both HS mode and FS mode). This bit deactivates current source pull-up function for SCL signal in HS mode. 0: Enable current source for SCL signal to support high-speed operation 1: Same driving for HS mode as FS mode	0x0000_0000

NOTE: Channel-width must be larger than swap size. That is,

When channel-width is set as byte, only bit-swap operation is supported.

When channel-width is set as halfword, then bit-swap and byte-swap are supported.

When channel-width is set as word, then bit-swap, byte-swap and half-word swap are supported.

Bus-channel width has no relation with channel-width and swap size.

Data for I2C function are transferred by SFR directly (the FIFO is not used). Therefore, I2C function does not support swap-operation.

UART data-transfer is based on byte (channel-width = 8). Therefore, UART function only support Bit-swap operation.

The data transfer for SPI function supports byte/ half-word/ word transfer. Therefore, SPI function supports bit/ byte/ halfword-swap operation.

Swap-operation

Bit-swap:

Bit-swap operation is based on byte, refer the following example:

Data[31:0] → (bit-swap) {data[24], data[25], data[26], data[27], data[28],data[29], data[30], data[31], data[16], data[17], data[18], data[19], data[20], data[21], data[22], data[23], data[8], data[9], data[10], data[11], data[12], data[13], data[14], data[15], data[0], data[1], data[2], data[3], data[4], data[5], data[6], data[7]}

Byte-swap:

Byte-swap operation is based on half-word, refer the following example:

Data[31:0] → (byte-swap) {data[23:16], data[31:24], data[7:0], data[15:8]}

Half-word-swap:

Halfword-swap operation is based on word, refer the following example:

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Data[31:0] → (halfword-swap) {data[15:0], data[31:16]}.

HSI2C supports multiple-swap operation, such as bit-swap and byte-swap can be available concurrently.

4.15.3.1.2 FIFO_CTL

- Address: Base Address + 0x0004, Reset Value = 0x0008_0083

Name	Bit	Type	Description	Reset Value
Tx_FIFO_trigger_level	[22:16]	RW	Specifies the trigger level to generate TXFIFO_ALMOST_EMPTY interrupt (Note: NOT TXFIFO_UNDERRUN or TXFIFO_OVERRUN interrupts). If the byte number in Tx-FIFO is less than the trigger level, the TXFIFO_ALMOST_EMPTY interrupt is generated. * Note: The trigger level is used for both interrupt and DMA-request. The maximum value is 16.	0x0000_0008
RSVD0	[15:11]	RW	Reserved	0x0000_0000
Rx_FIFO_trigger_level	[10:4]	RW	Specifies the trigger level to generate RXFIFO_ALMOST_FULL interrupt (Note: NOT RXFIFO_UNDERRUN or RXFIFO_OVERRUN interrupts). If the byte number in Rx-FIFO is larger than the trigger level, the RXFIFO_ALMOST_FULL interrupt is generated. * Note: The trigger level is used for both interrupt and DMA-req. The maximum value is 16.	0x0000_0008
TX_FIFO_RST	[3]	RW	Reset TX FIFO	0x0000_0000
RX_FIFO_RST	[2]	RW	Reset RX FIFO	0x0000_0000
TXFIFO_EN	[1]	RW	* Specifies the Tx FIFO enable signal: 0: Disables (FIFO-depth = 1byte) 1: Enables (FIFO-depth = 16 bytes)	0x0000_0001
RXFIFO_EN	[0]	RW	* Specifies the Rx FIFO enable signal: 0: Disables (FIFO-depth = 1byte) 1: Enables (FIFO-depth = 16 bytes)	0x0000_0001

4.15.3.1.3 TRAILING_CTL

- Address: Base Address + 0x0008, Reset Value = 0xFFFF_FFFF

Name	Bit	Type	Description	Reset Value
RSVD0	[31:24]	RW	Reserved	0x0000_00FF
TRAILING_CTL	[23:0]	RW	Specifies the count value from writing the last data in RX FIFO to flush trailing bytes in FIFO. * Note: The counter is based on PCLK.	0x00FF_FFFF

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4.15.3.1.4 INT_EN

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INT_SLAVE_ADDR_MATCH_EN	[15]	RW	* I2C slave address is matched: (for slave mode only) 0: Disables 1: Enables	0x0000_0000
INT_LOSE_ARB_MANUAL_EN	[14]	RW	* Lost arbitration during executing command: (for manual mode only) 0: Disables 1: Enables	0x0000_0000
INT_TRANSFER_DONE_MANUAL_EN	[13]	RW	* One BYTE (data or address) is TX-ed successfully, one BYTE data is RX-ed successfully or command (start or stop) is completed. (for manual mode only) * Note: In high-speed mode, completion of master-id-transfer does not assert this bit. 0: Disables 1: Enables	0x0000_0000
INT_TRANSFER_DONE_NOACK_MANUAL_EN	[12]	RW	* Transfer completed without ACK received (for manual mode only) * Note: After master-id (HS mode), this bit should be asserted. 0: Disables 1: Enables	0x0000_0000
INT_TIMEOUT_AUTO_EN	[11]	RW	* Time out during auto mode transaction: 0: Disables 1: Enables	0x0000_0000
INT_NO_DEV_AUTO_EN	[10]	RW	* No device responds the address (no device is selected by master). (for auto mode only) 0: Disables 1: Enables	0x0000_0000
INT_NO_DEV_ACK_AUTO_EN	[9]	RW	* Device does not return ACK during master TX. (for auto mode only) 0: Disables 1: Enables	0x0000_0000
INT_TRANS_ABORT_AUTO_EN	[8]	RW	* Lost arbitration during transaction (for auto_mode only) 0: Disables 1: Enables	0x0000_0000
INT_TRANSFER_D	[7]	RW	* Command is executed successfully. This bit is	0x0000_0000

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Name	Bit	Type	Description	Reset Value
ONE_AUTO_EN			automatically cleared by software read. (for auto mode only) 0: Disables 1: Enables	
INT_TRAILING_EN	[6]	RW	* Interrupt enable signal for trailing interrupt: 0: Disables 1: Enables	0x0000_0000
INT_RX_OVERRUN_EN	[5]	RW	* Interrupt enable signal for RX overrun: 0: Disables 1: Enables	0x0000_0000
INT_RX_UNDERRUN_EN	[4]	RW	* Interrupt enable signal for RX under run: 0: Disables 1: Enables	0x0000_0000
INT_TX_OVERRUN_EN	[3]	RW	* Interrupt enable signal for TX overrun: 0: Disables 1: Enables	0x0000_0000
INT_TX_UNDERRUN_EN	[2]	RW	* Interrupt enable signal for TX under run: 0: Disables 1: Enables	0x0000_0000
INT_RX_ALMOSTFULL_EN	[1]	RW	* Interrupt enable signal for RX FIFO: 0: Disables 1: Enables	0x0000_0000
INT_TX_ALMOSTEMPTY_EN	[0]	RW	* Interrupt enable signal for TX FIFO: 0: Disables 1: Enables	0x0000_0000

4.15.3.1.5 INT_STAT

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
INT_SLAVE_ADDR_MATCH	[15]	RW	Write 1 to clear this bit. READ: (for slave mode only) I2C slave address is matched. 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_LOSE_ARB_MANUAL	[14]	RW	Write 1 to clear this bit. READ: (for manual mode only) Lost arbitration during executing command 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_TRANSFER_D	[13]	RW	Write 1 to clear this bit.	0x0000_0000

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Name	Bit	Type	Description	Reset Value
ONE_MANUAL			READ: (for manual mode only) One BYTE (data or address) is TX-ed successfully, one BYTE data is RX-ed successfully or command (start or stop) is completed. 0: No interrupt occurred 1: Interrupt occurred * Note: In high-speed mode, completion of master-id-transfer does not assert this bit.	
INT_TRANSFER_DONE_NOACK_MANUAL	[12]	RW	Write 1 to clear this bit. READ: (for manual mode only) Transfer completed without ACK received 0: No interrupt occurred 1: Interrupt occurred * Note: After master-id (HS mode), this bit must be asserted.	0x0000_0000
INT_TIMEOUT_AUTO	[11]	RW	Write 1 to clear this bit. READ: (for auto mode only) Time out during auto mode transaction. 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_NO_DEV_AUTO	[10]	RW	Write 1 to clear this bit. READ: (for auto mode only) No device responds the address (no device is selected by master). 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_NO_DEV_ACK_AUTO	[9]	RW	Write 1 to clear this bit. READ: (for auto mode only) Device does not return ACK during master TX. 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_TRANS_ABORT_AUTO	[8]	RW	Write 1 to clear this bit. READ: (for auto_mode only) Lost arbitration during transaction 0: No interrupt occurred 1: Interrupt occurred * Note: In high-speed mode, completion of master-id-transfer does not assert this bit.	0x0000_0000
INT_TRANSFER_DONE_AUTO	[7]	RW	Write 1 to clear this bit. READ: (for auto mode only) Command is executed successfully. This bit is automatically cleared by software read. 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			* Note: In high-speed mode, completion of master-id-transfer does not assert this bit.	
INT_TRAILING	[6]	RW	Write 1 to clear this bit. If HSI2C does not reach Rx FIFO trigger level and does not receive data for "TRAILING_CNT" time, this interrupt is asserted, then the CPU must check the FIFO status and read out the rest. 0: No interrupt occurred 1: Interrupt occurred	0x0000_0000
INT_RX_OVERRUN	[5]	RW	Write 1 to clear this bit. READ: Rx FIFO overrun error. 0: No error occurred 1: Overrun error occurred.	0x0000_0000
INT_RX_UNDERRUN	[4]	RW	Write 1 to clear this bit. READ: Rx FIFO underrun error 0: No error occurred 1: Underrun error occurred	0x0000_0000
INT_TX_OVERRUN	[3]	RW	Write 1 to clear this bit. READ: Tx FIFO overrun error. 0: No error occurred 1: Overrun error occurred	0x0000_0000
INT_TX_UNDERRUN	[2]	RW	Write 1 to clear this bit. READ: Tx FIFO underrun error. 0: No error occurred 1: Underrun error occurred	0x0000_0000
INT_RXFIFO_ALMOST_FULL	[1]	RW	Write 1 to clear this bit. READ: 0: Data in RX FIFO is less than trigger level 1: Data in RX FIFO is more than trigger level	0x0000_0000
INT_TXFIFO_ALMOST_EMPTY	[0]	RW	Write 1 to clear this bit. READ: 0: Data in TX FIFO is more than trigger level 1: Data in TX FIFO is less than trigger level	0x0000_0000

NOTE:

1. I2C interrupt occurs when:

   Slave mode:

   The received slave address is matched with setting slave address in SFR I2C_ADDR. (HSI2C is selected by external I2C master)

   Master mode:

   Manual mode:

   Master lost arbitration

   Master completes one BYTE (either address or data) transfer but receives “Not ACK” (ACK bit is high)

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Master successfully accomplishes one BYTE transfer (ACK bit is low)

Auto mode:

No device acknowledges to the address

Master receives “Not ACK” during transaction (this is master TX mode, slave device doesn’t acknowledge to current byte transfer).

Transaction abort.

Transaction successfully completed.

Transaction timeout

4.15.3.1.6 FIFO_STAT

- Address: Base Address + 0x0030, Reset Value = 0x0100_0100

Name	Bit	Type	Description	Reset Value
RX_FIFO_EMPTY	[24]	R	* RX FIFO empty signal: 0: Non-empty 1: Empty	0x0000_0001
RX_FIFO_FULL	[23]	R	* RX FIFO full signal: 0: Non-full 1: Full	0x0000_0000
RX_FIFO_LEVEL	[22:16]	R	Data level in RX FIFO.	0x0000_0000
RSVD0	[15:9]	R	Reserved	0x0000_0000
TX_FIFO_EMPTY	[8]	R	* TX FIFO empty signal: 0: Non-empty 1: Empty	0x0000_0001
TX_FIFO_FULL	[7]	R	* TX FIFO full signal: 0: Non-full 1: Full	0x0000_0000
TX_FIFO_LEVEL	[6:0]	R	Data level in TX FIFO.	0x0000_0000

4.15.3.1.7 TXDATA

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TXDATA	[31:0]	W	Transmit data buffer for HSI2C	0xDEAD_BEAF

4.15.3.1.8 RXDATA

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RXDATA	[31:0]	R	Received data buffer for HSI2C.	0x0000_0000

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4.15.3.1.9 I2C_CONF

- Address: Base Address + 0x0040, Reset Value = 0x9803_C1FF

Name	Bit	Type	Description	Reset Value
AUTO_MODE	[31]	RW	* Auto flow control for I2C master: 1: Auto control. I2C master automatically controls the whole transaction. 0: Manual control. Software manually controls transaction.	0x0000_0001
ADDRMODE	[30]	RW	1: 10-bit-address mode 0: 7-bit-address mode	0x0000_0000
HS_MODE	[29]	RW	* High-Speed Mode: 0: Fast-mode 1: High-speed mode	0x0000_0000
FILTER_EN_SCL	[28]	RW	Enables SCL filter.	0x0000_0001
FILTER_EN_SDA	[27]	RW	Enables SDA filter.	0x0000_0001
RSVD0	[26:19]	RW	Reserved	0x0000_0000
FLT_CYCLE_SCL	[18:16]	RW	Glitch width to be removed for SCL.(number of clock cycles) If this field is 0, it works as if FILTER_EN_SCL were 0 (disabled).	0x0000_0003
FLT_CYCLE_SDA	[15:13]	RW	Glitch width to be removed for SDA.(number of clock cycles) If this field is 0, it works as if FILTER_EN_SDA were 0 (disabled).	0x0000_0006
RSVD1	[12:10]	RW	Reserved	0x0000_0000
STRCH_MANUAL	[9]	RW	Manually SCL stretch for slave. STRCH_EN should be 1 to make this bit effective.	0x0000_0000
STRCH_EN	[8]	RW	Enables slave SCL stretch	0x0000_0001
T_STOPTOIDLE	[7:0]	RW	STOP to IDLE state interval for I2C master.	0x0000_00FF

4.15.3.1.10 I2C_AUTO_CONF

- Address: Base Address + 0x0044, Reset Value = 0x0003_00FF

Name	Bit	Type	Description	Reset Value
MASTER_RUN	[31]	W	To start data transfer under I2C master mode, write "1" to this bit.	0x0000_0000
RSVD0	[30:18]	RW	Reserved	0x0000_0000
STOP_AFTER_TRANS	[17]	RW	For auto_mode only. If this bit is set, i2c master will assert STOP condition after entire transaction is complete, either succeed or fail.	0x0000_0001
READ_WRTIE	[16]	RW	* Type of transaction (For AUTO_MODE only). 1: Read (rx) 0: Write (tx)	0x0000_0001

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Name	Bit	Type	Description	Reset Value
TRANS_LEN	[15:0]	RW	Length of Transaction. (Number of Bytes to RX or TX).	0x0000_00FF

4.15.3.1.11 I2C_TIMEOUT

- Address: Base Address + 0x0048, Reset Value = 0x8000_00FF

Name	Bit	Type	Description	Reset Value
TIMEOUT_EN	[31]	RW	Enables timing out.	0x0000_0001
RSVD0	[30:16]	RW	Reserved	0x0000_0000
TOUT_COUNT	[15:0]	RW	Timing out for master RX/TX.	0x0000_00FF

4.15.3.1.12 I2C_MANUAL_CMD

- Address: Base Address + 0x004C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TX_DATA	[31:24]	RW	Specifies the data to be transmitted	0x0000_0000
RX_DATA	[23:16]	R	Specifies the received data	0x0000_0000
RX_ACK	[5]	RW	This is the Acknowledgement for master-RX. After software asserts READ_DATA, I2C master receives one BYTE from external i2c and sends one bit acknowledgement to external i2c.	0x0000_0000
READ_DATA	[4]	W	If you write "1" to this bit it forces I2C master to receive one BYTE data. The data will be then available in RX_DATA.	0x0000_0000
SEND_DATA	[3]	W	If you write "1" to this bit it forces I2C master to send one BYTE data. The data should be pre-written in TX_DATA.	0x0000_0000
SEND_STOP	[2]	W	If you write "1" to this bit it forces I2C master to send STOP condition.	0x0000_0000
SEND_RESTART	[1]	W	If you write "1" to this bit it forces I2C master to send RE-START condition.	0x0000_0000
SEND_START	[0]	W	If you write "1" to this bit it forces I2C master to send START condition.	0x0000_0000

NOTE: This Register is only valid in manual mode (AUTO_MODE is '0').

4.15.3.1.13 I2C_TRANS_STATUS

- Address: Base Address + 0x0050, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SLAVE_RW	[19]	R	* Current slave operation mode:	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			1: TX mode 0: RX mode	
STOP_COND	[18]	R	* Indicates stop condition on bus: 1: Stop condition on bus 0: No stop condition on bus	0x0000_0000
MASTER_BUSY	[17]	R	* Indicates whether the data is transferring or not in master mode: 1: The data is transferring 0: The data is not transferring	0x0000_0000
SLAVE_BUSY	[16]	R	* Indicates the data is transferring or not in slave mode: 1: The data is transferring 0: The data is not transferring	0x0000_0000
SLAVE_SCL_STRETCH	[15]	R	* Indicates SCL is stretching or not in slave mode: 1: SCL is stretching 0: SCL is not stretching	0x0000_0000
RSVD0	[14:0]	R	Reserved	0x0000_0000

NOTE: "MASTER BUSY": I2C master is busy executing transaction (for auto mode) or command (for manual mode). If auto mode, MASTER BUSY is high until all i2c transaction is accomplished. If manual mode, MASTER BUSY is high until current command is accomplished.

"SLAVE BUSY": high when I2C slave is receiving Address or when data is transferring.

4.15.3.1.14 I2C_TIMING_HS1

- Address: Base Address + 0x0054, Reset Value = 0x0F0F_0F00

Name	Bit	Type	Description	Reset Value
TSTART_SU_HS	[31:24]	RW	Start Condition setup timing. (Start condition after SCL rising-edge)	0x0000_000F
TSTART_HD_HS	[23:16]	RW	Start Condition Hold timing. (Start condition leads to SCL falling-edge)	0x0000_000F
TSTOP_SU_HS	[15:8]	RW	STOP condition setup timing. (STOP condition after SCL rising-edge)	0x0000_000F
TSDA_SU_HS	[7:0]	RW	SDA trigger timing (output setup timing) - If this field is lesser than or equal to (TSCL_L_HS/2), or greater than TSCL_L_HS, it has no effects and default trigger timing is used. Effective range is as following; (TSCL_L_HS/2) < TSDA_SU_HS <= TSCL_L_HS - To enhance setup timing in maximum frequency operation of high speed mode, it is recommended to use closer value or same value as TSCL_L_HS (maximum effective value).	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			- If the FLT_CYCLE_SCL is a little big value, it covers almost of half period of SCL low interval. And, by default, SDA triggering(changing) time is the half of TSCL_L_HS(FS). Without this bit field, SDA trigger time was NOT changeable, so that SDA is always triggered nearly SCL rising edge instead of the center of SCL low interval. TSDA_SU_HS can move SDA trigger time faster to the center of SCL low interval.	

4.15.3.1.15 I2C_TIMING_HS2

- Address: Base Address + 0x0058, Reset Value = 0x0F0F_FFFF

Name	Bit	Type	Description	Reset Value
TDATA_SU_HS	[31:24]	RW	Data setup timing	0x0000_000F
RSVD0	[23:16]	RW	Reserved	0x0000_000F
TSCL_L_HS	[15:8]	RW	SCL Low Period	0x0000_00FF
TSCL_H_HS	[7:0]	RW	SCL High Period	0x0000_00FF

4.15.3.1.16 I2C_TIMING_HS3

- Address: Base Address + 0x005C, Reset Value = 0x00FF_00FF

Name	Bit	Type	Description	Reset Value
CLK_DIV	[23:16]	RW	The PCLK divider factor when working in HS-mode.	0x0000_00FF
RSVD0	[15:8]	RW	Reserved	0x0000_0000
TSR_RELEASE	[7:0]	RW	RE-START condition.	0x0000_00FF

4.15.3.1.17 I2C_TIMING_FS1

- Address: Base Address + 0x0060, Reset Value = 0x0F0F_0F00

Name	Bit	Type	Description	Reset Value
TSTART_SU_FS	[31:24]	RW	Start Condition setup timing. (Start condition after SCL rising-edge)	0x0000_000F
TSTART_HD_FS	[23:16]	RW	Start Condition Hold timing. (Start condition leads to SCL falling-edge)	0x0000_000F
TSTOP_SU_FS	[15:8]	RW	STOP condition setup timing. (STOP condition after SCL rising-edge)	0x0000_000F
TSDA_SU_FS	[7:0]	RW	SDA trigger timing (output setup timing) - If this filed is lesser than or equal to (TSCL_L_FS/2), or greater than TSCL_L_FS, it has no effects and default trigger timing is used.	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			Effective range is as following; (TSCL_L_FS/2) <= TSDA_SU_FS <= TSCL_L_FS - It is recommended to use the reset value because it is not critical in fast speed mode.	

4.15.3.1.18 I2C_TIMING_FS2

- Address: Base Address + 0x0064, Reset Value = 0x0F0F_FFFF

Name	Bit	Type	Description	Reset Value
TDATA_SU_FS	[31:24]	RW	Data setup timing.	0x0000_000F
RSVD0	[23:16]	RW	Reserved	0x0000_000F
TSCL_L_FS	[15:8]	RW	SCL Low Period	0x0000_00FF
TSCL_H_FS	[7:0]	RW	SCL High Period	0x0000_00FF

4.15.3.1.19 I2C_TIMING_FS3

- Address: Base Address + 0x0068, Reset Value = 0x00FF_00FF

Name	Bit	Type	Description	Reset Value
CLK_DIV	[23:16]	RW	The PCLK divider factor when working in FS-mode.	0x0000_00FF
RSVD0	[15:8]	RW	Reserved	0x0000_0000
TSR_RELEASE	[7:0]	RW	RE-START condition.	0x0000_00FF

4.15.3.1.20 I2C_TIMING_SLA

- Address: Base Address + 0x006C, Reset Value = 0x0000_00FF

Name	Bit	Type	Description	Reset Value
I2C_TIMING_SLA	[15:0]	RW	Data hold timing for I2C slave. (data transition after SCL falling-edge) (For both auto and manual modes)	0x0000_00FF

NOTE: The value of TDATA_HD should be less than the value of 'SCL_perid/(2 × PCLK_perid)'.

4.15.3.1.21 I2C_ADDR

- Address: Base Address + 0x0070, Reset Value = 0xFF0F_FFFF

Name	Bit	Type	Description	Reset Value
MASTERID	[31:24]	RW	Master ID for I2C master. (for AUTO mode and HS mode only).	0x0000_00FF
EXTENDED_SLAVE_ADDR_MAS	[23:20]	RW	Extended Slave Address bit [13:10] for I2C master (for AUTO mode only).	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			This bit field is corresponding to "1110" 4bits of 11110xx of the first SLAVE address and is used only if CTL[30](EXTENDED_ADDR_MODE) is 1 and I2C_CONF[30](ADDRMODE) is 1. Note that MSB of 11110xx of the first SLAVE address had better be fixed to 1 but is also controllable by CTL[29](EXTENDED_ADDR_MSB). Note that LSB two bit (xx) of 11110xx of the first SLAVE address is I2C_ADDR[19:18] (see below bit field).	
SLAVE_ADDR_MAS	[19:10]	RW	Slave Address bit [9:0] for I2C master (for AUTO mode only).	0x0000_03FF
SLAVE_ADDR_SLA	[9:0]	RW	Slave Address bit [9:0] for I2C slave.	0x0000_03FF

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4.16 Serial Peripheral Interface (SPI)

4.16.1 Overview

The SPI (Serial Peripheral Interface) is the ARM PrimeCell PL022 SSP (Synchronous Serial Port). It is an APB slave block that enables synchronous serial communication with peripheral devices that have either a Motorola SPI compatible interface, Texas Instruments Synchronous Serial interface or National Semiconductor Microwire interface. The SSP can behave as a master or a slave interface and performs parallel-to-serial conversion on data written to an internal 16-bit wide and 8-deep transmit FIFO. It also performs serial-to-parallel conversion on the serial input data and buffers it a receive FIFO which is also 16bits wide and 8-deep. The SSP block asserts interrupts to request servicing of transmit and receive buffers to indicate an overrun or receive timeout condition in the receive FIFO.

A block diagram of the SPI is shown in *Figure 4-39*. The APB interface generates decoded write signals for registers in the device. The Register block contains the registers in the SSP. The control information written into the SSPCR0 and the SSPCPSR registers is synchronized to the SSPCLK clock domain through synchronization control logic located in the Register block and in the Clock Pre-Scaler.

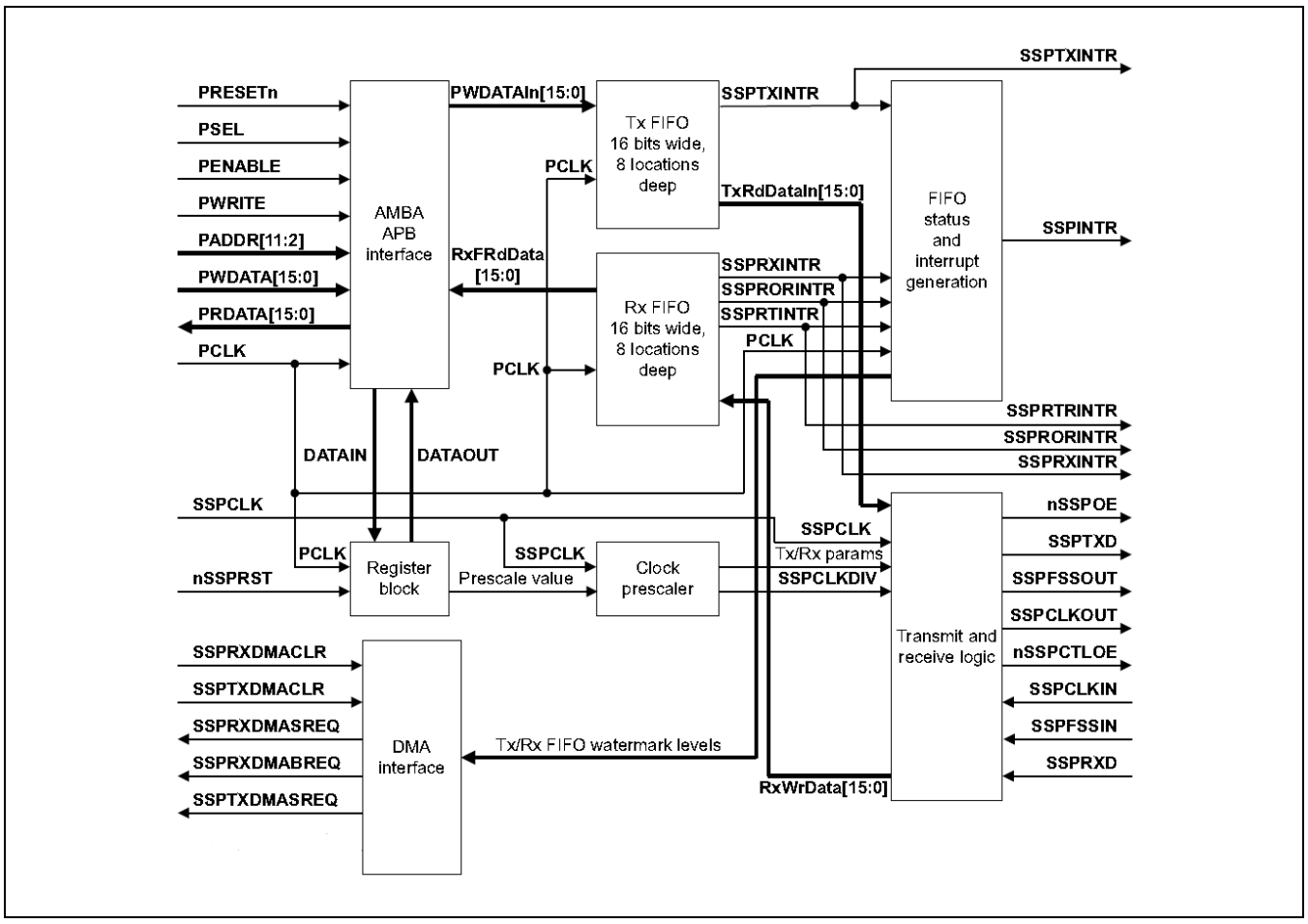


Figure 4-39 SPI Block Diagram

When the MS bit in the SSPCR1 register is cleared, the SSP behaves as a master, when set, the SSP behaves as a slave. In the Slave mode, the SSP acts as a receive-only slave when the SOD bit in the SSPCR1 register is set, when cleared it can also transmit. In the master mode, SSPCLK is first divided by the programmed pre-scale

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value and then by the SCR value to generate the SSPCLKOUT signal. In the slave mode, the slave interface makes use of the SSPFSSIN and SSPCLKIN signals from the external master. The Transmitter, in accordance with the transmission parameters programmed into the Control registers, converts data written in to the Transmit FIFO into a serial bit stream on the SSPTXD output line. The Receiver performs serial-to-parallel conversion on data received on the SSPRXD line. Received data is buffered in the 16-bit wide 8-deep receive FIFO. The Transmit FIFO and the Receive FIFO are 16-bits wide and 8-deep. They generate the service requests, SSPTXINTR and SSPRXINTR. The Receive FIFO also generates the Receive Overrun interrupt signal, SSPRORINTR and the Receive Timeout Interrupt signal, SSPRTINTR. The Interrupt generation block uses the SSPTXINTR, SSPRXINTR, SSPRORINTR and SSPRTINTR signals to generate the combined interrupt, SSPINTR.

4.16.2 Functional Description of SPI

The following sections describe the operation of the PrimeCell SSP:

- Interface reset
- Configuring the SSP
- Enable PrimeCell SSP operation
- Clock ratios
- Programming the SSPCR0 Control Register
- Programming the SSPCR1 Control Register
- Frame format
- Texas Instruments synchronous serial frame format
- Motorola SPI frame format
- Motorola SPI Format with SPO=0, SPH=0
- Motorola SPI Format with SPO=0, SPH=1
- Motorola SPI Format with SPO=1, SPH=0
- Motorola SPI Format with SPO=1, SPH=1
- National Semiconductor Microwire frame format
- Examples of master and slave configurations
- PrimeCell DMA interface

4.16.2.1 Interface reset

The PrimeCell SSP is reset by the global reset signal, PRESETn, and a block-specific reset signal, nSSPRST. An external reset controller must use PRESETn to assert nSSPRST asynchronously and negate it synchronously to SSPCLK. PRESETn must be asserted LOW for a period long enough to reset the slowest block in the on-chip system, and then taken HIGH again. The PrimeCell SSP requires PRESETn to be asserted LOW for at least one period of PCLK.

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4.16.2.2 Configuring the SSP

Following reset, the PrimeCell SSP logic is disabled and must be configured when in this state. It is necessary to program control registers SSPCR0 and SSPCR1 to configure the peripheral as a master or slave operating under one of the following protocols:

- Motorola SPI
- Texas Instruments SSI
- National Semiconductor.

The bit rate, derived from the external SSPCLK, requires the programming of the clock prescale register SPCPSR. See Clock prescale register, SSPCPSR.

4.16.2.3 Enable PrimeCell SSP operation

You can either prime the transmit FIFO, by writing up to eight 16-bit values when the PrimeCell SSP is disabled, or permit the transmit FIFO service request to interrupt the CPU. Once enabled, transmission or reception of data begins on the transmit, SSPTXD, and receive, SSPRXD, pins.

4.16.2.4 Clock ratios

There is a constraint on the ratio of the frequencies of PCLK to SSPCLK. The frequency of SSPCLK must be less than or equal to that of PCLK. This ensures that control signals from the SSPCLK domain to the PCLK domain are guaranteed to get synchronized before one frame duration:

$F_{SSPCLK} <= F_{PCLK}$

In the slave mode of operation, the SSPCLKIN signal from the external master is double-synchronized and then delayed to detect an edge. It takes three SSPCLKs to detect an edge on SSPCLKIN. SSPTXD has less setup time to the falling edge of SSPCLKIN on which the master is sampling the line.

The setup and hold times on SSPRXD, with reference to SSPCLKIN, must be more conservative to ensure that it is at the right value when the actual sampling occurs within the SSPMS. To ensure correct device operation, SSPCLK must be at least 12 times faster than the maximum expected frequency of SSPCLKIN.

The frequency selected for SSPCLK must accommodate the desired range of bit clock rates. The ratio of minimum SSPCLK frequency to SSPCLKOUT maximum frequency in the case of the slave mode is 12, and for the master mode, it is two.

To generate a maximum bit rate of 43.5Mbps in the master mode, the frequency of SSPCLK must be at least 87MHz. With an SSPCLK frequency of 87MHz, the SSPCPSR register must be programmed with a value of 2, and the SCR[7:0] field in the SSPCR0 register must be programmed with a value of 0.

To work with a maximum bit rate of 7.25Mbps in the slave mode, the frequency of SSPCLK must be at least 87MHz. With an SSPCLK frequency of 87MHz, the SSPCPSR register can be programmed with a value of 12, and the SCR[7:0] field in the SSPCR0 register can be programmed with a value of 0. Similarly, the ratio of SSPCLK maximum frequency to SSPCLKOUT minimum frequency is 254 x 256. See Control register 0, SSPCR0 and Clock prescale register, SSPCPSR

The minimum frequency of SSPCLK is governed by the following equations, both of which must be satisfied:

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F_{SSPCLK}(min) => 2 x $F_{SSPCLKOUT}$(max), for master mode

F_{SSPCLK}(min) => 12 x $F_{SSPCLKIN}$(max), for slave mode.

The maximum frequency of SSPCLK is governed by the following equations, both of which must be satisfied:

F_{SSPCLK}(max) <= 254 x 256 x $F_{SSPCLKOUT}$(min), for master mode

F_{SSPCLK}(max) <= 254 x 256 x $F_{SSPCLKIN}$(min), for slave mode.

4.16.2.5 Programming the SSPCR0 Control Register

See Control register 0, SSPCR0 for more information on the bit assignment of this register.

The SSPCR0 register is used to:

• program the serial clock rate

• select one of the three protocols

• select the data word size, where applicable.

The Serial Clock Rate (SCR) value, in conjunction with the SSPCPSR clock prescale divisor value, CPSDVSR, is used to derive the PrimeCell SSP transmit and receive bit rate from the external SSPCLK.

The frame format is programmed through the FRF bits, and the data word size through the DSS bits.

Bit phase and polarity, applicable to Motorola SPI format only, are programmed through the SPH and SPO bits.

4.16.2.6 Programming the SSPCR1 Control Register

See Control register 1, SSPCR1, for more information on the bit assignment of this register.

The SSPCR1 register is used to:

- select master or slave mode
- enable a loop back test feature
- enable the PrimeCell SSP peripheral

To configure the PrimeCell SSP as a master, clear the SSPCR1 register master or slave selection bit, MS, to 0. This is the default value on reset.

Setting the SSPCR1 register MS bit to 1 configures the PrimeCell SSP as a slave. When configured as a slave, enabling or disabling of the PrimeCell SSP SSPTXD signal is provided through the SSPCR1 slave mode SSPTXD output disable bit, SOD. You can use this in some multi-slave environments where masters might parallel broadcast.

To enable the operation of the PrimeCell SSP, set the Synchronous Serial Port Enable (SSE) bit to 1.

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Bit rate generation

The serial bit rate is derived by dividing down the input clock, SSPCLK. The clock is first divided by an even prescale value CPSDVSR in the range 2-254, and is programmed in SSPCPSR. The clock is divided again by a value in the range 1-256, that is 1 + SCR, where SCR is the value programmed in SSPCR0.

The following equation defines the frequency of the output signal bit clock, SSPCLKOUT:

$$F_{SSPCLKOUT} = \frac{F_{SSPCLK}}{\text{CPSDVR} \times (1 + \text{SCR})}$$

For example, if SSPCLK is 87MHz, and CPSDVSR = 2, then SSPCLKOUT has a frequency range from 169.921KHz to 43.5MHz.

4.16.2.7 Frame format

Each data frame is between 4-16 bits long, depending on the size of data programmed, and is transmitted starting with the MSB. You can select the following basic frame types:

- Texas Instruments synchronous serial
- Motorola SPI
- National Semiconductor Microwire

For all formats, the serial clock, SSPCLKOUT, is held inactive while the PrimeCell SSP is idle, and transitions at the programmed frequency only during active transmission or reception of data. The idle state of SSPCLKOUT is utilized to provide a receive timeout indication that occurs when the receive FIFO still contains data after a timeout period.

For Motorola SPI and National Semiconductor Microwire frame formats, the serial frame, SSPFSSOUT, pin is active-LOW, and is asserted, pulled-down, during the entire transmission of the frame.

For Texas Instruments synchronous serial frame format, the SSPFSSOUT pin is pulsed for one serial clock period, starting at its rising edge, prior to the transmission of each frame. For this frame format, both the PrimeCell SSP and the off-chip slave device drive their output data on the rising edge of SSPCLKOUT, and latch data from the other device on the falling edge.

Unlike the full-duplex transmission of the other two frame formats, the National Semiconductor Microwire format uses a special master-slave messaging technique that operates at half-duplex. In this mode, when a frame begins, an 8-bit control message is transmitted to the off-chip slave. During this transmission, the SSS receives no incoming data. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the requested data. The returned data can be 4-16 bits in length, making the total frame length in the range 13-25 bits.

4.16.2.7.1 Texas Instruments synchronous serial frame format

Figure 4-40 shows the Texas Instruments synchronous serial frame format for a single transmitted frame.

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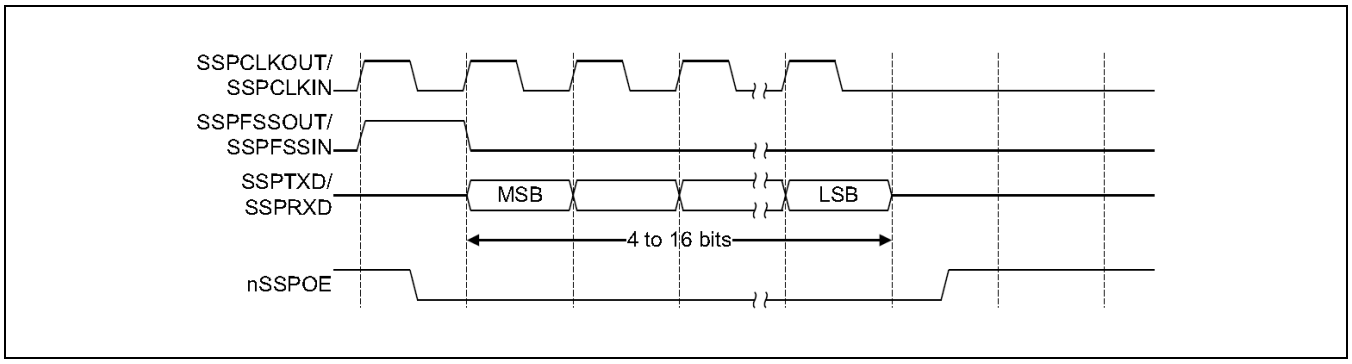


Figure 4-40 Texas Instruments synchronous serial frame format, single transfer

In this mode, SSPCLKOUT and SSPFSSOUT are forced LOW, and the transmit data line, SSPTXD is tristated whenever the PrimeCell SSP is idle. When the bottom entry of the transmit FIFO contains data, SSPFSSOUT is pulsed HIGH for one SSPCLKOUT period. The value to be transmitted is also transferred from the transmit FIFO to the serial shift register of the transmit logic. On the next rising edge of SSPCLKOUT, the MSB of the 4-bit to 16-bit data frame is shifted out on the SSPTXD pin. In a similar way, the MSB of the received data is shifted onto the SSPRXD pin by the off-chip serial slave device.

Both the PrimeCell SSP and the off-chip serial slave device then clock each data bit into their serial shifter on the falling edge of each SSPCLKOUT. The received data is transferred from the serial shifter to the receive FIFO on the first rising edge of PCLK after the LSB has been latched.

Figure 4-41 shows the Texas Instruments synchronous serial frame format when back-to-back frames are transmitted.

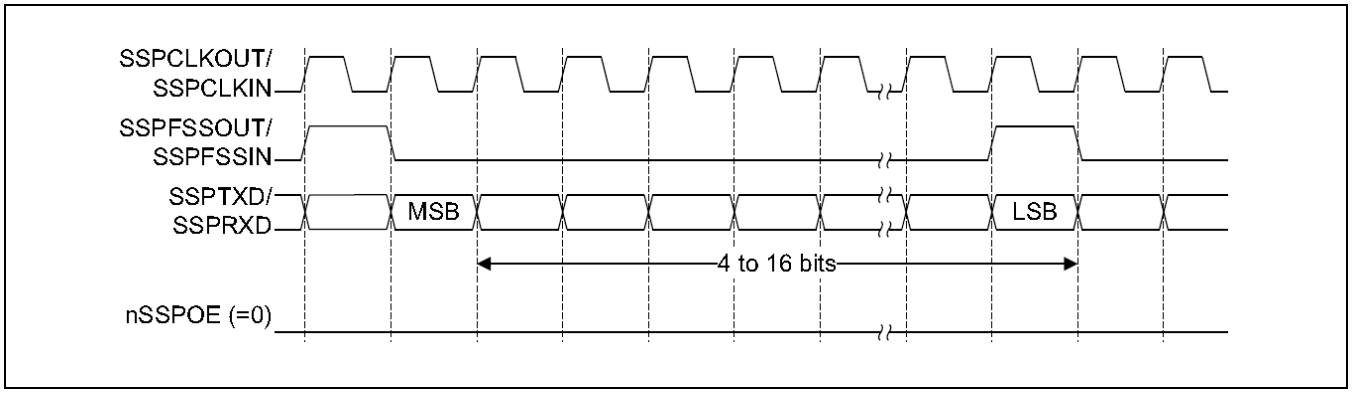


Figure 4-41 Texas Instruments synchronous serial frame format, continuous transfer

4.16.2.7.2 Motorola SPI frame format

The Motorola SPI interface is a four-wire interface where the SSPFSSOUT signal behaves as a slave select. The main feature of the Motorola SPI format is that you can program the inactive state and phase of the SSPCLKOUT signal using the SPO and SPH bits of the SSPSCR0 control register. See Control register 0, SSPCR0.

SPO, clock polarity

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When the SPO clock polarity control bit is LOW, it produces a steady state LOW value on the SSPCLKOUT pin. If the SPO clock polarity control bit is HIGH, a steady state HIGH value is placed on the SSPCLKOUT pin when data is not being transferred.

SPH, clock phase

The SPH control bit selects the clock edge that captures data and enables it to change state. It has the most impact on the first bit transmitted by either permitting or not permitting a clock transition before the first data capture edge.

When the SPH phase control bit is LOW, data is captured on the first clock edge transition.

When the SPH clock phase control bit is HIGH, data is captured on the second clock edge transition.

4.16.2.7.2.1 Motorola SPI Format with SPO=0, SPH=0

Figure 4-42 and *Figure 4-43* show single and continuous transmission signal sequences for Motorola SPI format with SPO=0, SPH=0. *Figure 4-42* shows a single transmission signal sequence for Motorola SPI frame format with SPO=0, SPH=0.

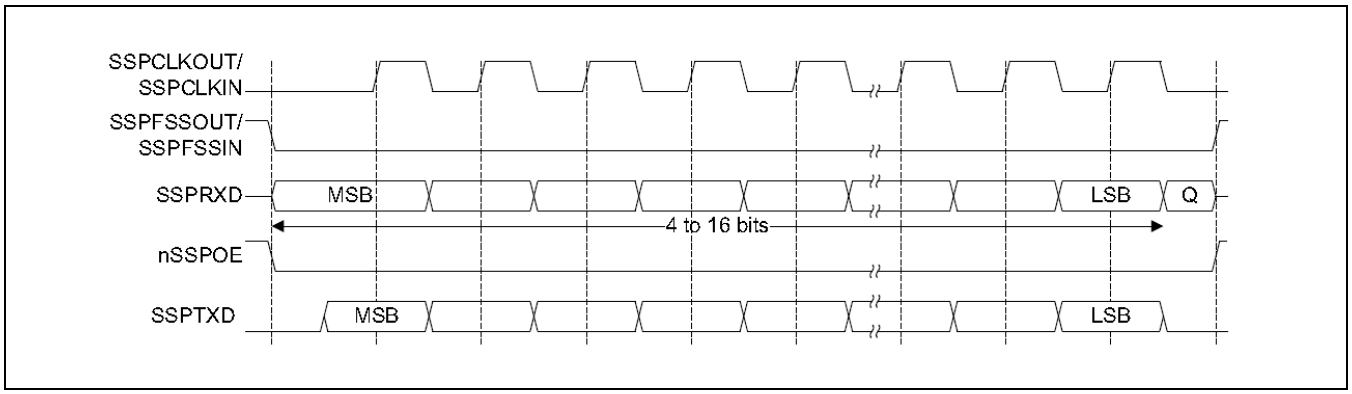


Figure 4-42 Motorola SPI frame format, single transfer, with SPO=0 and SPH=0

Figure 4-43 shows a continuous transmission signal sequence for Motorola SPI frame format with SPO=0, SPH=0

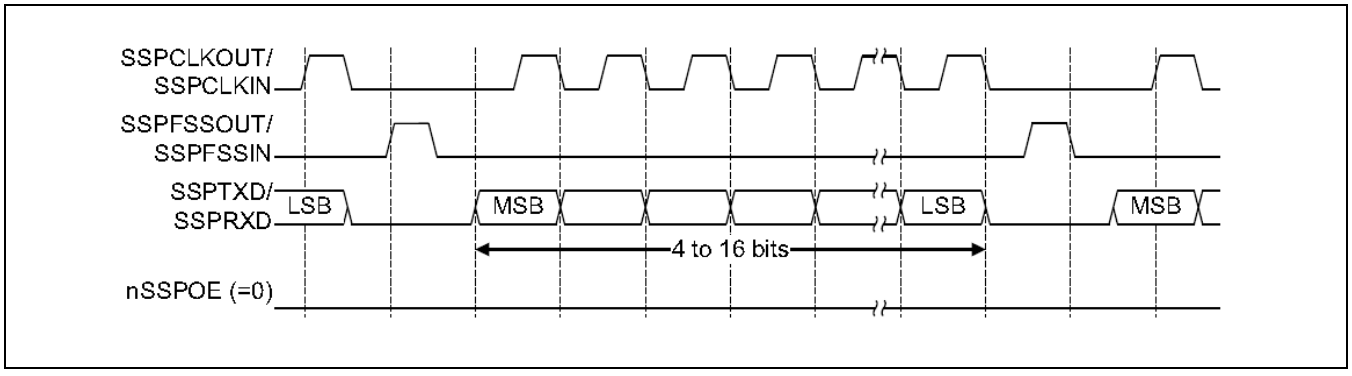


Figure 4-43 Motorola SPI frame format, continuous transfer, with SPO=0 and SPH=0

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.16.2.7.2.2 Motorola SPI Format with SPO=0, SPH=1

Figure 4-44 shows the transfer signal sequence for Motorola SPI format with SPO=0, SPH=1, and it covers both single and continuous transfers.

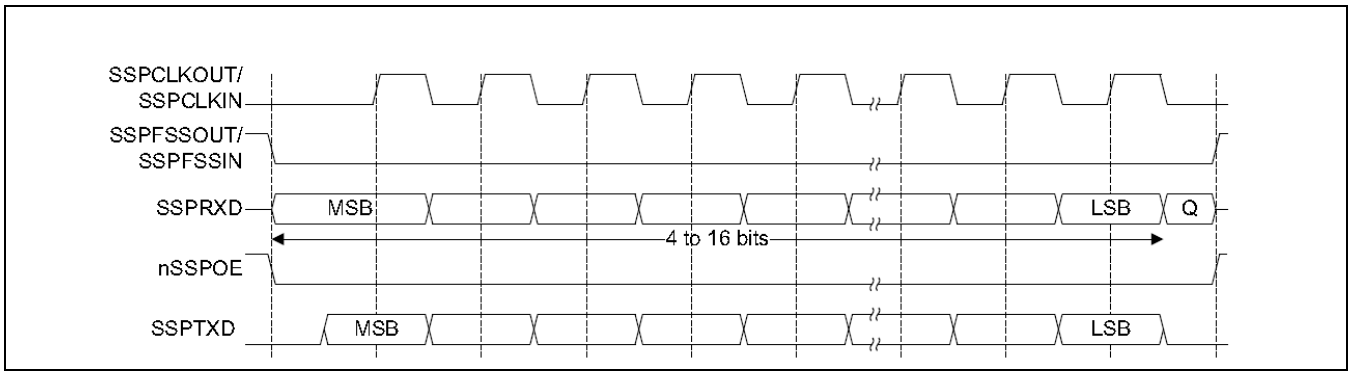


Figure 4-44 Motorola SPI frame format with SPO=0 and SPH=1, single and continuous transfers

4.16.2.7.2.3 Motorola SPI Format with SPO=1, SPH=0

Figure 4-45 and *Figure 4-46* show single and continuous transmission signal sequences for Motorola SPI format with SPO=1, SPH=0.

Figure 4-45 shows a single transmission signal sequence for Motorola SPI format with SPO=1, SPH=0.

In *Figure 4-45* Q is an undefined signal.

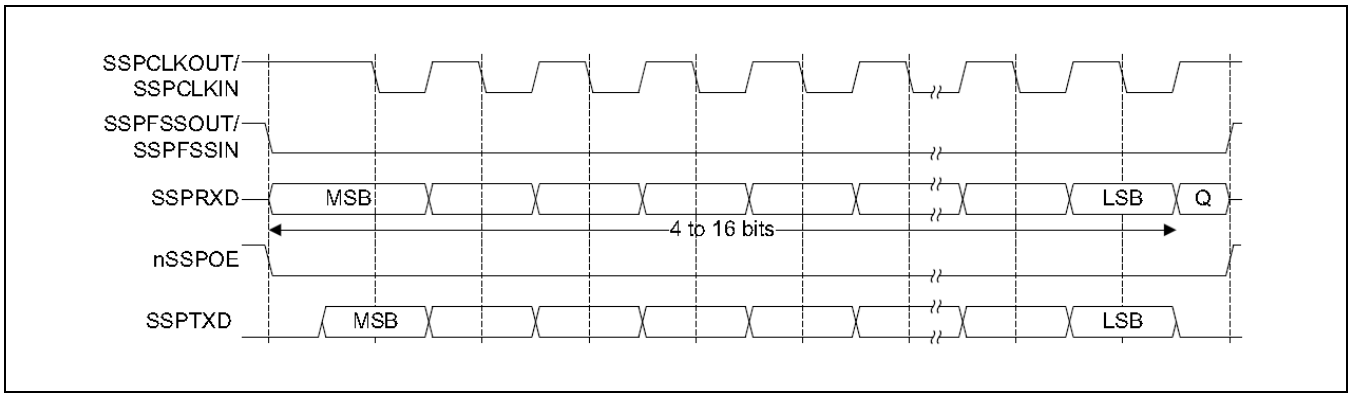


Figure 4-45 Motorola SPI frame format, single transfer, with SPO=1 and SPH=0

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 4-46 shows a continuous transmission signal sequence for Motorola SPI format with SPO=1, SPH=0

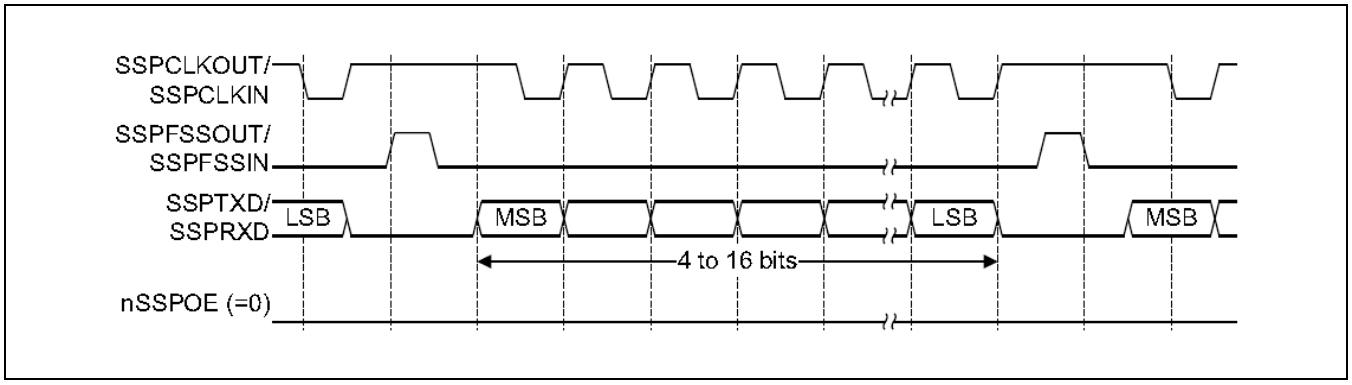


Figure 4-46 Motorola SPI frame format, continuous transfer, with SPO=1 and SPH=0

4.16.2.7.2.4 Motorola SPI Format with SPO=1, SPH=1

Figure 4-47 shows the transfer signal sequence for Motorola SPI format with SPO=1, SPH=1, and it covers both single and continuous transfers.

In *Figure 4-47*, Q is an undefined signal.

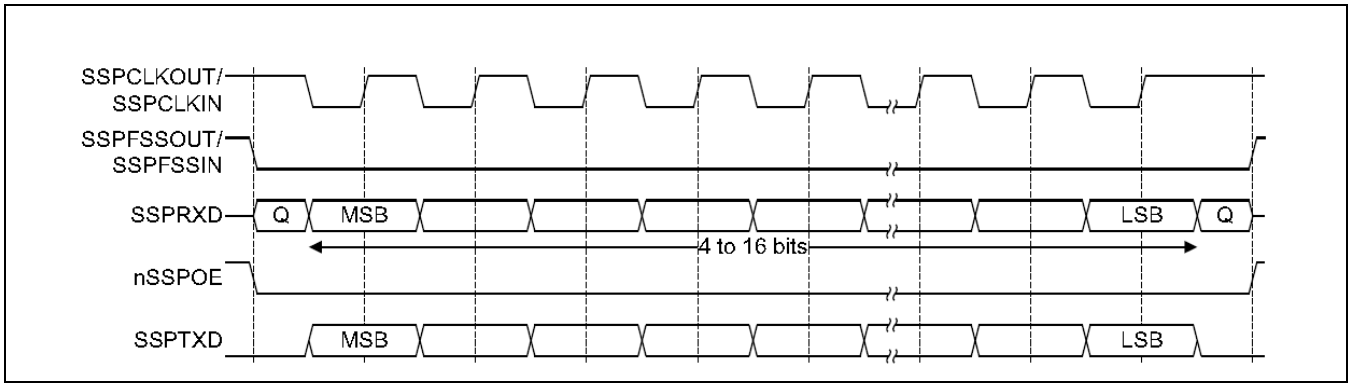


Figure 4-47 Motorola SPI frame format with SPO=1 and SPH=1, single and continuous transfers

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.16.2.7.3 National Semiconductor Microwire frame format

Figure 4-48 shows the National Semiconductor Microwire frame format for a single frame.

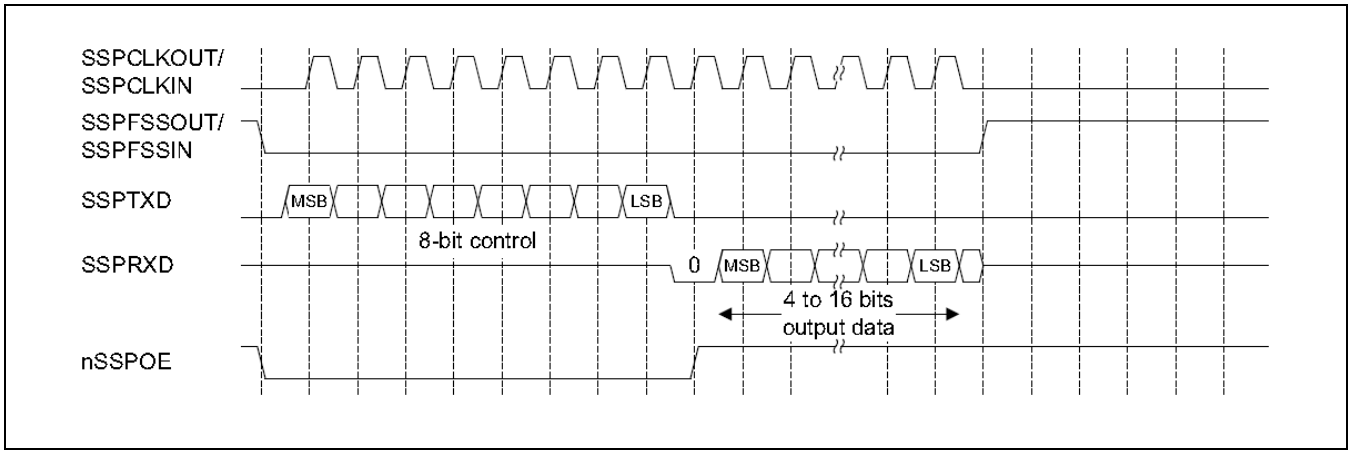


Figure 4-48 Microwire frame format, single transfer

Microwire format is very similar to SPI format, except that transmission is half-duplex instead of full-duplex, using a master-slave message passing technique. Each serial transmission begins with an 8-bit control word that is transmitted from the PrimeCell SSP to the off-chip slave device. During this transmission, the PrimeCell SSP receives no incoming data. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the required data. The returned data is 4 to 16 bits in length, making the total frame length in the range 13-25 bits.

NOTE: The off-chip slave device can tristate the receive line either on the falling edge of SSPCLKOUT after the LSB has been latched by the receive shifter, or when the SSPFSSOUT pin goes HIGH.

For continuous transfers, data transmission begins and ends in the same manner as a single transfer. However, the SSPFSSOUT line is continuously asserted, held LOW, and transmission of data occurs back-to-back. The control byte of the next frame follows directly after the LSB of the received data from the current frame. Each of the received values is transferred from the receive shifter on the falling edge SSPCLKOUT, after the LSB of the frame has been latched into the PrimeCell SSP.

Figure 4-49 shows the National Semiconductor Microwire frame format when back-to-back frames are transmitted.

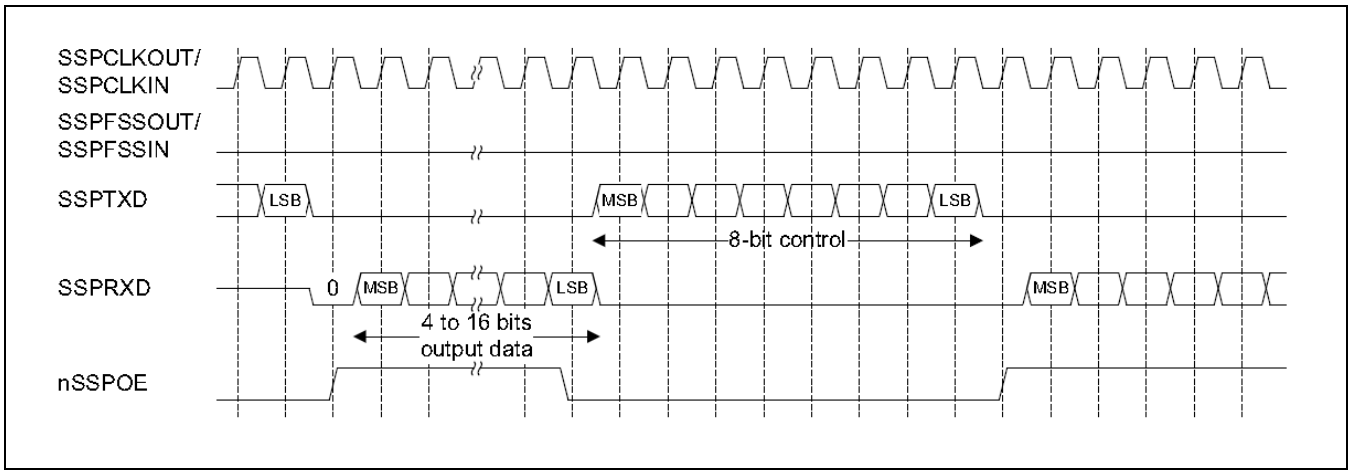


Figure 4-49 Microwire frame format, continuous transfers

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In Microwire mode, the PrimeCell SSP slave samples the first bit of receive data on the rising edge of SSPCLKIN after SSPFSSIN has gone LOW. Masters that drive a free-running SSPCKLIN must ensure that the SSPFSSIN signal has sufficient setup and hold margins with respect to the rising edge of SSPCLKIN.

Figure 4-50 shows setup and hold time requirements.

With respect to the SSPCLKIN rising edge on which the first bit of receive data is to be sampled by the PrimeCell SSP slave, SSPFSSIN must have a setup of at least two times the period of SSPCLK on which the PrimeCell SSP operates.

With respect to the SSPCLKIN rising edge previous to this edge, SSPFSSIN must have a hold of at least one SSPCLK period.

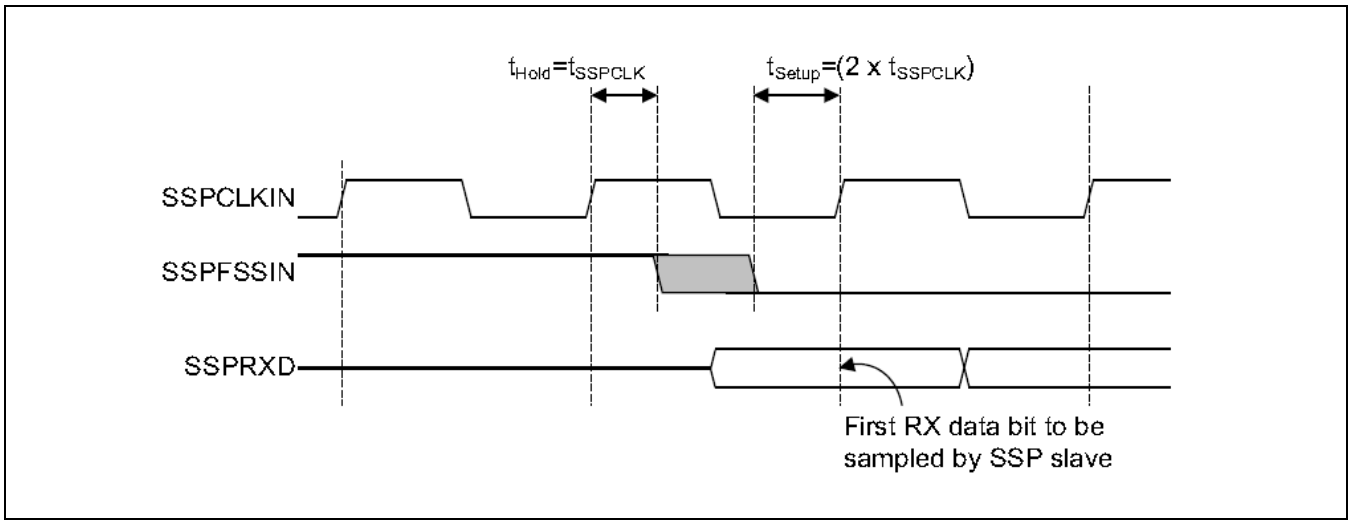


Figure 4-50 Microwire frame format, SSPFSSIN input setup and hold requirements

4.16.2.8 PrimeCell DMA interface

The PrimeCell SSP provides an interface to connect to the DMA controller. The PrimeCell SSP DMA control register, SSPDMACR controls the DMA operation of the PrimeCell SSP. See DMA control register, SSPDMACR.

In Rx DMA mode, single and burst transfers are supported with the register, RXIFLSEL of SSPCR1. In Tx DMA mode, only single transfer is supported.

Receive

The DMA interface includes the following signals, for receive:

SSPRXDMASREQ:

Single-character DMA transfer request, asserted by the SSP. This signal is asserted when the receive FIFO contains at least one character.

SSPRXDMABREQ:

Burst DMA transfer request, asserted by the SSP. This signal is asserted when the receive FIFO contains RXIFLSEL or more characters.

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SSPRXDMACLR:

DMA request clear, asserted by the DMA controller to clear the receive request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

Transmit

The DMA interface includes the following signals, for transmit:

SSPTXDMASREQ:

Single-character DMA transfer request, asserted by the SSP. This signal is asserted when there is at least one empty location in the transmit FIFO.

SSPTXDMACLR:

DMA request clear, asserted by the DMA controller, to clear the transmit request signals. If a DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

The burst transfer and single transfer request signals are not mutually exclusive. They can both be asserted at the same time. For example, when there is more data than the watermark level of four in the receive FIFO, the burst transfer request, and the single transfer request, are asserted. When the amount of data left in the receive FIFO is less than the watermark level, the single request only is asserted. This is useful for situations where the number of characters left to be received in the stream is less than a burst.

For example, if 19 characters must be received, the DMA controller then transfers four bursts of four characters, and three single transfers to complete the stream.

NOTE: For the remaining three characters, the PrimeCell SSP does not assert the burst request.

Each request signal remains asserted until the relevant DMA clear signal is asserted. After the request clear signal is deasserted, a request signal can become active again, depending on the conditions that previous sections describe. All request signals are deasserted if the PrimeCell SSP is disabled, or the DMA enable signal is cleared. *Table 4-10* describes DMA trigger points for the transmit and receive FIFOs

Table 4-10 DMA trigger points for the transmit and receive FIFOs

	Burst length	
Watermark level	**Transmit, number of empty locations**	**Receive, number of filled locations**
1/2	4	4

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Figure 4-51 shows the timing diagram for both a single transfer request, and a burst transfer request, with the appropriate DMA clear signal. The signals are all synchronous to PCLK.

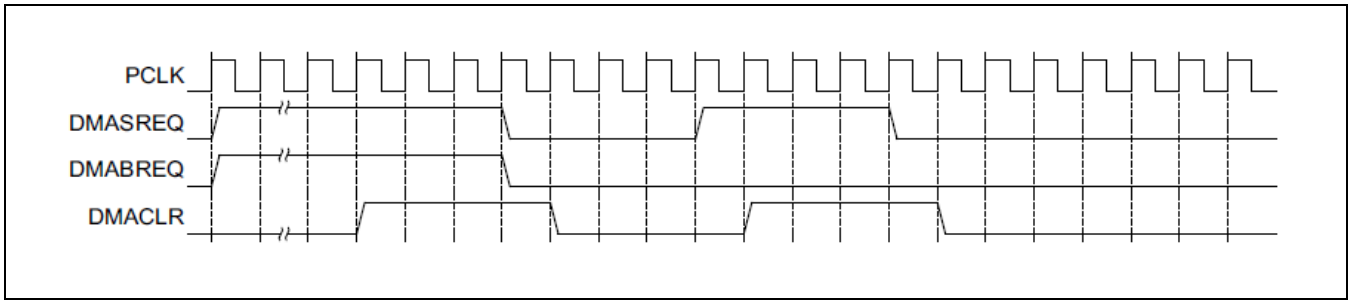


Figure 4-51 DMA transfer waveforms

4.16.3 Register Description

4.16.3.1 Register Map Summary

- Base Address: 0x4000D000

Register	Offset	Description	Reset Value
SSPCR0	0x0000	Control register 0	0x0000_0000
SSPCR1	0x0004	Control register 1	0x0000_0090
SSPDR	0x0008	SSPDR is the data register and is 16-bits wide.	0x0000_0000
SSPSR	0x000C	Status Register. Reads from this register return SSP status	0x0000_0003
SSPCPSR	0x0010	SSP Clock Pre-scale Register. This register specifies the factor by which SSPCLK is to be divided in the Clock pre-scaler. The division factor has to be an even number from 2 to 254. The least significant bit of the programmed number is hard-coded to zero. Data read back from this register has the least significant bit as zero if an odd number is written to this register.	0x0000_0000
SSPIMSC	0x0014	The SSPIMSC register is the interrupt mask set or clear register. On a read this register gives the current value of the mask on the relevant interrupt. A write of 1 to the particular bit sets the mask enabling the interrupt to be read. A write of 0 clears the corresponding mask	0x0000_0000
SSPRIS	0x0018	Raw Interrupt Status register. On a read this register gives the current raw status value of the corresponding interrupt prior to masking. A write has no effect	0x0000_0008
SSPMIS	0x001C	Masked Interrupt Status register. On a read this register gives the current masked status value of the corresponding interrupt. A write has no effect	0x0000_0000

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Register	Offset	Description	Reset Value
SSPICR	0x0020	Interrupt Clear register. On a write of 1, the corresponding interrupt is cleared. A write of 0 has no effect	0x0000_0000
SSPDMACLR	0x0024	DMA Control register	0x0000_0000
SSPPeriphID0	0x0FE0	Peripheral identification register bits [7:0]	0x0000_0022
SSPPeriphID1	0x0FE4	Peripheral identification register bits [15:8]	0x0000_0010
SSPPeriphID2	0x0FE8	Peripheral identification register bits [23:16]	0x0000_0004
SSPPeriphID3	0x0FEC	Peripheral identification register bits [31:24]	0x0000_0000
SSPCellID0	0x0FF0	PrimeCell identification register bits [7:0]	0x0000_000D
SSPCellID1	0x0FF4	PrimeCell identification register bits [15:8]	0x0000_00F0
SSPCellID2	0x0FF8	PrimeCell identification register bits [23:16]	0x0000_0005
SSPCellID3	0x0FFC	PrimeCell identification register bits [31:24]	0x0000_00B1

4.16.3.1.1 SSPCR0

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SCR	[15:8]	RW	Serial clock rate. The value SCR is used to generate the transmit and receive bit rate of the PrimeCell SSP. The bit rate is: FSSPCLK/{CPSDVR x (1+SCR)} where CPSDVSR is an even value from 2-254, programmed through the SSPCPSR register and SCR is a value from 0-255	0x0000_0000
SPH	[7]	RW	SSPCLKOUT phase (applicable to Motorola SPI frame format only). See Motorola SPI frame format	0x0000_0000
SPO	[6]	RW	SSPCLKOUT polarity (applicable to Motorola SPI frame format only). See Motorola SPI frame format	0x0000_0000
FRF	[5:4]	RW	* Frame format: 00: Motorola SPI frame format 01: TI synchronous serial frame format 10: National Microwire frame format 11: Reserved, undefined operation	0x0000_0000
DSS	[3:0]	RW	* Data Size Select: 0000: Reserved, undefined operation 0001: Reserved, undefined operation 0010: Reserved, undefined operation 0011: 4-bit data 0100: 5-bit data 0101: 6-bit data 0110: 7-bit data	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			0111: 8-bit data 1000: 9-bit data 1001: 10-bit data 1010: 11-bit data 1011: 12-bit data 1100: 13-bit data 1101: 14-bit data 1110: 15-bit data 1111: 16-bit data	

4.16.3.1.2 SSPCR1

- Address: Base Address + 0x0004, Reset Value = 0x0000_0090

Name	Bit	Type	Description	Reset Value
MFSS	[7]	RW	* Control FSSOUT manually. FSSOUT is AND gated by MFSSOUT: 0: FSSOUT is low. FSSOUT is manually controlled to be low 1: FSSOUT is not manually controlled (default) * NOTE: In master mode it operates on burst when MFSS is set "0" . In slave mode it cannot operate on burst	0x0000_0001
RXIFLSEL	[6:4]	RW	* Rx FIFO Interrupt level select: 001: 1 byte (default) 010: 2 byte 100: 4 byte Other: Reserved. Undefined operation	0x0000_0001
SOD	[3]	RW	Slave-mode output disable. This bit is relevant only in the slave mode(MS=1). In multiple-slave systems it is possible for an PrimeCell SSP master to broadcast a message to all slaves in the system while ensuring that only one slave drives data onto its serial output line. In such systems the RXD lines from multiple slaves could be tied together. * To operate in such systems the SOD bit can be set if the PrimeCell SSP slave is not supposed to drive the SSPTXD line: 0: SSP can drive the SSPTXD output in slave mode 1: SSP must not drive the SSPTXD output in slave mode	0x0000_0000
MS	[2]	RW	* Master or slave mode select. This bit can be modified only when the PrimeCell SSP is disabled(SSE=0): 0: Device configured as master(default)	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			1: Device configured as slave	
SSE	[1]	RW	* Synchronous serial port enable: 0: SSP operation disabled 1: SSP operation enabled	0x0000_0000
LBM	[0]	RW	* Loop back mode: 0: Normal serial port operation enabled 1: Output of transmit serial shifter is connected to input of receive serial shifter internally	0x0000_0000

4.16.3.1.3 SSPDR

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SSPDR	[15:0]	RW	SSPDR is the data register and is 16-bits wide. The entry in the receive FIFO(pointed to by the current FIFO read pointer) is accessed when SSPDR is read. As data values are removed by the PrimeCell SSP receive logic from the incoming data frame, they are placed into the entry in the receive FIFO, pointed to by the current FIFO write pointer. When SSPDR is written to, the entry in the transmit FIFO, pointed to by the write pointer, is written to. Data values are removed from the transmit FIFO one value at a time by the transmit logic. It is loaded into the transmit serial shifter, then serially shifted out onto the SSPTXD pin at the programmed bit rate. When a data size of less than 16 bits is selected, the user must right-justify data written to the transmit FIFO. Unused bits at the top are ignored by transmit logic. Received data less than 16 bits is automatically right-justified in the receive buffer. * Transmit/Receive FIFO: Read: Receive FIFO. Write: Transmit FIFO.	0x0000_0000

4.16.3.1.4 SSPSR

- Address: Base Address + 0x000C, Reset Value = 0x0000_0003

Name	Bit	Type	Description	Reset Value
BSY	[4]	R	* PrimeCell SSP busy flag: 0: SSP is idle 1: SSP is currently transmitting and/or receiving a frame or the transmit FIFO is not empty	0x0000_0000

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Name	Bit	Type	Description	Reset Value
RFF	[3]	R	* Receive FIFO full: 0: Receive FIFO is not full. 1: Receive FIFO is full.	0x0000_0000
RNE	[2]	R	* Receive FIFO not empty: 0: Receive FIFO is empty 1: Receive FIFO is not empty	0x0000_0000
TNF	[1]	R	* Transmit FIFO not full: 0: Transmit FIFO is full 1: Transmit FIFO is not full	0x0000_0001
TFE	[0]	R	* Transmit FIFO empty: 0: Transmit FIFO is not empty 1: Transmit FIFO is empty	0x0000_0001

4.16.3.1.5 SSPCPSR

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SSPCPSR	[7:0]	RW	SSP Clock Pre-scale Register. This register specifies the factor by which SSPCLK is to be divided in the Clock pre-scaler. The division factor has to be an even number from 2 to 254. The least significant bit of the programmed number is hard-coded to zero. Data read back from this register has the least significant bit as zero if an odd number is written to this register.	0x0000_0000

4.16.3.1.6 SSPIMSC

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TXIM	[3]	RW	* Transmit FIFO interrupt mask: 0: Transmit FIFO half empty or less condition interrupt is masked 1: Transmit FIFO half empty or less condition interrupt is not masked	0x0000_0000
RXIM	[2]	RW	* Receive FIFO interrupt mask: 0: Receive FIFO half full or less condition interrupt is masked 1: Receive FIFO half full or less condition interrupt is not masked	0x0000_0000
RTIM	[1]	RW	* Receive timeout interrupt mask: 0: Receive FIFO not empty and no read prior to timeout period interrupt is masked	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			1: Receive FIFO not empty and no read prior to timeout period interrupt is not masked	
RORIM	[0]	RW	* Receive overrun interrupt mask: 0: Receive FIFO written to while full condition interrupt is masked 1: Receive FIFO written to while full condition interrupt is not masked	0x0000_0000

4.16.3.1.7 SSPRIS

- Address: Base Address + 0x0018, Reset Value = 0x0000_0008

Name	Bit	Type	Description	Reset Value
TXRIS	[3]	R	Gives the raw interrupt state of the SSPTXINTR interrupt prior to masking	0x0000_0001
RXRIS	[2]	R	Gives the raw interrupt state of the SSPRXINTR interrupt prior to masking	0x0000_0000
RTRIS	[1]	R	Gives the raw interrupt state of the SSPRTINTR interrupt prior to masking	0x0000_0000
RORRIS	[0]	R	Gives the raw interrupt state of the SSPRORINTR interrupt prior to masking	0x0000_0000

4.16.3.1.8 SSPMIS

- Address: Base Address + 0x001C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TXMIS	[3]	R	Gives the transmit FIFO masked interrupt state of the SSPTXINTR interrupt after masking	0x0000_0000
RXMIS	[2]	R	Gives the receive FIFO masked interrupt state of the SSPRXINTR interrupt after masking	0x0000_0000
RTMIS	[1]	R	Gives the receive timeout masked interrupt state of the SSPRTINTR interrupt after masking	0x0000_0000
RORMIS	[0]	R	Gives the receive over run masked interrupt status of the SSPRORINTR interrupt after masking	0x0000_0000

4.16.3.1.9 SSPICR

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RTIC	[1]	W	Clears the SSPRTINTR interrupt	0x0000_0000
RORIC	[0]	W	Clears the SSPRORINTR interrupt	0x0000_0000

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4.16.3.1.10 SSPDMACLR

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
TXDMAE	[1]	RW	Transmit DMA Enable. DMA for the transmit FIFO is enabled if this bit is set to 1	0x0000_0000
RXDMAE	[0]	RW	Receive DMA Enable. DMA for the receive FIFO is enabled if this bit is set to 1	0x0000_0000

4.16.3.1.11 SSPPeriphID0

- Address: Base Address + 0x0FE0, Reset Value = 0x0000_0022

Name	Bit	Type	Description	Reset Value
SSPPeriphID0	[7:0]	R	Peripheral identification register bits [7:0]	0x0000_0022

4.16.3.1.12 SSPPeriphID1

- Address: Base Address + 0x0FE4, Reset Value = 0x0000_0010

Name	Bit	Type	Description	Reset Value
SSPPeriphID1	[7:0]	R	Peripheral identification register bits [15:8]	0x0000_0010

4.16.3.1.13 SSPPeriphID2

- Address: Base Address + 0x0FE8, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
SSPPeriphID2	[7:0]	R	Peripheral identification register bits [23:16]	0x0000_0004

4.16.3.1.14 SSPPeriphID3

- Address: Base Address + 0x0FEC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SSPPeriphID3	[7:0]	R	Peripheral identification register bits [31:24]	0x0000_0000

4.16.3.1.15 SSPCellID0

- Address: Base Address + 0x0FF0, Reset Value = 0x0000_000D

Name	Bit	Type	Description	Reset Value
SSPCellID0	[7:0]	R	PrimeCell identification register bits [7:0]	0x0000_000D

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4.16.3.1.16 SSPCellID1

- Address: Base Address + 0x0FF4, Reset Value = 0x0000_00F0

Name	Bit	Type	Description	Reset Value
SSPCellID1	[7:0]	R	PrimeCell identification register bits [15:8]	0x0000_00F0

4.16.3.1.17 SSPCellID2

- Address: Base Address + 0x0FF8, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
SSPCellID2	[7:0]	R	PrimeCell identification register bits [23:16]	0x0000_0005

4.16.3.1.18 SSPCellID3

- Address: Base Address + 0x0FFC, Reset Value = 0x0000_00B1

Name	Bit	Type	Description	Reset Value
SSPCellID3	[7:0]	R	PrimeCell identification register bits [31:24]	0x0000_00B1

Technical reference manual of SPI (PL022) can be found the link below.

http://infocenter.arm.com/help/topic/com.arm.doc.ddi0194g/DDI0194G_ssp_pl022_r1p3_trm.pdf

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4.17 Advanced Encryption Standard (AES)

S3FBP5A utilizes the AES to secure the personal biometric information and body signals. The AES supports electronic code block (ECB) and cipher block chaining (CBC) modes. It also operates with 128bit or 256bit key sizes.

4.17.1 Register Description

4.17.1.1 Register Map Summary

- Base Address: 0x40022000

Register	Offset	Description	Reset Value
AESCON	0x0000	AES Control	0x0000_0000
AESSTATUS	0x0004	AES Status	0x0000_0002
AESINDATA1	0x0010	AES Input Data [31:0]	0x0000_0000
AESINDATA2	0x0014	AES Input Data [63:32]	0x0000_0000
AESINDATA3	0x0018	AES Input Data [95:64]	0x0000_0000
AESINDATA4	0x001C	AES Input Data [127:96]	0x0000_0000
AESOUTDATA1	0x0020	AES Output Data [31:0]	0x0000_0000
AESOUTDATA2	0x0024	AES Output Data [63:32]	0x0000_0000
AESOUTDATA3	0x0028	AES Output Data [95:64]	0x0000_0000
AESOUTDATA4	0x002C	AES Output Data [127:96]	0x0000_0000
AESIV1	0x0050	AES IV [31:0]	0x0000_0000
AESIV2	0x0054	AES IV [63:32]	0x0000_0000
AESIV3	0x0058	AES IV [95:64]	0x0000_0000
AESIV4	0x005C	AES IV [127:96]	0x0000_0000
AESKEY1	0x0080	AES Key [31:0]	0x0000_0000
AESKEY2	0x0084	AES Key [63:32]	0x0000_0000
AESKEY3	0x0088	AES Key [95:64]	0x0000_0000
AESKEY4	0x008C	AES Key [127:96]	0x0000_0000
AESKEY5	0x0090	AES Key [159:128]	0x0000_0000
AESKEY6	0x0094	AES Key [191:160]	0x0000_0000
AESKEY7	0x0098	AES Key [223:192]	0x0000_0000
AESKEY8	0x009C	AES Key [255:224]	0x0000_0000

4.17.1.1.1 AESCON

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:6]	RW	*Reserved	0x0000_0000

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Name	Bit	Type	Description	Reset Value
AESKEY_SIZE	[5:4]	RW	*AES key size selection signal 00: 128bit key 01: Reserved 10: 256bit key 11: Reserved	0x0000_0000
Reserved1	[3:2]	RW	*Reserved	0x0000_0000
AES_CHAIN_MODE	[1]	RW	*AES chain mode selection signal 0: ECB mode 1: CBC mode	0x0000_0000
AES_OP	[0]	RW	*Encryption/decryption mode selection signal 0: Encryption 1: Decryption	0x0000_0000

4.17.1.1.2 AESSTATUS

- Address: Base Address + 0x0004, Reset Value = 0x0000_0002

Name	Bit	Type	Description	Reset Value
Reserved0	[31:3]	RW	*Reserved	0x0000_0000
AES_BUSY	[2]	R	*AES busy signal 0: Idle 1: Busy	0x0000_0000
AES_IN_READY	[1]	R	*AES input ready signal 0: AES input buffer is not empty 1: AES input buffer is empty, and the host is permitted to write the next block of data	0x0000_0001
AES_OUT_READY	[0]	RW	*AES output ready signal 0: AES output is not available 1: AES output is available for the host to retrieve	0x0000_0000

4.17.1.1.3 AESINDATA1

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESINDATA1	[31:0]	RW	*AES input data [31:0]	0x0000_0000

4.17.1.1.4 AESINDATA2

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESINDATA2	[31:0]	RW	*AES input data [63:32]	0x0000_0000

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4.17.1.1.5 AESINDATA3

- Address: Base Address + 0x0018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESINDATA3	[31:0]	RW	*AES input data [95:64]	0x0000_0000

4.17.1.1.6 AESINDATA4

- Address: Base Address + 0x001C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESINDATA4	[31:0]	RW	*AES input data [127:96]	0x0000_0000

4.17.1.1.7 AESOUTDATA1

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESOUTDATA1	[31:0]	R	*AES Output Data [31:0]	0x0000_0000

4.17.1.1.8 AESOUTDATA2

- Address: Base Address + 0x0024, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESOUTDATA2	[31:0]	R	*AES Output Data [63:32]	0x0000_0000

4.17.1.1.9 AESOUTDATA3

- Address: Base Address + 0x0028, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESOUTDATA3	[31:0]	R	*AES Output Data [95:64]	0x0000_0000

4.17.1.1.10 AESOUTDATA4

- Address: Base Address + 0x002C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESOUTDATA4	[31:0]	R	*AES Output Data [127:96]	0x0000_0000

4.17.1.1.11 AESIV1

- Address: Base Address + 0x0050, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
AESIV1	[31:0]	RW	*AES IV [31:0]	0x0000_0000

4.17.1.1.12 AESIV2

- Address: Base Address + 0x0054, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESIV2	[31:0]	RW	*AES IV [63:32]	0x0000_0000

4.17.1.1.13 AESIV3

- Address: Base Address + 0x0058, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESIV3	[31:0]	RW	*AES IV [95:64]	0x0000_0000

4.17.1.1.14 AESIV4

- Address: Base Address + 0x005C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESIV4	[31:0]	RW	*AES IV [127:96]	0x0000_0000

4.17.1.1.15 AESKEY1

- Address: Base Address + 0x0080, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY1	[31:0]	W	*AES Key [31:0]	0x0000_0000

4.17.1.1.16 AESKEY2

- Address: Base Address + 0x0084, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY2	[31:0]	W	*AES Key [63:32]	0x0000_0000

4.17.1.1.17 AESKEY3

- Address: Base Address + 0x0088, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY3	[31:0]	W	*AES Key [95:64]	0x0000_0000

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4.17.1.1.18 AESKEY4

- Address: Base Address + 0x008C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY4	[31:0]	W	*AES Key [127:96]	0x0000_0000

4.17.1.1.19 AESKEY5

- Address: Base Address + 0x0090, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY5	[31:0]	W	*AES Key [159:128]	0x0000_0000

4.17.1.1.20 AESKEY6

- Address: Base Address + 0x0094, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY6	[31:0]	W	*AES Key [191:160]	0x0000_0000

4.17.1.1.21 AESKEY7

- Address: Base Address + 0x0098, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY7	[31:0]	W	*AES Key [223:192]	0x0000_0000

4.17.1.1.22 AESKEY8

- Address: Base Address + 0x009C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
AESKEY8	[31:0]	W	*AES Key [255:224]	0x0000_0000

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4.18 True Random Number Generator (DTRNG)

The TRNG generates random numbers from thermal noise of a chip. The generated random numbers are non-deterministic sequence of bits and unpredictable. The TRNG module consists of several TRNG circuits which have different throughput.

4.18.1 Register Description

4.18.1.1 Register Map Summary

- Base Address: 0x40023000

Register	Offset	Description	Reset Value
TRNG_CLKDIV	0x0000	TRNG Clock Divider	0x0000_00C8
TRNG_CTRL	0x0004	TRNG Control	0x0000_0000
TRNG_POST_CTRL	0x0008	TRNG Post Processor Control	0x0000_0000
TRNG_FIFO_CTRL	0x000C	TRNG FIFO Control	0x0000_0000
TRNG_FIFO_0	0x0010	TRNG FIFO Control	0x0000_0000

4.18.1.1.1 TRNG_CLKDIV

- Address: Base Address + 0x0000, Reset Value = 0x0000_00C8

Name	Bit	Type	Description	Reset Value
Reserved0	[31:16]	RW	*Reserved	0x0000_0000
CLKDIV	[15:0]	RW	*The frequency of system clock is divided by CLKDIV to generate frequency of slow clock. If this value is 10, then the period of slow clock in TRNG is ten times longer than the period of system clock (AHB clock). The clock divider must be an even number. So, CLKDIV [0] bit is reserved to protect dividing odd number.	0x0000_00C8

4.18.1.1.2 TRNG_CTRL

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
RNGEN	[31]	RW	*Random number generator enable bit. This bit enables TRNG. After filled 32bits in FIFO, this bit is cleared automatically.	0x0000_0000
Reserved0	[30:0]	RW	*Reserved	0x0000_0000

4.18.1.1.3 TRNG_POST_CTRL

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
PPEN	[31]	RW	*Post processor enable bit. This bit enables the post processor. 0: Disables the post processor. The random bit bypasses the post processor in any value of PPSEL. 1: Enables the post processor.	0x0000_0000
Reserved0	[30:2]	RW	*Reserved	0x0000_0000
PPSEL	[1:0]	RW	*Post processor selector bit. 00: The random bit bypasses the post processor. 01: The random bit is processed by LFSR post processor. 10: The random bit is processed by VN(Von Neumann) post processor. 11: The random bit is processed by XOR post processor.	0x0000_0000

4.18.1.1.4 TRNG_FIFO_CTRL

- Address: Base Address + 0x000C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:9]	R	*Reserved	0x0000_0000
FIFOPTR	[5:0]	R	*FIFO pointer bits. This is a pointer to a flip-flop in TRNG FIFO to store next random bit. If this value is bigger than zero, FIFO[FifoPtr-1] is filled with a random bit. And FifoPtr is decremented by one until it becomes zero. 1~32 : A random bit is generated and stored in FIFO[FifoPtr-1] and FifoPtr is decremented by one.	0x0000_0000

4.18.1.1.5 TRNG_FIFO_0

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO_0	[31:0]	R	*TRNG FIFO [31:0]. LSB is the first random bit received.	0x0000_0000

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4.19 Ultra-low Power Samsung Reconfigurable Processor (ULP-SRP)

S3FBP5A utilizes Ultra-Low Power Samsung Reconfigurable Processor (ULP-SRP) as a high performance Digital Signal Processor (DSP). ULP-SRP is a reconfigurable processor that operates several modes of operation by software configuration. Re-configurability of ULP-SRP provides wide processing abilities for many applications from low to high complexity applications.

ULP-SRP features:

- Three different operating modes: VLIW, CGA2x2, and CGA3x3
- Memories: Instruction Memory, Data Memory, and Configuration Memory
- Direct Memory Controller (DMC)
- SRP Debugger
- AHB-Lite Slave Interface
- Two interrupt sources to Cortex-M4 core

Figure 4-52 shows the block diagram of ULP-SRP. ULP-SRP is a subsystem of S3FBP5A that supports AHB-Lite protocol. AHB-Lite Slave Interface in ULP-SRP is connected to AHB-Lite bus system. Cortex-M4 core utilizes ULP-SRP as a DSP subsystem through this interface. Two interrupt sources connected to Cortex-M4 core are available in ULP-SRP. One is debugging-interrupt used for prompt logging, another is done-interrupt used for informing Cortex-M4 of processing completion. Done-interrupt is also used for wake-up source of Cortex-M4.

ULP-SRP comprises SRP core which has a reconfigurable Coarse Grained Array (CGA). The CGA consists of nine functional units (FUs) that each FU contains a logical register file and is controlled by a Configuration Memory. A base processing is a Very Long Instruction Word (VLIW) mode processing using two FUs. The computational capacity of the core can be expanded by adding more FUs. This re-configurability can make processing platform capable of dynamically adjusting to the processing need according to the applications. There are three configurations for different computational capacities: the base processing (VLIW), the base processing plus two additional functional units (CGA2x2), and the base processing plus seven additional functional units (CGA3x3).

Data Memory Controller (DMC) is a simple bus system interfacing with SRP core, SRP memory, and AHB-Lite Slave Interface. Through DMC, SRP core accesses SRP memory, and AHB-Lite Slave Interface accesses memory-mapped registers in SRP core and SRP memory.

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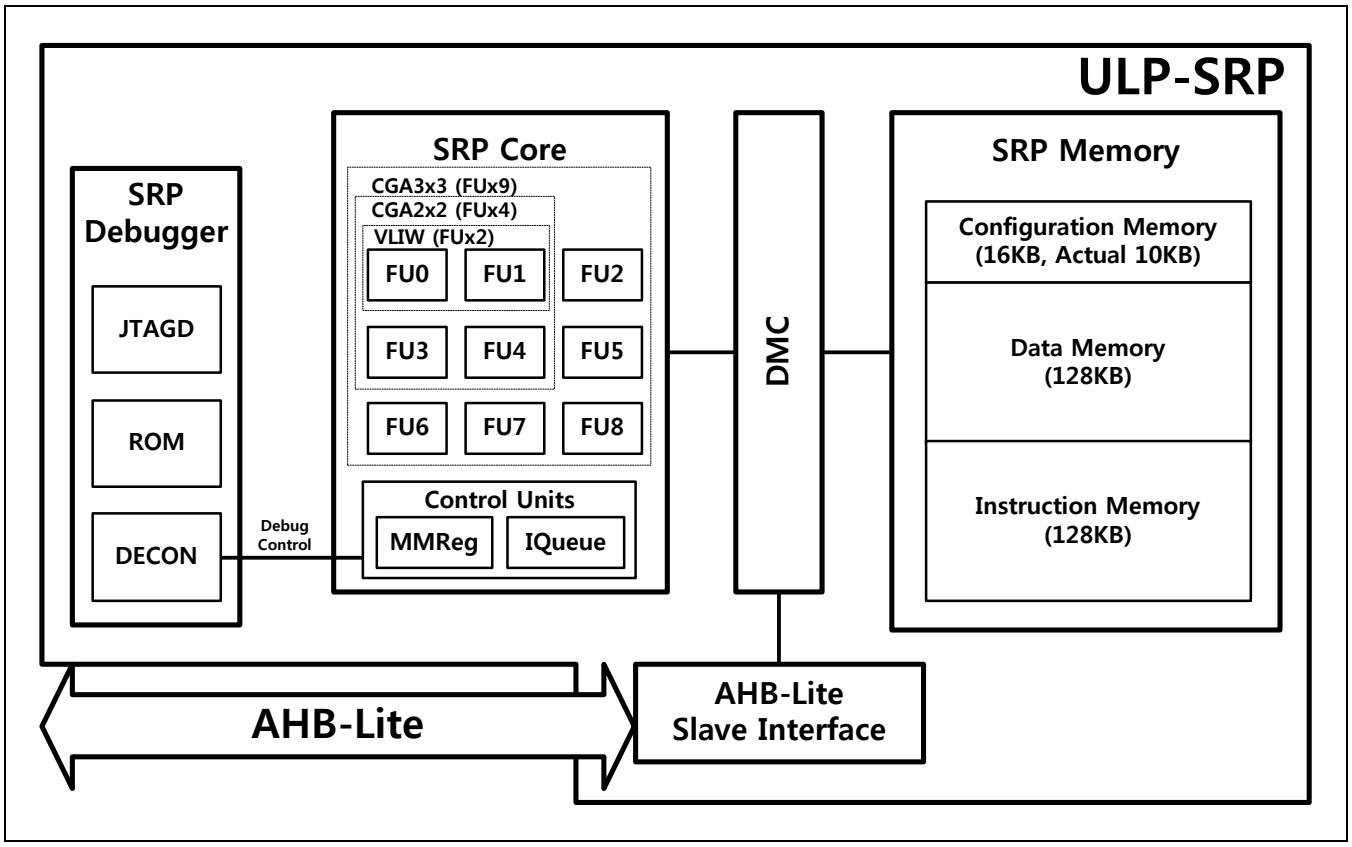


Figure 4-52 ULP-SRP Block Diagram

SRP memory consists of Instruction Memory, Data Memory, and Configuration Memory. 128KB Instruction Memory and 128KB Data Memory are used to store application code and user variables, respectively. Configuration Memory is used to store the variables for executing the pipeline operation in CGA mode. The actual size of Configuration Memory is 10KB and it is comprised of five 2KB SRAMs whose output is 128bit, i.e. they have depth of 128. When SRP core operates in CGA mode, maximum 576bit (72byte) data from Configuration Memory is needed per one clock period. Therefore 640bit (80byte) fetch from Configuration Memory occurs because the nearest number to 576 which can be expressed with multiple of 128, that is, five 128bit output SRAMs are fetched in parallel for 640bit data. However, Configuration Memory occupies 16KB memory space because only 80byte is used at every 128byte memory space. Physical SRAM is not assigned on the rest 48byte after 80byte.

4.19.1 Register Description

4.19.1.1 Register Map Summary (SRP Core)

- Base Address: 0x80F00000

Register	Offset	Description	Reset Value
SRP_ID	0x0000	SRP ID	0x0000_0000
CMEM_DYN_LOAD	0x0120	Configuration Memory Write Mode	0x0000_0000
SRPFRC_CFG	0x6000	SRP Free-running Counter Configuration	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

SRPFRC_CNTLOW	0x6010	Lower 32 bits of SRP Free-running Counter	0x0000_0000
SRPFRC_CNTHIGH	0x6014	Higher 32 bits of SRP Free-running Counter	0x0000_0000
SRP_PC	0x7000	SRP Program Counter	0x0000_0000
MCUIF_EVE	0x8000	Event Registers for MCU Interfacing	0x0000_0000
MCUIF_EVECLR	0x8004	Event Clear Registers for MCU Interfacing	0x0000_0000
MCUIF_PARAM0	0x8008	Parameter 0 Register for MCU Interfacing	0x0000_0000
MCUIF_PARAM1	0x800C	Parameter 1 Register for MCU Interfacing	0x0000_0000
MCUIF_PARAM2	0x8010	Parameter 2 Register for MCU Interfacing	0x0000_0000
MCUIF_PARAM3	0x8014	Parameter 3 Register for MCU Interfacing	0x0000_0000
MCUIF_RESULT	0x8018	Result Register for MCU Interfacing	0x0000_0000
MCUIF_LOG	0x801C	Log Register for MCU Interfacing	0x0000_0000

4.19.1.1.1 SRP_ID

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:8]	R	*Reserved	0x0000_0000
SRP_ID	[7:0]	R	*SRP ID	0x0000_0000

4.19.1.1.2 CMEM_DYN_LOAD

- Address: Base Address + 0x0120, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:1]	RW	*Reserved	0x0000_0000
CMEM_DYN_LOAD	[0]	RW	*Configuration Memory write mode bit. This bit determines the write access method for Configuration Memory. 0: Static mode which means core accesses Configuration Memory with automatic address increment. 1: Dynamic mode which means core accesses Configuration Memory with 32 bit address.	0x0000_0000

4.19.1.1.3 SRPFRC_CFG

- Address: Base Address + 0x6000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:9]	RW	*Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
SRPFRC_EN	[8]	RW	*Enable SRP Free-running Counter bit. 0: Disables SRP Free-running Counter. 1: Enables SRP Free-running Counter.	0x0000_0000
Reserved1	[7:0]	RW	*Reserved	0x0000_0000

4.19.1.1.4 SRPFRC_CNTLOW

- Address: Base Address + 0x6010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SRPFRC_CNTLOW	[31:0]	RW	*Lower 32 bits of SRP Free-running Counter. This is R/W register. Read this register: Read lower 32 bit counter value. Write this register: Set lower 32 bit counter initial value. SRP Free-running Counter will start counting with this value.	0x0000_0000

4.19.1.1.5 SRPFRC_CNTHIGH

- Address: Base Address + 0x6014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SRPFRC_CNTHIGH	[31:0]	RW	*Higher 32 bits of SRP Free-running Counter. This is R/W register. Read this register: Read higher 32 bit counter value. Write this register: Set higher 32 bit counter initial value. SRP Free-running Counter will start counting with this value.	0x0000_0000

4.19.1.1.6 SRP_PC

- Address: Base Address + 0x7000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SRP_PC	[31:0]	R	*Indicates SRP Program Counter	0x0000_0000

4.19.1.1.7 MCUIF_EVE

- Address: Base Address + 0x8000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:2]	RW	*Reserved	0x0000_0000
LOG_EVE	[1]	RW	*General purpose event register. This event register is a general purpose interrupt source of MCU core. MCU interrupt occurs when SRP core sets this register by software. User can also use this event register for prompt logging at debugging stage with	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			MCUIF_LOG register. 0: No event occurs. 1: Event occurs.	
USER_EVE	[0]	RW	*General purpose event register. This event register is a general purpose interrupt source of MCU core. MCU interrupt occurs when SRP core sets this register by software. 0: No event occurs. 1: Event occurs.	0x0000_0000

4.19.1.1.8 MCUIF_EVECLR

- Address: Base Address + 0x8004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:2]	W	*Reserved	0x0000_0000
LOG_EVECLR	[1]	W	*Event clear for LOG_EVE register. 0: No effect. 1: Clear LOG_EVE.	0x0000_0000
USER_EVECLR	[0]	W	*Event clear for LOG_EVE register. 0: No effect. 1: Clear USER_EVE.	0x0000_0000

4.19.1.1.9 MCUIF_PARAM0

- Address: Base Address + 0x8008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_PARAM0	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP, e.g. MCU can use this register for SRP program configuration.	0x0000_0000

4.19.1.1.10 MCUIF_PARAM1

- Address: Base Address + 0x800C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_PARAM1	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP e.g. MCU can use this register for SRP program configuration.	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.19.1.1.11 MCUIF_PARAM2

- Address: Base Address + 0x8010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_PARAM2	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP e.g. MCU can use this register for SRP program configuration.	0x0000_0000

4.19.1.1.12 MCUIF_PARAM3

- Address: Base Address + 0x8014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_PARAM3	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP e.g. MCU can use this register for SRP program configuration.	0x0000_0000

4.19.1.1.13 MCUIF_RESULT

- Address: Base Address + 0x8018, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_RESULT	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP e.g. SRP can use this register for returning result to MCU.	0x0000_0000

4.19.1.1.14 MCUIF_LOG

- Address: Base Address + 0x801C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
MCUIF_LOG	[31:0]	RW	*General purpose register. This register can be used for the interface between MCU and SRP e.g. SRP can use this register for prompt logging with LOG_EVE register.	0x0000_0000

4.19.1.2 Register Map Summary (SRP Control)

- Base Address: 0x4001A500

Register	Offset	Description	Reset Value
SRP_CTRL	0x0014	SRP_CTRL	0x0000_0009

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.19.1.2.1 SRP_CTRL

- Address: Base Address + 0x0014, Reset Value = 0x0000_0009

Name	Bit	Type	Description	Reset Value
CGA_MODE	[3]	RW	* SRP CGA Operation mode selection 1: CGA 3X3 mode operation 0: CGA 2X2 mode operation	0x0000_0001
VLIW_MODE	[2]	R	* SRP VLIW mode indicator (RO) 1: SRP is running on VLIW mode 0: SRP is running on CGA mode	0x0000_0000
DSP_HALTED	[1]	R	* SRP halt indicator (RO) 1: SRP is halted 0: SRP is not halted	0x0000_0000
DSP_HALT	[0]	RW	* SRP halt assertion 1: Assert SRP halt (SRP halted) 0: Release SRP halt (SRP starts to operate)	0x0000_0001

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20 Flash Controller

4.20.1 Overview

The flash controller consists of an internal 512KB Flash and 2KB Cache(SRAM). Flash controller is connected 2 AHB Bus, system bus for Flash IP and instruction/data code bus for Flash controller.

The flash controller supported 87MHz operation. For 87MHz operation, DCYCRDCON of SFR has to set 0x99F4.

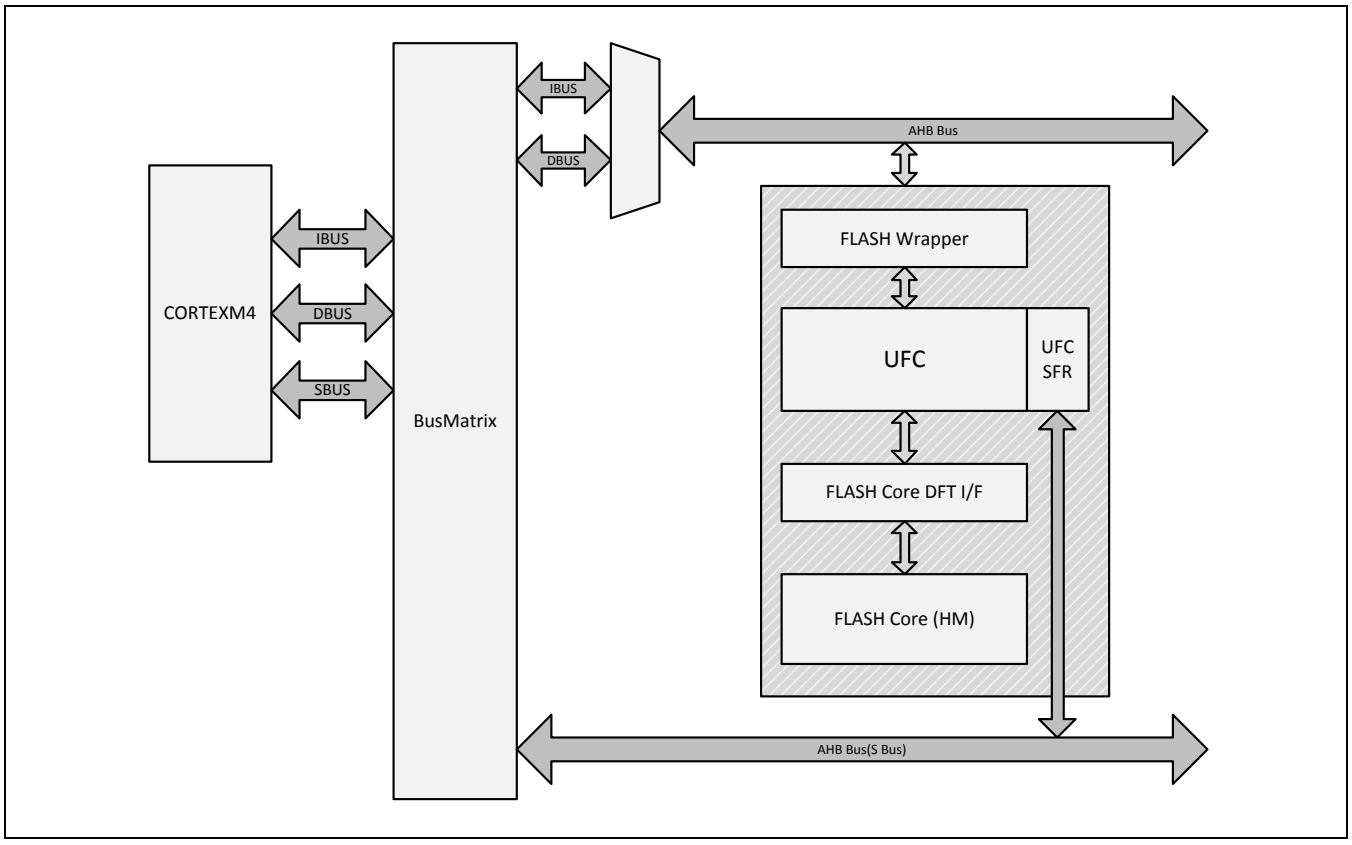


Figure 4-53 Flash controller connection inside system architecture

4.20.2 Key function

- 2X/4X prefetch mode for high-speed operation
- ECC function for improvement in flash cell reliability
- Cache for improvement in nor flash performance

Figure 4-54 shows the block diagram of Flash controller.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

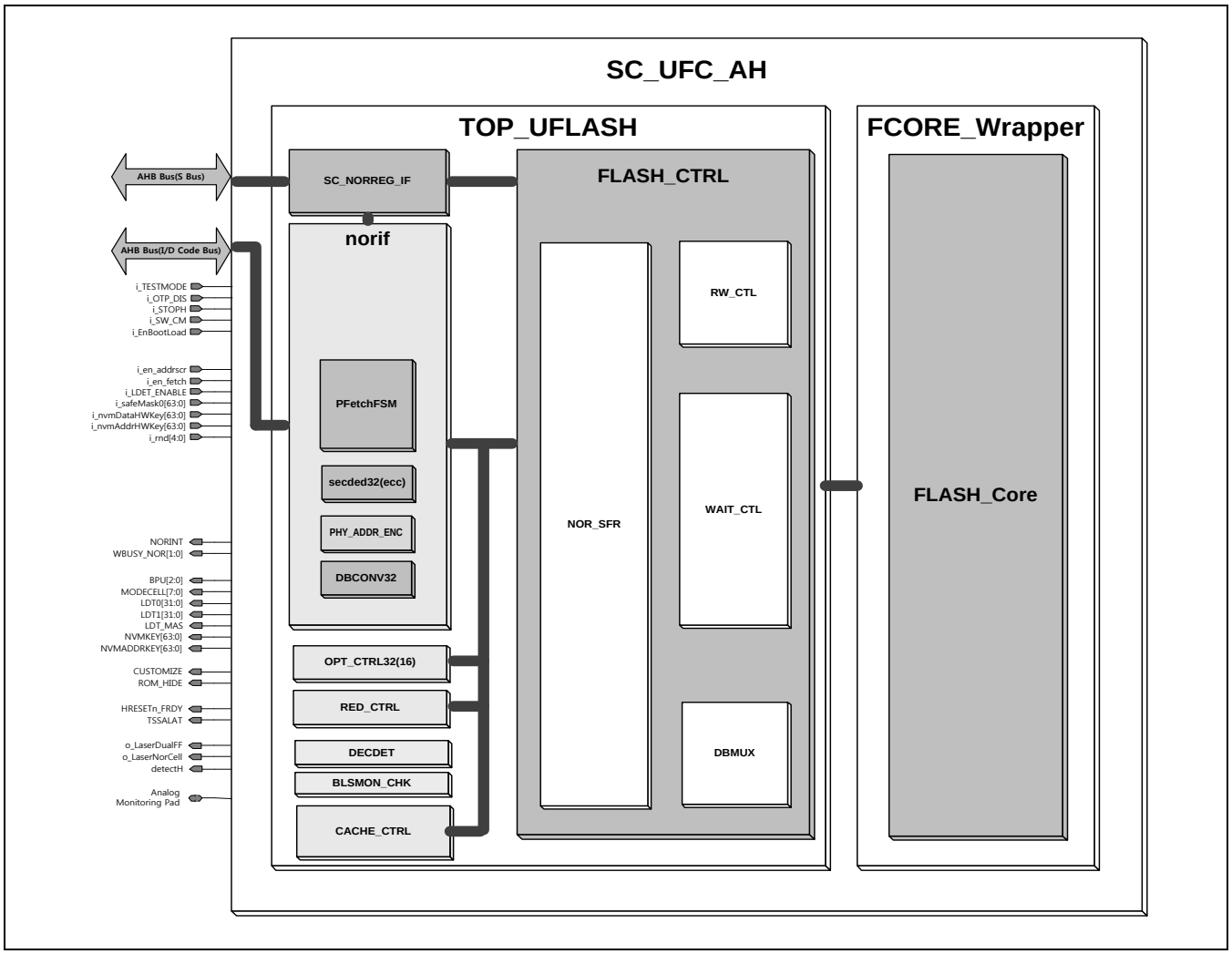


Figure 4-54 SC_UFC_AH Block Diagram

4.20.3 Functional description

4.20.3.1 Flash Memory Array

Flash memory is consists of main array and option cells.

- IF block is 3ea pages and include security and code protection information.
- DCT/LED/PROT is 1ea page. (1page = 512Byte)
- DCT block include option(trim) of flash core
- LED block include user option information
- PROT0~8 include user protection information

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

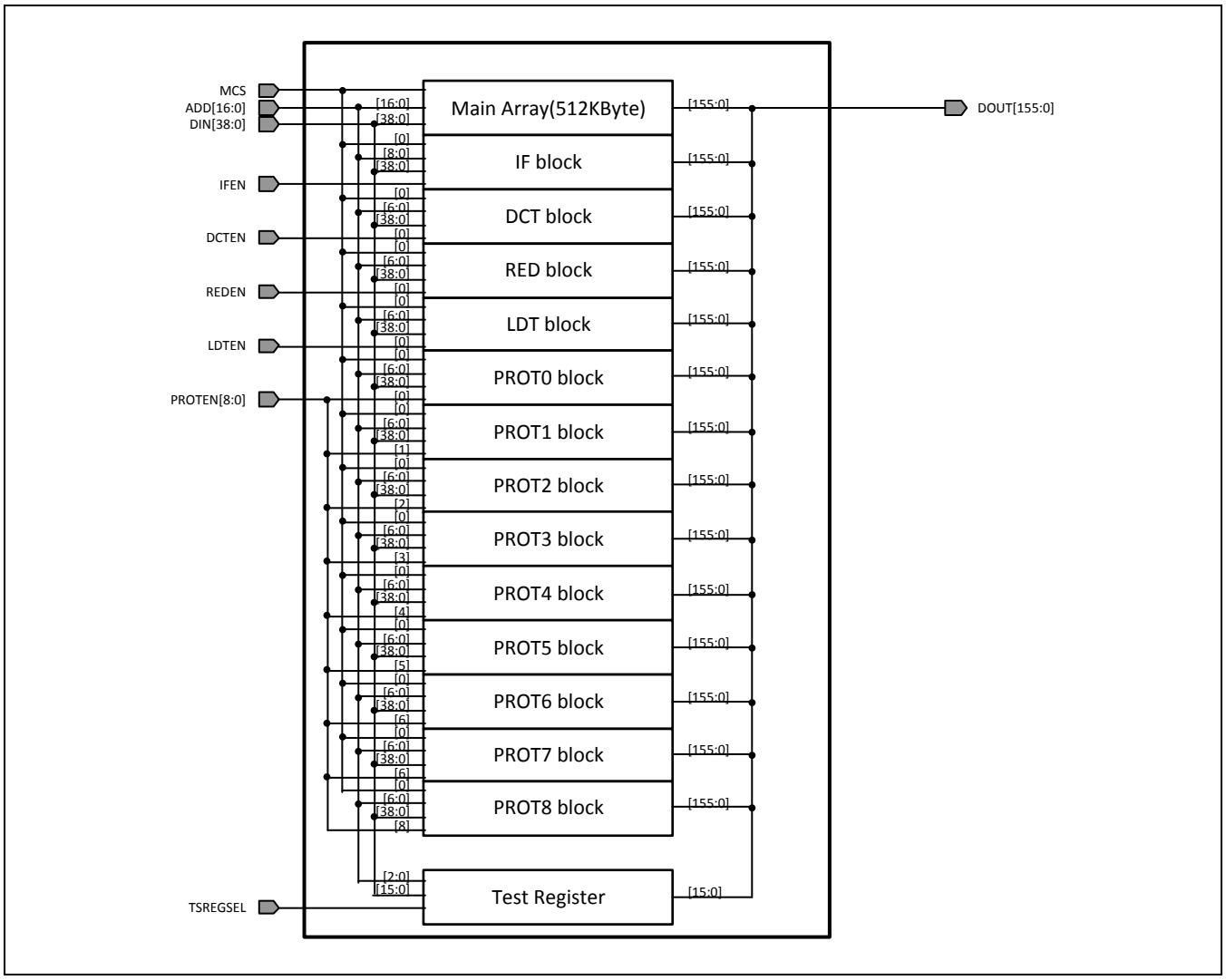


Figure 4-55 Flash Memory Array

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.3.2 Description of optional area access

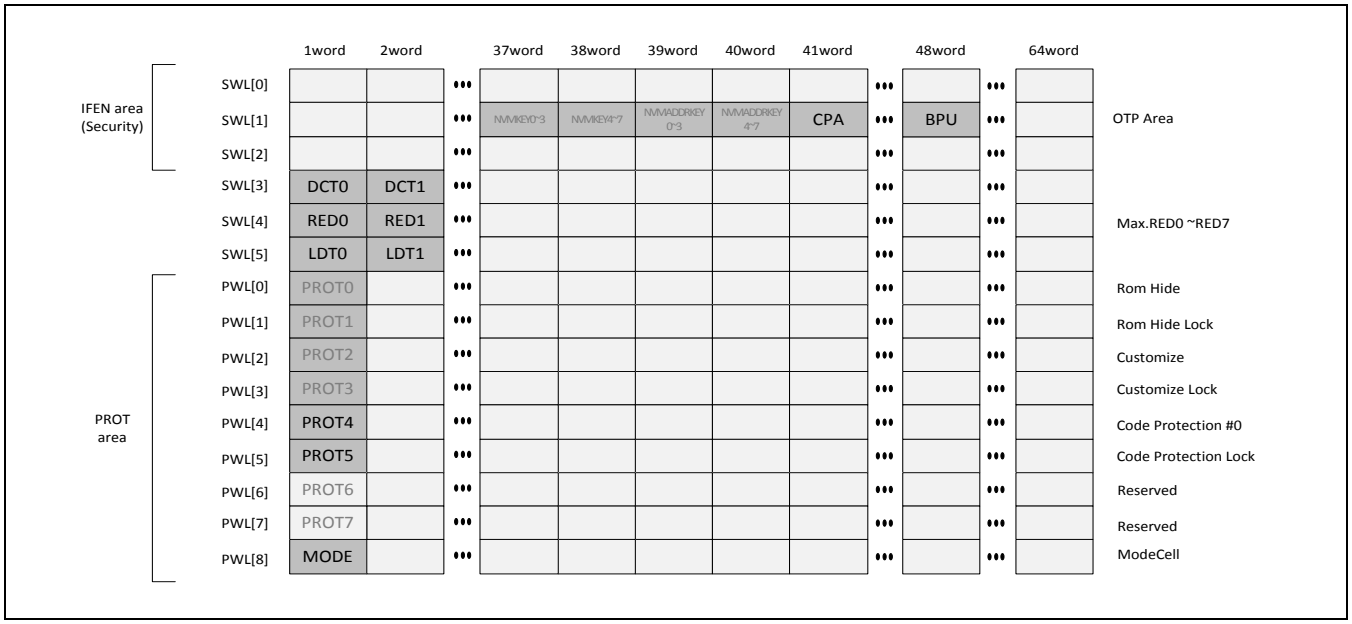


Figure 4-56 Option block configuration

4.20.3.2.1 Description of optional area access

4.20.3.2.1.1 Information area(security area) access

IF(security) Area	Test Mode	User Mode	Note
Security Area #0	ERS/PGM	Skipped ERS&PGM	
Security Area #1	ERS/PGM	Skipped ERS&PGM	
Security Area #2	ERS/PGM	ERS/PGM	

4.20.3.2.1.2 Loading data information in boot configuration(OTP AREA)

Usage	NVMKEY0~7[7:0]	CPA[15:0]	BPU[2:0]	Note
Adrress	SEC_BASE +OPT_adNVMKEY	SEC_BASE +OPT_adCPA	SEC_BASE +OPT_adBPU	
Valid bit	DIN[31:0]	DIN[15:0]	DIN[2:0]	

- CPA (Code Protection Area) can be assigned by sector address unit (Address[27:12])
- OPT_adCPA = 0x3E0
- Protection area = 0x00 ~ {~CPA[15:0], 12'hFFF}
- CPA = ~DIN[31:0] (ex.0xFFFFFFFD -> Protection area = 0x00 ~ 0x2FFF)
- Save the CAP value in password area 0x08~0x09 in security area

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.3.2.1.3 prot/LDT area access

Mode	PROT Area			LDT Area		
	Select signal	Valid Data Bit	Access Type	Select signal	Valid Data Bit	Access Type
Test Mode	PROTEN[7:0]	DIN[3:0]	Read/ERS/PGM	LDTEN	DIN[31:0]	Read/ERS/PGM
User Mode	PROTEN[7:0]	DIN[3:0]	Read/ERS/PGM	N/A	N/A	N/A

- if Lock bit is set, PROT area cannot be Erase/Program.
- Selection bit of PROTEN[7:0] / LDTEN / DCTEN / MODEEN is not available at a time
- LDTEN = LDTCON[0]
- LDT = ~DIN[31:0] (ex.0x55555555 -> MEM@LDT=0xAAAAAAAA)

4.20.3.2.1.4 dct area access

Mode	DCT Area		
	Select signal	Valid Data Bit	Access Type
Test Mode	DCTEN	DIN[31:0]	Read/ERS/PGM
User Mode	N/A	N/A	N/A

- DCTEN = LDTCON[2]

4.20.3.2.1.5 MODE AREA access

Mode	MODE Area		
	Select signal	Valid Data Bit	Access Type
Test Mode	MODEEN	DIN[26:16], DIN[11:0]	Read/ERS/PGM
User Mode	N/A	N/A	N/A

- MODEEN = LDTCON[4]
- MODE= ~DIN[26:0] (ex.0x55555555 -> MEM@MODE=0xAAAAAAAA)
- DIN[26:16] of MODE Area is assigned Option of UFC.
- DIN[26:16]={data wait extended opt[26:25], erase verify guard time[24:23],ecc disable[22], hslatchen[21], fecccon access[20], variable data access of literal pool area[19], prefetch disable[18], ecc mode[17:16]}

4.20.4 Supported operation

The flash controller supports write and erase operation. Write operation sets selected bits to 0. Erase operation sets selected bits to 1.

User can check the FRWCON register before the every write or erase operation. When FRWCON is not 0x0000, that means other function is processing the flash operation. User has to wait until the FRWCON is set to 0x0000.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.4.1.1 Write operation

Write operations must be performed on erased WU. it means the value of the WU must be 0xFFFFFFFF before write operation.

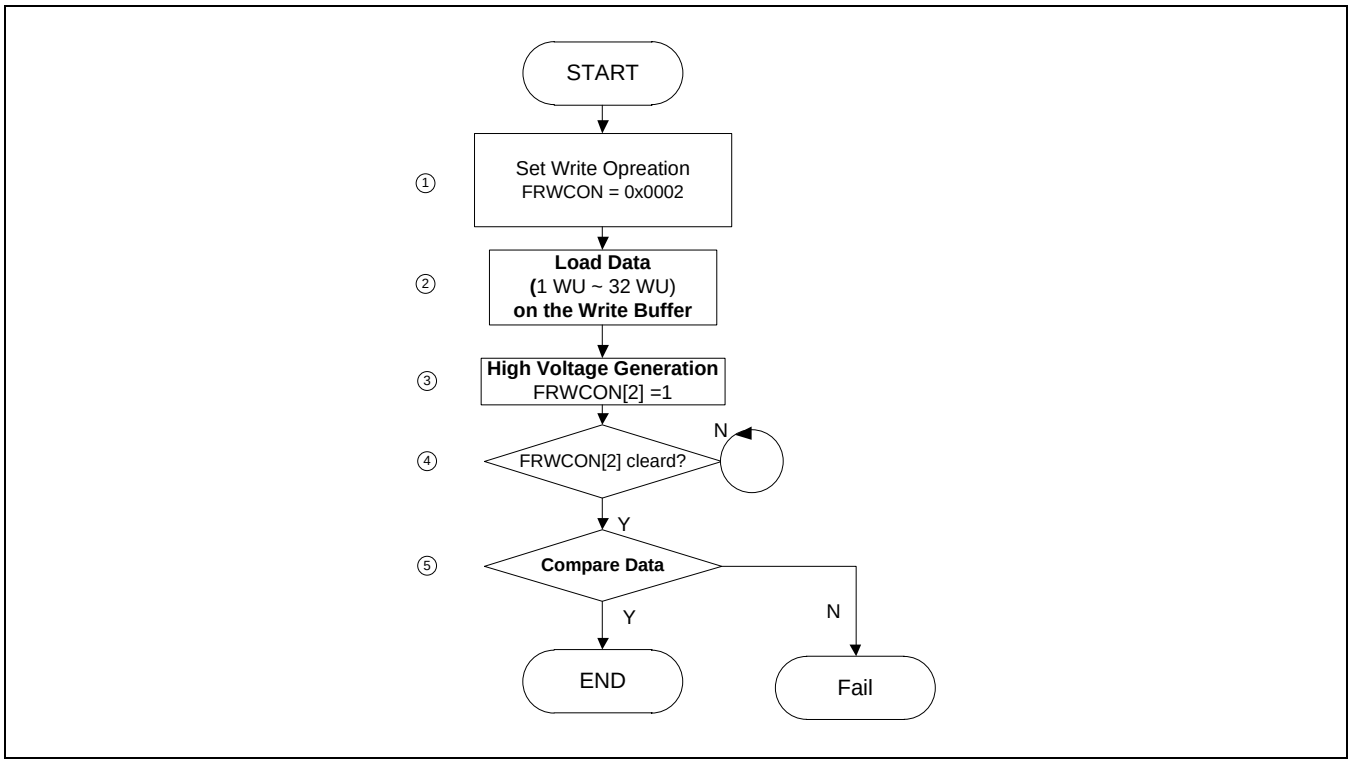


Figure 4-57 Write operation

1. Select write operation : FRWCON = 0x0002
2. Store RAM data to the targeted WU: At this time, the data is loaded on the write buffer from RAM.
3. Activate high voltage operation : FRWCON[2]=1
4. Wait for high voltage to end : until FRWCON=0x0000
5. It is recommended to Read all data and compare with the expected value.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.4.1.2 Erase operation

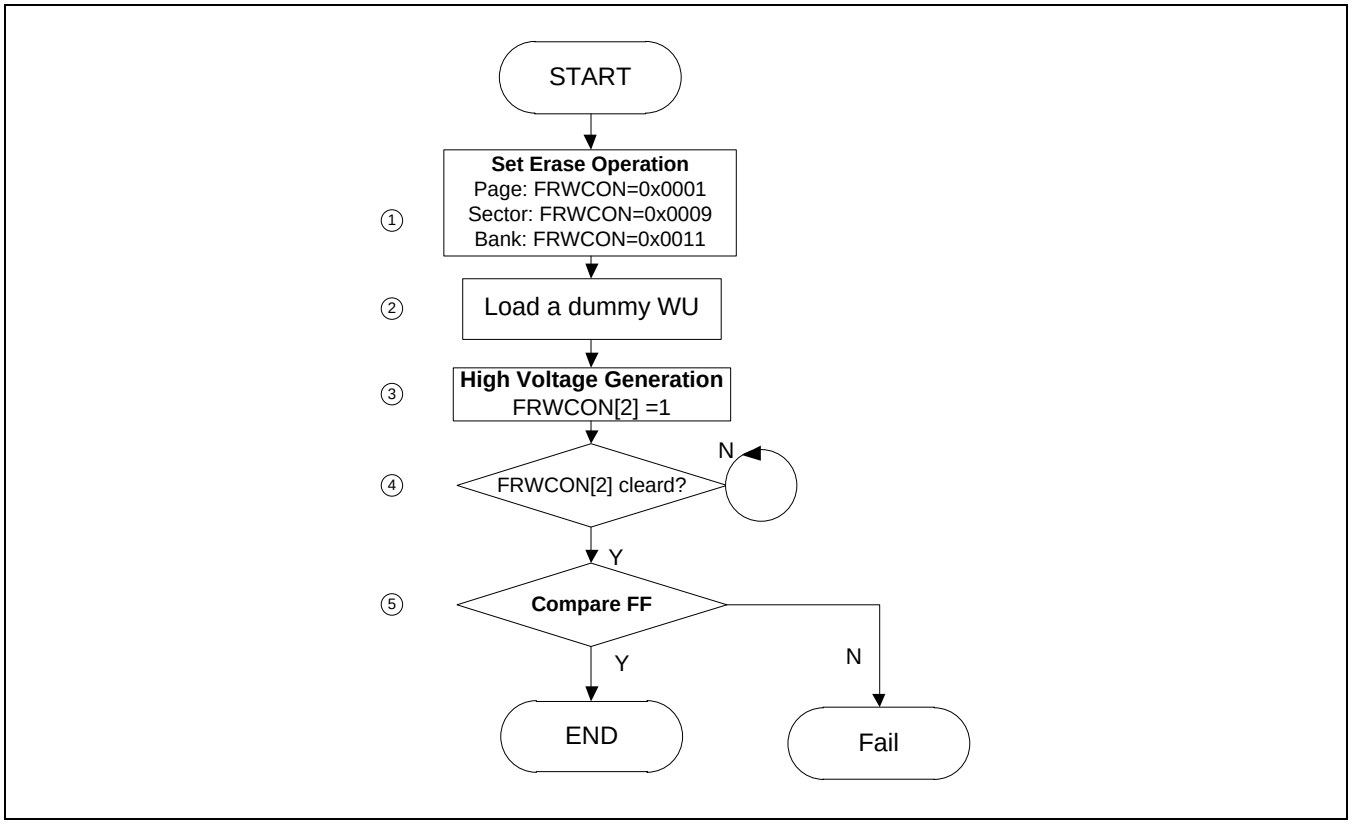


Figure 4-58 Erase operation

1. Select erase operation and the desired erase unit (FRWCON= (sector)0x0009 / (bank)0x0011 / (page) 0x0001)
2. Store dummy value to any WU within the target area: At this time, a dummy WU is loaded on the write buffer
3. Activate high voltage operation: FRWCON[2]=1
4. Wait for high voltage to end: until FRWCON=0x0000
5. It is recommended to read all data and compare with 0xFFFFFFFF

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.4.1.3 Option Cell Program Operation

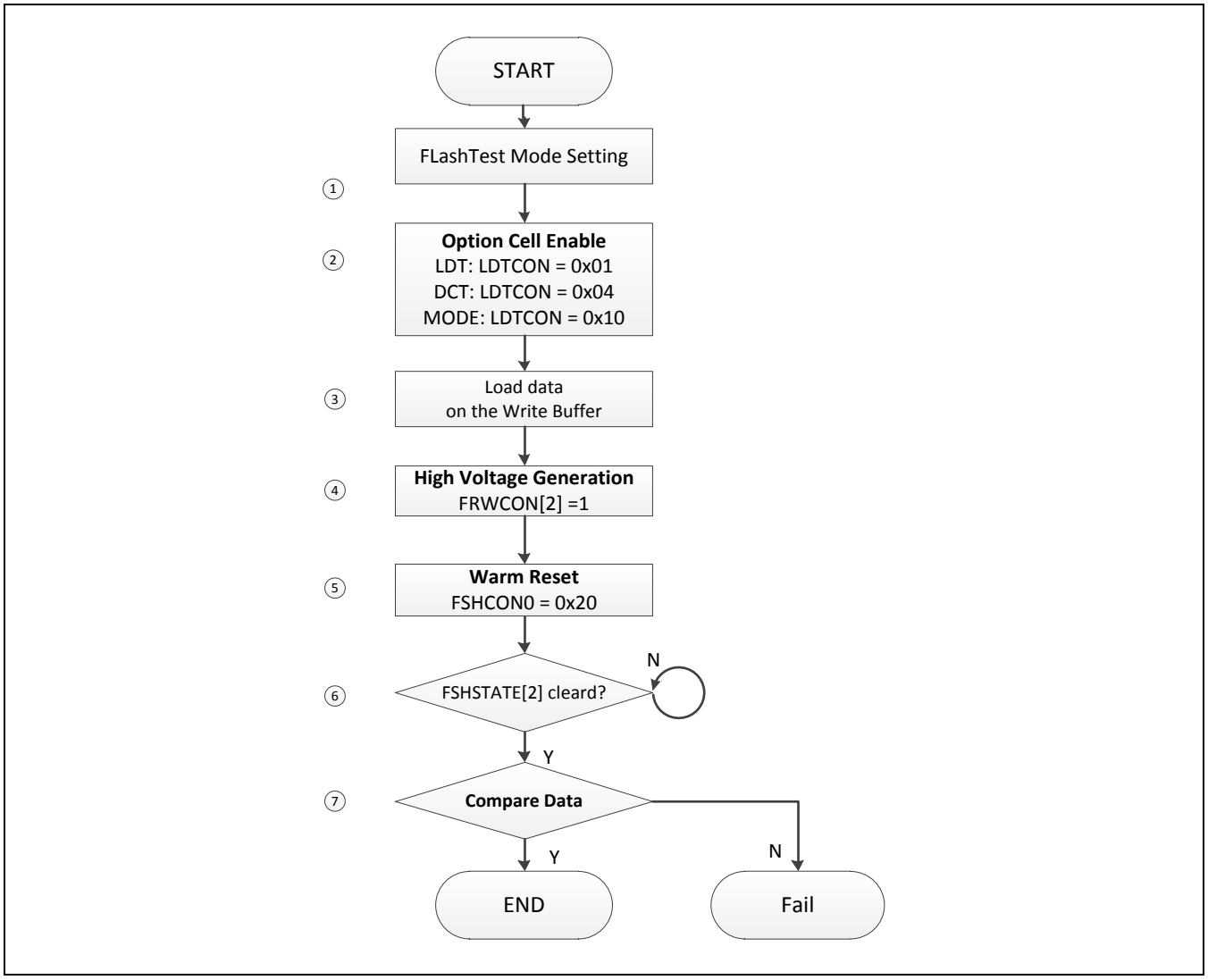


Figure 4-59 Option cell program operation

4.20.4.1.4 Interrupt control

After Erase/Write operation, Interrupt is generated by setting (Refer to Interrupt Controller)

4.20.5 Flash ip specification

Product		Reset Value
Organization	Memory Density	512KB (1 Matrix)
	Program page buffer size	256 Byte

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Product		Reset Value
	Data Input	39 bit (7bit for ECC)
	Data Output	156 bit (28 bit for ECC), X4
Power	VDDF	1.1V -5% +10%
	VDDHF	1.62V ~ 5.5V
Operating Temperature		-25℃ ~ 85℃
Endurance		500Kcycle
Data Retention		20 years
	Erase Unit	Page (512Byte) Sector (4KByte) Chip (128KByte)
Read Access Time	Endurance (<1K cycle for Code)	30ns
	Endurance (<500K cycle for Data)	80ns

4.20.6 Register Description

4.20.6.1 Register Map Summary

- Base Address: 0x40024000

Register	Offset	Description	Reset Value
FRWCON0	0x0000	FRWCON0 Register	0x0000_0000
FSHSTAT	0x0004	Flash Status register	0x0000_0000
PROTCON	0x0010	OPRSEL OR BLSCON	0x0000_0000
PROTMON	0x0014	BLSMON in ARM	0x0000_030F
DCYCRDCON	0x0020	DCYCREAD Control register	0x0000_0004
LDTCON	0x0030	LDTCON Register for TESTMODE ONLY	0x0000_0000
FSHCON0	0x0034	Flash Test Register #1	0x0000_0000
FECCCON	0x0038	Flash Test Register #2	0x0000_0000
FECCFADR	0x003C	ECC Fault Address	0x0000_0000
DCT0	0x0040	DCT0 Option register	0xFFFF_FFFF
DCT1	0x0044	DCT1 Option register	0xFFFF_FFFF
LDT0	0x0050	LDT0 Option register	0x0000_0000
LDT1	0x0054	LDT1 Option register	0x0000_0000
LDT2	0x0058	LDT2 Option register	0x0000_0000
LDT3	0x005C	LDT3 Option register	0x0000_0000
MODE	0x0060	MODECELL Option register	0x0000_0000
CPA0	0x0064	CPA0 Option register	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

4.20.6.1.1 FRWCON0

- Address: Base Address + 0x0000, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:16]	R	Reserved	0x0000_0000
ECC_Fail_Flag	[15]	RW	ECC Fail Flag 0: ECC data correct 1: if DOUT_ECC[6:0]!=7'hFF, ECC Fail flag is setted. When you want to check this bit again, you must set bit '0'.	0x0000_0000
ECC_DATA	[14:8]	R	ECC data bits(=DCOUT_ECC[6:0])	0x0000_0000
WriteVerify	[7]	RW	Write Verify_OFF control bit 0: No Action 1: Margin mode set(for Program(0x00) verify)	0x0000_0000
EraseVerify	[6]	RW	Erase Verify_ON Control bit 0: No Action 1: Margin mode set(for Erase(0xFF) verify)	0x0000_0000
FastOperation	[5]	RW	Write time selection 0: Normal Write Time(1.5ms) 1: Fast Operation The fast operation should be used for erase page and program	0x0000_0000
EraseModeselection	[4:3]	RW	Erase Mode Selection 00: Page Erase mode(256bytes) 01: Sector Erase mode (4Kbytes) 10: Mat Erase mode (Mat Erase) 11: forbidden	0x0000_0000
ProgramActivation	[2]	RW	Program Activation bit 0: No Operation 1: High Voltage Generation	0x0000_0000
EraseWriteSelection	[1:0]	RW	Erase and Write Selection bits 00: No operation 01: Erase 10: Write 11: forbidden	0x0000_0000

4.20.6.1.2 FSHSTAT

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:4]	R	Reserved	0x0000_0000
FSHSTAT	[3:0]	RW	Flash Status bits 0000: No error	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			0001: Option read completion after cold reset 0010: Option read completion after warm reset(Testmode only) 0011: No page buffer load 0100: more than 2 word page load 0101: no pgm or ers mode set 0110: write while write 0111: frwcon undefined set 1000:dual read mode detection(prefetch&dcycread)	

4.20.6.1.3 PROTCON

- Address: Base Address + 0x0010, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:12]	R	Reserved	0x0000_0000
CodeProtectionLock	[11:10]	W	Set this bit to 2'b01 : lock Code Protection enable bits 2'b 10: unlocking code protection enable bit 2'b 01: Locking Code Protection enable bit If User want to lock permanently code protection attribute in Flash area(PROTMON[5]= 0), user must follow the below sequences. 1) set these bits to 01 : Erase 2) set these bits to 01 : Program	0x0000_0000
CodeProtection0	[9:8]	W	Set 2'b10 : if user want to be writable flash area. 2'b 10: MAT0 Writable 2'b 01: MAT0 Write Protection If user want not to be writable flash area(PROTMON[4]= 0), user must follow the below sequences. 1) set these bits to 01 : Erase 2) set these bits to 01 : Program	0x0000_0000
Reserved	[7:0]	R	Reserved	0x0000_0000

4.20.6.1.4 PROTMON

- Address: Base Address + 0x0014, Reset Value = 0x0000_030F

Name	Bit	Type	Description	Reset Value
Reserved0	[31:11]	R	Reserved	0x0000_0000
CP_LOCK	[9]	R	Code Protection Lock Monitoring bit 0 : Flash code protection is locked	0x0000_0001

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			1: Flash code protection is unlocked.	
CP_SET	[8]	R	Flash write enable bit 0: Flash write protection(read only) 1: Flash writable	0x0000_0001
Reserved1	[7:4]	R	Reserved	0x0000_0000
Reserved2	[3:0]	R	Reserved	0x0000_000F

4.20.6.1.5 DCYCRDCON

- Address: Base Address + 0x0020, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
Reserved0	[31:16]	R	Reserved	0x0000_0000
Bank1DataWait	[15:12]	RW	NOR Flash Bank1 Data read wait cycle 0000: No wait 0001: 1 wait 0010: 2 wait 0011: 3 wait 0100: 4 wait 0101: 5 wait 0110: 6 wait 0111: 7 wait 1000: 8 wait 1001: 9 wait others : forbidden	0x0000_0000
Bank0DataWait	[11:8]	RW	NOR Flash Bank0 Data read wait cycle 0000: No wait 0001: 1 wait 0010: 2 wait 0011: 3 wait 0100: 4 wait 0101: 5 wait 0110: 6 wait 0111: 7 wait 1000: 8 wait 1001: 9 wait others : forbidden	0x0000_0000
Bank1CodeWait	[7:6]	RW	NOR Flash Bank1 Code read wait cycle 00: No wait 01: 1 wait 10: 2 wait (2x prefetch mode is not supported) 11: 3 wait (2x prefetch mode is not supported)	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
Bank0CodeWait	[5:4]	RW	NOR Flash Bank0 Code read wait cycle 00: No wait 01: 1 wait 10: 2 wait (2x prefetch mode is not supported) 11: 3 wait (2x prefetch mode is not supported)	0x0000_0000
CAC_FLUSH	[3]	RW	NOR Flash Cache Flush bit 0: Remain Cache 1: Cache Flush	0x0000_0000
CAC_EN	[2]	RW	NOR Flash Branch Cache enable bit 0: Cache disable 1: Cache enable	0x0000_0001
Noeffect	[1]	RW	NOR Flash 4X Prefetch control bit 0: Prefetch disable 1: Prefetch enable	0x0000_0000
_2XPrefetchenable	[0]	RW	NOR Flash 2X Prefetch control bit 0: Prefetch disable 1: Prefetch enable	0x0000_0000

4.20.6.1.6 LDTCON

- Address: Base Address + 0x0030, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:5]	R	Reserved	0x0000_0000
ModeCell	[4]	RW	ModeCell Selection bit 0: No action 1: Modecell enable	0x0000_0000
Reserved1	[3]	RW	Reserved1	0x0000_0000
DCTOption	[2]	RW	DCT Option Selection bit 0: No action 1: DCT option enable	0x0000_0000
Reserved2	[1]	RW	Reserved2	0x0000_0000
LDT0Option	[0]	RW	LDT0 Option selection bit 0: No action 1: LDT0 Option enable	0x0000_0000

4.20.6.1.7 FSHCON0

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:11]	R	Reserved	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
ChipEraseEnable	[10]	RW	Total chip erase enable bit 0: Chip Erase disable 1: Chip Erase enable	0x0000_0000
LDTSboxenable	[9]	RW	LDT Sbox enable bit 0: LDT permutation use only 1: LDT sbox & permutation Use	0x0000_0000
Write_data_inverting_selection	[8]	RW	Write data inverting selection bit 0: inverting disable 1: inverting enable	0x0000_0000
SALAT_measure_bit_for_trim	[7]	RW	SALAT measure bit for trim 0: SALAT inactive 1: SALAT active	0x0000_0000
HSATCHcontrol	[6]	RW	HSLATCH enable bit for EDS 0: HSLATHC disable(HSLATCH ⇒always 1) 1: HSLATHC enable(HSLATCH ⇒code: 1, data:0)	0x0000_0000
Option_value_update	[5]	RW	Option value update enable 0: disable 1: enable	0x0000_0000
Reserved1	[4]	R	Reserved	0x0000_0000
Read_DB_path_delay_measure	[3]	RW	Read data but path delay measure enable 0: no action 1: read data bus delay measure enable	0x0000_0000
Test_Register_access	[2]	RW	Test Register access enable 0: test register access disable 1: test register access enable	0x0000_0000
UserModeTestSelection	[1]	RW	User mode set enable 0: No action 1: User mode test enable	0x0000_0000
IFEN	[0]	RW	IFEN Enable bit 0: No action 1: IFEN enable	0x0000_0000

4.20.6.1.8 FECCCON

- Address: Base Address + 0x0038, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:15]	R	Reserved	0x0000_0000
ECCWritedata	[14:8]	RW	ECC write data bits	0x0000_0000
Reserved1	[7]	R	Reserved	0x0000_0000

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Name	Bit	Type	Description	Reset Value
ECC_fault_clear	[6]	RW	ECC fault clear bit(auto clear at next clock)	0x0000_0000
ECC_fault_type	[5:4]	R	ECC fault type monitoring bits	0x0000_0000
Reserved2	[3:2]	R	Reserved	0x0000_0000
ECCMain_data_access_type_selection	[1:0]	RW	ECC/Main Data access type selection bits	0x0000_0000

4.20.6.1.9 FECCFADR

- Address: Base Address + 0x003C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ECC_fault_address	[31:0]	R	ECC fault address	0x0000_0000

4.20.6.1.10 DCT0

- Address: Base Address + 0x0040, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
DCT0	[31:0]	RW	DCT0[31:0] option value	0xFFFF_FFFF

4.20.6.1.11 DCT1

- Address: Base Address + 0x0044, Reset Value = 0x0FFF_FFFF

Name	Bit	Type	Description	Reset Value
DCT1	[31:0]	RW	DCT1[31:0] option value	0xFFFF_FFFF

4.20.6.1.12 LDT0

- Address: Base Address + 0x0050, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDT0	[31:0]	RW	LDT0[31:0] option value	0x0000_0000

4.20.6.1.13 LDT1

- Address: Base Address + 0x0054, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDT1	[31:0]	RW	LDT1[31:0] option value	0x0000_0000

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4.20.6.1.14 LDT2

- Address: Base Address + 0x0058, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDT2	[31:0]	RW	LDT2[31:0] option value	0x0000_0000

4.20.6.1.15 LDT3

- Address: Base Address + 0x005C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDT3	[31:0]	RW	LDT3[31:0] option value	0x0000_0000

4.20.6.1.16 MODE

- Address: Base Address + 0x0060, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:29]	R	Reserved	0x0000_0000
UFC_OPTCELL10	[28]	R	ECC fail flag auto clear option 0: ECC fail flag auto clear disable 1: ECC fail flag auto clear enable	0x0000_0000
UFC_OPTCELL9	[27]	R	Performance Enhanced option in case of sequential read after idle 0: disable 1: enable	0x0000_0000
UFC_OPTCELL8	[26:25]	R	Data wait extended option 00: default_wait (2w/3w) 01: default_wait + 1 (3w/4w) 10: default_wait + 2 (4w/5w) 11: default_wait + 3 (5w/6w)	0x0000_0000
UFC_OPTCELL7	[24:23]	R	Erase verify read guard time decrease option 00: ref_vfycnt 01: ref_vfycnt -1 10: ref_vfycnt -2 11: ref_vfycnt -3	0x0000_0000
UFC_OPTCELL6	[22]	R	ECC Function disable option 0: ECC function enable 1: ECC function disable	0x0000_0000
UFC_OPTCELL5	[21]	R	HSLATCH attribute enable option 0: always code 1: On/Off by Code/Data access	0x0000_0000
UFC_OPTCELL4	[20]	R	FECCCON register access mode select option 0: No accessible in Umode 1: accessible in Umode	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
UFC_OPTCELL3	[19]	R	Data wait operating mode set option 0: wait by setting en_wait 1: wait by HPROT[0]	0x0000_0000
UFC_OPTCELL2	[18]	R	Prefetch disable option	0x0000_0000
UFC_OPTCELL1	[17]	R	Cache disable option bit 0: cache enable 1: cache disable	0x0000_0000
UFC_OPTCELL0	[16]	R	Post ecc mode enable 0: disable 1: post ecc mode enable	0x0000_0000
Reserved1	[15:9]	R	Reserved	0x0000_0000
LDT_MCELL	[11:8]	R	Option value for LDT master	0x0000_0000
MODECELL	[7:0]	R	MODECELL[7:0] option value	0x0000_0000

4.20.6.1.17 CPA0

- Address: Base Address + 0x0064, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[31:16]	R	Reserved	0x0000_0000
CPA0	[15:0]	R	CPA0 Option value (Code Protection #0)	0x0000_0000

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5 AFE

5.1 Overview

The AFEs (Analog Front-Ends) in S3FBP5A include ECG, PPG and BIA signal processing blocks along with five 12-bit SAR type ADCs and a high resolution 24-bit Sigma Delta ADC. It also has Power Management IPs (PMIPs) such as Buck converter and LDOs for power efficient operation. Each SAR ADC contains two separated data conversion channels. Among five SAR type ADCs, four SAR ADCs are used for data conversion of AFE signals and a remained SAR ADC is used for data conversion of die temperature and general purpose respectively. The typical operating voltage is 1.1V for ECG, PPG, BIA and SAR type ADCs and 3V for the Sigma Delta ADC, the Buck converter and Keep-alive LDOs.

The AFE for ECG measures voltage signal generated from heart. From the signal, various characteristics of heart pumping can be measured and predicted. The ECG AFE has low noise and wide dynamic range amplifiers and filters. The AFE for ECG also has integrated HPF (High Pass Filter) for cancellation of DC offset voltage occurred from human body, and a right leg driver for realization of high CMRR (Common Mode Rejection Ratio).

The AFE for PPG measures reflected or transmitted light when LEDs emit lights to blood vessels. From the reflected or transmitted lights, various characteristics such as oxygen saturation or heart rate can be measured. The PPG AFE has LED driver and PPG receiver sections. LED driver consists of 9 channels which user can select depending on situation and able to drive maximum 250mA. The PPG receiver has low noise Trans-Impedance Amplifier and input DC offset cancellation circuits.

The AFE for BIA predicts body impedance from measurement of voltage level in different body locations. When AC current is injected to body, fat in the body resists current flow. Using this characteristic, body impedance can be measured and used for analyzing body composition. The BIA AFE has current sourcing and body impedance sensing parts. The current sourcing part generates 1Hz to 250KHz AC currents and injects to human body. The body impedance sensing part measures voltage difference of human body and converts measured AC voltage to DC voltage level to predict body impedance.

S3FBP5A also has general purpose ADCs: 12-bit SAR ADC and 24-bit Sigma Delta ADC. They can be used for data conversion of other healthcare application such as GSR or Skin temperature. For example, GSR can be measured using current sourcing part in BIA and PPG receiver section. Skin temperature can be measured with the help of thermistor and Sigma Delta ADC.

S3FBP5A also has digital control logic to cooperate with analog AFEs. The AFE control logic has next four major functions: SAR-ADC Channel separation/Decimation, Five 16-bit FIFOs and one 24-bit FIFO, DMA controller of FIFO and Special function registers for AFE controller.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.2 Functions

- Suitable for low power Bandage type ECG acquisition system
- Enable to measure concurrently the electrode-tissue-impedance(ETI)
- Adoption of motion artifact suppression algorithm
- PPG acquisition system with LED driver
- BIA acquisition system with current driver
- GSR measurement
- Low powered 12 bit SAR type ADC
- Chopper circuit technique for Low noise signal processing
- General Purpose 24 Bit Sigma Delta ADC and SAR ADC.
- SAR-ADC Channel separation/Decimation
- Five 16-bit FIFOs and one 24-bit FIFO
- DMA controller of FIFO
- Special function registers for AFE controller

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Figure 5-1 illustrate AFE (Analog Front End) Block Diagram of S3FBP5A

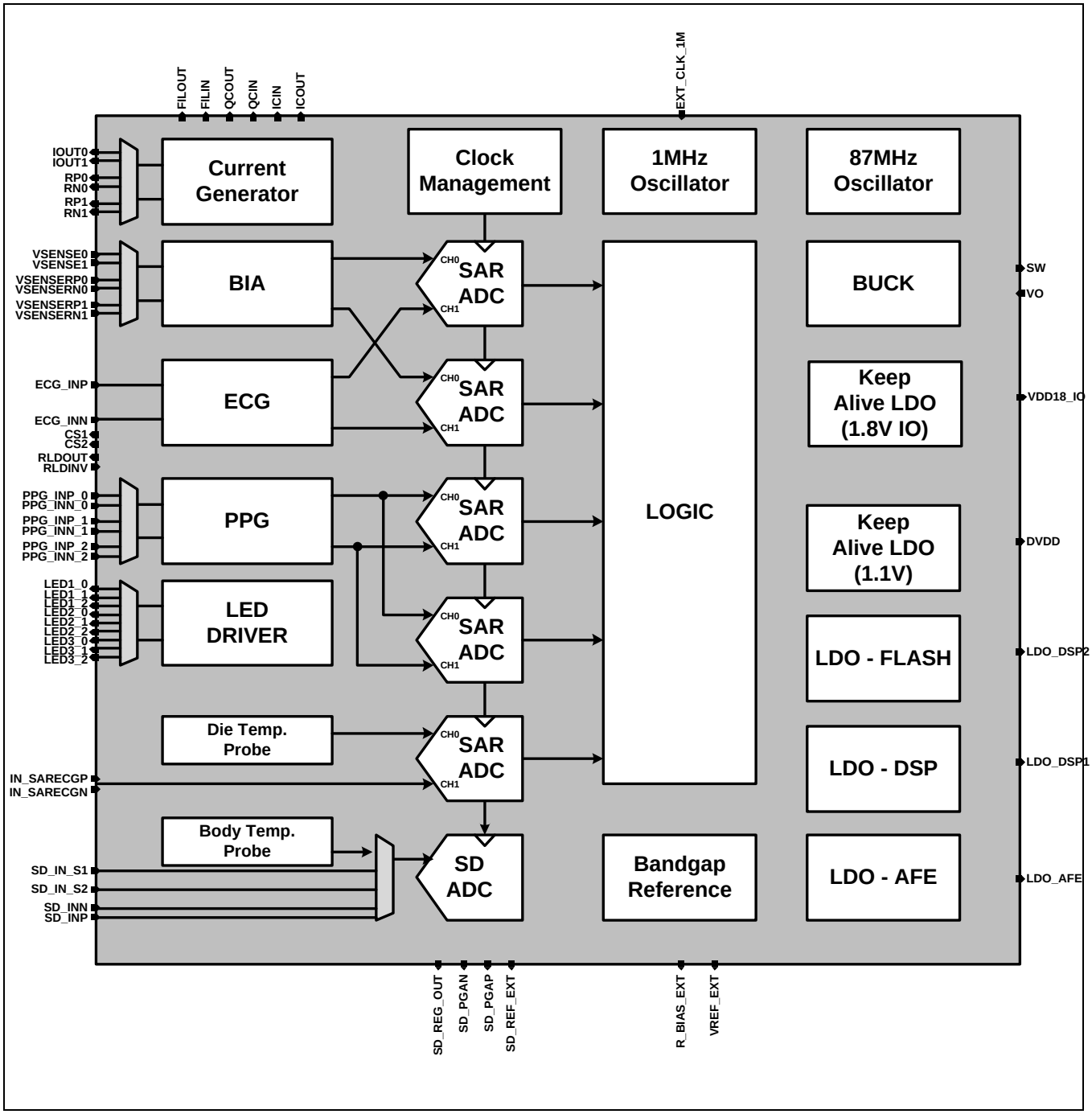


Figure 5-1 The Block Diagram of Analog Front End in S3FBP5A

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Figure 5-2 illustrates a block diagram of AFE control logic.

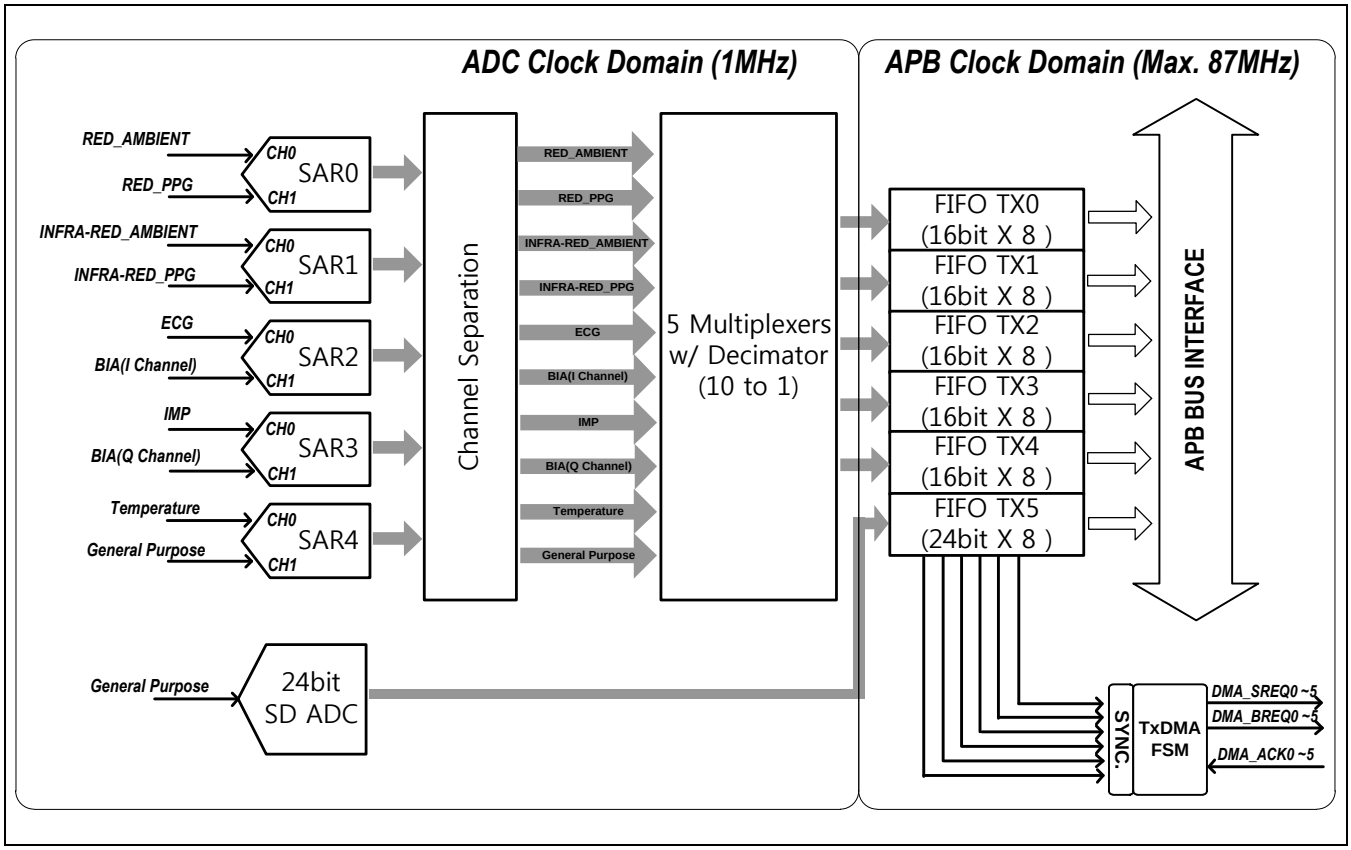


Figure 5-2 Block Diagram of the AFE Control Logic

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3 Functional Description

5.3.1 ECG

5.3.1.1 Overview

ECG Readout channel consists of electrocardiogram (ECG) acquisition block and electro tissue impedance (ETI) measurement block. ECG acquisition block has a high-pass filter (HPF), an instrumental amplifier (IA), a chopping spike filter (CSF) and programmable gain amplifier (PGA) with low pass filter(LPF).

HPF is used for filtering DC offset and enabling high input impedance level. DC offset cancellation is necessary for locating desired signals in allowed operating range and high input impedance is required to prevent input signal distortion. IA is a low noise pre-amplifier to increase input signal to certain level and to eliminate the 1/f noise using choppers. The gain range of IA is from 0 to 55 dB and the level can be controlled with 4-bit registers. It has inside choppers to eliminate the 1/f noise contribution and the chopping clock frequency is 2KHz. To eliminate chopping spike noise occurred during chopping operation, the CSF is applied. To provide optimized signal level for analog to digital conversion, the PGA and LPF are applied. The PGA has gain range of 0 to 15.6dB and the gain is controlled with 3-bit register. The LPF is applied for anti-aliasing purpose. The cut-off frequency range of LPF is between 75~250Hz and it can be controlled with 3-bit register.

The ETI block can measure impedance level between electrode and skin. The measured impedance level is used to estimate motion artifact signal if motion artifact detection algorithm is applied. The architecture of ETI block is almost same as the ECG block except HPF and chopping method. The ETI channel also has lead-off detection function to know whether the electrode is stick to user's skin or not. The ETI block in the S3FBP5A is normally not applied except the lead-off detection function is necessary.

The ECG Readout Channel has On-chip Right Leg Driver (RLD). The RLD is used to improve CMRR (Common Mode Rejection Ratio) of the ECG readout channel by applying bias voltage to the body. *Figure 5-3* shows the block diagram of ECG Readout Channel

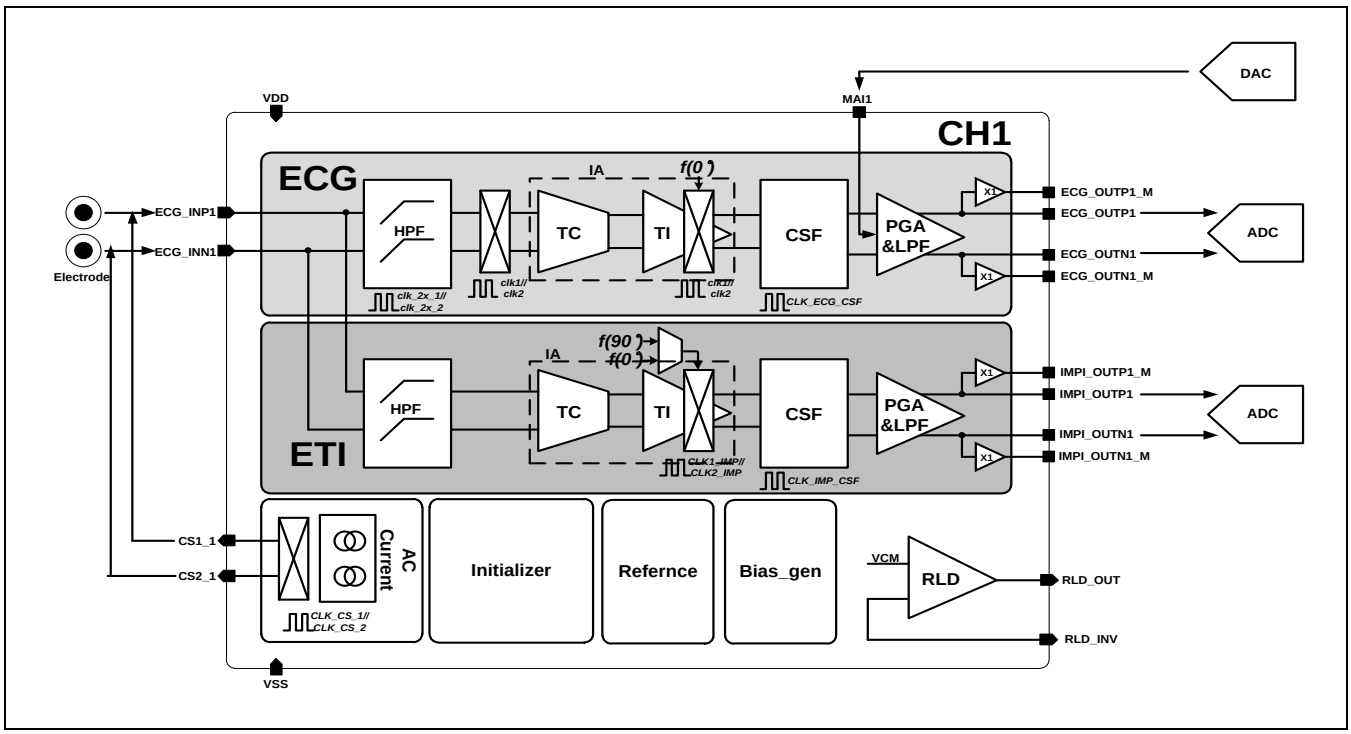


Figure 5-3 Block Diagram of ECG Readout Channel

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3.1.2 Register Description

5.3.1.2.1 Register Map Summary

- Base Address: 0x4001A000

Register	Offset	Description	Reset Value
ECG_GAIN	0x0000	ECG_GAIN	0x0000_0084
ECG_CONT	0x0034	ECG_CONT	0x0000_0000

5.3.1.2.1.1 ECG_GAIN

- Address: Base Address + 0x0000, Reset Value = 0x0000_0084

Name	Bit	Type	Description	Reset Value
ECG_IA_GAIN1	[7:4]	RW	ECG AFE IA Gain Control (V/V) 0: 1 1: 35.5 2: 70.8 3: 112.2 4:158.5 5: 177.8 6: 223.9 7: 251.2 8: 281.8 9: 316.2 10: 354.8 11: 398.1 12: 421.7 13: 446.7 14: 501.2 15: 562.3	0x0000_0008
Reserved0	[3]	RW	Reserved	0x0000_0000
ECG_GAIN1	[2:0]	RW	ECG AFE PGA Gain Control (V/V) 0: 1 1: 1.2 2: 1.6 3: 2 4: 3 5: 4 6: 5 7: 6	0x0000_0004

5.3.1.2.1.2 ECG_CONT

- Address: Base Address + 0x0034, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
RLD_EN	[6]	RW	Right Leg Driver enable control	0x0000_0000
ECG_IN_SHORT_CTRL	[5]	RW	ECG different Input short control	0x0000_0000
IA_TEST_EN	[4]	RW	IA Test enable control(low gain setting)	0x0000_0000
Reserved1	[3]	RW	Reserved	0x0000_0000
VB_IA_CONT1	[2:0]	RW	Bias voltage control for HPF output	0x0000_0000

5.3.2 PPG

5.3.2.1 Overview

The PPG applied in S3FBP5A can be used for heart rate and SpO2 (Pulse Oximetry) measurements. The PPG is composed of a LED driver and a receiver with mux. The LED driver is used to turn on led diodes. The current sinking circuits controlled by 7-bit LED Current Controller are applied to turn on LEDs such as IR, RED and Green diodes. LED light intensity is also controlled by 7-bit LED Current Controller and the controlled current range is from 0mA to 250mA with a step of 2.1mA. Nine external LEDs can be connected to the LED driver at the same time, and selection of LED to be turned on is performed by LED switches placed between LED current controller and external LEDs. LED switch controller selects switches to open and close. The LED driver also has connection fault detection function. The fault detector in the driver checks whether LED pin is shorted to VDD or not. If anyone of the LED pin is shorted to VDD, LED current driving operation is stopped. *Figure 5-4* shows the block diagram of the LED driver.

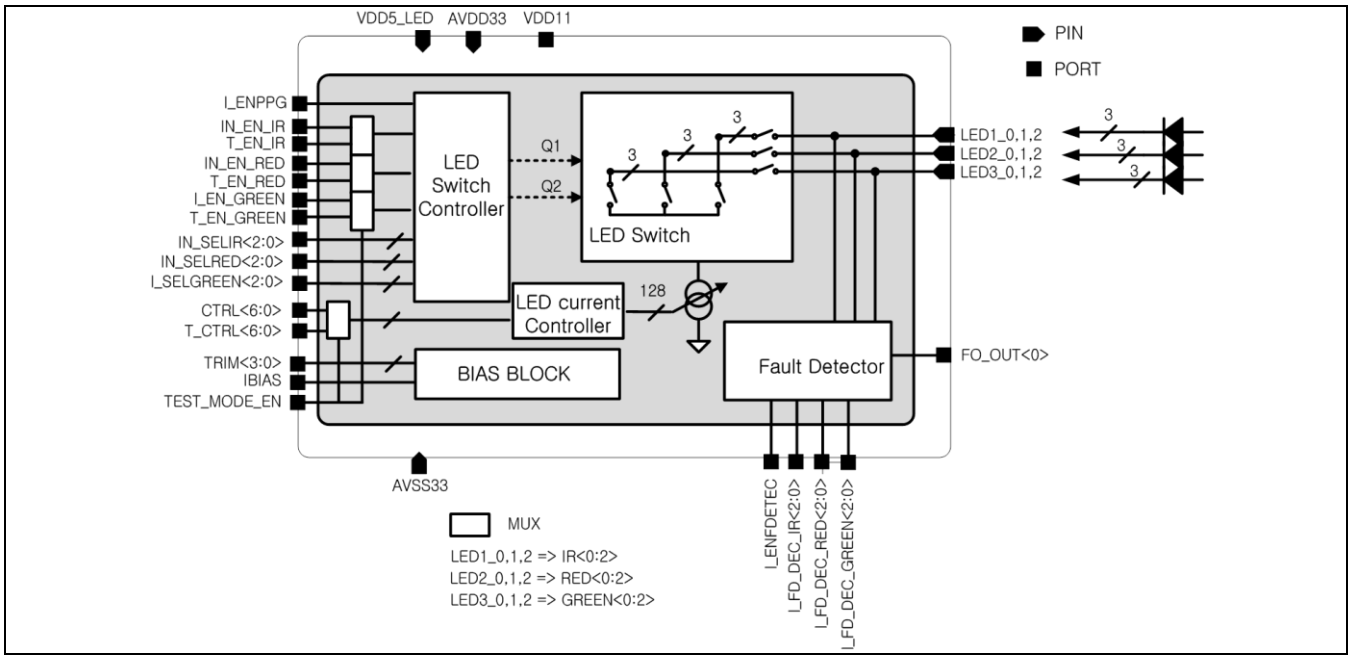


Figure 5-4 Block Diagram of LED Driver

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

The receiver consists of a differential Trans-Impedance Amplifier (TIA), a programmable gain amplifier (PGA), and a buffer. Current signal detected by photodiode is amplified by TIA and PGA for optimal data conversion. The TIA has 8 steps of resistor including 2.56 MΩ, 1.28MΩ, 640 kΩ, 320 kΩ, 160 kΩ, 80 kΩ, 40 kΩ, and 20kΩ. The noise source caused by ambient light can be minimized by DC compensator (DCC) in the TIA. The receiver has periodic ambient light sensing algorithm so that detected ambient light value is periodically cancelled by DCC block. The DCC has a cancellation range of 25uA caused by ambient light with a step of 25nA. The ambient light compensated and amplified signal in the TIA is transmitted to the PGA with low pass filter. In the PGA, the gain of signal is controlled and low-pass filtered for optimal data conversion. To ensure that the low-pass filter RC time constant has sufficiently high bandwidth, the feedback capacitor CF is programmable. Available CF values include: 10 pF, 20 pF, 30 pF, and 40 pF. Any combination of these capacitors can be used. *Figure 5-5* shows the Block Diagram of PPG Receiver.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

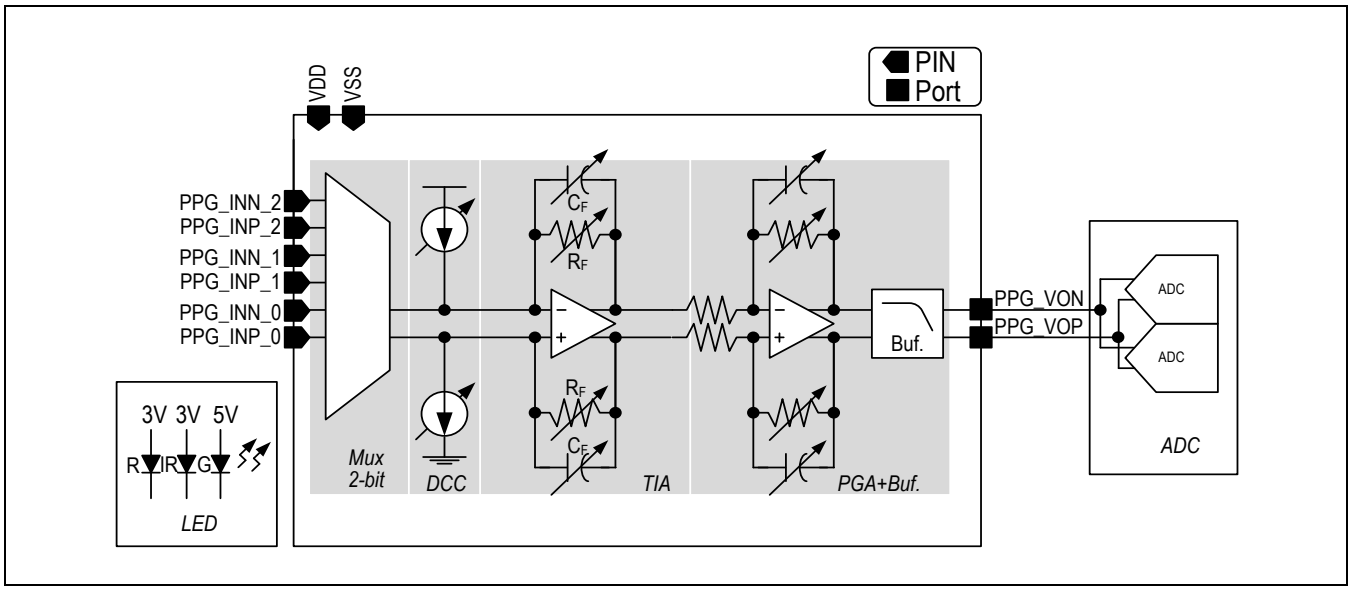


Figure 5-5 Block Diagram of PPG Receiver

Two SAR type ADCs are used to convert the analog domain signal to digital domain. For efficient data conversion, time-division method is applied to the PPG receiver. By controlling mux at the input of receiver, LED and ambient light sensing time can be divided. Figure 5-5 shows Sampling Procedure of PPG Receiver. For example, red LED signal is sampled on 1st ADC when red LED is on in the first time slot (R0). After then, the ambient signal is sampled on 2nd ADC when all LEDs are off(RA0). Similarly, IR LED signal is sampled on 1st ADC in the second time slot (IR0), and then the ambient signal is sampled on 2nd ADC when all LEDs are off(IRA0). The procedure is repeated every 1ms. The DCC value is updated with average to compensate ambient fluctuation every 256ms. *Figure 5-6* shows Sampling Procedure of PPG Receiver.

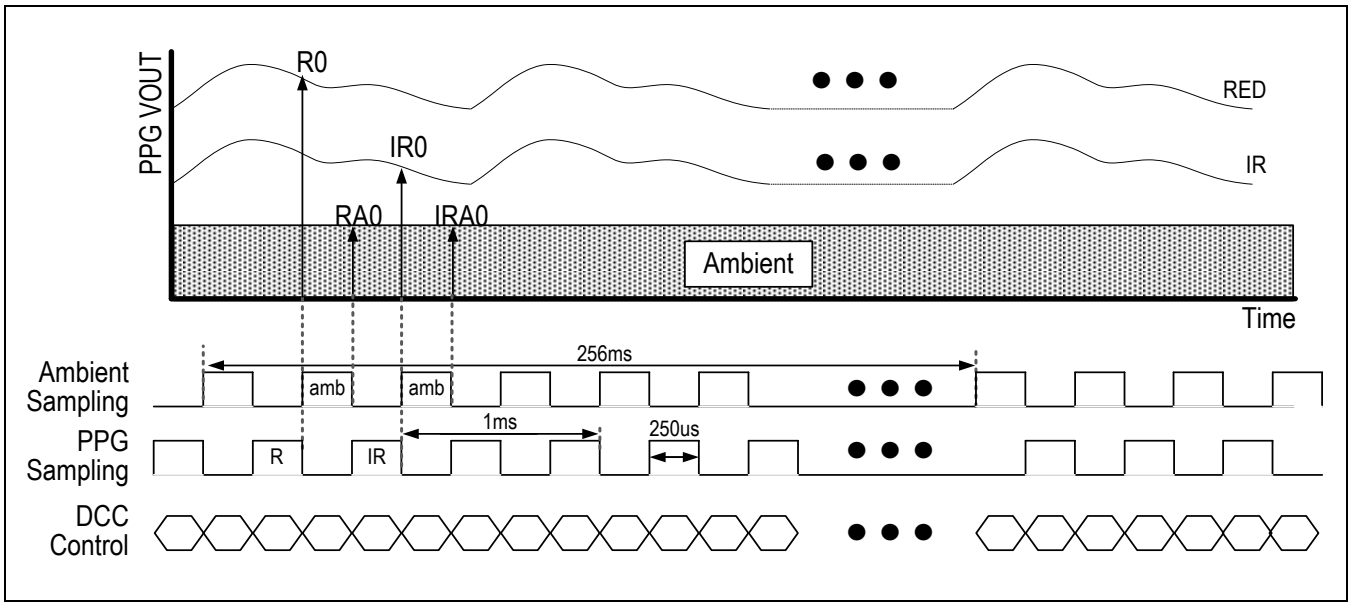


Figure 5-6 Sampling Procedure of PPG Receiver

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3.2.2 Register Description

5.3.2.2.1 Register Map Summary

- Base Address: 0x4001A200

Register	Offset	Description	Reset Value
PPG_MODE	0x0000	PPG_MODE	0x0000_0001
PPG_RST	0x0004	PPG_RST	0x0000_0000
PPG_PD_SEL	0x0008	PPG_PD_SEL	0x0000_0000
PPG_TIA_GAIN	0x000C	PPG_TIA_GAIN	0x0000_0024
PPG_PGA_GAIN	0x0010	PPG_PGA_GAIN	0x0000_0003
LED_MODE	0x005C	LED_MODE	0x0000_0001
LED_CH_SEL1	0x0064	LED_CH_SEL1	0x0000_0000
LED_CH_SEL2	0x0068	LED_CH_SEL2	0x0000_0000
LED_ISET_R	0x006C	LED_ISET_R	0x0000_0001
LED_ISET_IR	0x0070	LED_ISET_IR	0x0000_0001
LED_ISET_Green	0x0074	LED_ISET_Green	0x0000_0001
LED_PW	0x0078	LED_PW	0x0000_0000
LED_FLAG1	0x0090	LED_FLAG1	0x0000_0000
LED_FLAG2	0x0094	LED_FLAG2	0x0000_0000

5.3.2.2.1.1 PPG_MODE

- Address: Base Address + 0x0000, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7:3]	RW	Reserved	0x0000_0000
MODE2	[2]	RW	Green LED	0x0000_0000
MODE1	[1]	RW	Green & IR LED	0x0000_0000
MODE0	[0]	RW	RED & IR LED	0x0000_0001

5.3.2.2.1.2 PPG_RST

- Address: Base Address + 0x0004, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LED_T_EN_RED	[7]	RW	RED LED on (Testmode)	0x0000_0000
LED_T_EN_IR	[6]	RW	IR LED on (Testmode)	0x0000_0000
LED_T_EN_G	[5]	RW	Green LED on (Testmode)	0x0000_0000
LED_TEST_MODE_	[4]	RW	Testmode on	0x0000_0000

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Name	Bit	Type	Description	Reset Value
EN				
Reserved0	[3:1]	RW	Reserved	0x0000_0000
I_RSTN	[0]	RW	Internal Reset	0x0000_0000

5.3.2.2.1.3 PPG_PD_SEL

- Address: Base Address + 0x0008, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7:2]	RW	Reserved	0x0000_0000
PD_CH_SEL	[1:0]	RW	Photo Detector Channel Selection 11: NA 10: CH3 01: CH2 00: CH1	0x0000_0000

5.3.2.2.1.4 PPG_TIA_GAIN

- Address: Base Address + 0x000C, Reset Value = 0x0000_0024

Name	Bit	Type	Description	Reset Value
Reserved0	[7:6]	RW	Reserved	0x0000_0000
TIA_C	[5:4]	RW	TIA Cap 11: 40pF 10: 30pF 01: 20pF 00: 10pF	0x0000_0002
Reserved1	[3]	RW	Reserved	0x0000_0000
TIA_R	[2:0]	RW	TIA Resister (digital) 111: 2.56Mohm 110: 1.28Mohm 101: 640kohn 100: 320kohm 011: 160kohm 010: 80kohm 001: 40kohm 000: 20kohm	0x0000_0004

5.3.2.2.1.5 PPG_PGA_GAIN

- Address: Base Address + 0x0010, Reset Value = 0x0000_0003

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
TIA_R_A	[6:4]	RW	TIA Cap (analog)	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

PGA_A	[3:2]	RW	PGA (analog) 11: 1x 10: 1.5x 01: 2x 00: 2.5x	0x0000_0000
PGA	[1:0]	RW	PGA (digital)	0x0000_0003

5.3.2.2.1.6 LED_MODE

- Address: Base Address + 0x005C, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7:3]	RW	Reserved	0x0000_0000
FDCTRL	[2]	RW	0: 10us 1: 20us	0x0000_0000
EN_MASK	[1]	RW	Fault Detect Masking Enable	0x0000_0000
EN_FDETECT	[0]	RW	Fault Detect Enable	0x0000_0001

5.3.2.2.1.7 LED_CH_SEL1

- Address: Base Address + 0x0064, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
RED_CH_SEL	[6:4]	RW	RED LED Channel Selection	0x0000_0000
Reserved1	[3]	RW	Reserved	0x0000_0000
IRED_CH_SEL	[2:0]	RW	IR LED Channel Selection	0x0000_0000

5.3.2.2.1.8 LED_CH_SEL2

- Address: Base Address + 0x0068, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7:3]	RW	Reserved	0x0000_0000
RED_CH_SEL	[2:0]	RW	Green LED Channel Selection	0x0000_0000

5.3.2.2.1.9 LED_ISET_R

- Address: Base Address + 0x006C, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
R_ICTRL	[6:0]	RW	RED LED Current Initial Setting	0x0000_0001

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3.2.2.1.10 LED_ISET_IR

- Address: Base Address + 0x0070, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
IR_ICTRL	[6:0]	RW	IR LED Current Initial Setting	0x0000_0001

5.3.2.2.1.11 LED_ISET_Green

- Address: Base Address + 0x0074, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
G_ICTRL	[6:0]	RW	Green LED Current Initial Setting	0x0000_0001

5.3.2.2.1.12 LED_PW

- Address: Base Address + 0x0078, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7:2]	RW	Reserved	0x0000_0000
LED_PW	[1:0]	RW	00: 250 us 01: 200 us 10: 150 us 11: 100 us	0x0000_0000

5.3.2.2.1.13 LED_FLAG1

- Address: Base Address + 0x0090, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	R	Reserved	0x0000_0000
O_IR_FLAG	[6:4]	R	IR LED Fault Detect	0x0000_0000
O_RED_FLAG	[2:0]	R	RED LED Fault Detect	0x0000_0000

5.3.2.2.1.14 LED_FLAG2

- Address: Base Address + 0x0094, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	R	Reserved	0x0000_0000
O_GREEN_FLAG	[6:0]	R	Green LED Fault Detect	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3.3 BIA

5.3.3.1 Overview

BIA (Bioelectrical Impedance Analysis) is used to estimate body composition using the difference of conductivity of tissue. Conductivity is proportional to water and it is decreased when cell shape is closer to a round form (adipose tissue). The BIA measures body composition by applying a sinusoidal current into the body and measures impedance level of body using electrodes. The sinusoidal current is generated with DDS (Direct digital synthesizer) and a 6-bit DAC. A voltage-to-current converter applies this sinusoidal current into the body. The BIA includes I/Q demodulation circuits to obtain both the real and imaginary portions. *Figure 5-7* illustrate BIA block diagram

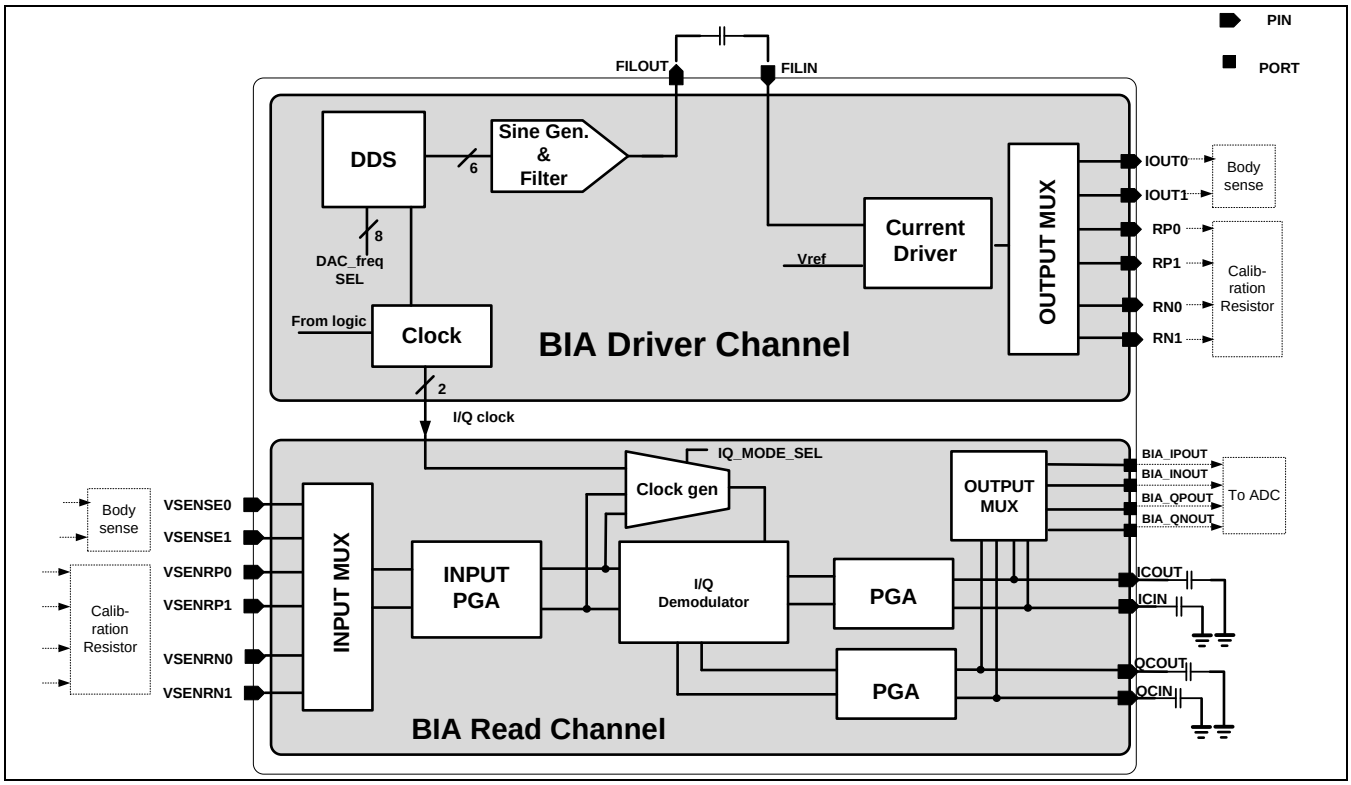


Figure 5-7 BIA Block Diagram

Body composition is calculated by measuring the impedance of the body and matching the result in a table. The BIA product company has its own matching table between this measured impedance and calculated body composition value. The table is usually based on age, sex, weight and so on. The BIA has two modes for body impedance measurement. One is Rectify mode and the other is I/Q demodulation mode. The BIA block injects current into the body at TX circuits and senses the converted voltage at RX block across the two measuring points.

As shown in the BIA Driver Channel of *Figure 5-7*, the current driving circuit injects currents to body. The direct digital synthesizer (DDS) generates 6 bit signals for DAC circuits and then the DAC makes the sinusoidal analog signal which has programmable frequency range from 1KHz to 255KHz. The output signal of the DAC is filtered by LPF, and a following external capacitor blocks the DC current and offset values contained in the signal. The filtered AC current drives reference resistor located at input of the output driver and the output driver finally injects currents into body. The tolerance of the driving current depends on the variation of source current from BGR(Band gap reference). But this can be compensated by applying identical type of resistor in the DAC. If the value of reference resistor used at input of the driver is reduced by 20%, then the flowing current must be increased by

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20% if the output sinusoidal voltage of DAC maintains same level. But, the output peak to peak voltage of the DAC will be also reduced by 20% due to the variation of identical type of resistor used in DAC. Finally, the resulting driving current will not be affected much by reference resistor value variation.

5.3.3.2 Functional Description

5.3.3.2.1 Rectify mode

In the Rectify mode shown in *Figure 5-8*, the read channel performs full wave rectification using self-generated clock signal. The rectified signal is low pass filtered and used as an input of 12-bit ADC. The low pass filter uses external large capacitors at its output pins. To remove the ripples of the output signals, over the value of uF capacitors are used.

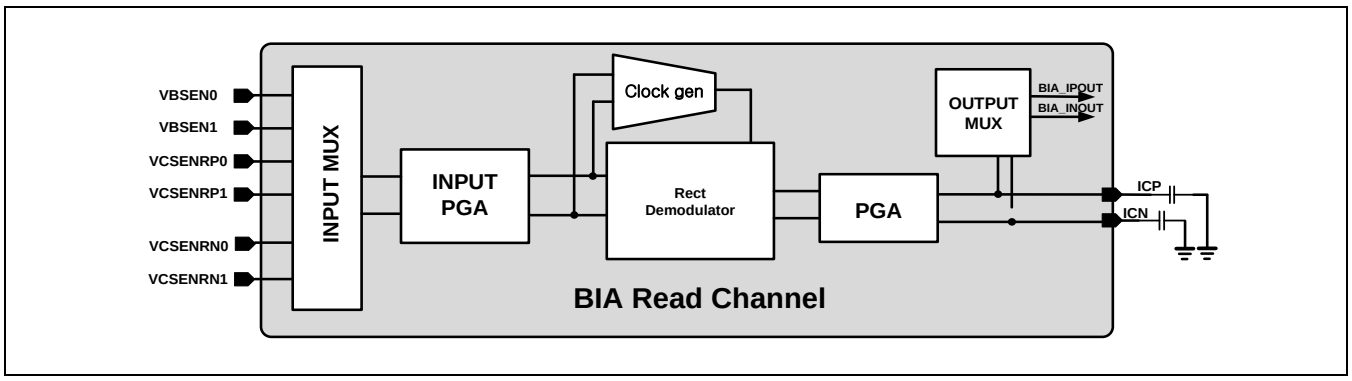


Figure 5-8 The Block diagram of BIA Rectify mode

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

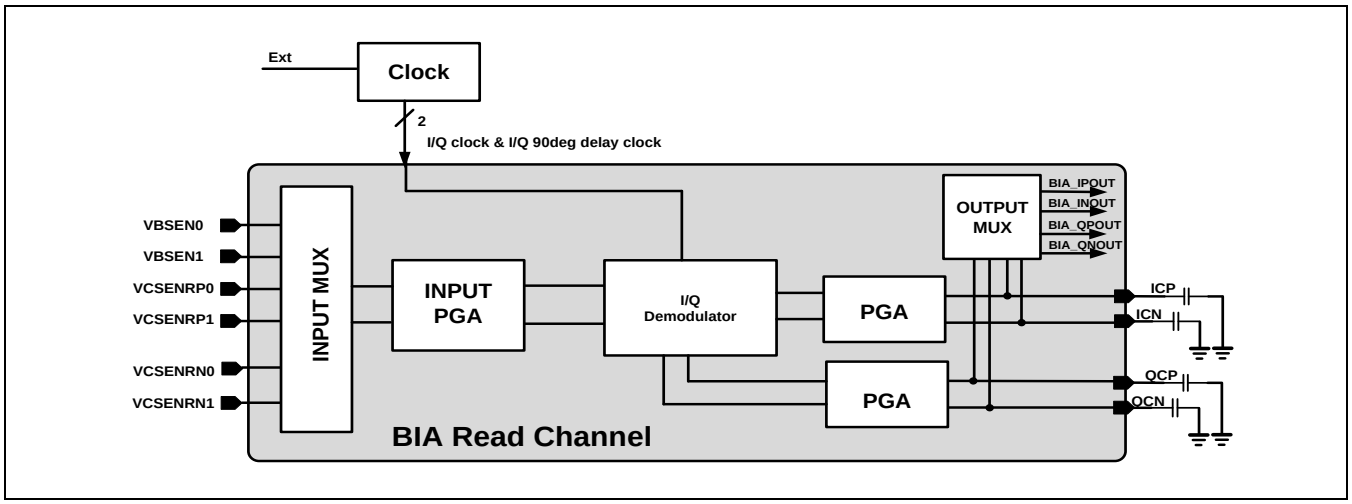


Figure 5-9 The Block diagram of BIA I/Q mode

5.3.3.2.2 I/Q demodulation mode

The BIA block has I/Q demodulator mode also to obtain both the real and the imaginary portions with a single frequency measurement. The current driving circuit injects currents to body as mentioned before. After processing the same path as rectify mode, the I/Q demodulator generates separate in-phase and quadrature-phase signals. These two signals are filtered with external large capacitors and delivered to ADC and the real and imaginary values of body impedances are calculated. The two clock signals for I/Q demodulator is generated from logic circuits. And the phase of I/Q clock can be varied by logical feedback.

5.3.3.2.3 Calibration

The calibration process is necessary to estimate body impedance. The four external Pins (RP0, RP1, RN0, RN1) are used for this calibration. For example, 500 ohm and 1.5Kohm resistors are used for the calibration. First, BIA driver channel injects current to 500 ohm and then DC value at 500ohm is recorded at the read channel. From this step, we can know the DC value of 500 ohm impedance. Second, BIA driver channel injects same current to 1.5Kohm resistor and then the DC value at 1.5Kohm is recorded in the read channel as same as the first step. Finally, the driver channel injects current to the body, then DC value is obtained from the read channel. If the DC value obtained from body is placed between DC value of 500ohm and 1.5kohm, impedance level of body can be estimated,

5.3.3.3 Register Description

5.3.3.3.1 Register Map Summary

- Base Address: 0x4001A300

Register	Offset	Description	Reset Value
BIA_TX_FIL	0x0000	BIA_TX_FIL	0x0000_0001
BIA_TX_FREQ	0x0004	* DDS Frequency Selection - DAC output frequency = DACFREQ<7:0> * Fclk / 1024 (ex: with Fclk=1MHz then FF = 255KHz)	0x0000_0032

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Register	Offset	Description	Reset Value
BIA_TX_AMPL	0x0008	BIA_TX_AMPL	0x0000_00A0
BIA_TX_SW	0x0010	BIA_TX_SW	0x0000_0001
BIA_RX_INPUT	0x0014	BIA_RX_INPUT	0x0000_0000
BIA_RX_AMP1	0x0018	BIA_RX_AMP1	0x0000_0004
BIA_RX_AMP2	0x001C	BIA_RX_AMP2	0x0000_0044

5.3.3.3.1.1 BIA_TX_FIL

- Address: Base Address + 0x0000, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7:2]	RW	Reserved	0x0000_0000
BIAFIL	[1:0]	RW	* DAC output LPF 3dB control 00: 100 Khz 01: 150 Khz 10 : 200 Khz 11 : 250 Khz	0x0000_0001

5.3.3.3.1.2 BIA_TX_FREQ

- Address: Base Address + 0x0004, Reset Value = 0x0000_0032

Name	Bit	Type	Description	Reset Value
DACFREQ	[7:0]	RW	* DDS Frequency Selection - DAC output frequency = DACFREQ<7:0> * Fclk / 1024, For example with Fclk=1MHz , FF = 255KHz	0x0000_0032

5.3.3.3.1.3 BIA_TX_AMPL

- Address: Base Address + 0x0008, Reset Value = 0x0000_00A0

Name	Bit	Type	Description	Reset Value
DACI	[7:6]	RW	* DAC output voltage range control 00 : 150mV 01 : 200mV 10 : 250mV 11 : 300mV	0x0000_0002
DAC	[5:0]	RW	* DAC output voltage control by code	0x0000_0020

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5.3.3.3.1.4 BIA_TX_SW

- Address: Base Address + 0x0010, Reset Value = 0x0000_0001

Name	Bit	Type	Description	Reset Value
Reserved0	[7:6]	RW	Reserved	0x0000_0000
DACSRN	[5:4]	RW	* Current driver switch connect to RN0 and RN1	0x0000_0000
DACSRP	[3:2]	RW	* Current driver switch connect to RN0 and RN1	0x0000_0000
DACSI	[1:0]	RW	* Current driver switch connect to IOUT0 and IOUT1 when 01 IOUT0 and VSENSE0 when 10	0x0000_0001

5.3.3.3.1.5 BIA_RX_INPUT

- Address: Base Address + 0x0014, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
Reserved0	[7:4]	RW	Reserved	0x0000_0000
VSENCTRP	[3:2]	RW	* Read Channel P input signal control 00 : Body impedence 01 : Body impedence reverse direction 11 : Offset outer register1 10: Offset outer register0	0x0000_0000
VSENCTRN	[1:0]	RW	* Read Channel N input signal control 00 : Body impedence 01 : Body impedence reverse direction 11 : Offset outer register1 10: Offset outer register0	0x0000_0000

5.3.3.3.1.6 BIA_RX_AMP1

- Address: Base Address + 0x0018, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
Reserved0	[7:5]	RW	Reserved	0x0000_0000
IQ_MODE_SEL	[4]	RW	* BIA Operation Mode Selection 1: IQ mode 0: Rectify mode	0x0000_0000
Reserved1	[3]	RW	Reserved	0x0000_0000
AMPCTRL	[2:0]	RW	* BIA input buffer Amp. Gain control 000 : 0 001: 0.5 010: 1.0 011: 1.5	0x0000_0004

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

			100: 2.0 101: 2.5 110: 3 111: 3.5V/V	

5.3.3.3.1.7 BIA_RX_AMP2

- Address: Base Address + 0x001C, Reset Value = 0x0000_0044

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
IAMPCTRL	[6:4]	RW	* BIA I-mode Amp. Gain control 000 : 0 001: 0.5 010: 1.0 011: 1.5 100: 2.0 101: 2.5 110: 3 111: 3.5V/V	0x0000_0004
Reserved1	[3]	RW	Reserved	0x0000_0000
QAMPCTRL	[2:0]	RW	* BIA Q-mode Amp. Gain control 000 : 0 001: 0.5 010: 1.0 011: 1.5 100: 2.0 101: 2.5 110: 3 111: 3.5V/V	0x0000_0004

5.3.4 GSR

5.3.4.1 Overview

It is well accepted that electrical characteristics of the skin has a strong correlation with the sympathetic nervous system of the human body. The skin conductance is an important indication of psychological or physiological arousal. For example, nervous system of a human body responds by releasing a flood of stress hormones, including adrenaline and cortisol if (s)he perceives a threat. The response of the nervous system leads sweat gland activity to increase, which in turn increases skin conductance level.

S3FBP5A supports GSR measurement via two electrodes placing on the skin. These two electrodes can be driven either by DC or by AC voltage source. Depending on resolution requirements, GSR value can be readout either by a 12bit low-power ADC or by a 24bit high-resolution ADC.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

5.3.4.2 Functional Description

- Case-1: Excitation with DC source

An embedded LDO is utilized to generate DC voltage signal applying to one electrode through a resistor of R. Another electrode is then biased by the ground potential through another resistor of R so that there is a constant electrical filed applied between the electrodes. Through the same 2 electrodes, either a 12bit or a 24bit ADC readouts differential (voltage) signal, and that correspond to the value of skin conductance (GSR). Depending on the DC supply voltage (VDD), resistor value (R), and resolved input voltage ($\mathrm{V_{ref}} \cdot \mathrm{ADCcode}/2^{\mathrm{N-1}}$) to ADC, the GSR value ($\mathrm{g_{GSR}}$) can be determined as follows:

$$\mathrm{g_{GSR}} = \frac{\mathrm{VDD}}{2\mathrm{R}} \cdot \frac{1}{\mathrm{V_{ref}} \cdot \mathrm{ADCcode}/2^{\mathrm{N-1}}} - \frac{1}{2\mathrm{R}}$$

where $\mathrm{V_{ref}}$ is ADC reference voltage, N is resolution of ADC, and ADC code is the measured ADC outputs. The 32 different values of VDD can be configured from 1.4V to 2.175V with 25mV/step.

Once the value of VDD along with the value of R is configured, the detectable GSR range depends on the performance of an ADC. The maximum detectable GSR value is limited by the ENOB of ADC, where ENOB stands for effective number of bit of the ADC, and it can be expressed as follows:

$$\mathrm{g_{GSR,max}} \approx \frac{\mathrm{VDD}}{2\mathrm{R}} \cdot \frac{1}{\mathrm{V_{ref}} \cdot 1/2^{\mathrm{ENOB-1}}}, VDD \gg \mathrm{V_{ref}} \cdot 1/2^{\mathrm{ENOB-1}}$$

On the other hand, the minimum detectable GSR value is limited by the maximum allowable ADC outputs (ADCcode,max).

$$\mathrm{g_{GSR,min}} = \frac{\mathrm{VDD}}{2\mathrm{R}} \cdot \frac{1}{\mathrm{V_{ref}} \cdot \mathrm{ADCcode,max}/2^{\mathrm{N-1}}} - \frac{1}{2\mathrm{R}}$$

Depending on the user preference, a high-resolution GSR readout can be achieved with the help of the 24bit ADC while power-efficient GSR readout can be realized with the 12bit ADC as the S3FBP5A supports wide degree of freedom for user to select the value of VDD, R, and N of ADC.

- Case-2: Excitation with AC source

In addition to that, S3FBP5A can drive two electrodes alternatively so that applied filed between the electrodes changes direction periodically. This feature not only prevents from electrodes polarization but effectively neutralizes charge accumulation inside the human body as the current flows change periodically as well. To achieve this, one electrode is driven by a square wave signal ($\mathrm{f}(0^o)$) and another electrode is driven by an out-of-phase square wave signal ($\mathrm{f}(180^o)$). The measured GSR value ($\mathrm{g_{GSR}}$) can be determined with the same principle specified in above equations but $\mathrm{g_{GSR}}$ is modulated at pulse rate. The pulse rate of the square wave can be selected among 2Hz, 4Hz, 8Hz, and 16Hz according to user preference. The sampled GSR value at the output of ADC can be simply demodulated by resampling the digital value accordingly.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

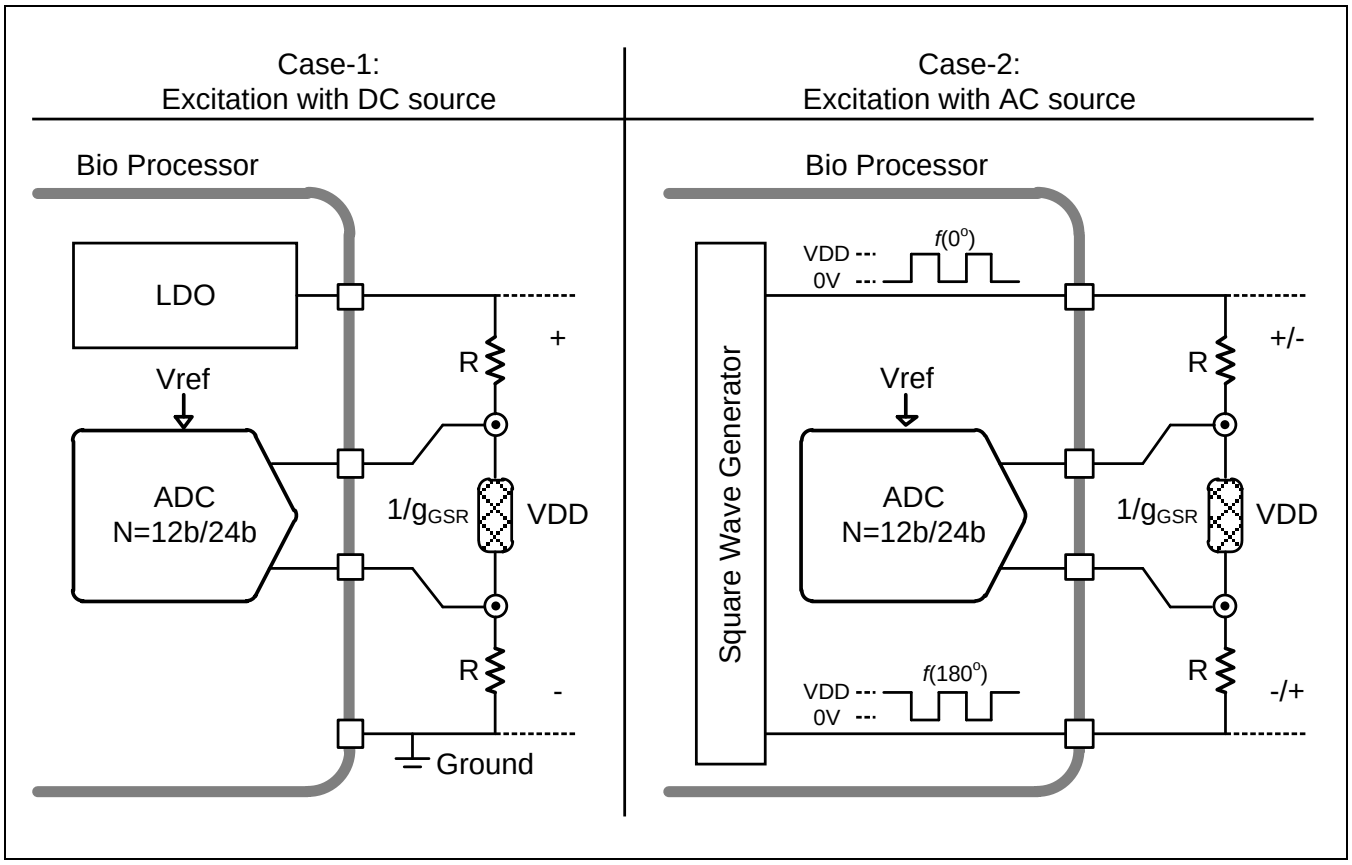


Figure 5-10 The Block diagram of GSR measurement

5.3.4.3 Register Description

5.3.4.3.1 Register Map Summary

- Base Address: 0x4001A400

Register	Offset	Description	Reset Value
PMIP_LDOKA18_V	0x0020	PMIP_LDOKA18_V	0x0000_0000

5.3.4.3.1.1 PMIP_LDOKA18_V

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDOKA18_VCTR	[4:0]	RW	* Keep-alive 1.8V LDO voltage selection 01110:1.4V 01111:1.425V 01100:1.45V 01101:1.475V 01010:1.5V 01011:1.525V	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

Name	Bit	Type	Description	Reset Value
			01000:1.55V 01001:1.575V 00110:1.6V 00111:1.625V 00100:1.65V 00101:1.675V 00010:1.7V 00011:1.725V 00000:1.75V 00001:1.775V 11110:1.8V 11111:1.825V 11100:1.85V 11101:1.875V 11010:1.9V 11011:1.925V 11000:1.95V 11001:1.975V 10110:2V 10111:2.025V 10100:2.05V 10101:2.075V 10010:2.1V 10011:2.125V 10000:2.15V 10001:2.175V	

5.3.5 Skin Temperature

The S3FB5A includes 2-type of GPADC. One is 12-Bit SAR type ADC and the other is 24-Bit format SD-ADC. This GPADC senses the node voltage between a fixed resistor (Rext) and a thermistor (Rtherm) to detect the temperature as shown in *Figure 5-11*. 8kohm@30℃ thermistor and 7.8kohm fixed resistor are recommended for accuracy. However, user increases the resister value to reduce the current consumption.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

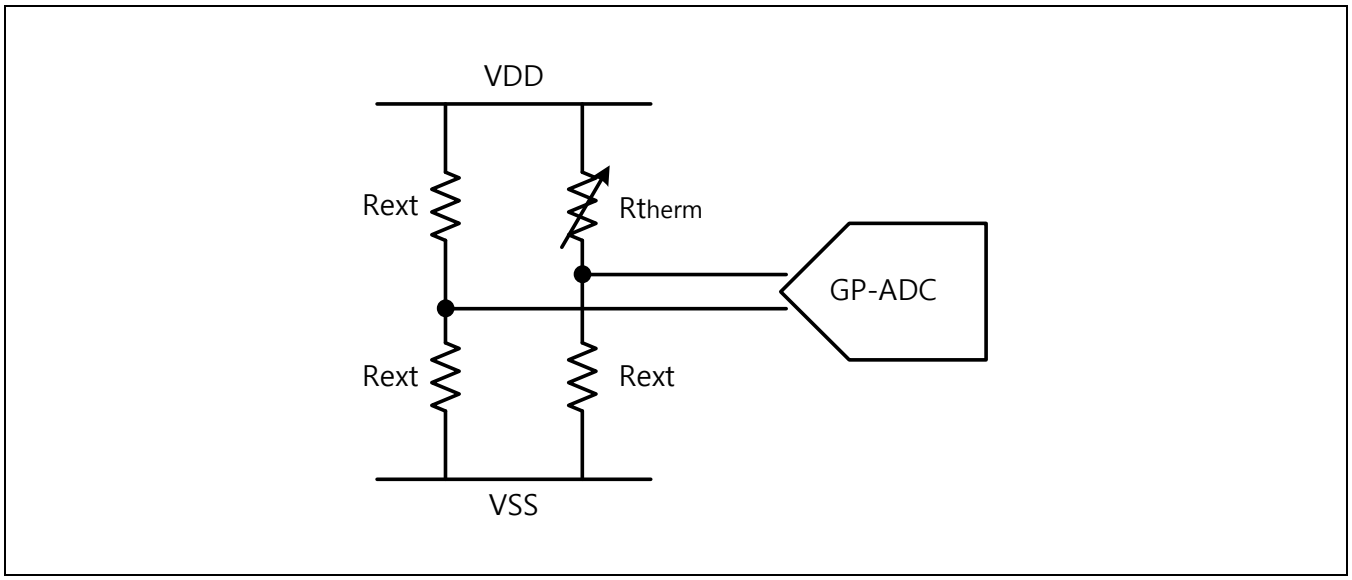


Figure 5-11 The Block diagram of Skin Temp. measurement

5.3.6 SAR ADC

5.3.6.1 Overview

The S3FBP5A contains five successive-approximation register (SAR) analog-to-digital converters (ADCs).The five SAR ADCs convert 10 analog inputs such as 4 PPG, 1 ECG, 1 IMP, 2 BIA, and 2 extra analog inputs to 12-bit digital data using time-multiplexing method. Input range of the ADC is from 0 to 2 times VDD (Power supply of ADC) by differential input, and the ADC operates in using internally generated synchronous clocks. Each SAR ADC can operate at 2KSPS. For the area reduction, the ADC adopts minimum sampling capacitor limited by KT/C noise. Typical 1.1V for power supply and reference voltage is used for the ADC. For minimal power consumption, each SAR ADC enters into the power down mode as soon as 12-bit data conversion is completed.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

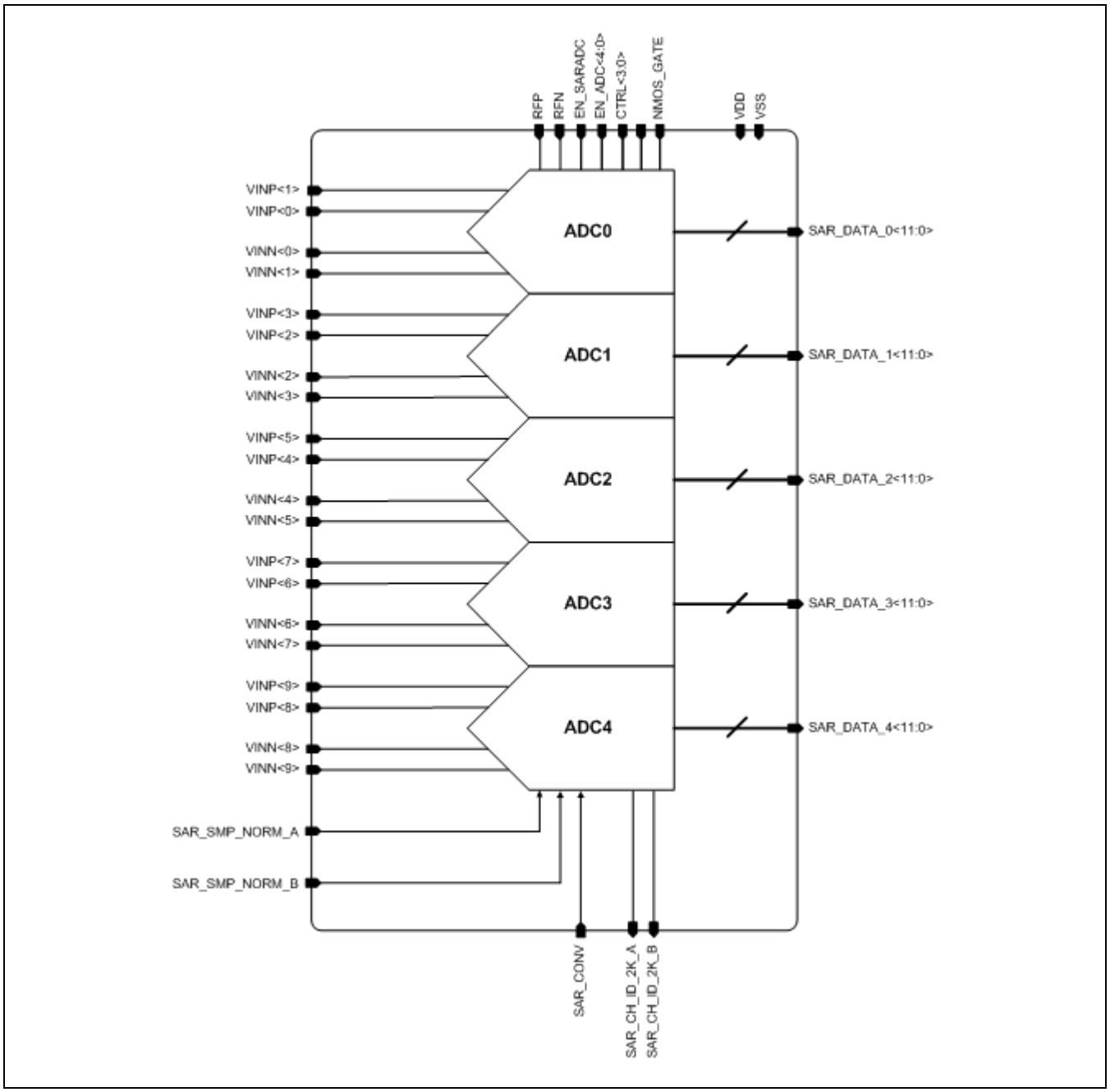


Figure 5-12 ADC Block Diagram

5.3.6.2 Functional Description

5.3.6.2.1 Timing Diagram

Each channel of the five SAR ADCs operates at Fs(Sampling Frequency)/2 independently. Successively sampled inputs in two channels of each ADC are converted into 12-bit digital data. The *Figure 5-13* shows the timing diagram. The ADC0, 2, 3 & 4 use an A-type sampling clock (SAR_SMP_NORM_A). Only ADC1 uses a B-type sampling clock (SAR_SMP_NORM_B) delayed by a half period of A-type sampling clock for PPG operation. The duty cycle of ADC sampling clock is 6.25%.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

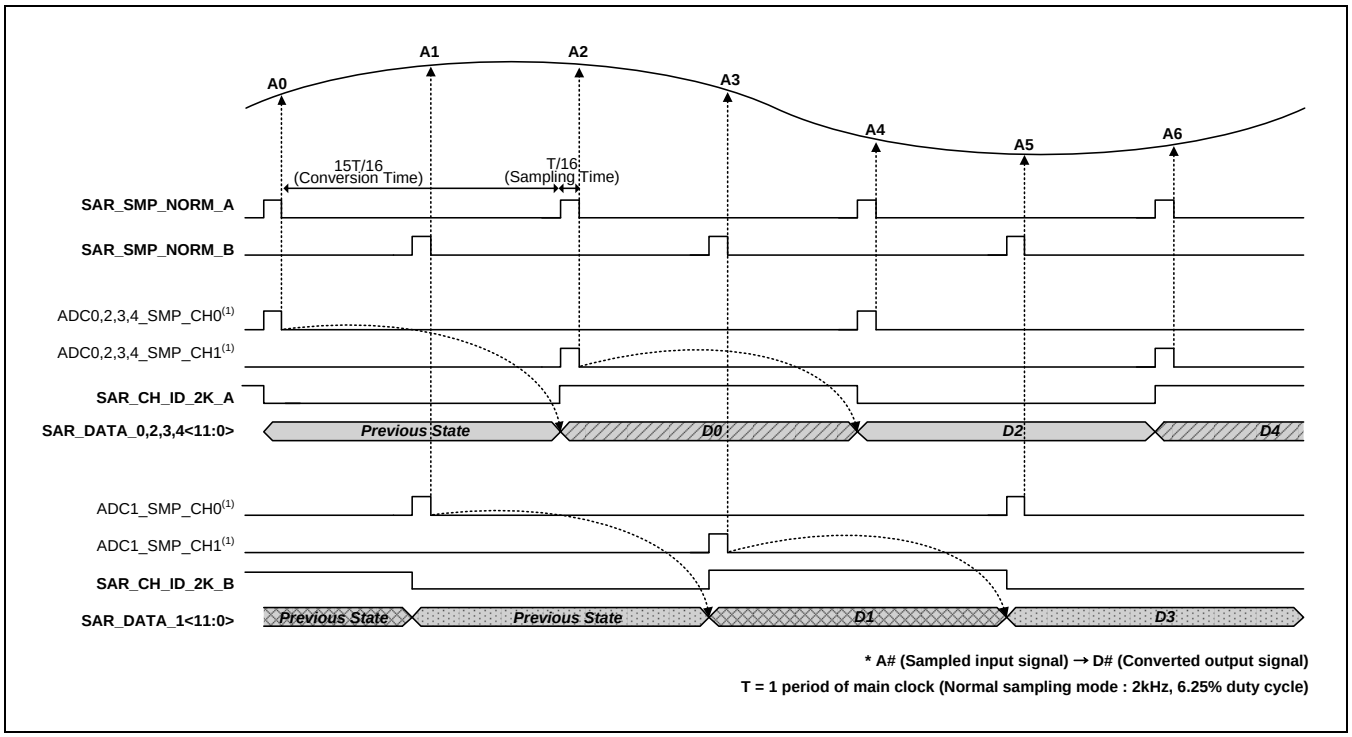


Figure 5-13 ADC Timing Diagram

NOTE: The internal sampling clocks used in each ADC.

5.3.6.2.2 ADC Input assignment

The below *Table 5-1* shows the input assignments of each SAR ADC. Four ADCs(From ADC0 to 3) are assigned for AFE data conversion and ADC4 is assigned for die temperature and general purpose.

Table 5-1 ADC Input Assignment

ADC	Channel		Port	Signal	Description
ADC0	CH0	PPG	VINP<0>	RAMB_T	RED LED ambient Signal
			VINN<0>	RAMB_C	
	CH1	PPG	VINP<1>	IRAMB_T	IRED LED ambient Signal
			VINN<1>	IRAMB_C	
ADC1	CH0	PPG	VINP<2>	RLED_T	RED LED Signal
			VINN<2>	RLED_C	
	CH1	PPG	VINP<3>	IRLED_T	IR LED Signal
			VINN<3>	IRLED_C	
ADC2	CH0	ECG	VINP<4>	ECG_T	ECG Signal
			VINN<4>	ECG_C	
	CH1	BIA	VINP<5>	I_T	BIA I signal
			VINN<5>	I_C	

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ADC	Channel		Port	Signal	Description
ADC3	CH0	IMP	VINP<6>	IMP_T	IMP Signal
			VINN<6>	IMP_C	
	CH1	BIA	VINP<7>	Q_T	BIA Q signal
			VINN<7>	Q_C	
ADC4	CH0	Die Temp	VINP<8>	TEMP_T	Die Temperature
			VINN<8>	TEMP_C	
	CH1	GP	VINP<9>	GP_T	General Purpose
			VINN<9>	GP_C	

5.3.7 High Resolution General Purpose Sigma Delta ADC

5.3.7.1 Overview

S3FBP5A supports the analog to digital conversion with high resolution for general purposes (GP) such as skin temperature sensing, fuel gauging and so forth. The prototype ADC is composed of delta sigma modulator, instrumentation amplifier (IA), input multiplex (MUX), regulator and decimation filter. The IA makes a DC input impedance of the modulator large, and then contributes to transferring relatively well the input signal to the modulator. It also has anti-aliasing filtering and gain controlling functions. The regulator delivers the stable power to the modulator. Its output voltage is used as a reference voltage for A/D conversion as well.

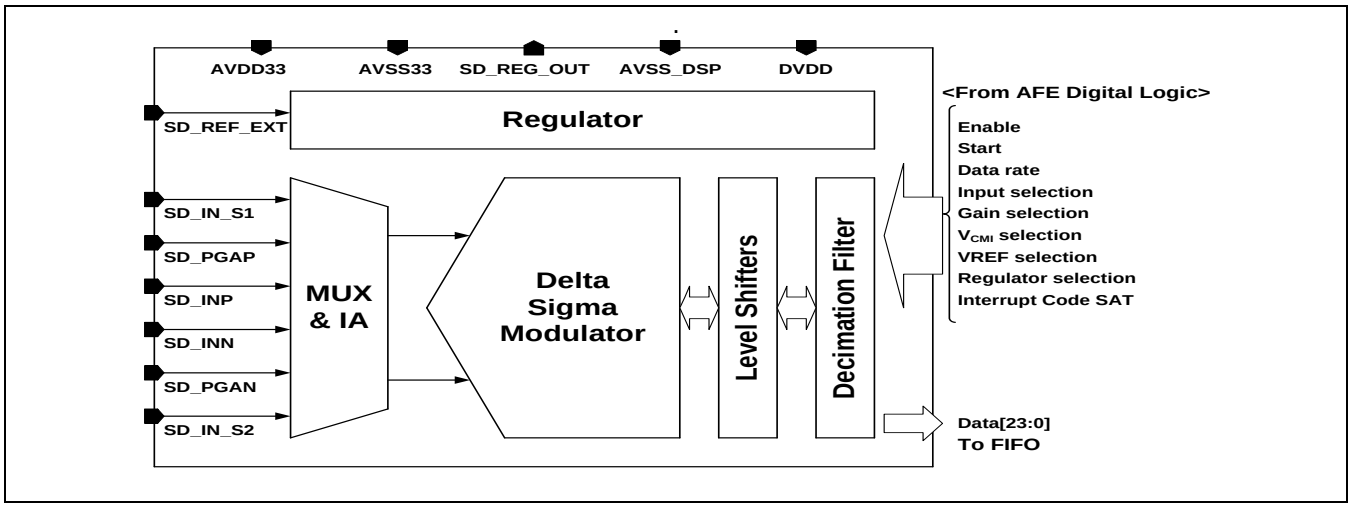


Figure 5-14 Block Diagram of General Purpose ADC

5.3.7.2 Frequency Response

Decimation filter consist of 1 SINC filter and 2 stage of FIR filter. The filters remove noise and down data rate to Fs.

Figure 5-15 illustrates Frequency Response of decimation filter.

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

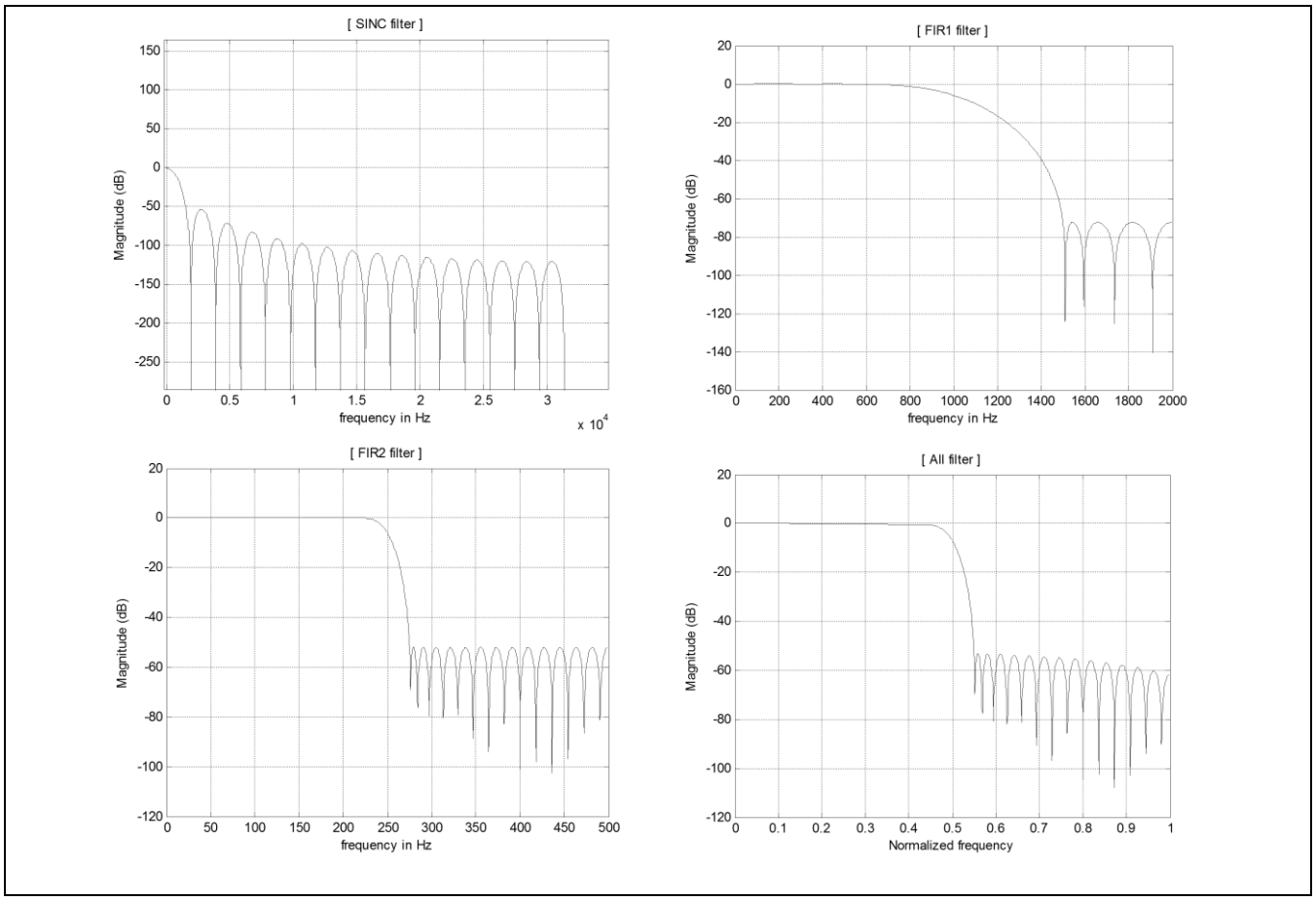


Figure 5-15 Frequency Response

Figure 5-16 illustrates Passband and Stopband frequency. *Figure 5-17* illustrates Passband ripple.

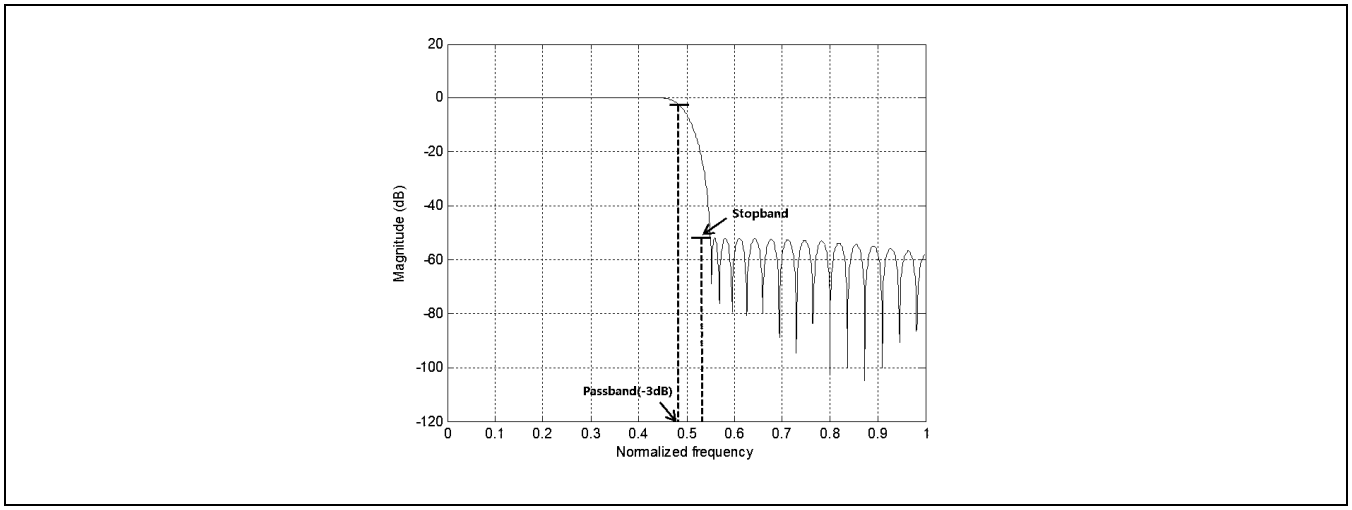


Figure 5-16 Passband and Stopband frequency

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

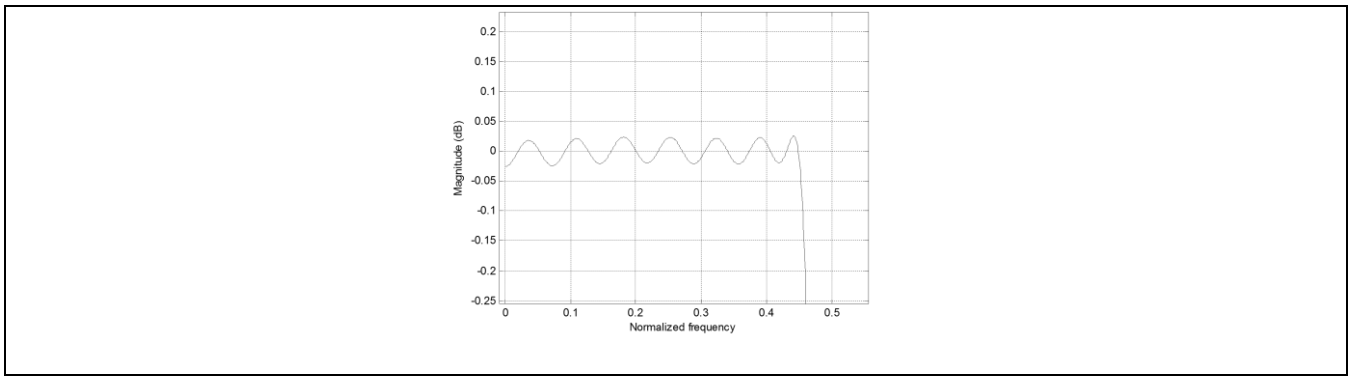


Figure 5-17 Passband ripple

5.3.8 FIFO Control

5.3.8.1 Overview

As shown in the *Figure 5-2*, S3FBP5A has 6 FIFOs, 5 FIFOs for 12bits SAR-ADCs and 1 FIFOs for 24bits Sigma-Delta ADC. Each FIFO has 8 depths. All FIFOs operate in PCLK domain. An input data for each FIFO can be selected among 10 ADC outputs by CH0_DATA_SEL[3:0]~CH5_DATA_SEL[3:0]. Each FIFO has 3 interrupt sources which are FULL, EMPTY and READY. Only one combined interrupt signal is generated from FIFOs and it can be cleared by writing "1" to INTR_CLR register. Interrupt masking is available using FIFO_INTR_MASK registers. Refer to the register table of RAFE Data control logic. Ready interrupt is generated when the number of data saved in FIFO is larger than the number of data set by FIFO_READY_SIZE_SET register. In addition, software reset of FIFOs is also supported by register control. DMA controller of FIFO

DMA controller in FIFO generates DMA request signals according to the register setting. The number of data which generates DMA request signal can be set by FIFO_READ_DATA_SIZE_SET register. After the controller gets acknowledge signal from uDMA, the request signals are cleared automatically.

5.3.8.2 Register Description

5.3.8.2.1 Register Map Summary

- Base Address: 0x4001A500

Register	Offset	Description	Reset Value
FIFO_STORED_DATA_NUMBER	0x0070	FIFO_STORED_DATA_NUMBER	0x0000_0000
FIFO_STATUS_READ	0x0074	FIFO_STATUS_READ	0x0000_0000
FIFO_INTR_MASK	0x0078	FIFO_INTR_MASK	0x0022_2222
FIFO_READ_DATA_SIZE_SET	0x007C	FIFO_READ_DATA_SIZE_SET	0x0088_8888
FIFO_READY_SIZE_SET	0x0080	FIFO_READY_SIZE_SET	0x0011_1111
DMA_REQUEST_SET	0x0084	DMA_REQUEST_SET	0x0000_0000
FIFO0_READ_DATA	0x0088	FIFO0_READ_DATA	0x0000_0000
FIFO1_READ_DATA	0x008C	FIFO1_READ_DATA	0x0000_0000
FIFO2_READ_DATA	0x0090	FIFO2_READ_DATA	0x0000_0000

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Register	Offset	Description	Reset Value
FIFO3_READ_DATA	0x0094	FIFO3_READ_DATA	0x0000_0000
FIFO4_READ_DATA	0x0098	FIFO4_READ_DATA	0x0000_0000
FIFO5_READ_DATA	0x009C	FIFO5_READ_DATA	0x0000_0000
ADC0_CH0_READ_DATA	0x00A0	ADC0_CH0_READ_DATA	0x0000_0000
ADC0_CH1_READ_DATA	0x00A4	ADC0_CH1_READ_DATA	0x0000_0000
ADC1_CH0_READ_DATA	0x00A8	ADC1_CH0_READ_DATA	0x0000_0000
ADC1_CH1_READ_DATA	0x00AC	ADC1_CH1_READ_DATA	0x0000_0000
ADC2_CH0_READ_DATA	0x00B0	ADC2_CH0_READ_DATA	0x0000_0000
ADC2_CH1_READ_DATA	0x00B4	ADC2_CH1_READ_DATA	0x0000_0000
ADC3_CH0_READ_DATA	0x00B8	ADC3_CH0_READ_DATA	0x0000_0000
ADC3_CH1_READ_DATA	0x00BC	ADC3_CH1_READ_DATA	0x0000_0000
ADC4_CH0_READ_DATA	0x00C0	ADC4_CH0_READ_DATA	0x0000_0000
ADC4_CH1_READ_DATA	0x00C4	ADC4_CH1_READ_DATA	0x0000_0000
SDADC_READ_DATA	0x00C8	SDADC_READ_DATA	0x0000_0000

5.3.8.2.1.1 FIFO_STORED_DATA_NUMBER

- Address: Base Address + 0x0070, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
DATA_NUM5	[23:20]	R	* The number of data stored in FIFO5	0x0000_0000
DATA_NUM4	[19:16]	R	* The number of data stored in FIFO4	0x0000_0000
DATA_NUM3	[15:12]	R	* The number of data stored in FIFO3	0x0000_0000
DATA_NUM2	[11:8]	R	* The number of data stored in FIFO2	0x0000_0000
DATA_NUM1	[7:4]	R	* The number of data stored in FIFO1	0x0000_0000
DATA_NUM0	[3:0]	R	* The number of data stored in FIFO0	0x0000_0000

5.3.8.2.1.2 FIFO_STATUS_READ

- Address: Base Address + 0x0074, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO_FULL5	[22]	R	* FIFO5 Full Interrupt	0x0000_0000
FIFO_EMPTY5	[21]	R	* FIFO5 Empty Interrupt	0x0000_0000
FIFO_READY5	[20]	R	* FIFO5 Ready Interrupt	0x0000_0000
FIFO_FULL4	[18]	R	* FIFO4 Full Interrupt	0x0000_0000
FIFO_EMPTY4	[17]	R	* FIFO4 Empty Interrupt	0x0000_0000
FIFO_READY4	[16]	R	* FIFO4 Ready Interrupt	0x0000_0000

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Name	Bit	Type	Description	Reset Value
FIFO_FULL3	[14]	R	* FIFO3 Full Interrupt	0x0000_0000
FIFO_EMPTY3	[13]	R	* FIFO3 Empty Interrupt	0x0000_0000
FIFO_READY3	[12]	R	* FIFO3 Ready Interrupt	0x0000_0000
FIFO_FULL2	[10]	R	* FIFO2 Full Interrupt	0x0000_0000
FIFO_EMPTY2	[9]	R	* FIFO2 Empty Interrupt	0x0000_0000
FIFO_READY2	[8]	R	* FIFO2 Ready Interrupt	0x0000_0000
FIFO_FULL1	[6]	R	* FIFO1 Full Interrupt	0x0000_0000
FIFO_EMPTY1	[5]	R	* FIFO1 Empty Interrupt	0x0000_0000
FIFO_READY1	[4]	R	* FIFO1 Ready Interrupt	0x0000_0000
FIFO_FULL0	[2]	R	* FIFO0 Full Interrupt	0x0000_0000
FIFO_EMPTY0	[1]	R	* FIFO0 Empty Interrupt	0x0000_0000
FIFO_READY0	[0]	R	* FIFO0 Ready Interrupt	0x0000_0000

5.3.8.2.1.3 FIFO_INTR_MASK

- Address: Base Address + 0x0078, Reset Value = 0x0022_2222

Name	Bit	Type	Description	Reset Value
MASK_FULL5	[22]	RW	* Mask FIFO5 full interrupt 1: Mask FIFO5 full interrupt 0: Unmask FIFO5 full interrupt	0x0000_0000
MASK_EMPTY5	[21]	RW	* Mask FIFO5 empty interrupt 1: Mask FIFO5 empty interrupt 0: Unmask FIFO5 empty interrupt	0x0000_0001
MASK_READY5	[20]	RW	* Mask FIFO5 ready interrupt 1: Mask FIFO5 ready interrupt 0: Unmask FIFO5 ready interrupt	0x0000_0000
Intr_Clr	[19]	WO	* FIFO Combined interrupt clear	0x0000_0000
MASK_FULL4	[18]	RW	* Mask FIFO4 full interrupt 1: Mask FIFO4 full interrupt 0: Unmask FIFO4 full interrupt	0x0000_0000
MASK_EMPTY4	[17]	RW	* Mask FIFO4 empty interrupt 1: Mask FIFO4 empty interrupt 0: Unmask FIFO4 empty interrupt	0x0000_0001
MASK_READY4	[16]	RW	* Mask FIFO4 ready interrupt 1: Mask FIFO4 ready interrupt 0: Unmask FIFO4 ready interrupt	0x0000_0000
MASK_FULL3	[14]	RW	* Mask FIFO3 full interrupt 1: Mask FIFO3 full interrupt	0x0000_0000

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Name	Bit	Type	Description	Reset Value
			0: Unmask FIFO3 full interrupt	
MASK_EMPTY3	[13]	RW	* Mask FIFO3 empty interrupt 1: Mask FIFO3 empty interrupt 0: Unmask FIFO3 empty interrupt	0x0000_0001
MASK_READY3	[12]	RW	* Mask FIFO3 ready interrupt 1: Mask FIFO3 ready interrupt 0: Unmask FIFO3 ready interrupt	0x0000_0000
MASK_FULL2	[10]	RW	* Mask FIFO2 full interrupt 1: Mask FIFO2 full interrupt 0: Unmask FIFO2 full interrupt	0x0000_0000
MASK_EMPTY2	[9]	RW	* Mask FIFO2 empty interrupt 1: Mask FIFO2 empty interrupt 0: Unmask FIFO2 empty interrupt	0x0000_0001
MASK_READY2	[8]	RW	* Mask FIFO2 ready interrupt 1: Mask FIFO2 ready interrupt 0: Unmask FIFO2 ready interrupt	0x0000_0000
MASK_FULL1	[6]	RW	* Mask FIFO1 full interrupt 1: Mask FIFO1 full interrupt 0: Unmask FIFO1 full interrupt	0x0000_0000
MASK_EMPTY1	[5]	RW	* Mask FIFO1 empty interrupt 1: Mask FIFO1 empty interrupt 0: Unmask FIFO1 empty interrupt	0x0000_0001
MASK_READY1	[4]	RW	* Mask FIFO1 ready interrupt 1: Mask FIFO1 ready interrupt 0: Unmask FIFO1 ready interrupt	0x0000_0000
MASK_FULL0	[2]	RW	* Mask FIFO0 full interrupt 1: Mask FIFO0 full interrupt 0: Unmask FIFO0 full interrupt	0x0000_0000
MASK_EMPTY0	[1]	RW	* Mask FIFO0 empty interrupt 1: Mask FIFO0 empty interrupt 0: Unmask FIFO0 empty interrupt	0x0000_0001
MASK_READY0	[0]	RW	* Mask FIFO0 ready interrupt 1: Mask FIFO0 ready interrupt 0: Unmask FIFO0 ready interrupt	0x0000_0000

5.3.8.2.1.4 FIFO_READ_DATA_SIZE_SET

- Address: Base Address + 0x007C, Reset Value = 0x0088_8888

Name	Bit	Type	Description	Reset Value
FIFO5_RESETB	[23]	RW	* FIFO5_RESETB 1: FIFO5 reset release	0x0000_0001

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Name	Bit	Type	Description	Reset Value
			0: FIFO5 reset	
TRIG_LEVEL5	[22:20]	RW	* Select the number of data stored in FIFO5 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL5+2).	0x0000_0000
FIFO4_RESETB	[19]	RW	* FIFO4_RESETB 1: FIFO4 reset release 0: FIFO4 reset	0x0000_0001
TRIG_LEVEL4	[18:16]	RW	* Select the number of data stored in FIFO4 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL4+2).	0x0000_0000
FIFO3_RESETB	[15]	RW	* FIFO3_RESETB 1: FIFO3 reset release 0: FIFO3 reset	0x0000_0001
TRIG_LEVEL3	[14:12]	RW	* Select the number of data stored in FIFO3 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL3+2).	0x0000_0000
FIFO2_RESETB	[11]	RW	* FIFO2_RESETB 1: FIFO2 reset release 0: FIFO2 reset	0x0000_0001
TRIG_LEVEL2	[10:8]	RW	* Select the number of data stored in FIFO2 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL2+2).	0x0000_0000
FIFO1_RESETB	[7]	RW	* FIFO1_RESETB 1: FIFO1 reset release 0: FIFO1 reset	0x0000_0001
TRIG_LEVEL1	[6:4]	RW	* Select the number of data stored in FIFO1 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL1+2).	0x0000_0000
FIFO0_RESETB	[3]	RW	* FIFO0_RESETB 1: FIFO0 reset release 0: FIFO0 reset	0x0000_0001
TRIG_LEVEL0	[2:0]	RW	* Select the number of data stored in FIFO0 to generate a DMA request. - The number of data which generates a DMA request is (TRIG_LEVEL0+2).	0x0000_0000

5.3.8.2.1.5 FIFO_READY_SIZE_SET

- Address: Base Address + 0x0080, Reset Value = 0x0011_1111

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Name	Bit	Type	Description	Reset Value
FIFO5_ENA	[23]	RW	* FIFO5 Enable/Disable control 1: FIFO5 Enabled 0: FIFO5 Disabled	0x0000_0000
FIFO_READ_NUM_SET5	[22:20]	RW	* Select the number of data stored in FIFO5 to generate the FIFO_READY5 interrupt. - The number of data which generates FIFO_READY5 interrupt is (FIFO_READ_NUM_SET5+1).	0x0000_0001
FIFO4_ENA	[19]	RW	* FIFO4 Enable/Disable control 1: FIFO4 Enabled 0: FIFO4 Disabled	0x0000_0000
FIFO_READ_NUM_SET4	[18:16]	RW	* Select the number of data stored in FIFO4 to generate the FIFO_READY4 interrupt. - The number of data which generates FIFO_READY4 interrupt is (FIFO_READ_NUM_SET4+1).	0x0000_0001
FIFO3_ENA	[15]	RW	* FIFO3 Enable/Disable control 1: FIFO3 Enabled 0: FIFO3 Disabled	0x0000_0000
FIFO_READ_NUM_SET3	[14:12]	RW	* Select the number of data stored in FIFO3 to generate the FIFO_READY3 interrupt. - The number of data which generates FIFO_READY3 interrupt is (FIFO_READ_NUM_SET3+1).	0x0000_0001
FIFO2_ENA	[11]	RW	* FIFO2 Enable/Disable control 1: FIFO2 Enabled 0: FIFO2 Disabled	0x0000_0000
FIFO_READ_NUM_SET2	[10:8]	RW	* Select the number of data stored in FIFO2 to generate the FIFO_READY2 interrupt. - The number of data which generates FIFO_READY2 interrupt is (FIFO_READ_NUM_SET2+1).	0x0000_0001
FIFO1_ENA	[7]	RW	* FIFO1 Enable/Disable control 1: FIFO1 Enabled 0: FIFO1 Disabled	0x0000_0000
FIFO_READ_NUM_SET1	[6:4]	RW	* Select the number of data stored in FIFO1 to generate the FIFO_READY1 interrupt. - The number of data which generates FIFO_READY1 interrupt is (FIFO_READ_NUM_SET1+1).	0x0000_0001
FIFO0_ENA	[3]	RW	* FIFO0 Enable/Disable control 1: FIFO0 Enabled 0: FIFO0 Disabled	0x0000_0000
FIFO_READ_NUM_SET0	[2:0]	RW	* Select the number of data stored in FIFO0 to generate the FIFO_READY0 interrupt. - The number of data which generates FIFO_READY0 interrupt is (FIFO_READ_NUM_SET0+1).	0x0000_0001

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5.3.8.2.1.6 DMA_REQUEST_SET

- Address: Base Address + 0x0084, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
DMA_EN5	[20]	RW	* DMA5 Request Enable 1: DMA5 Enable 0: DMA5 Disable	0x0000_0000
DMA_EN4	[16]	RW	* DMA4 Request Enable 1: DMA4 Enable 0: DMA4 Disable	0x0000_0000
DMA_EN3	[12]	RW	* DMA3 Request Enable 1: DMA3 Enable 0: DMA3 Disable	0x0000_0000
DMA_EN2	[8]	RW	* DMA2 Request Enable 1: DMA2 Enable 0: DMA2 Disable	0x0000_0000
DMA_EN1	[4]	RW	* DMA1 Request Enable 1: DMA1 Enable 0: DMA1 Disable	0x0000_0000
DMA_EN0	[0]	RW	* DMA0 Request Enable 1: DMA0 Enable 0: DMA0 Disable	0x0000_0000

5.3.8.2.1.7 FIFO0_READ_DATA

- Address: Base Address + 0x0088, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO0_READ0	[15:0]	R	* FIFO0 Read data	0x0000_0000

5.3.8.2.1.8 FIFO1_READ_DATA

- Address: Base Address + 0x008C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO1_READ1	[15:0]	R	* FIFO1 Read data	0x0000_0000

5.3.8.2.1.9 FIFO2_READ_DATA

- Address: Base Address + 0x0090, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO2_READ2	[15:0]	R	* FIFO2 Read data	0x0000_0000

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5.3.8.2.1.10 FIFO3_READ_DATA

- Address: Base Address + 0x0094, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO3_READ3	[15:0]	R	* FIFO3 Read data	0x0000_0000

5.3.8.2.1.11 FIFO4_READ_DATA

- Address: Base Address + 0x0098, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO4_READ4	[15:0]	R	* FIFO4 Read data	0x0000_0000

5.3.8.2.1.12 FIFO5_READ_DATA

- Address: Base Address + 0x009C, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
FIFO5_READ5	[23:0]	R	* FIFO5 Read data (Sigma-delta ADC output)	0x0000_0000

5.3.8.2.1.13 ADC0_CH0_READ_DATA

- Address: Base Address + 0x00A0, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC0_CH0_RDATA	[11:0]	R	* ADC0 Channel0 Read Data	0x0000_0000

5.3.8.2.1.14 ADC0_CH1_READ_DATA

- Address: Base Address + 0x00A4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC0_CH1_RDATA	[11:0]	R	* ADC0 Channel1 Read Data	0x0000_0000

5.3.8.2.1.15 ADC1_CH0_READ_DATA

- Address: Base Address + 0x00A8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC1_CH0_RDATA	[11:0]	R	* ADC1 Channel0 Read Data	0x0000_0000

5.3.8.2.1.16 ADC1_CH0_READ_DATA

- Address: Base Address + 0x00AC, Reset Value = 0x0000_0000

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Name	Bit	Type	Description	Reset Value
ADC1_CH1_RDATA	[11:0]	R	* ADC1 Channel1 Read Data	0x0000_0000

5.3.8.2.1.17 ADC2_CH0_READ_DATA

- Address: Base Address + 0x00B0, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC2_CH0_RDATA	[11:0]	R	* ADC2 Channel0 Read Data	0x0000_0000

5.3.8.2.1.18 ADC2_CH1_READ_DATA

- Address: Base Address + 0x00B4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC2_CH1_RDATA	[11:0]	R	* ADC2 Channel1 Read Data	0x0000_0000

5.3.8.2.1.19 ADC3_CH0_READ_DATA

- Address: Base Address + 0x00B8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC3_CH0_RDATA	[11:0]	R	* ADC3 Channel0 Read Data	0x0000_0000

5.3.8.2.1.20 ADC3_CH1_READ_DATA

- Address: Base Address + 0x00BC, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC3_CH1_RDATA	[11:0]	R	* ADC3 Channel1 Read Data	0x0000_0000

5.3.8.2.1.21 ADC4_CH0_READ_DATA

- Address: Base Address + 0x00C0, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC4_CH0_RDATA	[11:0]	R	* ADC4 Channel0 Read Data	0x0000_0000

5.3.8.2.1.22 ADC4_CH1_READ_DATA

- Address: Base Address + 0x00C4, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
ADC4_CH1_RDATA	[11:0]	R	* ADC4 Channel1 Read Data	0x0000_0000

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5.3.8.2.1.23 SDADC_READ_DATA

- Address: Base Address + 0x00C8, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
SDADC_RDATA	[23:0]	R	* Sigma Delta ADC Read Data	0x0000_0000

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5.3.9 PMIP

5.3.9.1 Overview

S3FBP5A also has PMIPs such as buck type DC-DC converter (buck converter) and LDOs to manage required power operations. The PMIPs in S3FBP5A are composed of a 3V to 1.3V buck converter for high efficient voltage conversion and five LDOs to provide various analog and digital powers. The buck converter operating in PFM (Pulse Frequency Modulation) mode is applied to minimize power efficiency loss due to input and output voltage difference. The input power source of the buck converter can be any kind of 3V range power such as coin battery or PMIC. The buck converter generates1.3V from 3V power source and provide the powers to three different LDOs such as 1.1V LDO for AFEs(LDO3), 1.1V LDO for digital circuits(LDO1) and 1.1V LDO for Flash memory(LDO2). To provide low ripple output voltage and minimize high frequency operation, a 4.7uH inductor and 10uF capacitor are applied. In the case of LDOs, 1uF capacitors are used to minimize high frequency noise and fluctuation of voltage. S3FBP5A has two keep alive LDOs:1.1V LDO for keep-alive logic and I/O in digital domain and 1.8V LDO for keep-alive 1.8V region I/O. Both LDOs receive powers from external 3V power source same as the buck converter.

To provide reference voltage for the buck converter and LDOs, a band gap reference (BGR) circuit is applied. The BGR circuit generates required reference voltage and currents. The BGR circuits and Keep-alive LDOs always turn on first whenever an external power source is attached. For example, the BGR circuit turns on first and generates reference voltages and currents. After then, keep-alive LDOs wake up and start providing powers to keep-alive logic and I/Os. DC-DC converter after receiving power OK signal from the Keep-alive LDO starts operation and providing powers to LDOs generating 1.1V powers to AFEs and digital circuits. *Figure 5-18* shows the block diagram of the PMIPs.

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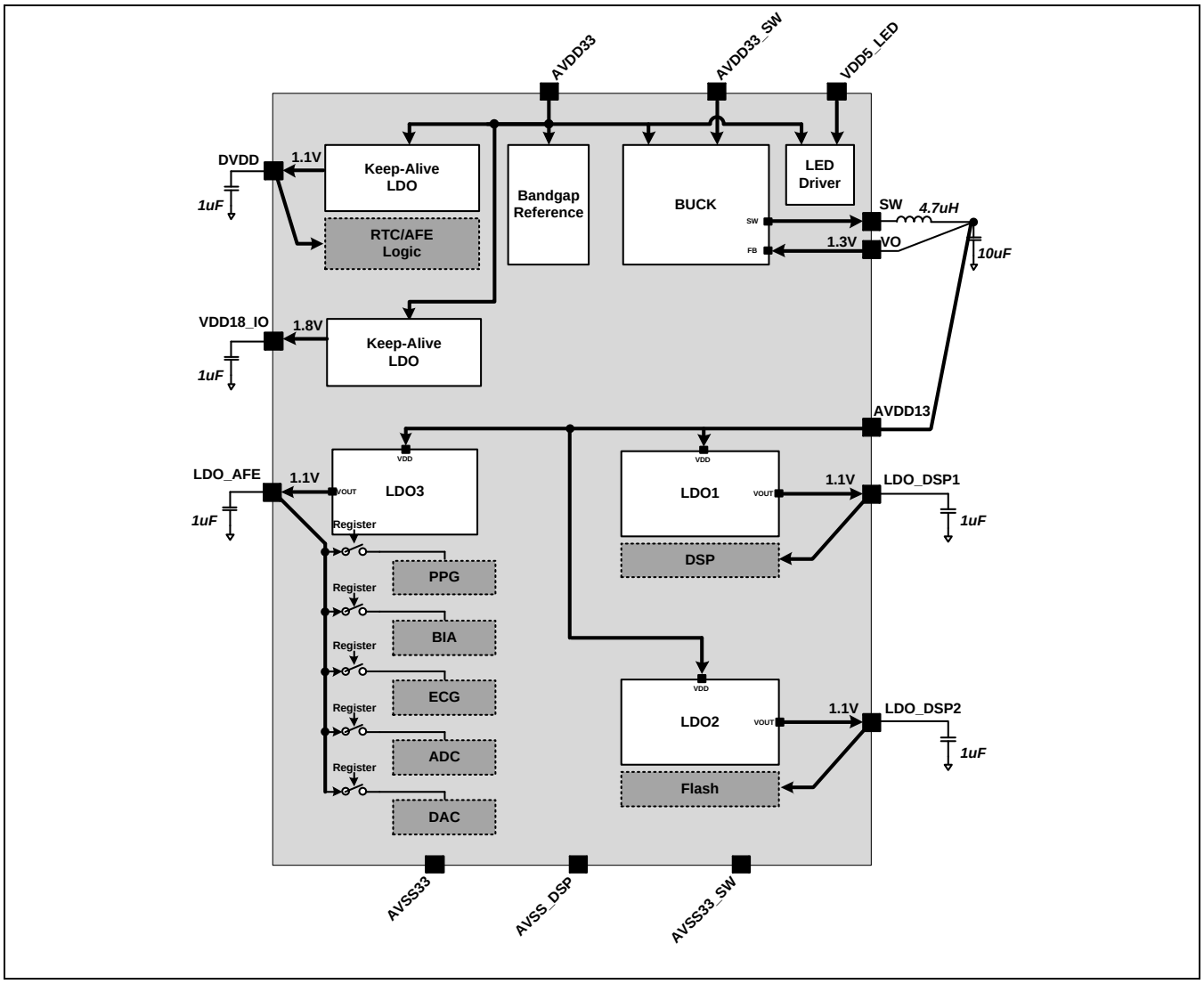


Figure 5-18 Block Diagram of PMIPs

5.3.9.2 Register Description

5.3.9.2.1 Register Map Summary

- Base Address: 0x4001A400

Register	Offset	Description	Reset Value
PMIP_LDO_OFF	0x0010	PMIP_LDO_OFF	0x0000_0040
PMIP_LDO1_V	0x0014	PMIP_LDO1_V	0x0000_0018
PMIP_LDO2_V	0x0018	PMIP_LDO2_V	0x0000_0018
PMIP_LDO3_V	0x001C	PMIP_LDO3_V	0x0000_0018
PMIP_LDOKA18_V	0x0020	PMIP_LDOKA18_V	0x0000_0000
PMIP_PWR_SW	0x002C	PMIP_PWR_SW	0x0000_003F

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Register	Offset	Description	Reset Value
PMIP_UVLO	0x0030	PMIP_UVLO	0x0000_0067

5.3.9.2.1.1 PMIP_LDO_OFF

- Address: Base Address + 0x0010, Reset Value = 0x0000_0040

Name	Bit	Type	Description	Reset Value
Reserved0	[7:4]	RW	Reserved	0x0000_0004
LDO1_PULL_DN	[3]	RW	* LDO pull down 1: Enable 0: Disable	0x0000_0000
LDO2_PULL_DN	[2]	RW	* LDO pull down 1: Enable 0: Disable	0x0000_0000
LDO3_PULL_DN	[1]	RW	* LDO pull down 1: Enable 0: Disable	0x0000_0000
Reserved1	[0]	RW	Reserved	0x0000_0000

5.3.9.2.1.2 PMIP_LDO1_V

- Address: Base Address + 0x0014, Reset Value = 0x0000_0018

Name	Bit	Type	Description	Reset Value
Reserved0	[7:5]	RW	Reserved	0x0000_0000
LDO1_VCTRL	[4:0]	RW	* LDO voltage selection 11111: 1.275V ,11110: 1.25V 11101: 1.225V ,11100: 1.2V 11011: 1.175V ,11010: 1.15V 11001: 1.125V ,11000: 1.1V (default) 10111: 1.075V ,10110: 1.05V 10101: 1.025V ,10100: 1.0V 10011: 0.975V ,10010: 0.95V 10001: 0.925V ,10000: 0.9V 01111: 0.875V ,01110: 0.85V 01101: 0.825V ,01100: 0.8V 01011: 0.775V ,01010: 0.75V 01001: 0.725V ,01000: 0.7V 00111: 0.675V ,00110: 0.65V 00101: 0.625V ,00100: 0.6V 00011: 0.575V ,00010: 0.55V 00001: 0.525V ,00000: 0.5V	0x0000_0018

5.3.9.2.1.3 PMIP_LDO2_V

- Address: Base Address + 0x0018, Reset Value = 0x0000_0018

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
Reserved0	[7:5]	RW	Reserved	0x0000_0000
LDO2_VCTRL	[4:0]	RW	* LDO voltage selection 11111: 1.275V ,11110: 1.25V 11101: 1.225V ,11100: 1.2V 11011: 1.175V ,11010: 1.15V 11001: 1.125V ,11000: 1.1V (default) 10111: 1.075V ,10110: 1.05V 10101: 1.025V ,10100: 1.0V 10011: 0.975V ,10010: 0.95V 10001: 0.925V ,10000: 0.9V 01111: 0.875V ,01110: 0.85V 01101: 0.825V ,01100: 0.8V 01011: 0.775V ,01010: 0.75V 01001: 0.725V ,01000: 0.7V 00111: 0.675V ,00110: 0.65V 00101: 0.625V ,00100: 0.6V 00011: 0.575V ,00010: 0.55V 00001: 0.525V ,00000: 0.5V	0x0000_0018

5.3.9.2.1.4 PMIP_LDO3_V

- Address: Base Address + 0x001C, Reset Value = 0x0000_0018

Name	Bit	Type	Description	Reset Value
Reserved0	[7:5]	RW	Reserved	0x0000_0000
LDO3_VCTRL	[4:0]	RW	* LDO voltage selection 11111: 1.275V ,11110: 1.25V 11101: 1.225V ,11100: 1.2V 11011: 1.175V ,11010: 1.15V 11001: 1.125V ,11000: 1.1V (default) 10111: 1.075V ,10110: 1.05V 10101: 1.025V ,10100: 1.0V 10011: 0.975V ,10010: 0.95V 10001: 0.925V ,10000: 0.9V 01111: 0.875V ,01110: 0.85V 01101: 0.825V ,01100: 0.8V 01011: 0.775V ,01010: 0.75V 01001: 0.725V ,01000: 0.7V 00111: 0.675V ,00110: 0.65V 00101: 0.625V ,00100: 0.6V 00011: 0.575V ,00010: 0.55V 00001: 0.525V ,00000: 0.5V	0x0000_0018

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5.3.9.2.1.5 PMIP_LDOKA18_V

- Address: Base Address + 0x0020, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
LDOKA18_VCTR	[4:0]	RW	* Keep-alive 1.8V LDO voltage selection 01110:1.4V 01111:1.425V 01100:1.45V 01101:1.475V 01010:1.5V 01011:1.525V 01000:1.55V 01001:1.575V 00110:1.6V 00111:1.625V 00100:1.65V 00101:1.675V 00010:1.7V 00011:1.725V 00000:1.75V(Default) 00001:1.775V 11110:1.8V 11111:1.825V 11100:1.85V 11101:1.875V 11010:1.9V 11011:1.925V 11000:1.95V 11001:1.975V 10110:2V 10111:2.025V 10100:2.05V 10101:2.075V 10010:2.1V 10011:2.125V 10000:2.15V 10001:2.175V	0x0000_0000

5.3.9.2.1.6 PMIP_PWR_SW

- Address: Base Address + 0x002C, Reset Value = 0x0000_003F

Name	Bit	Type	Description	Reset Value
PWR_EN_BIA	[5	RW	* Set to Internal LDO Mode 0: BIA Power disable	0x0000_0001

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Name	Bit	Type	Description	Reset Value
			1: BIA Power enable	
PWR_EN_PP	[4]	RW	* Set to Internal LDO Mode 0: PPG RX Power disable 1: PPG RX Power enable	0x0000_0001
PWR_EN_READ	[3]	RW	* Set to Internal LDO Mode 0: ECG Power disable 1: ECG Power enable	0x0000_0001
PWR_EN_ADC	[2]	RW	* Set to Internal LDO Mode 0: ADC Power disable 1: ADC Power enable	0x0000_0001
PWR_EN_SUB	[1]	RW	* Set to Internal LDO Mode 0: DAC Power disable 1: DAC Power enable	0x0000_0001

5.3.9.2.1.7 PMIP_UVLO

- Address: Base Address + 0x0030, Reset Value = 0x0000_0067

Name	Bit	Type	Description	Reset Value
Reserved0	[7]	RW	Reserved	0x0000_0000
UVLO_EN	[6]	RW	* UVLO On/Off 1: Enable 0: Disable	0x0000_0001
Reserved1	[3]	RW	Reserved	0x0000_0000

5.3.10 Clock Management

5.3.10.1 Overview

There are 2-clock sources. One is 1MHz clock for AFE Logic and analog operation, the other is 87MHz for DSP and MCU. Both clocks can be set by register and flash codes. If power is on, the 1MHz oscillator operates directly and generates 32kHz clock source using divider. This clock connects alive control logic and generates ECG and ADC clock source. The 87MHz clock source is controlled by MCU. SAR ADC clocks generated from 1MHz Oscillator. Clock generator presents clocks for AFE(Analog Front End) Circuits including, Reset clock(64Hz), Chopping clocks(2kHz), Chopping Spike Filter clocks(4kHz), and AC current source clocks(8kHz).

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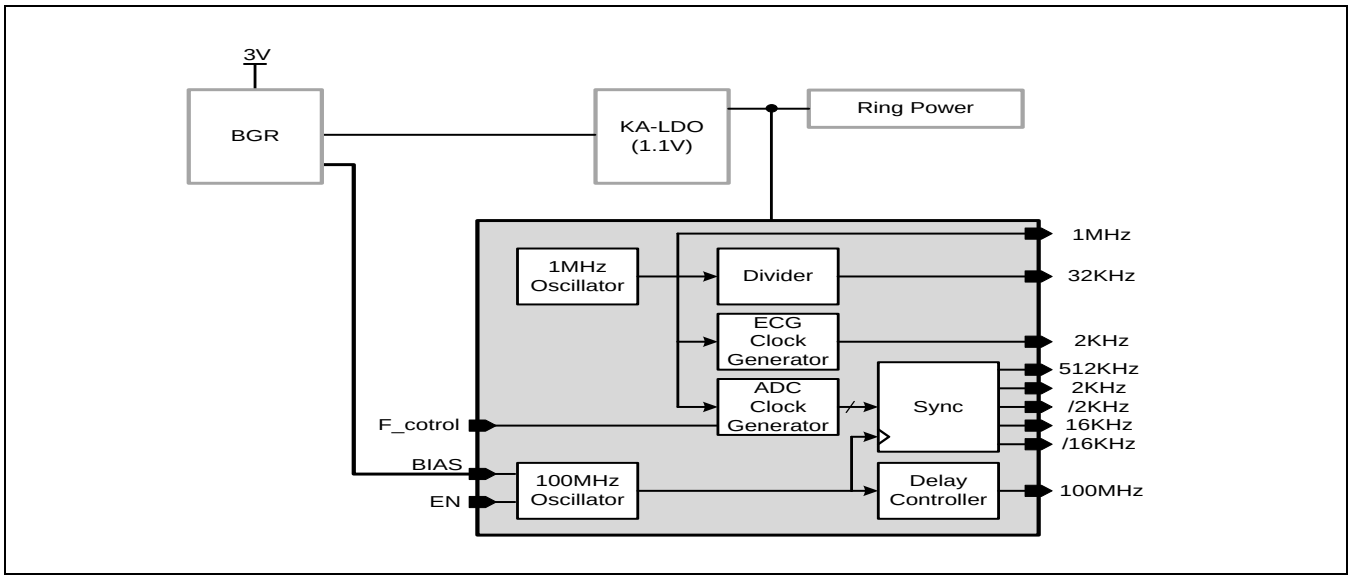


Figure 5-19 Clock Generator Outputs

5.4 Register Description

5.4.1 Register Map Summary

- Base Address: 0x4001A000

Register	Offset	Description	Reset Value
AFE_EN1	0x0000	AFE_EN1	0x0000_0003
AFE_EN2	0x0004	AFE_EN2	0x0000_0005
ADC_EN	0x001C	ADC_EN	0x0000_001F
APB_RESET	0x005C	APB_RESET	0x0000_000F
APB_CLK	0x0060	APB_CLK	0x0000_000F
APB_CH_DATA1	0x0064	APB_CH_DATA1	0x0000_0020
APB_CH_DATA2	0x0068	APB_CH_DATA2	0x0000_0064
APB_CH_DATA3	0x006C	APB_CH_DATA3	0x0000_0008
APB_DECM1	0x0070	APB_DECM1	0x0000_0000
APB_DECM2	0x0074	APB_DECM2	0x0000_0000
APB_DECM3	0x0078	APB_DECM3	0x0000_0000
GPADC_CTRL1	0x007C	GPADC_CTRL1	0x0000_0004
GPADC_CTRL2	0x0080	GPADC_CTRL2	0x0000_0000

5.4.1.1 AFE_EN1

- Address: Base Address + 0x0000, Reset Value = 0x0000_0003

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
Reserved0	[7:6]	RW	Reserved bits	0x0000_0000
EN_PPG	[5]	RW	0: PPG RX disable 1: PPG RX enable	0x0000_0000
EN_LED_DRV	[4]	RW	0: LED Driver disable 1: LED Driver enable	0x0000_0000
EN_BIA_CS	[3]	RW	0: BIA TX disable 1: BIA TX enable	0x0000_0000
EN_BIA_READ	[2]	RW	0: BIA RX disable 1: BIA RX enable	0x0000_0000
EN_IMP	[1]	RW	0: ECG Impedance Channel disable 1: ECG Impedance Channel enable	0x0000_0001
EN_ECG	[0]	RW	0: ECG disable 1: ECG enable	0x0000_0001

5.4.1.2 AFE_EN2

- Address: Base Address + 0x0004, Reset Value = 0x0000_0005

Name	Bit	Type	Description	Reset Value
Reserved0	[7:4]	RW	Reserved bits	0x0000_0000
EN_SDADC	[3]	RW	0: SDADC disable 1: SDADC enable	0x0000_0000
EN_REF	[2]	RW	0: Voltage and Current Reference disable 1: Voltage and Current Reference enable	0x0000_0001
EN_DAC	[1]	RW	0: DAC for ECG Channel disable 1: DAC for ECG Channel enable	0x0000_0000
EN_ADC	[0]	RW	0: SAR-ADC disable 1: SAR-ADC enable	0x0000_0001

5.4.1.3 ADC_EN

- Address: Base Address + 0x001C, Reset Value = 0x0000_001F

Name	Bit	Type	Description	Reset Value
Reserved0	[7:5]	RW	Reserved bits	0x0000_0000
EN_ADC_C	[4:0]	RW	* ADC enable 0: ADC1~ADC5 disable 1: ADC1 enable 2: ADC2 enable 4: ADC3 enable 8: ADC4 enable 16: ADC5 enable	0x0000_001F

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5.4.1.4 APB_RESET

- Address: Base Address + 0x005C, Reset Value = 0x0000_000F

Name	Bit	Type	Description	Reset Value
FIFO_RESETB	[3]	RW	* RAFE FIFOs Software Reset 1: Assert reset 0: Release reset	0x0000_0001
BIA_RESETB	[2]	RW	* BIA Logic Software Reset 1: Assert reset 0: Release reset	0x0000_0001
PPG_RESETB	[1]	RW	* PPG Logic Software Reset 1: Assert reset 0: Release reset	0x0000_0001
ECG_RESETB	[0]	RW	* ECG Logic Software Reset 1: Assert reset 0: Release reset	0x0000_0001

5.4.1.5 APB_CLK

- Address: Base Address + 0x0060, Reset Value = 0x0000_000F

Name	Bit	Type	Description	Reset Value
FIFO_CK_EN	[3]	RW	* RAFE FIFOs Clock Gating Control 1: Clock Enable 0: Clock Disable	0x0000_0001
BIA_CK_EN	[2]	RW	* BIA Logic Clock Gating Control 1: Clock Enable 0: Clock Disable	0x0000_0001
PPG_CK_EN	[1]	RW	* PPG Logic Clock Gating Control 1: Clock Enable 0: Clock Disable	0x0000_0001
ECG_CK_EN	[0]	RW	* ECG Logic Clock Gating Control 1: Clock Enable 0: Clock Disable	0x0000_0001

5.4.1.6 APB_CH_DATA1

- Address: Base Address + 0x0064, Reset Value = 0x0000_0020

Name	Bit	Type	Description	Reset Value
CH1_DATA_SEL	[7:4]	RW	* FIFO1 Channel Data Selection 0: ADC0 Channel 0 1: ADC0 Channel 1 2: ADC1 Channel 0 3: ADC1 Channel 1 4: ADC2 Channel 0 5: ADC2 Channel 1 6: ADC3 Channel 0 7: ADC3 Channel 1 8: ADC4 Channel 0	0x0000_0002

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			9: ADC4 Channel 1 10~15: N/A	
CH0_DATA_SEL	[3:0]	RW	* FIFO0 Channel Data Selection 0: ADC0 Channel 0 1: ADC0 Channel 1 2: ADC1 Channel 0 3: ADC1 Channel 1 4: ADC2 Channel 0 5: ADC2 Channel 1 6: ADC3 Channel 0 7: ADC3 Channel 1 8: ADC4 Channel 0 9: ADC4 Channel 1 10~15: N/A	0x0000_0000

5.4.1.7 APB_CH_DATA2

- Address: Base Address + 0x0068, Reset Value = 0x0000_0064

Name	Bit	Type	Description	Reset Value
CH3_DATA_SEL	[7:4]	RW	* FIFO3 Channel Data Selection 0: ADC0 Channel 0 1: ADC0 Channel 1 2: ADC1 Channel 0 3: ADC1 Channel 1 4: ADC2 Channel 0 5: ADC2 Channel 1 6: ADC3 Channel 0 7: ADC3 Channel 1 8: ADC4 Channel 0 9: ADC4 Channel 1 10~15: N/A	0x0000_0006
CH2_DATA_SEL	[3:0]	RW	* FIFO2 Channel Data Selection 0: ADC0 Channel 0 1: ADC0 Channel 1 2: ADC1 Channel 0 3: ADC1 Channel 1 4: ADC2 Channel 0 5: ADC2 Channel 1 6: ADC3 Channel 0 7: ADC3 Channel 1 8: ADC4 Channel 0 9: ADC4 Channel 1 10~15: N/A	0x0000_0004

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5.4.1.8 APB_CH_DATA3

- Address: Base Address + 0x006C, Reset Value = 0x0000_0008

Name	Bit	Type	Description	Reset Value
CH4_DATA_SEL	[3:0]	RW	* FIFO4 Channel Data Selection 0: ADC0 Channel 0 1: ADC0 Channel 1 2: ADC1 Channel 0 3: ADC1 Channel 1 4: ADC2 Channel 0 5: ADC2 Channel 1 6: ADC3 Channel 0 7: ADC3 Channel 1 8: ADC4 Channel 0 9: ADC4 Channel 1 10~15: N/A	0x0000_0008

5.4.1.9 APB_DECM1

- Address: Base Address + 0x0070, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
CH3_DECM	[7:6]	RW	* FIFO3 Channel Decimation Rate Selection 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000
CH2_DECM	[5:4]	RW	* FIFO2 Channel Decimation Rate Selection 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000
CH1_DECM	[3:2]	RW	* FIFO1 Channel Decimation Rate Selection 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000
CH0_DECM	[1:0]	RW	* FIFO0 Channel Decimation Rate Selection 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000

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5.4.1.10 APB_DECM2

- Address: Base Address + 0x0074, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value
PPG_IR_RED_DECM	[5:4]	RW	* PPG IR RED data Decimation Rate Selection (Input data to PPG Logic) 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000
ECG_IMP_DECM	[3:2]	RW	* ECG IMP Channel Decimation Rate Selection (Input data to ECG Logic) 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000
CH4_DECM	[1:0]	RW	* FIFO4 Channel Decimation Rate Selection 0: No decimation (1KHz sampled data) 1: 1/2 decimation (512KHz Sampled data) 2: 1/4 decimation (256KHz Sampled data) 3: N/A	0x0000_0000

5.4.1.11 GPADC_CTRL1

- Address: Base Address + 0x007C, Reset Value = 0x0000_0004

Name	Bit	Type	Description	Reset Value
SDADC_START	[4]	RW	* A/D conversion is performed only when 'SDADC_START' goes high.	0x0000_0000
SDADC_DFILTER	[3:2]	RW	* Data rate selection for digital filter 0: 1KHz sampled data 1: 500Hz sampled data 2: 250Hz sampled data 3: 125Hz sampled data	0x0000_0001
SDADC_SEL_INPUT	[1:0]	RW	* Input selection of GPADC 0: Differential input with SD_INP and SD_INN 1: Single-ended input with SD_IN_S1 2: Single-ended input with SD_IN_S2 3: Internal input to sense the die temperature	0x0000_0000

5.4.1.12 GPADC_CTRL2

- Address: Base Address + 0x0080, Reset Value = 0x0000_0000

Name	Bit	Type	Description	Reset Value

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Name	Bit	Type	Description	Reset Value
SDADC_GAIN_PGA	[6:4]	RW	* PGA gain setting 0~7: 1, 2, 3, 4, 6, 7, 11 & 16	0x0000_0000
RESERVED	[3]	RW	RESERVED	0x0000_0000
SDADC_SEL_VCM	[2:0]	RW	* Opposite input voltage at single ended mode 0: 4/32•VREF 1: 7/32•VREF ... 6: 22/32•VREF 7: 25/32•VREF	0x0000_0000

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.

6 Application Circuits

Figure 6-1 shows the example case of application circuit using S3FBP5A

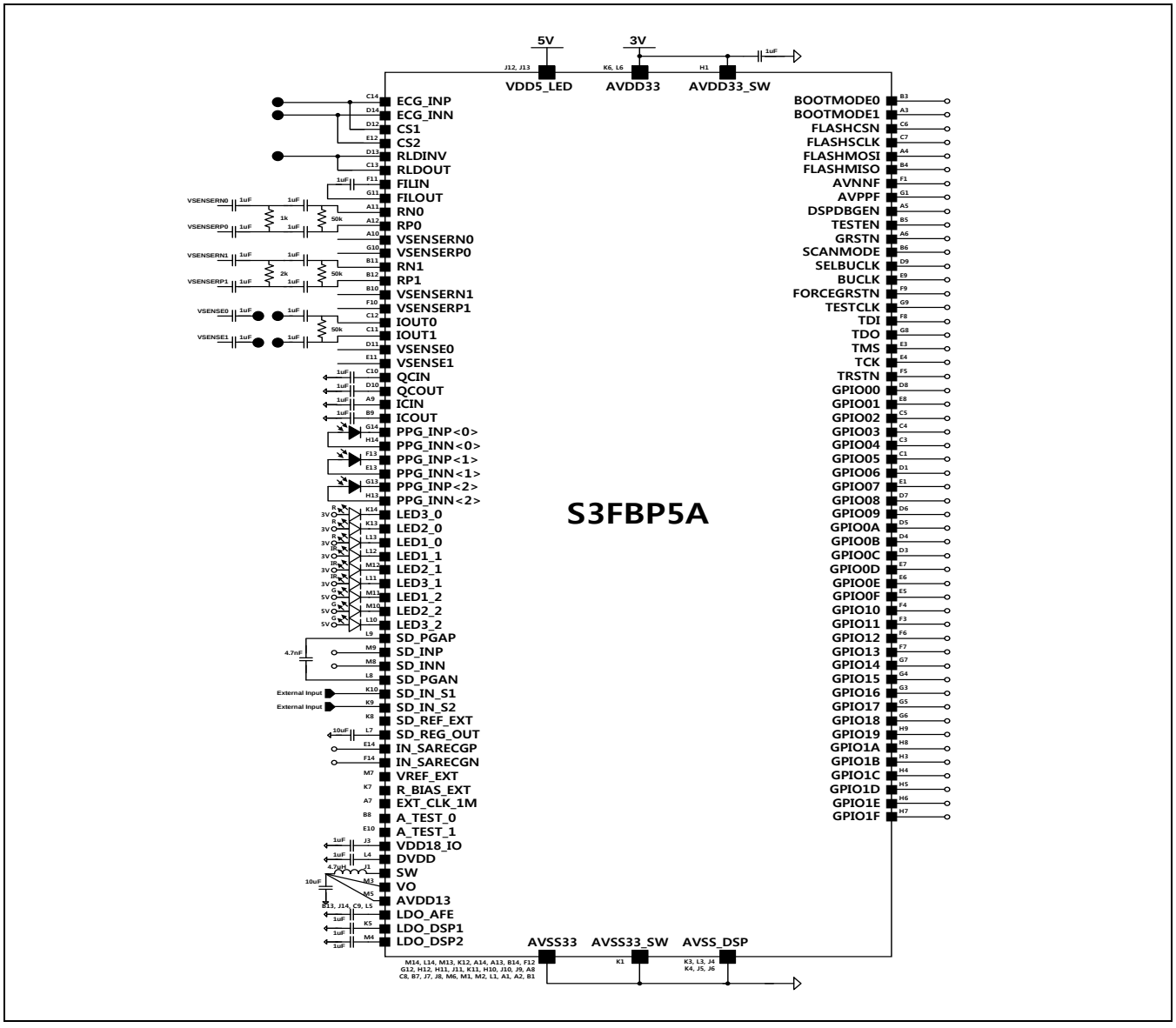
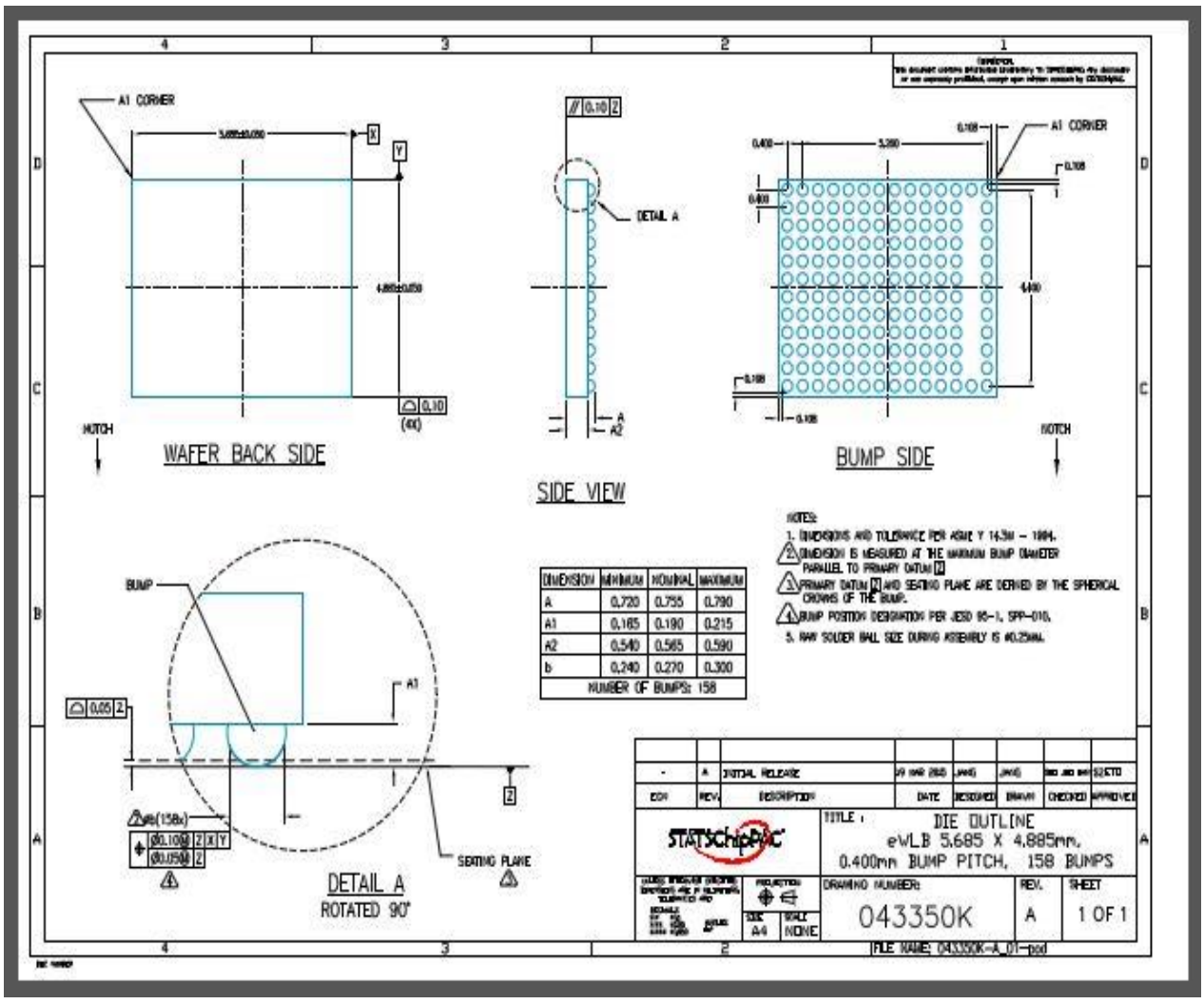


Figure 6-1 Example Case of Application Circuit using S3FBP5A

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7 Mechanical Data



DIMENSION	MINIMUM	NOMINAL	MAXIMUM
A	0.720	0.755	0.790
A1	0.165	0.190	0.215
A2	0.540	0.565	0.590
b	0.240	0.270	0.300
NUMBER OF BUMPS: 158			

Figure 7-1 158-FOWFP-5.68 ×4.88 × 0.75mm PACKAGE OUTLINE

Preliminary product information describes products that are in development, for which full characterization data and associated errata are not yet available. Specifications and information herein are subject to change without notice.